
Owner's Technical Specification

Part B

Rumah and Nairyah IPPs

Revision 6

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SCHEDULE H	TURN OVER PROCEDURE AND TOP LIST
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SCHEDULE R	FRP PROJECT SPECIFICATION MANAGEMENT
SCHEDULE X	QUALITY ASSURANCE AND QUALITY CONTROL SPECIFICATIONS

1. INTRODUCTION

1.1 Purpose and Structure of the Document

This document, the Owner's Technical Specification Part B (OTS Part B) defines requirements for the engineering, procurement and construction of each of Rumah and Nairyah IPPs that the Contractor shall comply with, which are in addition to those specified in OTS Part A.

In addition to the technical requirements of this section, the selection of the technology, the design and the arrangement of the Power Units shall be based on the site conditions as described in OTS Part A.

The Plant shall be designed, manufactured and configured in such a way as to optimise Plant performance, thus providing a competitive least cost electricity generation taking into account the design and operation requirements of this OTS, Part A and Part B.

The Plant shall be designed, manufactured, installed and be suitable in every respect for continuous operation at the Guaranteed Net Contracted Power Capacity $1,800 \text{ MW} \pm 5\%$ (GNCPC) at part loads, under the anticipated transient operation as well as the climatic conditions and the site environmental restrictions and shall exhibit high availability and reliability. Please refer to section 6.4 for the Availability Reliability and Maintainability statement. In order to demonstrate the availability and reliability, the Contractor shall perform a RAM study.

The Plant shall be designed to permit unconstrained operation over the full range of ambient conditions. The Plant shall have operationally flexible power capacity that will play a key role in providing flexible power generation.

All the sections below (this document- Part B) define the Owner's requirements for the configuration of the Plant in addition to that specified in OTS Part A. OTS Part A shall generally have the priority over OTS Part B. In case of any requirement specified in OTS Part B found to be in contradiction with the requirements stated in OTS Part A, the conservative and stringent requirement of them will prevail in addition to the compliance of the requirement stated in Part A.

2. PROJECT DESCRIPTION

Saudi Power Procurement Company ("SPPC", the principal buyer) is developing a new power plant on a build, own and operate ("BOO") basis.

The Rumah and Nairyah IPPs shall comprise all facilities necessary for the generation of power as specified in this OTS Part B and OTS Part A.

2.1 Not Used

2.2 Location and Layout

The location of the allocated area and the Site can be seen in the overall layout given in Annex 1.2-1 and Annex 1.2-2 of Part III (Drawings and Diagrams) of the Part A.

2.3 Scope of work

The scope of works for the Project shall cover the design, engineering, procurement, manufacturing, factory testing, transport to site, erection, construction, commissioning and performance testing, including site investigation, site development and all related civil works, management and supervision, insurance and bonds during construction.

New gas fired CCGT power plant of total capacity of $1800 \pm 5\%$ MW; together with connection of the power plant to the grid substation, associated infrastructure and facilities and ensuring plant's Decarbonization Readiness.

Only reserved space for future flue gas Duct and HRSG stack flange connection is required for Decarbonization.

The Contractor is free to design the Plant as they deem appropriate as long as design conditions and technical/performance requirements specified in the OTS Part A and OTS Part B are complied with.

The Contractor shall include in their scope, all the equipment, works and services necessary for complete, safe and reliable operation and maintenance of the Plant, even if certain essential works are not expressly stated in this OTS Part B and OTS Part A.

2.3.1 Preparatory works

Preparatory works, such as, the following shall be included:

- Data collection regarding interface points
- Site preparation, e.g. clearing, dewatering, backfilling with compaction/consolidation, grading, levelling, earth moving, soil improvement, etc. including access areas
- Demolishing, rerouting, strengthening and removal of existing vegetation, infrastructures, buildings, roads and structures, as applicable.
- Site drainage and sewage drainage systems
- Liaison with third parties that have access to Site.
- Removing, strengthening and rerouting of existing canals, roads etc.
- Wadi diversion if required.

Please refer to section 7 of this document and OTS Part A for further details.

2.3.2 Studies and Surveys

Contractor shall perform the site surveys and studies, which shall include, but not limited to the following:

- Site survey.
- Topographical survey.
- Investigation subsoil conditions and study of geological formation and seismic conditions of the area.

- Geotechnical and Geophysical Survey work;
- Hydrology and Flood risks;
- Meteorological studies;
- Hydraulic calculations - for pipelines steady state and ponds (if applicable);
- Logistic Survey for construction and O&M stage
- Any other survey and studies required for project completion.

Also refer to relevant sections of this document and OTS Part A for further details.

2.3.3 Infrastructure Works

- Construction and/or improvement of temporary and permanent access roads to site as well as new access roads on-site;
- Construction of bridges, culverts, diversions as required to cross the existing road for routing Intake and discharge conveyor (if applicable);
- Robust pre-treatment system,
- Temporary and permanent parking areas
- Temporary and permanent roads and lighting
- Temporary and permanent site fencing, including gates
- Environment monitoring boreholes as required during Construction and operation stage
- Removal of existing temporary site offices, if applicable
- Removal and relocation of existing facilities like roads, cables, O/H lines and pipes crossing the Site, etc. including replacement materials (if required)
- Temporary Offices during construction for Owner / Owner's Representative fully furnished and serviced with a telephone system, internet access and cleaning
- Construction of permanent boundary walls and/or fence and internal fencing and Gates all including lighting
- Flood mitigation work (if applicable);
- Modifications of existing infrastructure, as required.
- Scour protection if required;
- Permanent furnished office for SPPC and Permanent accommodation compound for O&M staffs are out of EPC scope;

Please refer to section 7 of this document for further details.

2.3.4 Construction Facilities

- Temporary fully furnished offices for Owner and their representatives, as described in the Temporary Site Facilities Specification, Contractor, lay-down areas, warehouses, workshops for site construction and pre-fabrication purposes, vehicles, mobile equipment, provision of ambulance, temporary power generators, temporary air compressors, as well as supply of fuel, temporary firefighting and alarm detection system, temporary first aid facilities, temporary housing, as far as applicable.
- Site services as required for the construction and commissioning, such as supplies of electric power (incl. lighting), potable water, construction water, water for commissioning, telecommunication, first aid facilities, security system, etc.
- Demolition and removal of temporary construction facilities after Commercial Operation Date including restoration of surface areas.

- Design, construction and maintenance of the EPC's own living area, offices and construction temporary facilities shall follow EPC contractor's own need.

Please refer Section 3 of this document for further details.

2.3.5 Miscellaneous Project Requirements

The following services/facilities which are general for the Plant and not related to specific equipment/facilities shall form part of the Contractor's scope.

- Engineering
- Documentation
- Restoring damages to existing property
- Painting and corrosion protection
- HAZOP and HAZID studies.
- Suitable preservation and protection system for prolonged shutdown of the Plant as per OEM recommendation
- Safety Integrity Level study (SIL)
- Master Equipment list preparation as per O&M Contractor CMMS template requirement
- Taking approval as required from the other stake holders & 3rd parties as required.

2.4 Site & Meteorological Conditions

The minimum design requirements for the Plant are briefly described below . The following description is, however, subject to verification by the Contractor based on their own investigations as to the most severe climatic conditions which should be used as a basis for the final design of the Plant. However the Contractor shall not apply design conditions less onerous than those described below.

Climatic Conditions for Nariyah IPPs

For Reference Site Conditions refer to OTS Part A, Part II of the RFP, Nariyah Technical Specifications, Section 4.1.4.1, Table 4.1-2 Reference Site Conditions (RSC).

For Design Climatic Conditions, Rainfall, Design Wind Speed and Direction, and Dust refer to OTS Part A, Part II of the RFP, Nariyah Technical Specifications, Section 4.1.4, Table 4.1-1 Design Climatic Conditions.

Site climate conditions provided for Nariyah site as indicated in Saudi Aramco Engineering Standard SAES-A-112 Meteorological and Seismic Design Data for Riyadh have been Used as the nearest available to the site. Seismic design shall be in accordance with the latest version of the Saudi Building Code (SBC 301).

Climatic Conditions for Rumah IPPs

For Reference Site Conditions refer to OTS Part A, Part II of the RFP, Rumah Technical Specifications, Section 4.1.4.1, Table 4.1-2 Reference Site Conditions (RSC).

For Design Climatic Conditions, Rainfall, Design Wind Speed and Direction, and Dust refer to OTS Part A, Part II of the RFP, Rumah Technical Specifications, Section 4.1.4, Table 4.1-1 Design Climatic Conditions.

Site climate conditions provided for Rumah site as indicated in Saudi Aramco Engineering Standard SAES-A-112 Meteorological and Seismic Design Data for Riyadh have been used as the nearest available to the Site. Seismic design shall be in accordance with latest revision of Saudi Building Code (SBC 301).

The given values for rainfall shall be verified with local authorities/ data and related codes for 1 in 100 years of rainfall in mm/hr and the same shall be considered for design and construction.

All correction curves relating to climatic conditions used in the Project should, as a minimum, have ranges encompassing the climatic conditions described in this Section.

2.5 Codes and Standards

The Contractor shall ensure that the engineering, design, construction, testing, etc. of all equipment, facilities, components and systems which they propose shall form part of the Project, are in accordance with internationally recognised standards and codes in their latest edition applicable at the start of each relevant work stream. EPC shall ensure that the standards and codes shall be used in accordance with the Section 4.1.14 of Part II (Technical Specification) of the RFP (OTS Part A).

All Design, construction work shall be carried out as per ESIA requirement.

All Civil, structural and architectural work shall be carried out in line with local codes, standards and regulations.

The following Standards are mandatory and must be applied:

- Local Laws and Regulations of Saudi Arabia
- Royal Commission regulations, codes and standards
- Saudi Arabian Government High Commission for Industrial Security (HCIS), Security Directives (SEC) (including No. 1 to 19) and Safety and Fire Protection Directives (SAF) (including No. 1 to 25). For purposes of these directives, the Rumah and Nairyah are considered “vital” facilities.
- Security Directives
 - SEC-01 General Requirements for Security Directives
 - SEC-02 Security Fencing
 - SEC-03 Security Gate
 - SEC-04 Security Lighting
 - SEC-05 Security Systems at Industrial Facilities
 - SEC-06 Security Devices
 - SEC-07 Power Supply
 - SEC-08 Security Communications & Data Networks
 - SEC-09 Structures Housing Security Equipment
 - SEC-10 Railway Operations in Industrial Facilities
 - SEC-11 Personal Identification Verification Cards
 - SEC-12 Cybersecurity¹⁾
 - SEC-13 Facilities with Marine Interface
 - SEC-14 Security Project Management
 - SEC-15 Security Management at Industrial Facilities
 - SEC-16 Pipeline & Pipeline Corridors
 - SEC-17 Electrical Power Substations
 - SEC-18 Security for Industrial Cities
 - SEC-19 Residential Compounds for Industrial Facilities
- Safety and Fire Protection Directives
 - SAF-01 General Requirements for Safety & Fire Protection Directives
 - SAF-02 Environmental, Health and Safety (EHS) Management
 - SAF-03 Buildings for Industrial Facilities

- SAF-04 Fire Protection Systems and Equipment
- SAF-05 Industrial Drainage
- SAF-06 Plant Layout, Spacing, and Access
- SAF-07 Onshore and Nearshore Wellsite Safety
- SAF-08 Storage Tanks and Storage Vessels
- SAF-09 Fireproofing in Plants
- SAF-10 Pressure Piping, Pressure Vessels, and Transportation Pipelines
- SAF-11 Emergency Shutdown, Isolation, and Depressuring
- SAF-12 Electrical Safety
- SAF-13 Work Permits, Isolation of Energy, and Hazardous Work
- SAF-14 Manufacture, Transportation, Storage, and Use of Explosive Materials
- SAF-15 Private Industrial Fleet Vehicle Filling Stations
- SAF-16 Bulk Plants and Air Fueling Support Facilities
- SAF-17 Offshore Production Facilities
- SAF-18 Mines and Mineral Processing Plants
- SAF-19 Power Plants and Associated Facilities
- SAF-20 Pre-Incident Planning and Management of Emergencies
- SAF-21 Personal Protective Equipment
- SAF-22 Portable Electric and Electronic Devices
- SAF-23 Flammable/Combustible Liquids in Portable Containers
- SAF-24 Hazardous Materials in Portable Containers
- SAF-25 Incident Reporting and Investigations
- National Center for Environmental Compliance (NCEC) Standards, as a minimum,
- Meteorological and Environmental Protection Administration (MEPA) as a minimum,
- The Technical Requirements of Saudi Electricity Company
- The Saudi Arabian Grid Code Requirements, Electricity & Cogeneration Regulatory Authority (ECRA) Regulations,
- The Saudi Arabian Building Code
- Essential Cybersecurity Controls (ECC-1:2018) by National Cybersecurity Authority
- SEC Codes and Standards for Electrical and I&C equipment inside Rumah and Nairyah 380 kV GIS Substation
- For SAF 19, supply piping inside drain and guard piping is not considered based on GT OEM standard design. The lubrication, control and seal oil system including piping are designed based on the GT OEM Standard.
- For SAF 19, stationary combustion engines and gas turbines are designed based on OEM standard, i.e. NFPA-37 is used as reference only.

Internationally Recognized Standards and Codes:

AASHTO	American Association of State Highway and Transportation Officials
ACI	American Concrete Institute
AGMA	American Gear Manufacturers Association
AIJ	Architectural Institute of Japan
AISC	American Institute of Steel Construction
AISE	Association of Iron and Steel Engineers
AISI	American Iron and Steel Institute
AMCA	Air Movement and Control Association
ANSI	American National Standards Institute
API	American Petroleum Institute
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigeration and Air Conditioning Engineers ASME American Society of Mechanical Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing Materials
AWS	American Welding Society
AWWA	American Water Works Association
BSI	British Standards Institution
CIRIA	Construction Industry Research and Information Association
DIN	Deutsches Institut fuer Normung
HCIS	High Commission of Industrial Security
HEI	Heat Exchange Institute
HIS	Hydraulic Institute Standards
IBC	International Building Code
ICE	Institution of Civil Engineer
ICEA	Insulated Cable Engineers Association
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IPCEA	Insulated Power Cable Engineers Association
ISA	International Society of Automation
ISO	International Standardisation Organisation
JEC	Japanese Electrotechnical Committee
JEMA	The Japan Electrical Manufacturers Association
JIS	Japanese Industrial Standards
NACE	National Association of Corrosion Engineers
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
SSPC	Society for Protective Coatings

TEMA	Tubular Exchanger Manufacturers Association
VDE	Verband Deutscher Elektrotechniker (German Society of Electrical Engineers)
VDI	Verband Deutscher Ingenieure (German Society of Engineers)
VGB	Technische Vereinigung der Grosskraftwerksbetreiber E.V. (Society of big power plants operator)

Approval will be required for applying other equivalent standards. The Contractor proposing such shall provide rationale and evidence of equivalence to support their case for approval.

If in case any of the requirement specified in local codes and standards found to be in contradiction with the requirements stated in the International codes and standards the most conservative and stringent requirement of them will prevail.

SBC code shall be followed for Wind and seismic load calculations.

The latest Codes and Standards edition, including Saudi Arabian Grid code, at the moment of Bid date (15 Aug 2024) is applied.

2.6 Plant Interfaces

Refer to section 2 of OTS Part A.

3. PROJECT MANAGEMENT REQUIREMENTS

The scope of works covered by the EPC Contractor in terms of Project Management Strategy shall include, but not be limited to the general activities exposed hereby. For more detailed information, please refer to Appendix W of the EPC Contract in addition to the requirements in below sections.

If there is any conflict, between the OTS requirements for the above-mentioned plans and Appendix W of the EPC Contract, Appendix W of the EPC Contract requirements shall prevail.

The project management procedures outline provided by the EPC contractor is provided in Appendix W of the EPC Contract. However, the procedures shall be developed to meet the Owner Project Management requirements in this section and Appendix W of the EPC Contract of this document.

3.1 Project Management

3.1.1 Project Management and Coordination Procedures

Contractor shall be responsible for the management and control of all phases of its work and that of its sub-contractors and suppliers.

Contractor shall, within thirty (30) days after the (Limited) Notice to Proceed Date, submit to the Owner for approval a comprehensive set of project management and co-ordination procedures, which describe both the Contractor's project management strategy covering the different phases of the project, the project organisation including partners and major subcontractors as well as all project management related procedures, including co-ordination and communication with the Owner. Based on and in accordance with the requirements stated in Appendix W of the EPC Contract of OTS and EPC Contract, these procedures shall cover the items detailed below. The project management plan shall reach at least "approved with comments" status prior commencement of mobilization to site.

For further details regarding requirement towards the EPC Contractor's Project Management Plan and its content please refer to the Appendix W of the EPC Contract.

EPC Contractor's project management organization including sub-contractors (the status and responsibilities of staff shall be included) integral part of the Project Management plan.

3.1.2 Project Control System

The Scope of Project Control Services to be performed by Contractor shall include, but not be limited to, functions as described under the following major headings:

- Project Control Procedures and Documentation
- Project Planning and Scheduling
- Project Progress and Performance Measurement
- Progress Reporting
- Project Schedule Risk Analysis

Please refer to the Appendix W of the EPC Contract for further details regarding the Owner's requirements towards EPC Contractor's Project Control System.

3.1.2.1 Project Planning and Scheduling

The Contract Programme shall be the overall schedule of key dates agreed prior to award of the EPC Contract and included in the EPC Contract. The schedule shall not be altered except as otherwise agreed by Owner and shall be broken down into a detailed network of activities, clearly showing the following phases:

Project milestones (contractual milestones, critical milestones, Interface and Owner's milestones, payment milestones)

- Detailed Design/Engineering Deliverable Schedule
- Procurement Management Schedule
- Manufacturing and Equipment/Materials Delivery Schedule
- Construction Schedule (including Commissioning detailed schedule level 4 (to be delivered 2 months before first commissioning test start)
- Weekly and daily commissioning schedule to be delivered since start date of commissioning test
- access to the site for preliminary and construction activities; availability of services (water, distillate oil, electricity); key sequences involving other parties on terminal points; completion of all work relating to terminal points; start of testing and commissioning; tests on completion; reliability run; performance tests; takeover of the power unit).

Prior to the EPC Contract signature, it shall be issued and agreed a Level 2 "Overall Project Master Schedule" based on the previously indicated.

After the EPC Contract signature and within the first 30 days after NTP date, it shall be issued for approval the Level 3 "Detailed Project Master Schedule". This Schedule shall be developed using Critical Path Method (CPM) network on Primavera P6 Software and shall include the detailed activities to be executed as part of the Engineering, Procurement, Construction and Commissioning phases. Level 3 schedule shall comply with the dates contained on the Level 2 Schedule and always fulfil with the Contractual dates.

Level 3 Schedule will be considered, once approved by Owner, as the Planning Driving Document to comply with. Once the Owner has approved it, it will be frozen as the Project Baseline.

Please refer to the Appendix W of the EPC Contract for further details regarding the Owner's requirements towards EPC Contractor's Planning and Scheduling duties.

3.1.2.2 Project Progress and Performance Measurement

Contractor shall develop and maintain a formalised progress measurement system based on the Project Baseline Schedule and the weighting and progress rules indicated in the Appendix W of the EPC Contract.

The purpose of this Procedure is to provide a consistent method for measuring and reporting the physical progress and thereby providing a basis for checking and analysing progress, productivity and forecasting.

It shall be Contractor's responsibility to measure his own progress in line with and at the levels of detail defined within this Specification.

3.1.2.3 Progress Reviewing and Reporting

During the execution of the Contract details of progress of the work shall be reported and presented in accordance with Owner's formats included in the Appendix W of the EPC Contract of this Specification. The purpose of this activity shall be enabling the review and status of the actual status of the Project.

The Contractor shall prepare and submit for approval a Monthly Progress Report following the index included in the Appendix W of the EPC Contract of this document. In addition, during Construction and Commissioning phase a Weekly Progress Report, it shall be issued.

A Project Final Close-out Report shall be prepared and submitted for approval upon completion of the Work.

Please refer to the Appendix W of the EPC Contract for further details regarding the Owner's requirements towards EPC Contractor's Progress Review and reporting.

3.1.2.4 Project Schedule Risk Analysis

Contractor shall conduct a Schedule Risk Analysis on a quarterly basis using a computerized system to predict probability of Project completion as per target. The analysis and the supporting data shall be presented to the Owner.

Please refer to the Appendix W of the EPC Contract for further details regarding the Owner's requirements towards EPC Contractor's Project Schedule Risk Analysis.

3.1.3 Project Communication Management System

Starting from the (Limited) Notice to Proceed up to the end of the Warranty Period all parties to the Project (in particular, EPC Contractor, Owner, Owner's Representatives) shall consequently use a Project Communication & Document Management System (PC&DMS): "Think project".

This system shall, in general, manage the following tasks:

- Archiving and distribution of all Project-related documents (engineering and non-engineering)
- Collaboration between parties to the Project and the related workflows, e.g., during design review (e.g., uploading, reviewing, commenting, approving, revising of documents)
- Project correspondence (by e-mail), incl. connection to Outlook (it shall be made sure that all Project related correspondence is stored at one central place: PC&DMS). This connection should allow to bring over preliminary correspondence pieces exchanged prior to the start of the PC&DMS.

The PC&DMS shall be implemented (set-up) in such a way that the following Project basic requirements are satisfied:

- Project-specific documents attributing (numbering)
- Project-specific correspondence attributing (numbering)
- Project-specific correspondence distribution matrix
- Project-specific security (access control).

Please refer to Appendix W of the EPC Contract for detailed requirements are regarding PC&DMS. The PC&DMS shall be ready for use at the date of Contract signing or Limited Notice to Proceed, whichever occurs first).

Contractor shall be responsible for consequent and obligatory use of the PC&DMS, incl. proper attributing and upload of documents and correspondence, training all users as well as any maintenance activities, if required. All costs related to the use of the PC&DMS, including but not limited to the software license, hardware, costs for system administration, training, user support, shall be borne by Contractor.

There will be Owner's PC&DMS' Administrator overlooking all activities related to the use of PC&DMS who will be in close contact with the Contractor's Document Controller.

Contractor shall provide an uninterrupted dedicated high speed Internet access for every PC&DMS user group (e.g., in particular document controllers, design review team, Project Management team) allowing PC&DMS to manage even an extreme workload without delay. Only in case of sudden technical problems (e.g., sudden interruption in internet service) and urgent cases (e.g., urgent distribution of documents) the Contractor may use other communication means. After the normal situation has been restored, all parties shall continue using the PC&DMS and parties are responsible to retrospectively upload all documents into the PC&DMS which were transmitted offline.

Owner reserves the right to use the PC&DMS also for managing confidential documents without access rights to any other party. All information/data stored in the PC&DMS are property of the Owner. At the end of the project,

Contractor shall provide the Owner with two sets of offline archives (read-only USB archive with full search, tracking and reporting capability).

Please refer to the Appendix W of the EPC Contract for further details regarding the Owner's requirements towards EPC Contractor's Project Communication and Document Management System.

3.1.3.1 Communications

All correspondence, communications and submissions relating to the EPC Contract shall be in English.

The Contractor shall have a Site Manager fluent in the English Language for routine administration with the Owner/Owner's Representative at the site.

For all documents, goods, materials, maintenance equipment, spare parts, instruments, etc. to be handed over to the Owner, the Contractor shall obtain a signed receipt by an authorised representative of the Owner that the goods, etc., have been received in a satisfactory condition as required under the EPC Contract.

For further details related to requirements regarding Key Personnel and Contractors Personnel, please refer to Appendix P of EPC Contract.

3.1.3.2 Diaries for Daily Work Measurements

The Contractor shall keep suitable diaries in an approved form to which at all times the Owner/Owner's Representative shall have access. The Contractor shall record all work progress as completed on the day on which the records are made, together with full particulars and details of all obstructions or modified or extra works or incidents, and the number of men employed in each of the several positions of the work in progress.

The diaries shall be available at all times for Owner/Owner's Representative to verify and record, and he shall sign all entries therein which he verifies, and have full liberty himself to make entries therein but no such signature or entry shall imply acceptance of our agreement with any area of work material used. The diaries shall, when signed by both parties, be deemed to be a true record of the work executed and shall become the property of Owner.

Please refer to the Appendix W of the EPC Contract for further details regarding the Owner's requirements towards EPC Contractor's reporting.

3.2 Site Management

3.2.1 General

The Contractor's representative on the site or his nominated deputy shall be given full responsibility to enter into negotiations regarding points arising out of erection and setting to work so that the work may be expedited with as few delays as possible.

Until the whole of the Work has been taken over by the Owner, the Contractor shall be entirely responsible (except as is provided by the EPC Contract) for the Work.

On completion of the work the Site shall be left clean and tidy to the satisfaction of Owner. Any damage done by the Contractor or its sub-contractors shall be made good at the Contractor's expense. The Contractor shall landscape the Site where required.

The carrying out of all Work included in the EPC Contract is to be supervised by a sufficient number of qualified representatives of the Contractor and full facilities and assistance are to be afforded by the Contractor for the Owner and the Owner's Representative to check the Work. The Contractor shall be required to carry out dimensional checks at various stages of completion as mutually agreed and these will be witnessed by the Owner and/or the Owner's Representative at his option.

Temporary storage during installation of metallic materials have to ensure that proper preservation is in place and to provide control against corrosion. Specific care to be taken to ensure corrosion protection is place for site works.

3.2.2 Temporary Works

Temporary works must meet the requirements as described in BS 5975:2019, Code of practice for temporary works procedures and the permissible stress design of false work, as 'providing an "engineered solution" that is used to support or protect either an existing structure or the permanent works during construction, or to support an item of plant or equipment, or the vertical sides or side-slopes of excavation during construction operations on-site or to provide access. It is used to control stability, strength, deflection, fatigue, geotechnical effects and hydraulic effects within defined limits.' It might or might not remain in place after the works. Following local codes, standards and regulations are mandatory.

For Temporary Maritime works if applicable, it must meet the requirement and conditions of BS-6349, Maritime Works that may be used to Design temporary works for marine construction, in particular floating structures, pile gates, formwork, access platforms, scaffold, cradles, jetties, Breakwater, silt curtains, dredging, the effect of waves and varying water levels, offshore Geotechnical conditions and more.

Construction facilities for the site shall include temporary housing camp(s) and offices, laydown areas, warehouses, workshops for site construction purposes, health infrastructure (as necessary), vehicles, mobile equipment, and all other facilities required during construction covering both requirements imposed by local laws and regulations, recommendations of IFC (international Finance Corporation as well as Owner's, and EPC's requirements.

Temporary infrastructure work during construction is the EPC's responsibility and shall include:

- construction of on-site access roads,
- construction of a water storage and distribution system, as required for the construction and installation of the Project,
- construction of electricity distribution systems on-site, as required for the construction and installation of the Project,
- construction of communication system (telephone, internet on-site as required for the construction and installation of the Project),
- construction of Temporary site Facilities in line with the Temporary Site facilities Specifications, and
- construction of firefighting systems, compressed air supply, and distribution systems, potable water systems, sanitary systems, underground cable ducts and trenches, yard lighting, construction fencing to enclose the areas, parking, and other support facilities.

Owner shall review the Temporary Construction work design and method statement and it is Contractor's responsibility to satisfy the Owner and close the comments made by the Owner.

EPC shall be responsible to obtain all necessary permits and approvals needed for the temporary works and shall ensure they are available prior commencing the construction and/or at completion of the temporary facilities as stipulated by the local laws.

3.2.3 Access

3.2.3.1 Obstruction to Access

All work at the Site shall be carried out in such a manner as not to obstruct the operation of any other Contractor on the Site and operation of the power plants.

The Contractor shall also allow the Owner, and the Owner's Representative and any third party nominated by the Owner unhindered access to all parts of the Work in progress and where normal access is obstructed he shall provide temporary alternative access at its own cost.

For Safety and Security reasons, Contractor shall know at any time which personnel is working on Site and/or visiting the Site by means of implementing adequate Security and Access control. For further details, refer to Appendix Z – HSE and Social Management Plan.

The Contractor shall provide fencing and barriers to segregate the construction areas from operational areas within the Plant.

3.2.3.2 Access Road and Site Roads

Temporary access roads (and their subsequent removal) required by the Contractor from the permanent road system for construction erection and demolition purposes shall be provided by the Contractor at his own expense and the location and form of construction shall be agreed with the Owner prior to construction. The location of temporary access roads shall be determined in relation to the requirements of other contractors working on site and the Contractor shall comply with the Owner's instructions on these matters. Curb Stones should be applied where applicable.

3.2.3.3 Clear Road Access

The Contractor shall take precautions to keep all temporary and permanent roadways clear of any spillage from its traffic to the satisfaction of the Owner's Representative. All such spillage that occurs shall be cleared immediately at the Contractor's expense, including excessive earth brought in on wheels or tracks of the traffic, or other materials.

3.2.3.4 Access for the equipment

Permanent provision for trouble free access to the manholes in the equipment shall be provided as per Owner's requirement for the O&M purpose.

3.2.3.5 Contractor's Laydown Area

Contractor to nominate a suitable land for laydown area within the available land at the location. The scope of preparation of land, construction, maintenance and demobilization of the Laydown area shall be within the Contractor's scope.

3.2.4 Transport to Site

The Contractor shall make its own arrangements for the supply to site of all materials, plant and resources necessary for the execution of the Works. The Contractor shall accordingly employ agents as necessary to secure the appropriate authorizations from the Authorities for entry into the Saudi Arabia.

The disposal of excess material excavated soil, disposal of demolished items shall be within the scope of Contractor based on their due diligence and ESIA requirement.

The amount of the EPC Contract price will be deemed to include all costs associated with the importation and clearing of such goods and materials and the Contractor shall be responsible for all costs and arrangements necessary in order to transport all plant and materials to site safely, economically and on programme, including off-loading and transporting to the required positions on site and shall obtain all necessary approvals from the relevant highway/maritime authorities and others as necessary.

The Contractor shall advise the Owner of the movement of abnormal loads and shall provide full details of such loads to the Engineer. In addition, the Contractor shall identify any limits on the weight and/or physical size of plant that can be delivered to the site and shall be responsible for any additional costs associated with modification or upgrading required to the transport route to make it suitable for abnormal loads.

The transport route to site shall be agreed with the relevant authorities and shall take account of local restrictions. The method of transportation shall be suitable for the materials, plant and resources carried without causing nuisance from dust, spillage, etc. or damage to the existing roads etc. The delivery of materials etc. shall not be permitted outside site working hours.

The Contractor shall inform himself fully as to all transport facilities, requirements and loading gauges to ensure that all equipment as packed for transport shall conform to these limitations of Services During Construction Period

3.2.5 Site Services During Construction Period

3.2.5.1 General

Following EPC Contract award, the Contractor shall provide, as a minimum, the following services during the Plant construction period:

- At laydown areas and work faces, temporary fencing, lighting and guarding and all other materials and services necessary for the safety and security of persons and property;
- Permanent marks from which the principal centre lines will be established will be provided by Contractor in agreement with the Owner together with level marks at suitable places on the Site. These beacons and benchmarks shall be the sole points of reference for setting out and levelling purposes and shall be carefully maintained by the Contractor. The Contractor shall provide for the sole use of the Owner's Representative all equipment required for the checking of the setting out of the Works. Equipment shall be complete with all accessories, all to the satisfaction of the Owner's Engineer, and shall be maintained in good working order. The Contractor shall provide the Owner's Representative with all assistance necessary for checking the setting out of the Works;
- Temporary roads, parking areas, etc.
- Warehouse
- Compressed air supply
- Telephones and fire-fighting equipment
- Waste disposal facilities
- Sanitary facilities and sewage disposal (septic tank and fecal suction truck will be used for sewage treatment of construction facility area); and
- Site security e.g. Gate House, badge access system, CCTV, key and master key system
- Temporary Storage Facilities for spares when the warehouse is not operational

Notwithstanding the above, the Owner reserves the right to request the Contractor to provide any additional services reasonably required to safely and effectively complete the Project until the whole of the Works has achieved Commercial Operation.

3.2.5.2 Electricity Supply

The Contractor shall arrange the supply of electricity for construction purposes.

The supply should be made available at one fixed point on the Site. The Contractor shall be responsible for providing and maintaining the whole of the installation.

All design, location, utilization and maintenance of the installation must receive the approval of the Owner or his authorized representative, who may require the disconnection or alteration of any parts, which in his opinion may be dangerous. All local electricity regulations and ordinances shall be strictly observed in the distribution and use of the supply.

3.2.5.3 Water Supply

All water supplies as required by the Contractor during the construction and Commissioning stage need to be procured generated or produced by the Contractor itself or their Subcontractors.

3.2.6 Temporary Site Offices for Owner

The Contractor shall establish, after Notice to Proceed, Temporary Site Facilities as specified in the SCHEDULE Q of this document.

3.2.7 Contractor's Site Buildings

The Contractor shall provide and maintain an office on or near the Site and other buildings as necessary for the due performance of the Work. The office shall be open at all reasonable hours to receive instructions and other communications in the form of manual delivery, post, telephone, telefax and email.

The Contractor shall be responsible for arranging the provision of telephone and telefax lines external from the Site.

3.2.8 Site Labour

The Contractor shall include for the provision of all necessary supervisory, skilled and unskilled labour for erecting, testing, commissioning, and maintaining the Work together with all necessary transport and accommodation where required, and other services necessary to support his supervisory and labour force.

3.2.9 Sanitary and Health Control

The Contractor shall implement a programme of health education to ensure that all appropriate personnel working on the site are aware of health risks associated with living and working together with necessary protective measures.

All persons connected with the Work shall be obliged to use the sanitary facilities provided and maintained on the site by the Contractor.

The Contractor shall be responsible for the removal from the Site and ultimate disposal of his refuse in accordance with any local regulations.

The facilities provided and the standards maintained shall be in accordance with the WHO Guidelines as a minimum.

If appropriate, special precautions shall be taken by the Contractor to keep the incidence of mosquito borne and other diseases to a minimum. These shall include spraying with approved insecticides the insides of all buildings under his control and avoiding the formation of pools of water and other likely mosquito breeding places within and adjacent to the Works.

3.2.10 Clearing of Site and Maintenance

Excluding any munitions, the Contractor will be responsible for the removal of any remaining temporary or permanent facilities from the Site under permits and licenses to be provided by others.

During construction, the Contractor shall keep the site reasonably clean and tidy and remove all waste material to the satisfaction of the Owner. On completion of the work the site shall be left clean and tidy to the satisfaction of the Owner. All waste removal activities must comply with the relevant legislation.

3.2.11 Waste Disposal

The Contractor shall be responsible for ensuring that all waste including contaminated or hazardous soil or sub soil arising from the Works is deposited, treated, kept, disposed of and carried in accordance with the provisions of all applicable environmental regulations. The burning of waste shall not be permitted. All costs for the disposal and transport of waste are deemed to be included in the EPC Contract price.

The Contractor shall ensure that the methods employed in handling, storing and disposal of all dangerous or noxious substances are such that pollution or contamination of the Site or surrounding areas shall not occur. Waste materials shall not be dumped in the sea or on the Site but shall be deposited in an approved off-site pit or disposal area or otherwise disposed of in accordance with the approved environmental procedures. The Contractor shall be responsible for provision of the approved disposal facilities, including obtaining of all necessary permissions.

3.3 Quality Assurance and Quality Control

The detailed requirements for this section can be found within Quality Assurance and Quality Control Specifications annexed as Appendix-X. For purpose of this specification, the following equipment and materials are main equipment and materials:

No.	Equipment List			
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1	Gas Turbine			
2	Steam Turbine			
3	Gas Turbine Generator and Auxiliary Systems			
4	Steam Turbine Generator and Auxiliary Systems			
5	HRSB pressure parts (drum \modules\coils\headers)			
6	GT Step-Up and Unit Auxiliary Transformers			
7	ST Step-Up and Unit Auxiliary Transformers			
8	Turbine steam bypass dump device/ valve			
9	Generator Circuit Breaker (GCB)			
10	HRSB Feed Water& Condensate Extraction Pumps			
11	All medium voltage (>1000V) motors, drives and driven pumps			
12	Alloy Steel Piping, Pre-Fabrication (the P11, P22, P91 alloy steel) and feedwater system piping			
13	HP Alloy Steel Valves & all other valves of class 900 and above			
14	MV Switchgear			
15	Steam Turbine condenser and Heat Exchangers			
16	Any Other Major Equipment/Systems of the Plant (to be agreed with the Owner)			

EPC contractor shall submit the "As Built" baseline data (Vibration, Thermography scanning, oil analysis and etc.), to be used as reference for any condition monitoring plan (for equipment derating).

All turbine enclosure shall be tested for the capability to contain CO₂. Concentration levels should meet NFPA standards and be capable of preventing combustion for at least 30 minutes. The integrity can be tested by

releasing CO₂ and monitoring the concentration decay or by carrying out fan door tests to evaluate their capability to contain CO₂ as per the design and in line with NFPA guidelines. For avoidance of doubt, this is not applicable to steam turbine enclosure.

3.4 Training

3.4.1 Introduction

The Contractor shall include for the provision of training for the operation and maintenance staff.

Training is to cover all aspects of Plant, equipment and system, and operation and maintenance requirements, as required to run the Plant in a safe and efficient manner. Training shall take place early enough to allow operation staff to participate fully in the commissioning process.

The training package and programme shall be agreed with the Owner at least six months before the start of commissioning according to 3.4 requirements. Details of the training proposed, including DCMS OEM training, shall be provided.

For further details regarding the requirements please also refer to the Appendix W of the EPC Contract.

3.4.2 General

The Contractor shall develop and implement a Training Programme for the Owner's designated personnel that shall be site based. This Contractor training shall complement and be integrated with Owner's long-term training programme in order to optimise the total training package.

The overall objectives of the Contractor's Training Programme shall be to:

- Deliver thorough training to introduce, instruct and supervise the Owners technical staff to the essential aspects of the Plant for proper, safe and efficient operation and achieve a high standard of awareness, knowledge and understanding of the Plant and Plant systems.
- Provide an effective contribution to the overall training and development of the Owner's personnel.

The Contractor shall ensure adequate training in all aspects of its codes of practice and safety rules including the training of the Owner's staff engaged in plant commissioning. Training of the Owner's staff in the application of the Owner's safety rule system will be provided by the Owner.

The timing of Contractor's training activities shall be co-ordinated with the Owner's training activities. The aim is for the Owner's personnel to be trained early enough such that they can fully participate in the commissioning process under the direction of the Contractor's commissioning engineers.

3.4.3 Owner's Personnel

The Contractor shall liaise directly with the management of the Owner regarding all training issues.

In the light of the Owner's recruitment objectives, it is understood that:

- All senior staff for Operation and Maintenance of the Plant will have experience received on similar plants.
- All junior staff will have reached such a degree of competence to allow understanding and practical application in their specialised field.

All expenses for Owner's and Operator's personnel e.g. living accommodation, travelling expenses, visa, daily allowance etc, will be at Owner's cost.

The Contractor shall provide trainers, training materials and training aids. Training materials shall be in both hard & traceable soft copy.

3.4.4 On-Site Training Programme

Within 180 days of the Notice to Proceed, the Contractor shall prepare and submit to Owner for review and acceptance a complete Training Programme based on the guidelines and requirements specified in this section. Prior to this submission, the Contractor will have discussed the Training Programme thoroughly with the Owner and will have supplied a preliminary training programme.

The initial Contractor training on-site for the majority of the Owner's personnel shall be delivered prior to the start of commissioning of the Plant. The objective of this training shall be to equip the Plant staff to undertake commissioning activities safely and effectively, under the direction of the Contractor's commissioning team and allow them to participate in the production of Commissioning Documentation and development of the Commissioning Recording System.

The total Production, Operations, Maintenance and Engineering staff complement to be trained on-site will comprise; managers, engineers/team leaders; operations technicians and maintenance technicians.

Further experience of the Plant shall be gained during commissioning, prior to the Plant staff taking over Plant operation. The objective of this training shall be to achieve further development of the Owner's personnel. The Contractor shall allow for the additional coaching and assessment of the Owner's staff during this period by additional training personnel, to be supplied by the Owner. These additional training resources will also be available to assist in the commissioning activities.

Each element of the on-site training programme shall concentrate on specific systems/ plant areas and shall include but not be limited to the following components:

- Task steps;
- Standard performance for each step;
- Tools and equipment;
- Safety requirements;
- Resources (e.g. manufacturer's manuals, drawings and plant documents).

3.4.5 On the Job Training and Work Experience

The Contractor shall accommodate operations and maintenance technicians and engineers/ team leaders being placed with the construction Contractor 's commissioning team to gain work experience and knowledge about the Plant and its systems. The way in which these staff integrate into the construction organization and when they commence their secondment shall be agreed by both parties.

However, such personnel shall operate and maintain the plant during these periods under the full direction, instruction and supervision of the Contractor.

3.4.6 Language Requirements

The language for training shall be in English.

3.4.7 Trainer's Qualifications

Trainers shall be suitably qualified, through experience and education, to teach their respective subject matter.

Trainers and instructors shall be fluent in the English language where it is appropriate to deliver training in English.

3.4.8 Training Period/ Programme

For the purposes of this Specification, the Training Period shall be as defined in the jointly agreed training programme based on the Training Scheme and such training shall be completed at latest prior to Taking Over.

Contractor on-site training sessions shall be conducted on a full-time basis over time scales within the training period to be detailed in the Contractor's technical proposal. The Contractor's Training Programme shall be defined as the combination of all the training activities carried out within the Training Period.

3.4.9 Evaluation of Training

The Owner will review and audit the Training Programme and individual performance at any time for Contractor's compliance before accepting the successful completion of any part of the Training Programme.

3.4.10 Competence Assessment

Trainees will undergo competency assessment at two points within the Training Period. The Owner will first undertake a competency assessment prior to commencement of commissioning and secondly at the completion of training.

The Contractor shall keep comprehensive records of all the personnel who have been trained in a format agreed by Owner and these records shall become the property of the Owner, to be used in the final assessment of staff.

3.4.11 Facilities and Training Materials/Aids

The Contractor shall provide at site suitable facilities to support the Owner and Contractor's jointly agreed training programme.

The Contractor shall provide for training all materials, textbooks, manuals and other materials required for the Contractor's Training Programme. Classroom equipment such as overhead projectors, white/blackboards, VCR and monitor shall be provided by the Contractor.

Training aids, such as films, slides, interactive videos, software packages materials and supplies considered necessary by the Contractor to support, maintain and successfully supplement the Contractor's Training Programme shall be furnished for the Training Period by the Contractor.

Instructions manuals, provided by equipment suppliers, for operations and maintenance shall form part of the basic reference material, in addition to any other references as required.

The Contractor shall arrange for representatives from selected major equipment suppliers to provide specific instruction concerning the construction, engineering, operation and maintenance of the plant and equipment provided under the EPC Contract, including civil works.

3.4.12 Review Process

The Owner reserves the right to review and discuss with the Contractor all training materials/ aids, course outlines, training schedules, steps, procedures, guidelines and other training matters.

4. N.A.

5. GENERAL TECHNICAL REQUIREMENTS

5.1 General Requirements

The Contractor will be obliged to utilise Good Practice in preparing the design of the Plant. Good Practices means those practices, methods, equipment, specifications and standards of safety and performance (as may change from time to time) employed by experienced international contractors or operators in the electricity generation industry engaged in the same type of undertaking under the same or similar circumstances and conditions which, in the exercise of reasonable judgement in light of the facts known at the time the judgement was made, are considered good, safe and prudent practice commensurate with standards of safety, performance, dependability, efficiency and economy. Equipment proposed shall be from reputable manufacturers with sufficient experience accumulated on the particular units and equipment proposed.

The Contractor, if unable to perform the engineering, design and construction supervision activities within their own resources, shall engage reputable engineering consultant(s) with specific international experience to engineer, design and supervise erection and commissioning of the Project.

The engineering, construction, testing and commissioning of the Project shall be in-line with the documents/drawings submitted by the Contractor and approved by Owner. Modifications, if required, to the approved documents/drawings and corresponding change in construction, testing and commissioning of the Project shall be brought to Owner's notice and such modifications shall be carried out only after the written approval of Owner. Wherever there is a considerable change in the scope of work, the same shall be mutually discussed and agreed between Owner and the Contractor through a Change Notice document. The Change Notice document shall specify the proposed technical modifications, reasons for the same, its commercial impacts, related schedule of activities along with timelines and responsibilities of all the parties involved.

5.1.1 Scope of Engineering Services

All required engineering services in order to meet the project requirements shall be executed.

The engineering and coordination shall be done for all equipment and services, described in this Specification and required for fulfilling the Works.

The Contractor shall prepare and submit at his cost the engineering activities necessary for the design and execution of the works of the Plant.

The Contractor shall engage a first class Tier 1 engineering office with relevant engineering services experience and references in similar international projects to perform the complete project engineering services, including but not limited to:

- Basic and detailed engineering,
- Technical evaluation of the equipment to be purchased,
- Vendors engineering review (P&ID, data sheet, lay out, calculations) and integration, including but not limited to overall piping stress analysis and DCS integration
- Attendance to the DCS FAT ensuring the validation of proper communication with all PLCs to be integrated,
- Technical Advisors (TA) on site.

The first-class Tier 1 engineering office proposed by the Contractor shall be approved by the Owner prior to its selection, however Tier-1 Engineering office shall be selected from below list:

- Black & Veatch
- Ramboll
- WSP
- CH2M Hill
- Tata Consulting Engineers
- Sargent & Lundy
- EA

- IDOM
- AFRY

The contractor shall comply with the codes and standards specified in the Part A and Part B of the OTS.

The following list, relating to the engineering activities to be carried out by the Contractor, is not exhaustive or limiting:

- Functional breakdown of the project

The EPC Contractor shall submit, before starting the engineering activities as such the functional breakdown of the project and the services of the general engineering activities, along with the corresponding design documents.
- Surveys and investigations at the site and at the workplace, including but not limited to:
 - Various surveys of the site and verification of its physical condition including soil, sub-soil, air pollution, raw water analysis, noise levels and various risks
 - Soil and foundation studies
 - Surveys and verification at the site of the exact position and characteristics of equipment and networks at the limit of the batteries to which the Contractor shall provide his connections
 - Checking of information appearing in the documents and drawings supplied by the Project Owner to the Contractor.
- General process engineering activities

The Contractor shall be responsible for the definition and engineering of the general process and the special processes employed for reaching the objectives fixed, handling the functions of the Plant and ensuring the best availability, reliability and maintainability.

For this purpose, the Contractor shall not limit himself to assembling the parts supplied by his Subcontractors, but he shall:

- Study the process employed as a whole
- Prepare the mass, heat and electrical energy reports of the general process for the different cases of normal and degraded operation modes that can be envisaged
- Design the different sub-assemblies of the Plant matching the definition of their normal and extreme operating conditions and characteristics
- Analyse the consequence of the operation of the entire Plant and its different components due to variation in the operation of a component or its possible failure due to wear or malfunctioning and employ the means and actions necessary for preventing the resulting consequences that may adversely affect the operation and the availability of the entire Plant, the strength of the equipment, the safety of the personnel and the protection of the environment. This analysis of the consequence should be done via structured reviews such as HAZOP, HAZID, HAZCON, SAFOP etc. The outcome of such reviews shall clearly demonstrate that risk associated with the integrity of plant and safety of the personnel is maintained within ALARP.
- Designing and dimensioning the links between different sub-assemblies of the Plant to achieve full integration with the maximum reliability, availability and maintainability.

The Contractor shall prepare for review the following drawings and documents (list not exhaustive):

- Studies and conceptual documentation
- Specifications including the equipment
- Process flow diagrams (PFD's) including utilities
- Piping and Instrumentation Diagrams (P&ID's) including utilities
- Functional diagrams and control flowcharts including utilities
- System description with functional technical notes including utilities

- Equipment List Asset Register (list of all maintainable equipment along with their key attributes such as equipment typology; failure class code)
- Checking the heights and NPSH of the pumps
- List of lines (diameter, operating conditions and design classes)
- List of connection points, with operating and design conditions
- Pipes and fittings classification
- Effluents studies (balances, characteristics at the terminals points)
- Layout plans (overall installation and detailed plans per unit/area/building)
- Single wire electrical diagrams and protection specifications
- Fully integrated control-command configuration and architecture
- PLCs and regulation flowcharts
- Layout and installation of the control room and the administration building
- Construction BOQ
- All Civil, structural and Architectural drawings, documents for all the facilities and utilities
- Method of Statements
- Drawings and Design document for Decommissioning and Demolition Activities
- All the drawings and the documents associated with fire detection, firefighting and safety
- Review of hazardous areas drawings· General Commissioning Plan and the specific testing and commissioning procedures and protocols fully integrated Plant Operating procedures as well as integration of OEM O&M documentation in a single consistent documentation
- Spare parts interchangeability records (as built)

OEM manuals (as built) Plant identification system and tagging for each component shall be provided. The same system shall be adopted for the whole Plant equipment.

A consistent uniform documentation for the entire Plant is not possible due to the thorough standardisation of systems and system components. Therefore, the documentation standards of manufacturers will be used.

However as far as possible a uniform drawing title block, header and a uniform based document numbering system will be used. Should a supplier specific not allow this, then a corresponding document cover sheet will be added. The individual system documentation will be integrated by the Contractor into one uniform structured documentation for the entire Plant.

All Plant components (mechanical equipment, piping special equipment such as valves, traps, vents, drains, electrical, instruments etc) must be identified on PID in order to have a common Plant and spare part designation system. Furthermore, all signals from black boxes to DCS system shall be identified using the same coding system.

General installation engineering activities, engineering of connections or links to the external networks of the Plant Engineering activities for the integration of the main equipment

The EPC Contractor shall ensure the design and detailing (including for embedded parts) of GT/ST/Transformer foundations shall fully comply OEM requirements.

Engineering activities by discipline

Special studies including but not limited to:

- reliability, maintainability and availability (RAM Study)
- ergonomic studies
- noise generated by the Contractor's equipment

- specific studies for preventing particular dangers during both construction, operation and maintenance

These studies concern notably the means to be employed for avoiding the dangers. They include: HAZOP, HAZCON, HAZAN, HAZID, risk analysis and Safety Instrumented Function (SIF) allocation.

The Contactor shall consider the consequences of damage on equipment or on the environment in hit SIF allocation in relation to Safety, environment, reputation, production / business

The Contractor shall keep the Owner up to date with regards to the reports and action sheets of all these studies. Each SIF allocation shall be traced to the relevant P&ID, system description and other relevant documents.

The Contractor shall present a strategic Program, including all Design cycle and gates to be agreed with the Owner, such as HAZOP, HAZCON, constructability reviews, operability reviews, maintainability reviews, and design reviews to be performed in the presence of the Owner. HAZOP or HAZAN analysis shall be performed for the whole Plant in the same place and led by a Third party with the involvement of all the key suppliers and parties involved in the engineering and design. In no case will the Owner accept split of HAZOP or HAZAN analysis per main packages.

The contractor shall demonstrate how work at heights, coactivity, etc are being eliminated or limited)

- Special engineering activities relating to fire detection and protection
- Logistic study
- Material handling

The engineering and design works should be performed in accordance with the minimum Engineering and Design Management plan requirements defined in this document and its annexes.

With respect to Vendors Documentation, EPC Contractor shall provide:

During engineering stage, main documents for review (data sheet, general arrangement, P&ID, electrical and I&C diagrams, performance curves, among others). Final Dossier for each Equipment, including the following documents:

- a) Engineering File
- b) Quality Dossier
- c) Maintenance and Instructions Manual

a) Engineering File

On completion of Manufacturing, the Supplier shall include in the Engineering File, not limited to, the following documents:

- Equipment general description
- Design technical data
- Description of manufacturing and erection procedures
- General drawings including P&IDs layouts and sectional layouts and arrangement drawings of connections and accessories
- Detail construction drawings, indicating list of parts and materials
- Complete mechanical calculations
- Table of maximum allowable loads for connections
- Foundation and anchor bolts, loads and dimensions
- Nameplate drawing
- List of consumers and consumables, including lubricant and lube oil consumers, and utility consumption list.
- List of instruments

- List of terminal points
- List of valves and accessories
- List of spares parts and special tools.
- Storage and maintenance instructions before commissioning

b) Quality Dossier or Manufacturing File

On completion of the supply, the Supplier shall provide a final manufacturing file, including the final issue of the documents included in the list of project documents that as a minimum, contains the following documents:

- Inspection Points Programmes completed. This document, with all the points duly signed and stamped by the Supplier and with the Purchaser's signature and stamp at the points he has witnessed, will be sufficient as record of inspection activities
- Copy of the procedures submitted for approval
- Copy of quality certificates, reception reports, test reports, test protocols, etc, indicated in the Programme as to be issued at each point. This shall include Welding documentation must be included (WPS, PQR's, Welders Certificates, etc..) and Calibration certificates of test instruments and devices
- Inspection & test reports of all the equipment including type test reports wherever applicable
- Documented reports on significant deviations that have occurred, including the raised Non-conformance reports of such equipment or component, if any
- The final manufacturing quality dossier shall be arranged as per the component/equipment approved ITPL requirements with clear traceability from the pre-manufacturing documentation review up till pre-shipment/packaging inspection stages
- All non-conformity reports raised by manufacturer, EPC Contractor, Owner or third-party inspectors etc. shall be made part of the final manufacturing quality dossiers
- Release certificate
- Manufacturer's Final Quality Certificate

On completion of the erection, the EPC Contractor shall provide a final Site Work Conformance File (SWCF), including the final issue of the documents included in the list of project erection and commissioning procedure that as a minimum, contains the following documents:

- Approved Inspection and test plan for all the items/components or systems erected at the project site.
- Copy of approved erection and commissioning procedures for each individual equipment's and components or system as applicable.
 - Copy of quality certificates, reception reports, test reports, test protocols, etc, indicated in the Programme as to be issued at each point. This shall, also include Welding documentation (WPS, PQR's, Welders Certificates, etc..) and Calibration certificates of test instruments and devices.
 - Documented reports on significant deviations that have occurred, including the raised Non-conformance reports of such equipment or component, if any. The above documents shall also include the critical equipment's site preservation and storage reports.
 - Turn Over package reports
 - Commissioning protocols and Cold/Hot commissioning reports
- Finally, the structure of the SWCF file shall be clearly specified in the project SWCF procedure, which is approved by the Owner.

c) Maintenance and Instruction Manuals

This document shall contain all the information required for the operation and maintenance of equipment during the operation phase, as well as all the procedures required to carry out any necessary repairs during the lifetime of the equipment.

Manuals shall include, but not limited to, the following:

- Equipment storage requirements
- Erection instructions with drawings. Commissioning and initial test instructions
- Operating instructions, including but not limited to safety precautions and operating limits
- Maintenance procedures, routine adjustments, preventive maintenance schedules and any predictive maintenance recommendations
- Parts illustrations, including parts lists adequate for the purpose of identifying and ordering replacement parts and designation of the number required per component and lists of recommended spare parts
- Detailed descriptions of the functions of each principal component of a system
- Safety precautions
- Preservation requirements for equipment covering initial preservation measures, and during operations which may arise because of different operational schemes that put equipment / units in idle conditions for extended period. Examples include provision of nitrogen blanketing, flushing lines etc.
- Provision must be provided for preservation of equipment which is susceptible to corrosion in extended periods of standby / idle. The preservation strategy for assets susceptible for short 1-6 months, medium 6-12 months, long term >12months should be specified in the O&M manual but NDZ (notice to deviate from 0 MW) should remain at <6hrs as a minimum. For equipment supply and storage on site prior to construction specifically rotating plant, humidity shall be controlled to <30%RH at all times, passive or active systems are acceptable but provision for monitoring humidity should be provided

5.1.2 Equipment Identification System/Labelling

The Contractor shall apply the Reference Designation System for Power Plants (RDS-PP) equipment numbering system for identifying plant and equipment. The system shall be clear, concise and systematic and shall be applied by the Contractor and its sub-contractors consistently across all the Plant areas and shall be easily incorporated into an Asset Management System. Equipment identification in the External Facilities shall be in accordance with the existing installations.

Instruction plates, nameplates and labels shall be provided for all equipment, valves, cabinets and panels of the Plant giving particulars of duty, size, serial number and full information for identification and operation in English language only. Where considered appropriate for safety/emergency purposes (such as caution plates and emergency exists etc.), the Contractor shall agree with the Owner and provide as part of its Work scope a reasonable number of instruction plates, nameplates and labels in both the English language and Arabic.

Labels shall be of sufficient size to carry a full description of the plant item and a unique item alpha-numeric identification. Abbreviations in the plant description shall only be used with prior approval of the Investor.

Labels shall be fitted such that they are not readily lost or broken during routine operation and maintenance.

Any labels damaged or defaced during installation or Plant start-up shall be replaced by the Contractor prior to Taking Over by the Investor.

5.1.3 Plant Access Provisions

Platforms shall be provided around equipment as required for maintenance work, testing, inspection and safe operation. Platforms shall be sized to facilitate safe ergonomic operation of manual valves and equipment between waist and shoulder level of a standard man. Relief valves shall be grouped together and an access platform provided. Operating valves which cannot be serviced from a platform shall be operated by other approved means.

All platforms to be visited on a daily basis shall be provided with stairways except where a permanent access ladder is the only feasible means of access. All interior and exterior platforms, gratings, checker plate, stairways and handrails shall be galvanised.

Platforms shall be constructed using galvanised open steel flooring in a normal condition and a suitable grade of GRP or stainless grating in areas of high salinity, in accordance to the internationally recognized standards and shall cater for the relevant loadings for maintenance. Frequently accessed platform shall not be less than 1 m wide and those platforms shall be extended up to the equipment, valves and instrumentation which they serve.

All edges of floors, platforms and walkways shall be provided with curbs or kicking strips.

Particular attention shall be paid to internal and external access in order to facilitate inspection, cleaning and maintenance. For GT enclosure inside, OEM standard is acceptable.

5.1.4 Temporary Site Offices for Owner

The Contractor shall build, after Notice to Proceed, Site office facilities for the Owner, Owner's Engineers and Owner's Representative, as described in SCHEDULE Q.

5.1.5 Standardization and Interchangeability

To minimise spares holding and enhance maintenance and operation, plant items performing similar functions e.g. valves, switchgear shall, as far as practicable, be standardised and obtained from a minimum number of suppliers. Similarly, the design of major plant items shall be consistent to maximise interchangeability of components.

For Gas turbine package, vendors of instruments and control system will be selected from Supplier's standard list of reputable vendors to realize high safe, reliable and optimal unit operation.

Type/model of the instruments will be selected as per Supplier's design practice.

5.1.6 Availability and Reliability

Plant maintenance philosophies and the selection of equipment shall reflect the need for a Plant with a high availability.

The Contractor shall ensure that their detailed design complies with the Availability, Reliability and Maintainability requirements during the course of design implementation.

5.1.7 Requirements of Materials

Materials selected by the Contractor shall be proven to be suitable to all relevant process and environmental conditions prevailing at the Site, and sufficient for the complete Plant life cycle.

Components of the Plant which are subject to abrasion or potential high rates of wear including heat transfer surfaces shall be selected from proven grades of high corrosion/erosion resistant materials (copper alloys, nickel alloys, stainless steels and titanium).

GRP water supply pipes shall be manufactured in accordance with AWWA C 950. The pipe shall have a minimum stiffness of 2500 N/m² in accordance with AWWA C 950 and shall include an internal and external corrosion barrier.

Stainless-steel piping shall be passivated in accordance with ASTM A380 after shop and site fabrication has been completed as required. Socket welded is acceptable for 2" and below including stainless steel materials.

Where materials are subject to an elevated risk of corrosion (e.g. due to stagnant water, high temperatures, aggressive conditions or presence of crevices) additional protection shall be applied e.g. by coating or cathodic protection.

5.1.8 Site Storage of Chemicals

Storage facilities for all operational chemicals shall be provided. At least 7 days (shall be considered for 30 days keeping in view consumption, long lead delivery and availability) storage shall be provided for chemical for the operation of the Power Facility. The tank shall be equipped with level measurement for remote monitoring.

Storage facilities for hazardous chemicals shall be subject to review and comment by Owner at the design stage of the Project.

The chemicals supplied for this Plant shall be of food grade /non-carcinogenic if required. All chemicals used in the project shall be in line with the EIA study recommendations.

All chemical handling equipment shall be bundled such that any chemical spillage is contained. Each storage tank shall be equipped with a bund having a volume of at least 110% of the capacity of the tank. If hydrochloric acid is used, the acid bulk storage tank shall be equipped with adequate fume extraction facilities.

For chemical spill management, secondary containment shall be considered for truck unloading so to avoid spillage. Chemical storage area shall be provided with fire detection and sprinkler type protection system as per NFPA 101 and 400.

5.1.9 Hazardous Areas

Areas where a potential flammable atmosphere may exist shall be classified to the relevant sections of the Institute of Petroleum (IP) and IEC codes, or other internationally recognised standard. Suitably certified equipment shall be used in the designated hazardous areas adopting the installation practices in line with applicable IEC or other internationally recognized codes. To the maximum extent practical, equipment requiring operator attention and/or electrical equipment shall not be installed in hazardous areas.

The Contractor is required to demonstrate in the Plant Design that adequate segregation of electrical equipment from hazardous areas has been made, and that no hazardous area extends beyond the Site boundary fencing.

The Contractor shall appoint an independent specialist to undertake a HAZOP study to demonstrate the safety and integrity of the Plant design. Owner shall be invited to attend all HAZOP meetings. The results of the HAZOP shall be made available to Owner and relevant Competent Authority prior to construction.

HAZOP studies shall be carried out for entire plant including power plant and all subsystems at the appropriate time during the design and the construction phase to ensure timely implementation of HAZOP study results, if any, without affecting the project schedule.

The Contractor shall appoint an IECEx or other internationally recognized standard certified/authorized 3rd party for review, inspection and certification of all hazardous area equipment and installations at site. The aforesaid 3rd party shall be proposed by the Contractor and shall be approved by the Owner prior to its selection.

5.1.10 Security Systems

The Plant design shall include suitable security systems such as security fences. Facilities for security personnel such as a guard house shall also be included.

There shall also be building access control systems and building intrusion detection. The buildings to be fitted with this equipment shall be determined during the detailed design phase.

The Plant is to be protected from both casual and malicious intrusion. The protective measures which the Contractor shall implement shall include:

- perimeter security fence;
- CCTV system to monitor the perimeter and provide operational overviews and details of the site;
- intruder detection to buildings;
- access control system to site and buildings;
- gate intercom system to allow visitor access; and
- mechanised site access gate.

These measures shall be fully integrated to allow an automatic response to events to aid operator decisions.

The main security gate shall be provided with large screen display with multi camera monitoring.

5.1.11 Restoring Damages to Existing Property

Damage to any property belonging to Owner, the Government or others, due to accident or exposure caused by the Contractor's activities during and after construction activities, or during plant operation, shall be restored by the Contractor at their cost. The Contractor shall be held responsible for any damage caused by the Contractor's activities to the adjoining properties.

No excavation shall be started without the presence of the representative of the respective authorities of the existing buried/on ground/aboveground services in the concerned area. Written permission shall be required from the concerned parties before the start of any excavations. The Contractor shall make arrangements to expose, support and protect the related existing services and shall include in their scope, the works involved in overcoming the obstacles caused by the existing services by deeper excavations, necessary temporary supports, complete relocation or any other means which may deem fit and are approved by the respective Owners or the Government.

5.2 Steelwork

Steel sections shall be new and shapes shall conform to American, British, European, Japanese Standards, GB Standards or equivalent international standards subject to approval by the Owner during detailed Engineering stage. All structural steel sections shall be hot rolled with a minimum grade A572 in accordance with ASTM standards and the AISC manual for steel construction. Contractor can use built up section higher than 400mm as per the design requirement subject to the QA/QC of the built up section is approved by the Owner. In this case the utilization ratio for built up members shall not be more than 90% for design Pre-engineered steel structures shall be approved by the Owner/Owner's Engineer. All pre-engineered constructions shall be in accordance with the specification of steel works, especially the description of material, etc.

For HRSG following materials shall be applied:

A572 : HRSG main frames.

ASTM A36 : Casing/ducting, diverter, by-pass and exhaust stack, secondary structures (i.e. platforms, stairways, etc.)

All bolts in elevated steelwork connections shall be minimum grade ASTM A325 or A490 high strength bolts to suit the required design. All holding down bolts shall be minimum grade ASTM A307. Finishes etc. shall be selected such, to eliminate galvanic corrosion. Bolt finishes shall conform with the finishes applied to the steelwork elements which they connect. Welded connections will comply with the AWS D1.1 of the American Welding Society.

Non Destructive Inspection shall fulfill the requirements of AWS D1.1 of the American Welding Society.

5.2.1 Material

All materials in contact with seawater and brine if applicable shall be of suitable design and shall have proven suitability in comparable installations. Corrosion due to inadequate combination of different metals must be avoided.

Where materials are subject to an elevated risk of corrosion (e.g. due to stagnant water high temperatures, aggressive conditions or presence of crevices) additional protection shall be applied e.g. by coating or cathodic protection.

5.3 Surface Preparation and Painting

5.3.1 General

The whole preparation and paint system shall be suitable for the operating environment specified. In general, all paintings have to be shop finish painted.

Though proprietary items may be used in their standard finish, subject to the approval of the Investor care shall be taken in the application of this principle, to ensure that cosmetic, and more importantly safety and operational, colour clashes do not occur. For specified applications, adequate supervision shall be provided for all stages of preparation, application and testing.

The Contractor shall provide painting specifications and application methods that are suitable for the exposure conditions in accordance with the international standards and painting manufacturer's recommendations. The painting specification shall include the paints to be applied, thinners or any additives that may be required, tools and instruments required to prepare the surfaces, the application of protective coats, quality control, quality assurance and personnel safety.

During detail design, GT OEM should prepare painting specification for relevant supply package which needs to follow painting class and durability requirements as per section 5.3.1 of this document, and submit it for approval to the Owner.

All steelworks shall be protected in accordance with EN ISO 12944 and EN ISO 14713 or equivalent. Galvanising work shall be performed by means of the hot dip galvanizing process, unless otherwise specified and shall comply with the following standards: EN ISO 14713 and ISO 1461.

The coating durability shall be based on a 25 years of Design life (PPA term) painting system for outdoor and also for 25 years of Design life (PPA term) painting system for Indoor. The ambient environment shall be classified in the C4 for outdoor installations and C3 categories for indoor atmospheric corrosion as a minimum or as per Section 5.3.12 whichever is stringent. Where more corrosive environment will prevail, EPC shall do proper due diligence and provide the solution accordingly subject to Owner's approval.

All switchyard structures shall be Hot Dipped Galvanized with C4 category for 25 years of Design life(PPA term) or as per Section 5.3.12 whichever is stringent.

After all machining, forming and welding has been completed, all steelwork surfaces shall be thoroughly cleaned of rust, scale, welding slag or spatter and other contaminations prior to any painting.

The surface of all non-ferrous parts usually painted, shall be cleaned, rubbed down, stopped, filled and given one priming coat of paint in preparation for the under and finishing coats.

Any damage occurring to any part of a painting scheme shall be made good to the same standard of corrosion protection and appearance as that originally employed. Any finish coat applied on site shall be considered for decorative purposes only.

Interior structural steel and equipment and materials with a shop applied finish paint system shall only be touch-up painted at site.

The following articles contain various lists of equipment and materials, identifying the painting work typically required for each kind of item. These lists should be used to establish the general philosophy for the scope of the painting work and any additions or changes to equipment as the detailed design and procurement are completed should be treated accordingly. Applied coatings shall have a design life to first maintenance of minimum 25 years from PCOD (duration between manufacturing/painting until PCOD need to be added on the top of 25 years).

Following codes and standards shall be followed for concrete and metal corrosion protection work.

- ISO 1461, Hot dip galvanized coatings on fabricated iron and steel articles - Specifications and test methods
- ISO 10474, Steel and steel products - Inspection documents
- ISO 9223, Corrosion of metals and alloys – Corrosivity of atmospheres – Classification, determination and estimation
- ISO 9224, Corrosion of metals and alloys — Corrosivity of atmospheres — Guiding values for the corrosivity categories
- ISO 9225, Corrosion of metals and alloys – Corrosivity of atmospheres – Measurement of environmental parameters affecting corrosivity of atmospheres
- ISO 9226, Corrosion of metals and alloys – Corrosivity of atmospheres – Determination of corrosion rate of standard specimens for the evaluation of corrosivity
- ISO 12944 (all parts), Paints and varnishes — Corrosion protection of steel structures by protective paint systems
- ISO 14713-1, Zinc coatings – Guidelines and recommendations for the protection against corrosion of iron and steel in structures – Part 1: General principles of design and corrosion resistance
- SSPC-Steel Structure Painting Council
- ASTM A780 – Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
- ACI 222R-01 – Protection of Metals in Concrete Against Corrosion
- ACI 318 – Building Code Requirements for Structural Concrete
- BS EN 1504 – Repair of Concrete Structure
- ASTM C920 – Standard Specification for Elastomeric Joint Sealants
- BS 8118 – Structural use of aluminium. Code of practice for design
- EN ISO 21809 – part 3 – Field joint coatings - Amendment 1: Introduction of mesh-backed coating systems
- BS EN 12501 – Protection of metallic materials against corrosion. Corrosion likelihood in soil.
- BS EN 13195 – Offshore construction works, Delivery, Castings, Cold-working, Water transport engineering, Corrosion, Inspection, Wrought alloys, Forgings, Rivets, Aluminium, Shipbuilding, Aluminium alloys, Rolled products
- ASTM – A116, A153, A702, A491, F2611, F1712, A121, F1910, F1665, A702, F1083, F900, etc. for Fences,
- NACE SP0108-2008 - Standard Practice, Corrosion Control of Offshore Structures by Protective Coatings
- NACE SP0297-2012 - Standard Practice Maintenance Painting of Electrical Substation Apparatus Including Flow Coating of Transformer Radiators
- NACE SP0207-2007 – Standard Practice Performing Close-Interval Potential Surveys and
- DC Surface Potential Gradient Surveys on Buried or Submerged Metallic Pipelines
- ISO 9224 - Corrosion of metals and alloys — Corrosivity of atmospheres — Guiding values for the corrosivity categories

All steelworks shall be blast cleaned (to SA 2-1/2 or SA 3 as applicable (Swedish Standard if submitted in English language or ISO 8501 whichever is stringent). Paint systems applied to surfaces which may at any time be submerged below water shall not be susceptible to debonding or other degradation through cathodic action.

The components and/or equipment shall be mechanically and /or chemically cleaned during the following stages of the Contract.

- Cleaning in workshop
- Cleaning before painting and/or corrosion protection (application of prime coat)
- Cleaning before erection and during installation.

Cleaning of fabricated component items shall be carried out after fabrication and final heat treatment or welding at manufacturer's works or at site, as appropriate.

For cleaning in workshop and before painting mechanical cleaning as opposed to alternative chemical cleaning is the preferred method for works cleaning except where this is precluded by design or access considerations.

Machined surfaces shall be protected during the cleaning operations.

In the event of the surfaces not being cleaned to the purchaser's satisfaction, such parts of the cleaning procedures or agreed alternatives as are deemed necessary to overcome the deficiencies shall be carried out at the supplier's sole expense.

For reclining small areas, hand cleaning by wire brushing may be permitted. Wire brushes used on austenitic steel bristles.

Austenitic stainless steels, copper and aluminium alloys, cast iron, bimetallic and metallic/plastic items, and components fabricated by spot welding or riveting shall not be chemically cleaned. All weld areas shall be suitably stress relieved before chemical cleaning.

5.3.2 Chemical Cleaning

Chemical cleaning shall be done for boiler pressure parts components at site. The detailed procedure for chemical cleaning as well as the system for which chemical cleaning is required shall be submitted by the contractor for Owner's approval.

5.3.3 Steam blowing

Main steam, hot and cold reheat piping and any other piping system as necessary shall be subjected to steam blowing. The procedure and the system for which steam blowing is required will be submitted by the EPC Contractor for Owner's approval.

5.3.4 Mechanical cleaning at manufacturer's works

Mechanical cleaning shall be carried out by abrasive blasting.

5.3.4.1 Surface condition

The Metal surfaces shall be clean and free of mil scale, rust, dirt, grease and any other deleterious matter.

Where metal surfaces are to be painted the surface profiles shall conform to the painting specification requirements. Where this does not apply surfaces shall have a surface texture not coarser than Grade 80 abrasive paper.

5.3.4.2 Abrasives

Abrasives containing silica, silicates or slag residues shall not be used for water/steam side surfaces of plant except for cleaning sand castings, where hydro blasting with sand may be used.

For austenitic materials only, abrasives containing 98% or more of alumina, Al₂O₃, shall be used.

5.3.4.3 Removal of abrasive and debris

After cleaning, abrasive and debris shall be thoroughly removed for components.

5.3.5 Alternative chemical cleaning at manufacturer's works

The procedure shall comprise pre-treatment acid treatment to achieve cleanliness equivalent to that specified for mechanical cleaning. The procedure to be adopted must meet with the purchaser's approval.

5.3.6 Protection at manufacturer's works

As soon as all items have been cleaned and within four hours of the subsequent drying, they shall be given suitable anti-corrosion protection.

All water and steam side surfaces shall be protected by the application of approved water-soluble corrosion inhibitors, or vapour phase inhibitors that can be subsequently removed by site water washing or steam blowing.

The gas side of steam generating plant items shall be protected by the application of temporary protective that do not require to be removed before commissioning, but which are removed during initial firing. However, the designed temporary painting system shall be suitable for long term storage at the site corrosive atmosphere as applicable.

The rate of application of volatile corrosion inhibitors shall be at least 10 grams per square metre or 35 grams per cubic metre, whichever is the greater, except for pipes up to 300 mm diameter for which the minimum application rates shall be 5 grams per square metre.

Immediately after the protective treatment has been applied all vessels and pipes shall be suitably sealed off by discs or caps or approved alternatives to prevent ingress from the surroundings. Cylindrical plugs shall not be driven into the ends of pipes. These protective covers shall not be removed until immediately before final connection is made to the associated equipment.

5.3.7 Weather conditions

Painting shall be done only when the surface temperature is above 5°C. Surface temperature must be at least 3°C above dew point to ensure that condensation does not occur on the surface.

Reasonable protection against precipitation shall be exercised for the painting of outdoor parts.

Precautions shall also be taken against solar radiation to ensure that the specified dry film thickness of priming or finish coats is obtained.

Any prime coat exposed to excess humidity, rain, dust etc., before drying, shall be permitted to dry and the damaged area of primer shall be removed and the surface prepared and primed again.

Sheltered or unventilated horizontal surfaces on which dew may collect require more protection, and to achieve this an additional top coat of paint shall be applied.

The temperature quoted as "normal" in the "Paint System Tables" refers to the average local climatic conditions.

5.3.8 Surface preparation

In preparing any surface to be coated, all loose paint, dirt, grease, rust, scale, weld slag or spatter or any other extraneous material shall be removed and defects repaired, so as to obtain a clean, dry, even surface to receive the priming or finishing coat (s) as called for in the painting schedules. Sharp edges should be rounded, especially when tank linings have to be applied.

All machined surfaces, including flange faces, shall be suitably covered to prevent damage during surface preparation.

All surfaces should be blast cleaned whenever possible. Surface preparation methods

Bare steel surfaces should be prepared by one of the methods described below in order of preference and in accordance with Swedish Standard SIS 05 59 00 or Steel Structures Painting Council, SSPC, Vis 1, or DIN 55928, section 4.

(a) *White metal blast cleaning:* Sa 3 or SSPC - SP 5

Sa 3 Blast cleaning to bare metal mill scale, rust and foreign matter must be removed completely. Subsequently, the surface is cleaned with vacuum cleaner, clean dry compressed air or a clean brush. It must then have a uniform metallic colour and correspond in appearance to the prints designated Sa 3.

(b) *Near white metal blast cleaning* Sa 2 1/2 or SSPC - SP 10

Sa 2 1/2. Very thorough blast cleaning. Mill scale, rust and foreign matter shall be removed to the extent that the only traces remaining are slight imperfections in the form of spots or stripes. Subsequently, the surface is cleaned with a vacuum cleaner, clean dry compressed air or a clean brush. It must then correspond in appearance to the prints designated Sa 2 1/2.

Mechanical cleaning should only be used when procedures (a) and (b) are not practicable.

(c) *Near white metal blast cleaning P Sa 2 1/2 or DIN 55928*

Very thorough blast cleaning. Very adhesive coatings remain. From all other surface mill scale and rust are to be removed to such an extent that the only traces remaining are slight imperfections in the form of spots or stripes. Further treatment see Sub b).

The adhesivity of residual coatings in the transition zone has to be tested even after the application of the primer.

(d) *Very thorough mechanical scraping and wire brushing St 3*

St 3 very thorough scraping and wire-brushing - machine brushing - grinding - etc. are to be preferred. Surface preparation as for St 2. But much more thoroughly. After the removal of dust, the surface must have a pronounced metallic sheen and correspond to the prints designated St. 3.

(e) *Thorough scraping and wire brushing: St 2*

St 2 Thorough scraping and wire-brushing - machine brushing - grinding - etc. The treatment shall remove loose mill scale, rust and foreign matter. Subsequently, the surface is cleaned with a vacuum cleaner, clean dry compressed air or a clean brush. It should then have a faint metallic sheen. The appearance must correspond to the prints designated St 2.

SIS 055900	DIN 55928 Part 4	Surface preparation method	BS 4232 only for blasting	SSPC-Vis
Sa 2 1/2	Sa2		Second quality	Near white SP10
Sa 3	Sa 3	Blasting acc to items (a), (b), (c)	First Quality	White metal SP 5
-	PSa 2 1/2		-	-
St 2	St 2	Hand / or power tool derusting	-	Hand tool cleaning SP 2
St 3	St 3	Acc to items (d) and (e)	-	Power tool cleaning SP3
-	F1	Flame jet cleaning	-	Flame cleaning SP4
-	Be	Pickling	-	Pickling

Steel structures to be blast cleaned have to be free of pitting and other severely corroded places in accordance with B.S. 4232 and SIS 055900.

The abrasives used for blast-cleaning shall be graded flint, grit, shot or silica sand and shall be such that they will produce an average keying profile on the blast-cleaned surface of not more than 40 microns.

An air pressure of 7 bar (g) at the nozzle shall be used.

After blast-cleaning, all accumulated grit, sand, dust, etc., must be removed leaving the surface clean, dry and free of mill scale, rust grease and other foreign matter.

In the event of rusting after completion of the surface preparation, the surface must be cleaned again in the manner specified.

Oil, grease, soil, cement, salts, acids or other corrosive chemicals shall be cleaned from steel surfaces, by the use of solvents, emulsions or cleaning compounds. The final wiping shall be with clean solvent and clean rags or brushes. There shall be no detrimental residue left on the surface.

Primed areas which suffer damage must be spot blasted on site to a degree of cleanliness P Sa 2 1/2 before touching up.

Protective coating must be applied as quickly as possible after the completion of surface preparation no matter what cleaning method has been used.

No blast-cleaned surface shall be allowed to remain uncoated overnight.

Wood surfaces shall be sanded clean. All nail holes shall be puttied and sanded before priming.

Concrete: If a protective coating is required, concrete shall be allowed to cure before painting.

5.3.9 Preparation of coating materials

All containers shall remain un-opened until required for use.

Primers and paints which have livered, gelled or otherwise deteriorated shall not be used.

The oldest primer or paint of each kind shall be used first.

All ingredients in any container shall be thoroughly mixed before use and shall be agitated frequently during application to keep the primer in suspension.

Primer or paint mixed in the original container shall not be transferred until all settled pigment is incorporated into the body of liquid.

Mixing in open containers shall be done in a well-ventilated area.

Primer or paint shall be mixed in a manner ensuring the breakdown of all lumps, complete dispersion of pigment and uniform composition.

Two-component primers shall be mixed in accordance with the manufacturer's instructions.

Thinners shall not be added to primers or paints unless necessary for proper application according to the manufacturer's instructions.

When use of thinners is permitted, it must be added to the primer or paint during mixing.

5.3.9.1 Health and safety of work

The supplier has to check all painting work to be carried out according to the specification of the paint supplier further to all relevant prescriptions and regulations concerning the health and safety of work.

The paint supplier has to present a written specification including at least the flash point of the paints, ventilation requirements, handling precautions such as inhalation, eye and skin protection, and first aid procedure, storage requirements, spill or leak procedure, fire precaution, waste disposal.

5.3.9.2 Methods

Quality of the surface to be painted or coated has to be tested acc. to DIN 55928 and DIN 8202.

Temporary corrosion protections are to be completely removed prior to applying the definite one, in accordance with DIN 55928.

All prime coatings shall be applied by brush or airless spray or a combination of these methods, as approved by the coating manufacturer.

All doors, windows, stairways, handrails (if painted), bolts, flanges and equipment supports shall be finish painted by brush.

Spray guns should not be used outside in windy weather or near surfaces of a contrasting colour unless the latter is properly protected.

All cold-spray painting shall be done using standard equipment in accordance with accepted standards and methods.

Care has to be taken not to connect spraying devices for nitro and bakelite paints simultaneously to oil based paints.

Paint applied to items that are not be painted shall be removed at the supplier's expense, leaving the surface clean, unstained and undamaged.

5.3.9.3 Dry film thickness (dft)

To the maximum extent practicable the coats shall be applied as a continuous film of uniform thickness and free of pores. Overspray, skips, runs, sags and drips should be avoided. The different coats shall not be of the same colour.

5.3.9.4 Consumption of paints

This has to be evaluated according to DIN 53220. The paints shall be tested as per recognized International code.

Each coat of paint shall be allowed to harden before the next is applied. For epoxy paint the hardening time normally is 12-14 hours. Suppliers' recommendations regarding hardening time of epoxy paints must be followed.

Particular attention must be paid to full film thickness at edges.

The minimum total dry film thickness of the paint systems shall be as recommended in the following tables below. The dft is given in microns (millionths of a metre).

5.3.10 Protective coatings and paint systems

The colour coding for identification of pipelines should comply with standard practice.

The type and number of protective coats for any item requiring painting are to be in accordance with DIN 55928 and are to be at least of a quality as shown in the attached tables "Paint Systems"

Alternative to the 'paint system' specified, are to be presented on the schedule Departure from Specification, as indicated elsewhere.

Generally, all parts shall receive the specified prime coat (s) at the supplier's works to ensure that no corrosion occurs during transport to the site and storage at the site.

Parts which cannot be damaged during transport shall receive the full number of coats.

5.3.11 Galvanizing

Galvanizing works shall conform in all respect to ASTM or B.S. 729, B.S. 3083 and

B.S.C.P. 2008 and to DIN 50976 whatever requires the higher quality and shall be performed by the hot dip process, unless otherwise specified.

It is essential that details of steel members and assemblies which are to be hot-dip galvanized should be designed in accordance with B.S 4479 or ASTM.

Vent-holes and drain-holes should be provided to avoid high internal pressures and air-locks during immersion, which may cause explosions, and to ensure that molten zinc is not retained in pockets during withdrawal.

Careful cleaning of welds is necessary before welded assemblies are dipped. The welds and the surrounding metal should be cleaned separately, preferably be blast-cleaning, because the usual preliminary pickling cannot be relied on to remove the welding slag.

All defects of the steel surface including cracks, surface laminations, laps and folds shall be removed in accordance with B.S. 4360. All drilling, cutting, welding, forming and final fabrication of unit members and assemblies shall be completed, where feasible, before the structures are galvanized. The surface of the steelwork to be galvanized shall be free from paint, oil, grease and similar contaminants in accordance with DIN 55928, part 4 and DIN 50976. The weight of zinc coating per unit area has to be noted in the manufacturing documents in accordance with DIN 50976.

The minimum average coating weight shall be as specified in Table 1 of B.S. 729 or Table 2, DIN 50976, whatever requires higher quality.

Structural steel items shall be initially grit-blasted to B.S. 4232, second quality, (Sa 2 1/2) or by pickling in a bath and the minimum average coating weight on steel sections 5 mm thick and over shall be 900 g/m².

For Hot dipped galvanization work ASTM 123 can be followed if it complies with C4 corrosion protection for Very High durability for the outdoor ambient condition.

On removal from the galvanizing bath, the resultant coating shall be smooth, continuous, free from gross surface imperfections such as bare spots, lumps, blisters and inclusions of flux, ash or dross.

Galvanized contact surfaces to be joined by high-tensile friction-grip bolts shall be roughened before assembly so that the required slip factor (defined in B.S. 3294, part 1 and B.S. 4604, part 1) is achieved. Care shall be taken to ensure that the roughening is confined to the area of the mating faces.

Bolts, nuts and washers, including general grade high-tensile friction grip bolts (referred to in B.S. 3139, and B.S.4395 part 1) shall be hot dip galvanized and subsequently centrifuged (according to B.S. 729). Nuts shall be tapped up to

0.4 mm oversize after galvanizing and the threads oiled to permit the nuts to be finger-turned on the bolt for the full depth of the nut. No lubricant, applied to the projecting threads of galvanized high-tensile friction-grip bolt after the bolt has been inserted through the steelwork, must be allowed to come into contact with the mating faces of the steelwork. A local remelting of the galvanized parts to achieve the nuts to be finger turned on the bolt is possible in accordance with DIN 50976.

Protected slings must be used for offloading and erection. Galvanized work which are to be stored at the works or on site shall be stacked so as to provide adequate ventilation to all surfaces to avoid wet storage staining (white rust).

Small areas of the galvanized coating damaged in any way shall be restored in accordance with DIN 55928, part A and DIN 50976 by:

- Cleaning the area of any weld slag rust and other impurities and by thorough wire brushing to give a metallic clean surface.
- application of suitable number of coats of zinc-rich paint containing more than 90 % w/w of zinc in dried film. The dry film thickness shall exceed at least 50 % the thickness of the desired galvanization. In case of application of a low melting point zinc alloy repair rod, the rods shall be in accordance with DIN1707, the thickness of the alloy shall be at least as of the desired galvanization.

The restored area shall not exceed 1 % of the galvanized surface.

Surface restoration of parts in contact with drinking water is not allowed and the quality of the galvanization is to be in accordance with DIN 2444.

After fixing, bolt heads, washers and nuts shall receive two coats of zinc- rich paint. Connections between galvanized surfaces and copper, copper alloy or aluminum surfaces shall be protected by suitable preferably hydrophobe tape wrappings to the Owner's approval.

5.3.11.1 Sprayed Metal Coatings

Corrosion protection may also be achieved by spraying of suitable metals as zinc and/or aluminum on the surfaces of structures. For special cases tin, copper, lead can be used as well. Methods of surface preparation have to conform to B.S. 2569 or to DIN 8567. A proper treatment of the surface followed by an immediate spraying is to apply to ensure adhesion of the sprayed metal. The surface has to be clean, free of impurities, rust, mill scale and rough enough to have binding properties to ensure good enticulation with the sprayed layer. Suitable roughness can be achieved by blast cleaning according to BS 4232 or DIN 8567. Welds are to be cleaned and prepared with special care. All surfaces to be treated have to be dry and accessible.

Application of coatings, requirements for thickness, adhesion, composition of coating metals, and subsequent treatment have to conform to BS 2569 DIN 8565 and 8567 or as per OEM manufacturing specification.

Testing of the spray coated layer are to be carried out in accordance with DIN 8565.

The EPC Contractor has to specify the type, composition and thickness of the sprayed metal and of the sealing coating according to DIN 8565 including the corresponding warranties and tests if sprayed metal coating will be applied.

5.3.11.2 Safety of work

All precautions connected with this type of application of corrosion protection have to be in accordance with German regulation DVS 2307, page 1.2.

Sprayed, unfused coating of metals and metallic compounds applied by combustion gas flame, plasma arc, detonation and similar processes, and the preparation of components, spraying techniques, sealing, finishing and inspection shall be according to B.S. 4761.

The hot galvanized surface has to be cleaned before the application of the coats to remove corrosion products, dirt, dust, grease.

The cleaning can be achieved by

- brush off
- washing with 1 - 1.5 % ammonia water with up to 0.1 % detergent added and followed by wet grinding using e.g. scotch brite to turn the foam to grey colour,
- steam blasting,

5.3.12 Minimum painting requirement, unless governed by the corrosion class:

System-No.	Surface location	Temp. °C	Surface preparation	Coating systems	No. of coats	Generic type	Dry thickness (OFT) per coat um	Total DFT μm
1	Structural steel works, piping, vessels, tanks INDOOR	up to 120	SA 2.5	Primer Finish	1 1	Zinc-Epoxy Epoxy High Solid	80 80	80 160
2	Structural steel works, piping, vessels, tanks OUTDOOR	up to 120	SA 2.5	Primer Intermediate Finish	1 1-2 1	Complying with durability as per section 5.3.1	80 80 80	80 80 80 240
3	Piping, tanks, etc. INDOOR and OUTDOOR, Insulated	up to 120	SA 2.5	Primer	1	Zinc-Epoxy	50	50
4	Pumps, motors, other equipment OUTDOOR	up to 120	SA 2.5	Primer Intermediate Finish	1 1 1	Zinc-Epoxy Epoxy High Solid 2-Como.Polvurethane	80 110 50	80 110 50 240
5	Pumps, motors, other equipment INDOOR	up to 120	SA 2.5	Primer Finish	1	Zinc-Epoxy Epoxy	80	80

					2	High Solid	50	100 180
6	Piping, reactors OUTDOOR Insulated	> 120	SA 2.5	Primer	1	Zinc Ethysilicate	75	75

Additional 1x Finish Coat 2-Comp. Polyurethane, 50 µm, when exposed to UV and colour retention is required or when exposed to weathering

System- No.	Surface location	Temp. °C	Surface preparation	Coating system	No. of coats	Generic type	Dry film thickness (DFf) per coat	Total DFT µm
7	Stacks	< 120	SA 2.5	Primer	1	Zinc Ethysilicate	75	75
	OUIDOOR							
		< 200		Finish	2	Silicone Acrylic	50	100
								175
*8	Steel surfaces	200 - 450	SA 2.5	Primer	1	Zinc Ethysilicate	75	75
	Uninsulated							
				Finish	2	Silicone Aluminium	25	50
9	Galvanized surfaces	up to 120	Mechanical cleaning from contaminants and zinc salts by means of washing or steam jetting or sweep- blasting with fines and	When Finish Coat is required, such as sea climate with chloride exposure	1	Epoxy High Solid*	125	125
10	Steel surfaces permanently in contact with water	Medium temp. °C up to 60	SA 2.5	Prime and Finish Coat in One	1	Glass flake reinforced High Solid Epoxy	500	500

5.3.13 Painted Surfaces

All outdoor onshore steel work shall be painted (except for Switchyard, which requires HDG) with C4 corrosion protection for very high durability up to the PPA term from IAT. Cladding shall be galvanized and painted. Indoor steelwork shall be painted as per corrosion protection class mentioned above. Surfaces to be painted include exposed surfaces for the internal parts of all building structures and the exposed surfaces of all indoor equipment, piping, valves, and hangers, unless excluded hereinafter. The following lists include the major surfaces of indoor materials (Unless noted otherwise) and equipment to be painted. Surfaces of materials and minor equipment not listed but requiring painting for protection shall also be painted.

- Breeching not insulated and lagged
- Concrete masonry units (interior and exterior)
- Concrete bare wall, floors, sidewalks, curbs, and paving including in chemical storage and water treatment areas
- Miscellaneous supports for tray, bus duct, conduit, lighting, electric unit heaters, etc
- Field erected tanks (interior and exterior)
- Reservoirs
- Air receivers
- Shop fabricated tanks (if not insulated)
- Turbine generator
- Interior structural steel
- Galvanized Metal wall panels and flashings

5.3.14 Paint Colours

The colours of the primer, intermediate and finishing coats of a paint system shall be easily distinguishable from each other and the materials used shall be suitable for the method of application employed and preferably be supplied by one manufacturer who shall ensure that all coatings are compatible.

Using the Contractor's proposal as a basis, the final colours and identification systems for the equipment, piping etc. shall be in accordance with accepted practice and with the agreement of the Owner and off taker.

Where materials are subject to an elevated risk of corrosion (e.g. due to stagnant water or presence of crevices) additional protection shall be applied e.g. by coating or cathodic protection.

5.4 Corrosion Protection and Misc. Requirements

Where dissimilar metals are in contact or close proximity and corrosion may occur through electrolytic action or differences in electrical potential, protection shall be afforded by electroplating, suitable gaskets, cathodic protection or other means.

Chromium plated parts shall not be used in any damp or corrosive atmosphere.

All surfaces shall be adequately protected in transit, and any damage shall be renovated immediately on off-loading and on completion of erection.

After cleaning and inspection but before the plant leaves the Contractor's works, the machined surfaces of steel and ironwork shall be covered and protected with a preserving fluid according to the related applicable standards otherwise to the Owner's approval.

All external steel screw fixings shall be supplied in the hot dipped spun galvanized condition, stainless steel, or sherardized to comply with BS 4921, Class 1 or Class 2 or equivalent with passivation treatment. The EPC Contractor shall provide with his bid the civil design criteria he intends to follow, in which details regarding his proposed methods of corrosion protection for reinforced concrete and steel structures are given and procedures described.

All splash zones and concrete surface in contact with aggressive atmosphere shall be painted with epoxy coating.

5.5 Noise Levels

All plant shall run without undue vibration and with the least practicable amount of noise.

Noise levels from the Plant shall not exceed the limits identified in section 4.1.9.5 of the OTS Part A.

5.6 Insulation

Insulation of the Plant shall generally be in accordance with the Contractor's standard procedure. Notwithstanding this, the provisions described below shall apply to the Contractor's design.

All exposed portions of the Plant which are subject to a temperature exceeding 60°C and are likely to come into direct contact with operational and maintenance and other personnel shall be provided with approved thermal insulation so that the temperature on the outside of the insulation does not exceed 60°C when the ambient temperature is 35°C.

All heat insulation materials shall be chemically inert and shall remain so in the event of being saturated with water. The Contractor shall ensure that insulation materials shall be prevented from any possible formation of Hexavalent Chromium contamination for all types of turbines, generators and other balance of plant equipment/pipeline. All materials shall be asbestos free. Thermal insulation of pipe work and equipment shall comply with the relevant recommendations of BS 5970 and BS 5422. All insulation material shall be in accordance with BS 3958 or equal Standard as applicable with regard to definition, physical characteristics and tests. OEM's standard codes shall be allowed.

All insulation adjacent to hot surfaces shall be capable of withstanding the maximum surface temperature obtainable in service without deterioration or breakdown of the material. The maximum permissible temperature, which the insulation will withstand in continuous service, shall be stated.

Thermostatic steam traps shall be insulated in accordance with the manufacturer's instructions.

The type, thickness and heat loss of insulation, reinforcing and protection shall be subject to approval and shall be stated.

In addition:

- Valves shall be insulated with suitable valve boxes that can be removed/replaced for/following maintenance or preferably insulated with valve blankets that are held in position with Velcro straps or similar;
- The hot piping, subject to the expansion during the operation shall be insulated according to the fixed / movable support philosophy and with enough overlap to maintain necessary expansion of pipe,
- Vapor barriers should be applied on the insulation to extend the lifetime of the cladding material
- The cladding should be capable to easy remove and reinstalling of the material during/after maintenance or inspection
- Wire shall be avoided wherever possible as a fixing material for insulation as it tends to cut into the insulation. Bands shall be the preferred method;
- Insulation of bends and elbows using straight pipe insulation cut into sections as required to obtain a reasonable smooth exterior shall be avoided by the Contractor. Preformed insulation shall be used wherever possible;
- The Contractor shall ensure that a permanent small hole is provided in flange lagging to act as a leakage indicator;
- All outdoor insulation shall be sealed by the Contractor using suitable compounds;
- The Contractor shall state the amount of overlap on any cladding, particularly with respect to corrugated sheeting;
- Piping insulation which runs on, for example, HRSG platform floors shall be capable of withstanding treading on by operators and maintenance staff.

5.7 Piping and Valves

5.7.1 Pipes System

The design code for the metallic piping shall be ASME B31.1 Power piping standard. Boiler piping shall be in accordance with ASME Section I - Boiler Code.

The EPC Contractor shall provide a piping specification for each piping system which shall describe the following:

- Service description
- Pipework material
- Design pressure and temperature
- Operating pressure and temperature
- Pipe wall thicknesses including corrosion allowance
- Materials
- Pipe connection arrangements
- Flange ratings and material groups
- Principal valve types and materials
- Hydrostatic test pressure

The corrosion allowance to be considered for piping shall be as follows:

- Carbon steel: 1.5 mm
- Alloy steel: 1 mm (0 mm can be allowed as long as design life criteria can be met)
- Stainless steel: 0 mm
- Plastic: 0 mm

The piping thicknesses shall be sufficient to withstand the design pressure at the design temperature during the plant's life cycle.

The minimum thicknesses to be used in the process piping shall be the following:

- Carbon steel piping
 - DN ≤ 6" Schedule STD
 - DN ≥ 8" Schedule 20
- Alloy steel piping
 - DN ≤ 2" Schedule 80
 - DN ≥ 2½" Schedule STD
- Stainless steel piping Schedule 10S

Piping design of the main fluid systems shall be developed assuming velocity of fluids determined by the EPC Contractor taking into account pressure drop of the fluid in the lines, possible erosion, water hammer effects, noise emission and any other possible effects.

Piping shall be designed within the following flow velocities, unless otherwise agreed with the Owner.

Service	Maximum Velocity
Superheated steam	< 65 m/s
Saturated steam	< 45 m/s
Bypass Line Upstream & Downstream of Bypass Valve	< 100 m/s
Feedwater	Suction < 2.5 m/s Discharge < 4 m/s (Normal Operation) Discharge < 6 m/s (transient operating condition, such as bypass, start-up/shutdown)

Service	Maximum Velocity
Condensate Water (normal operation)	Suction < 2 m/s Discharge < 4 m/s
Circulating water	2 to 3 m/s
General service water	Suction < 2 m/s Discharge < 3.6 m/s
Natural gas	< 30 m/s
Compressed air and other gases	10 to 25 m/s

Piping systems larger than DN50 shall be stress analysed as required and the scope of analysis shall meet the requirements of ASME B31.1.

In Gas turbine package, the piping, piping support, piping hangers, valves, fittings, welding of pipe works and etc. of the proposed generator and its auxiliaries will be in accordance with JIS(Japanese Industrial Standards) or manufacturer's standard. The stress analysis is not necessary and will not be conducted for the piping of the generator and its auxiliaries.

All pipe systems shall be arranged, installed, supported and provided with all necessary means of venting, draining and expansion subject to the approval of the Owner in accordance with the Contract before erection commences. The space between pipes shall be sufficient to allow easy repair and/or replacement of insulation or repaint.

The pipework layout shall be designed so that items of equipment and sections of pipework can be removed from the pipelines without major disturbance to the adjacent pipework, additionally piping shall be arranged so that full access is provided for the operation and maintenance of equipment, and that removal or replacement of equipment can be achieved with the minimum dismantling of piping. Particular care shall be taken to ensure that pipework thrusts are not transmitted to machinery or associated apparatus.

Dead legs shall be avoided but where this is not possible provision shall be made for flushing the pipework. Changes in pipe bore sizes shall be by the use of proprietary fittings or fabricated sections to avoid sudden changes.

All nuts, bolts, washers, flanges, gaskets, flanged tied adaptors, drain valves, special connection pieces, supporting hangers, brackets or clips and temporary supports for the pipework, together with all terminal point connection materials shall be supplied and installed.

All flanged joints shall be fitted with suitable insertion ring gaskets which shall be carefully adjusted concentric to the bore of the pipe so that no undue strain is placed on any of the bolts or flanges of the pipe when bolting up.

The Contractor shall be responsible to issue a bolting torque procedure in accordance with the applicable Standards and on the case of equipment's, fulfilling also the recommendations of the suppliers.

When the conditions of the system exceed $P \geq 40$ bar or $T \geq 425^\circ\text{C}$, two drainage and vent valves shall be included. Additionally, drains in steam lines will be provided with double isolation valve if design pressure > 22 barg, whatever the temperature.

All the saturated steam pipes shall have a 1% slope when cold and 0.5% slope when hot for draining.

All drains or vents shall be routed to a collection point.

All steam pipe work shall be carefully designed to ensure that it is fully drainable. Drain and vent valves shall be supplied as necessary.

For steam lines, drain design shall consider the guidelines included in ASME TDP-1 Standard providing recommended practices for the prevention of water damage to steam turbines used in Electric Power Generation.

The steam pipe work shall be carefully designed with regard to anchor points and expansion provisions to ensure that any thrust transmitted to the turbine inlet flanges is kept to a minimum. Adequate pipe supports and anchors shall be supplied for all piping.

5.7.2 Materials

The piping specification provided by Contractor will define the selected material as well as the applicable standards and identification code for the different system services.

The materials used for the manufacture of the metal piping must be in accordance with the ASME or ASTM standards.

The material of all the elements (pumps, valves, strainers, etc.) shall be suitable for connecting to the corresponding piping system to which they belong.

As a general rule pipe and valve sizes 1/4", 1-1/4", 3-1/2", 5", 7" and 22" will not be used.

Steel flanges shall be in accordance with ASME B16.5 for DN ≤ 24", and ASME B16.47 Series A or AWWA C207 for DN > 24".

Bolts, nuts, washers for flange joints shall be hot dip galvanized as applicable, except for bolts/nuts to be used in high temperature (bichromated).

The gaskets of flange joints will be suitable for each fluid and design requirements.

Pipes and fittings shall be joined by welding in accordance with the following criteria:

DN > 2": Butt weld (BW)

Hydraulic pipework shall be of stainless steel, tubing high pressure grade ASTM A213-TP316L.

5.7.3 Valves

The design, fabrication and testing of the valves shall be in accordance with the applicable Codes and Standards (ASME B16.34, API 6D, etc.) The class of the valves shall be compatible with the design service requirements.

The EPC Contractor shall provide a valve specification (data book or valve class) for the valves which shall describe the following:

- Valve characteristics
- Design and operation conditions
- Construction data
- Materials
- Actuator
- Other requirements

The Valve List shall include the detailed identification (KKS, P&ID, line, system, fluid, type, ND, rating/schedule, ends, etc.), design data, insulation and heat tracing data, painting data (system and colour), critical valve, manufacturer, and origin of the valve.

Valves, valve drives and accessories, which are of a similar make, size and type, shall be standardized as much as possible in order to facilitate interchangeability and maintenance and limit the stock of spare parts.

All valve bodies shall be of the same nominal size as the adjacent piping, unless otherwise required. The internal diameter of valve ends adjacent to the pipework shall be the same as the internal diameter of the connecting pipe (full bore type).

Valves shall be designed for long life tightness and smooth non-sticking operation at all operating conditions. They shall be wear resistant as possible, reliable and approved in practice.

Valves shall be installed maintainable from the outside (field repairable).

After disassembly of actuator (regardless of manual or motor actuator), valve position must be readily identifiable.

For all globe and check valves, the direction of flow shall be clearly stamped on the body of the valve.

For butterfly valves, preferred direction of flow shall be stamped on the body of the valve.

Gate valves shall not be used in flow regulation and throttling service.

In vacuum service and wherever otherwise necessary to prevent the entry of air, valves shall be provided with packings appropriate for vacuum service (without external sealing water).

Each equipment item will be provided with isolation valves.

Instrument isolation valves shall be ball or globe valves.

Welded valves of size 2-1/2" and above shall be provided with butt welding ends prepared as per ASME B16.25 and welded valves 2" or below shall have socket welded ends as per ASME B16.11.

Control valve sizing shall be made in accordance with ANSI/ISA S75.01 standard and the control valve Contractor's sizing computer program. The Contractor shall be responsible for the proper sizing, selection, and application of all control valves, control valve components including trim or plug design, actuators, valve materials, valve accessories, and other such factors directly under the control of the Contractor.

Proper control valve selection shall ensure that the required energy can be dissipated without exceeding the maximum vibration levels in the piping system and without exceeding the wear properties of the trim material. Contractor shall select control valve trim design that facilitates maximum reduction of control valve induced vibration and trim wear.

The opening percentages (referred to 100% full stroke) shall be between:

- a) Minimum flow >10%
- b) Normal Flow 30-60% for linear curves. 30-75% for iso-percentages.
- c) Maximum flow <75% for linear curves. <85% for iso-percentages

When the valve is also used for isolation purposes, class V (ANSI/FCI 70-2) in general and Class VI for "Tight shut off" (TSO) application shall be provided.

Except for turbine control valves, each critical and/or high pressure control valve station shall be provided with bypass valves and motorized valves on feedwater and steam lines.

All regularly operated isolation valves and control valves shall be accessible from a permanent floor or access platform.

Safety and relief valves design shall be in accordance with the applicable standards: API RP 520 Parts I and II, API 521 and ISO 4126. Valves shall be grouped together and an access platform provided.

For Gas turbine package, GT OEM standardized equipment, and the control valves with accessories are not general service ones but dedicated ones. Hence, the control valves with accessories will follow GT OEM's standard.

In Gas turbine package, the valves are designed according to GT OEM standard for following systems.

- Generator Cooling Water System
- Generator Seal Oil System
- Generator H₂ & CO₂ Gas System
- Generator H₂ & CO₂ Gas Supply System

5.8 Welding

5.8.1 General

Welders employed on any work under the EPC Contract, either at works or on site, shall be proved to be competent to carry out the type and standard of welding required prior to commencing work. All welders shall be qualified in accordance with the governing design code or, where no appropriate standard is thereby indicated, with Section IX, "Welding and Brazing Qualifications of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code" or approved equivalent.

In Gas turbine package, the piping, piping support, piping hangers, valves, fittings, welding of pipe works and etc. of the proposed generator and its auxiliaries will be in accordance with JIS(Japanese Industrial Standards) or manufacturer's standard, provided that reference standards are equivalent to International standard such as ISO for welding procedure qualification and NDT requirement.

In the case of pressure parts and pressure part attachment welds, each approved welder shall have a unique identification number which shall be indicated on all welds made by him and on all permanent test records applicable to those welds including radiographs.

All welders engaged on either shop or site welding shall be under the supervision of a competent welding engineer who shall demonstrate, to the Owner's approval according to the related applicable standards, a complete system for the control and treatment of consumables, deployment of welders, use of approved procedures, heat treatment, non-destructive testing, including spectrographic material checks on weld and parent metals, repairs and general quality control. Also, the welding Engineer shall monitor all the qualified welders at site or manufacturing shop for their performance, repair rate, re-training and re-qualification in consent with Owner.

5.8.2 Welder's Tests

Sample welds shall be carried out on specimens of similar shape, thickness and chemical analysis to the material to be welded, and in a position which shall represent as far as possible the worst conditions under which actual welding shall take place, as laid down by the approved Standard/Code. The welders shall be considered approved to weld after the positive result of the radiographs of the sample welds.

The method of preparation of the test pieces and the standards to be used in examination of them shall be according to the agreed Standard or Codes, or as may be agreed with the Owner. All expenses of every kind in connection with welder's tests shall be borne by the Contractor.

5.8.3 Welding Procedures

All welding procedures including those proposed in repair situations shall be qualified as required by the governing design code or, where no appropriate standard is thereby indicated, by Section IX of the ASME Code or approved equivalent and all qualified procedures shall be submitted to the Owner for approval prior to the commencement of any welding. This applies in full for boiler, water steam cycle and all BOP piping, where the required welding procedures shall be submitted as "For Information". For rest of the plant equipment, the requirement shall be dealt on case-to-case basis.

Gas turbine interconnecting piping will be shop welded according to ASME B31.1 and all relating non-destructive testing will be performed according ASME B31.1. Test documentation related to these welds shall be submitted to Owner.

5.8.4 Preheating

Where necessary, the work shall be heated prior to welding in an approved manner, and the temperature maintained throughout the operation. The metal temperature shall be checked by means of suitable thermocouples or temperature indicating crayons.

For 9 Cr material welds, the use of thermocouples is mandatory and the preheating process must be recorded in all the welds. The use of torches for the Preheating of material grade 2 1/4 Cr and 9Cr is it not allowed.

5.8.5 Post Weld Heat Treatment

Regardless of the requirements of the specific code or specifications applicable to the design and/or construction of the plant and associated pipework, it is a mandatory requirement of this specification that all welds shall be subject to appropriate post weld treatment with the following exceptions:

- Austenitic stainless steels
- Plain carbon steels containing less than 0.25 per cent carbon at thicknesses up to and including 30 mm
- Low alloy steels containing $\frac{1}{2}$ per cent molybdenum and $1\frac{1}{4}$ per cent chromium, $\frac{1}{2}$ per cent molybdenum up to and including thicknesses of 12.5 mm provided that:
 - A hardness traverse across the weld zone gives no single result greater than 240 HV10
 - No fillet weld leg is greater than 8 mm
 - A minimum preheat of 200°C is used for welding

Plain carbon steels shall contain no more than 0.25% carbon.

All post weld heat treatments of materials and components for use at temperatures in excess of 400°C shall be so designed as to guarantee maximum creep properties in the parent materials and weldments.

The soaking times at the selected heat treatment temperatures shall be:

- Low alloy steels not containing vanadium, 5 minutes per mm of thickness with a minimum time of 120 minutes
- Low alloy steels containing vanadium and high (11 to 15 per cent) chromium steels containing vanadium, 180 minutes regardless of thickness

In the event that any post weld heat treatment is carried out at a temperature in excess of the original tempering temperature of the parent materials, a re-assessment of the design strength limit stresses for the materials will be required. This may, in the absence of acceptable mechanical test data for the specific condition, require re-determination of the relevant mechanical properties of the materials.

The requirements for post weld heat treatment do not affect or replace in any way the requirements for maintenance of welding preheats or post welding heating for the purposes of diffusing hydrogen from the weldment.

The PWHT (post weld heat treatment holding temperatures), time and heating and cooling rates must fulfil the requirements of the applicable Part of the ASME Code depending of the type of component. Before the start of any PWHT the charts must be signed by the Owner or his Representative

5.8.6 Inspection of Welding

The extent of weld inspection, the final weld quality, and the method of examination shall be as stated in this OTS or the agreed Standards or Codes. The Contractor shall maintain, both at works and on site, adequately indexed records of all welding, weld inspection and repairs, which shall be available to the Owner, together with facilities for viewing of X-ray photographs.

The views taken by X-ray shall be to a procedure and resolution agreed with the Owner and shall incorporate the appropriate image quality indicator. In case that the Inspection and Manufacturing Code allows the use of two different NDT methodologies the Owner reserves the right to choose the most appropriate one in his understanding.

However, for the Butt welds of thickness of 60 mm and above shall be tested either by X-Ray or by Phased Array Ultrasonic Testing (PAUT) method instead of Gamma rays due to its limitation. If PAUT method is selected, demonstration block as same as welding & material being tested is necessary in addition to code required standard calibration blocks.

After the completion of the PWHT on the Gr92, Gr91, Gr22 and Gr11 an hardness test must be performed as follows:

- 100 % on Base Metals, Weld and Heat Affected Zones (3 indentations in each area)

The maximum and minimum acceptable values for each material must be agreed with the Owner before the start of the works,

On the 9Cr materials, the NDT Inspections must be performed after 48h after the conclusion of the PWHT

Adequate safe storage facilities and appropriate safety regulations shall be provided for gamma-ray units, if used.

5.8.7 Weld Repairs

Where repairs on major items of plant are found to be necessary by welding, dimensioned sketches of the excavated areas shall be prepared and the proposed repair procedure submitted to the Owner before any repair is commenced. Under no circumstances shall the repair be commenced until the agreement of the Owner has been obtained. The areas shall be subjected to a suitable non-destructive test and on completion of the weld will again be non-destructively tested. In the event that the repair requires all or any part of the item to be stress-relieved, the non-destructive tests shall be repeated on any part deemed to have been affected by the stress relief. The welds repairs are allowed just on the cases that is accepted able by the Manufacturing Code. Supplier shall document all the major and minor welding repairs areas carried out on all the weldable castings applicable for turbine casings, valves, equipment's if any, and these documents including the post repair weld stress relieving chart and NDT records shall be a part of the manufacturing quality dossier. The major welding repair shall be carried out only with Owner approval.

In the case of weld cracking, the Owner may require a full investigation to be made of the cause of the cracking prior to any approval of repair procedures. The cost of all such investigations and reports is to be wholly borne by the Contractor.

Weld repair of cast iron parts is not permitted under any circumstances.

5.8.8 Welding Electrodes and Consumables

Before any welding is commenced, the Contractor shall submit a list of all electrodes and consumables to be used to the Owner, as well as and a Welding Material Storage and Handling Procedure.

Electrodes and consumables shall comply with the agreed National and International Standard and shall be suitable for the materials to be joined, the quality, size and type of weld and the welding position.

In each case, the appropriate BS or American Society for Testing and Materials (ASTM) designation together with the proprietary brand name shall be given.

5.9 Protection of Moving Parts

All moving parts where accessible to operation personnel shall be protected and suitably guarded to meet all relevant regulations. All guards shall be designed to facilitate easy removal.

5.10 Balancing

All rotating parts of the machinery shall be statically and dynamically balanced as per the relevant design and testing code unless otherwise agreed in writing by the Owner.

In Gas turbine package, generator rotor is balanced by the most effective method being considered its characteristic. Since generator rotor runs through several critical speeds up to the rated speed, generator rotor is an elastic rotor not a rigid rotor. In such case, the high speed balancing (i.e. dynamic balance) is most effective. Therefore, statically balance is not applicable.

5.11 Lubrication

Any components requiring manual lubrication shall be provided with greasing nipples or other suitable arrangement of type mounted on a panel and identified.

A remotely mounted electrically operated lubricator of approved type shall be provided to serve components, if any, requiring continuous lubrication by external mechanical means.

The Contractor shall supply all grease and oil for the start-up, commissioning of the Plant up to Take Over, except as otherwise noted herein. The Contractor shall supply a list of all recommended lubricants to the Owner prior to start-up.

All bearing surfaces shall be properly lubricated before the Plant is operated.

5.12 Hazardous Areas

Areas where a potential flammable atmosphere may exist shall be classified to the relevant sections of IEC codes or other internationally recognised standard. Suitably certified equipment shall be used in the

designated hazardous areas. To the maximum extent practical equipment requiring operator attention and/or electrical equipment shall not be installed in hazardous areas.

The Contractor shall demonstrate in its Proposal adequate segregation of electrical equipment from hazardous areas and that no hazardous area extends beyond the Plant boundary fencing.

5.13 Maintenance

5.13.1 General

The Contractor shall provide complete details of maintenance and spare parts requirements, including the routine maintenance and the major inspection and overhaul programmes needed in order to achieve the required Plant availability and performance. The maintenance requirement documentation shall include, but not be limited to:

- Lists of spare parts, consumable items and special tools for each type of maintenance activity i.e. operational, scheduled and strategic groupings;
- Frequency between planned maintenance activities;
- Duration of each major inspection and maintenance activity in terms of man-hours and target elapsed time, inclusive of supervisory levels and with details of required labour grades;
- Complete detailed description of each maintenance activity necessary to carry out maintenance.

The planned maintenance outage programme shall include for statutory inspections that need to be carried out in accordance with the appropriate local regulations.

5.13.2 Maintenance Facilities

The Contractor shall provide sufficient access to and around all equipment, in accordance with good utility design practice, to allow effective maintenance and removal; such access provisions shall include, for example, ensuring that sufficient space and access for safe gas turbine air intake filter replacements using lifting equipment to be provided by the Contractor exists without requiring the use of temporary scaffolding or other temporary means of access. Wherever possible, trouble free access shall be provided for manholes in the equipment, the use of the use of temporary scaffolding or other temporary means of access shall be avoided as far as possible. General arrangement drawings will be provided clearly identifying the outline of all major plant equipment, their weights and associated floor loading capacity and lay down location.

Sand, dust and the salty Site environment represent hazards for outdoor situated plant, those parts of the Plant where maintenance activities will be performed shall be provided with permanent arrangements for slinging or handling during maintenance and overhaul. Plant requiring regular routine maintenance that does not have direct or mobile crane access shall be provided with monorail and hoists for items that may not be reasonably manipulated by a single average person. Monorail or forklift access shall be provided for moving heavy equipment and for equipment that requires regular dismantling for maintenance and repair. The lay down areas for all major plant and around each enclosure shall have adequate space for all maintenance activities including space for plant covers and components removed for access, direct heavy transport/trailer access and direct mobile crane access.

Provision shall be made for lifting gear for the maintenance and replacement of the plant components. The Steam Turbine building shall have overhead cranes with main and auxiliary hooks sized to enable removal of the heaviest item for maintenance. Special devices, slings, runway beams, trolleys and other lifting tackle necessary for the maintenance of the plant shall be provided by the Contractor.

The marking and test specification of Safe Working Load values on lifting equipment shall be in SI units.

Mobile plant such as fork-lift trucks, cranes, sewage tankers etc. required for the operation and maintenance of the Plant shall be provided by the Owner to a recommended list to be provided by the Contractor. All mobile plant required for EPC purposes, such as during construction and commissioning, shall be provided by the Contractor. Crane worthy area to be made wherever access for maintenance is required.

5.13.3 Laydown Area

The Contractor shall provide sufficient indoor laydown area in each turbine building or outdoor in the surrounding of each powerhouse. The area should be such a way that generator rotor removal and turbine major overhaul can be performed simultaneously. Following areas shall be considered for laydown purpose or as per OEM recommendation whichever as a maximum:

- Area for seven 20 foot containers (for the storage of insulation, tools, miscellaneous parts and as office container), with electrical power supply, telecommunication (in- and outdoor if necessary);
- Area of 200m² for the storage of noise enclosure building material, power supply, accessible for forklift.
- Area of 200m² for the storage of pipes, valves, burners, turbine parts and material testing equipment, with power supply and accessible for forklift.
- Area of 150m² for abrasive blasting, with power supply, accessible for forklift and 10t trailer; and
- Area of 50m² dark room for UV ray material testing, accessible for forklift.

5.14 Spare Parts and Tools

The Contractor shall provide all spares, tools and consumables necessary for discharging his responsibilities in carrying out the Works, including commissioning, and until PCOD (PAD). As part of his scope, the Contractor shall also provide all gas turbine, steam turbine, generators, all major equipment (including but not limited to HRSG, pump, Heat exchanger, gas compressor)-related special tools in order to enable performance of all maintenance activities including Major Overhauls during PPA term.

All special tools necessary for the overhaul, maintenance and adjustment of the Facility and its equipment and facilities shall be provided by the Contractor. Gas Turbine, Steam Turbine, GT and ST Generator OEMs to consider the special tools required for Rotor removal, Rotor Turning and Rotor Standing Pedestal during Major Maintenance. All Electrical and I&C software packages to be included in the special tools.

5.14.1 Recommended Spares for Maintenance

The EPC Contractor shall declare before substantial completion of each Plant Unit the list of spares held for the purposes above and provide such spares to the Owner without any additional cost at the COD of the last Plant Unit to reach substantial completion.

The EPC Contractor shall provide with its proposal a priced list of spares (including country of origin), which he recommends the Owner should possess. These recommended spares should be classified as follows:

- (a) Maintenance Spares and Consumables, these are items for which the EPC Contractor anticipates that demands will arise in normal operation of the Plant.
- (b) Overhaul Spares and Consumables, these are items for which the EPC Contractor anticipates that demands will arise during the programmed minor and major overhauls. The EPC Contractor shall provide prices and estimated quantities required for each item for both minor and major overhauls.
- (c) Spare Parts' country of origin
- (d) Strategic/ Breakdown Spares, these are items for which the EPC Contractor anticipates that demands may arise through breakdowns which could jeopardize the availability or safety of the Plant. The EPC Contractor shall state the proposed stock holdings for such items together with the prices of each.

All spares shall be directly interchangeable with the corresponding parts installed in the Plant and shall meet the requirements of the specification of the Plant. Spares shall be warranted for a period equivalent to the warranty period for the original Plant.

The Spare Parts as per above EPC Contractor's recommendations shall be available at site prior to 90 days of COD of first Unit. Any spare parts used during the commissioning/ testing and/ or warranty period by EPC (to fulfil their obligations related to defect liability and warranty) shall be replenished promptly within 30 to 90 days by the EPC contractor at their cost, including procurement, shipment and payment of applicable taxes/ customs duty.

5.14.2 Overhaul Spares and Consumables

These are items for which the Contractor anticipates that demands will arise during the programmed minor and major overhauls. The Contractor shall provide prices and estimated quantities required of each item for both minor and major overhauls except gas turbine.

5.14.3 Strategic/Breakdown Spares

These are items for which the Contractor anticipates that demands may arise through breakdowns which could jeopardize the availability or safety of the plant. The Contractor shall state the proposed stock holdings for such items together with the prices of each except GT

All spares shall be directly interchangeable with the corresponding parts installed in the plant and shall meet the requirements of the specification of the original plant.

Spares shall be warranted for an equivalent period to the original plant.

Spares list shall be in the format of a schedule including the following data:

- Description;
- Identification;
- Price;
- Delivery (Weeks).

The Contractor shall replace at its own expense all spares taken from the station stores and used during construction, commissioning, testing and warranty (Defects Notification Period).

5.14.4 Special Tools

All special tools necessary for the overhaul, maintenance and adjustment of the Facility and its equipment and facilities shall be provided by the Contractor. Generator OEM to consider the special tools required for Generator Rotor removal, Rotor Turning and Rotor Standing Pedestal during Major Maintenance. All Electrical and I&C software packages to be included in the special tools.

5.15 Equipment Use for Commissioning and Testing

The Contractor, pursuant to its obligations in the EPC Contract, shall identify the required equivalent operating hours (EOH) for commissioning and testing for each item of plant to achieve all intermediate milestones up to the PCOD respectively.

5.16 Workshop and Stores Equipment

5.16.1 Workshop

Suitable workshop facilities shall be provided complete with amenities. The workshops shall be suitably sized to carry out all routine off-plant servicing, as well as breakdown maintenance. The facility shall be divided into two (2) areas, area 1 being suitable for heavier mechanical work, welding, etc., area 2 being a "clean" area suitable for electrical and instrumentation work. Contractor shall provide (as option) following workshop equipment in the Proposal:

Item No.	Description	Q'ty
1	Lathe, 300 mm centre height, 3000 mm centre distance	1
2	Lathe, 165 mm centre height, 1000 mm centre distance	1
3	Universal Milling Machine, 1370 x 320 mm	1
4	Shaping Machine, 610 mm stroke	1
5	Radial Drill, 2515 mm radius	1

Item No.	Description	Q'ty
6	Pedestal Drill, up to 16 mm	1
7	Tool Grinder, 2 x 300 mm wheels	1
8	Tool Grinder, 2 x 150 mm wheels	1
9	Rough Grinder, 2 x 300 mm wheels	1
10	Pipe Bender, 150 mm	1
11	Hand Operated Plate Shears	1
12	Hacksaw, 300 mm dia. cap.	1
13	Hydraulic Press, 50 t	1
14	Oxy-Acetylene Profile Cutter, 2000 x 1000 mm	1
15	Oxy-Acetylene Cutter/Welder	2
16	Electric Welding Set, 20/200 A	4
17	Argon Arc Welding Set	2
18	TIG Welding Machine, 16 kVA	2
19	Gas Arc Gouging Equipment	1
20	Marking-off Table 2000 x 1000 mm	1
21	Industrial Type Vacuum Cleaner	5
22	Greasing Machine	1

The workshop facility shall have lorry access to a loading bay and loading and unloading facilities. The overhead handling equipment shall be capable of placing equipment in any area of the workshop facility. Where work such as grinding, welding, or testing with gases may be carried out, suitable air/gas extraction equipment shall be fitted.

5.16.2 Workshop Electrical and I&C Workshop Equipment

Following Electrical and I&C workshop equipment shall be provided (as option) in the Proposal:

Item No.	Description	Q'ty
1	Bench Drill for drills 4 - 19 mm	1
2	Cordless Electric Drill	1
3	Impact Drill	1
4	Pedestal Grinder, 150 mm wheel	1

Item No.	Description	Q'ty
5	Work Benches	2
6	Work Bench Vices, size 150 mm	2
7	Soldering Guns / Heat Guns	2
8	Dial Gauge Indicators	2
9	Compound Dial Gauge	2
10	Vernier Caliper, 150 mm	1
11	Vernier Caliper, 300 mm	1
12	Digital Insulation Resistance Tester, 250/500/1000V	1
13	Ductors/Micro-ohmmeter	1
14	Earth Resistance Tester	1
15	Transformer Oil Test Sets	1
16	Transistor Circuit Checker	1
17	Digital Timer	1
18	Phase Rotation Indicators	1
19	Digital Tachometer	1
20	AC Ammeters, 0-1-5 A	1
21	DC Ammeters, 0-10-10 A with suitable shunt	1
22	DC Voltmeter, 0-150-300 V	1
23	AC Voltmeter, 0-250-500 V	1
24	Wattmeters, 110V, 1/5 A	2
25	Sets of Electricians Hand Tools	1
26	3-Phase Frequency Meter, range 55-60 Hz, 0-600V	1
27	Barometer	1
28	Digital Multimeter	1
29	Precision Multimeter	1
30	3-Phase Tester	1

Item No.	Description	Q'ty
31	Ohmmeter	1
32	Cramping Set	1
33	Clip-On Ammeter	1
34	Frequency Meter	1
35	Electronic Work Station	1
36	Hand-Pulled Trolley/Workbench	1
37	Micrometer Set	1
38	Pressure Calibrator	1
39	Vacuum Gauges and Manometers Sets	1
40	Deadweight Tester	1
41	Thermometer Set (Precision Type)	1
42	Dry Bulb/Wet Thermometer Set	1
43	Optical/Digital RPM Meter	1
44	Calibration Set for Turbine Supervisory Instruments Sensors	1
45	Portable Sound Level Meter SPL and dBA	1
46	Calibration Equipment for Particulate Materials	1
47.	Calibration Equipment for Conductivity Meter	1
48.	Portable Vibration Meter	1
49	Hart Communicators / or equivalent (if applicable)	2

5.16.3 Stores

A suitable store building shall be provided. The store building shall be equipped with HVAC equipment as described in. The storage facility shall be sized to accommodate a full range of spare parts for the Plant, as recommended by the Contractor, and indicated in its offer. The stores shall also retain spares to carry out recommended routine and scheduled servicing of the Plant. These items shall include, but not be limited to:

- General mechanical spares (fasteners, pipe fittings, etc.)
- Electrical equipment (lamps, fuses, etc.)
- Domestic goods
- Gases, suitably housed
- Operational equipment

Small items shall be binned in suitably shelved racks. Larger items shall be stored in an adequate area with lifting equipment to load and unload as necessary. All necessary bins, racks, shelves and lifting equipment shall be provided, to the Owner's approval.

Suitable heavy vehicle access shall be provided to the building with the necessary loading and unloading facilities.

The total area and arrangement for the stores building shall be agreed with the Investor.

5.17 Mobile Plant and Equipment

Except where stated otherwise in this specification, mobile plant such as fork-lift trucks, cranes, sewage tankers etc. required for the operation and maintenance of the Facility shall be provided by the Owner to a recommended list provided by the Contractor. All mobile plant required for EPC purposes, such as construction and commissioning, shall be provided under the scope of the EPC Contract.

5.18 Laboratory

The Contractor shall provide a fully equipped laboratory suitable for carrying out all testing requirements for the day-to-day operation of the Facility including, but not limited to:

- HRSG feed water;
- Steam;
- Potable water, including hydrocarbon content;
- Drain water, including hydrocarbon content;
- Fuel testing;
- Moisture analysis for lubricating and transformer oils.
- Fire detection and fighting system

The laboratory shall also contain suitable office accommodation for chemists.

The fully air-conditioned laboratory shall be equipped with counter flanges for demineralized water and for potable water supply from the local distribution systems. Operator desk and respective furniture shall be included along with suitable storage facilities.

Chemical and analysis of wastewater shall be collected and drained separately.

The Contractor will present the list of laboratory test and measurement devices and furniture to the owner for approval. Laboratory analysis benches will have a suction fan to absorb toxic gases.

There will be separate and suitable storage cabinets to prevent chemicals from interacting. All chemicals to be used in the laboratory will have traceability with the certificate of analysis.

Analysis devices that will meet ISO standards and are compatible with international certification will be selected.

Analysis devices suitable for field measurements will be available in the laboratory (PH, Conductivity Meter, SDI, Hydrometer, Validation Devices).

Automatic sprinkler system protection shall be required for all new laboratories handling chemicals as per NFPA13, Cl:22.8.1.

5.19 GT OEM proposal

GT OEM proposal is included as below.

BIDDER						
No.	Date Sent by Bidder	Bidder	Type of Question	Volume	Document Reference & Section	MPW reply (2024/7/30)
46a			General	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5. GENERAL TECHNICAL REQUIREMENTS (Whole section)	Supplier understands that requirements in this section are generally related to whole plant and the requirements should be considered by EPC contractor. Since, supplier supplies pre-engineered and standardized equipment, such as GT/Generator and associated auxiliaries, supplier understands that the requirements in this section are not intend for supplier's equipment. On the other hand, regarding the each requirement in this section, Supplier would like to list the following main examples of difference between these requirements and Supplier's standard design.
46b			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.1 Pipes System The EPC Contractor shall provide a piping specification for each piping system which shall describe the following: - Service description - Pipework material - Design pressure and temperature - Operating pressure and temperature - Pipe wall thicknesses including corrosion allowance - Materials - Pipe connection arrangements - Flange ratings and material groups - Principal valve types and materials - Hydrostatic test pressure	Supplier will submit required information for EPC's work. But, regarding the pre-engineered equipment, such as GT enclosure, documents and data related inside of the equipment, which will not be used for EPC's work, will not be submitted.
46c			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.1 Pipes System The corrosion allowance to be considered for piping shall be as follows: - Carbon steel: 1.5 mm - Alloy steel: 1 mm (0 mm can be allowed as long as design life criteria can be met) - Stainless steel: 0 mm - Plastic: 0 mm	In supplier's design, the corrosion allowance is not considered as fixed value. Instead, supplier select piping thickness which is more thicker than calculated thickness, and the difference between selected and calculated thickness is considered as margin for corrosion. For reference, as shown in followings, pipe thickness at enhanced cooling air compressor outlet is calculated for another project (detail engineering stage), and it has following margin. [e.g.1: Enhanced cooling air compressor outlet piping] - Pipe size: 150A - Material: Carbon steel - Calculated thickness (Required): 3.3mm - Selected thickness (Actual): 5.3mm <- minus tolerance and thinning by bend are considered. - Margin: 5.3 - 3.3 = 2.0 > 1.5mm (OTS requirement) Since corrosion allowance is not considered as fixed value, some parts of piping has lower margin than OTS requirement (refer to followings). But, supplier has rich experience with supplier's calculation method. [e.g.2: TCA Cooler inlet piping] - Pipe size: 450A - Material: Alloy steel - Calculated thickness (Required): 15.5mm - Selected thickness (Actual): 16.3mm <- minus tolerance is considered (non bended part) - Margin: 16.3 - 15.5 = 0.8 < 1.0mm (OTS requirement)
46d			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.1 Pipes System Piping shall be designed within the following flow velocities, unless otherwise agreed with the Owner. - Natural gas < 30 m/s - Compressed air and other gases < 10 to 25 m/s	Supplier has selected OEM standard velocity based on OEM's experience. For example, based on OEM standard, fuel gas flow velocity upstream of FGH is restricted at 30 m/s, but fuel gas flow velocity downstream of FGH is restricted at 40 m/s.

46e			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.1 Pipes System All pipe systems shall be arranged, installed, supported and provided with all necessary means of venting, draining and expansion subject to the approval of the Owner in accordance with the Contract before erection commences. The space between pipes shall be sufficient to allow easy repair and/or	Inside GT enclosure, FG/FO/PA unit, MFCP/WTP/ECA compressor skid, etc. and supplier's connecting piping such as lube system, cooling air system etc. are based on supplier's pre-engineered design. So, supplier request the Owner to accept supplier's pre-engineered design without approval.
46f			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.2 Materials As a general rule pipe and valve sizes 1/4", 1-1/4", 3-1/2", 5", 7" and 22" will not be used.	The sizes of 1-1/4", 5" and 22" are used in supplier's standard design. For example, 22" is used at connecting piping between GT and GT cooling air cooler, and the pipe size 22" cannot be changed because it is decided based on cooling air flow design calculation.
46g			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.2 Materials Bolts, nuts, washers for flange joints shall be hot dip galvanized as applicable, except for bolts/nuts to be used in high temperature (bichromated).	Supplier will use stainless steel bolts in control air system and lube oil system partially as OEM standard design.
46h			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.3 Valves Valves, valve drives and accessories, which are of a similar make, size and type, shall be standardized as much as possible in order to facilitate interchangeability and maintenance and limit the stock of spare parts.	Supplier does not select valve size and type considering facilitate interchangeability and maintenance and limit the stock of spare parts.
46i			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.3 Valves All valve bodies shall be of the same nominal size as the adjacent piping, unless otherwise required. The internal diameter of valve ends adjacent to the pipework shall be the same as the internal diameter of the connection row. http://www.flubb.com/blogs	In many cases, valve sizes and adjacent piping sizes are not same nominal sizes, since sizing method of valve and piping are different as follows; Valve: Calculation of Cv or Cg value Piping: Calculation of flow velocity
46j			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.3 Valves Valves shall be installed maintainable from the outside (field repairable). After disassembly of actuator (regardless of manual or motor actuator), valve position must be readily identifiable.	Fuel gas and fuel oil flow control valves are not allowed to be disassembled on site by valve vendor, since the valves are electrical and mechanical precision products. So, the valves must be sent back to valve vendor for maintenance.
46k			Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.3 Valves Instrument isolation valves shall be ball or globe valves.	Instrument isolation valves shall be gate or globe valves.

46l		Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.3 Valves	The opening percentages (referred to 100% full stroke) shall be between: a) Minimum flow >10% b) Normal Flow 30-60% for linear curves, 30-75% for iso-percentages. c) Maximum flow <75% for linear curves, <85% for iso-percentages.	The operating range of control valves are based on recommended range of valve vendors and supplier's experience.
46m		Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.7.3 Valves	When the valve is also used for isolation purposes, class V (ANSI/FCI 70-2) in general and Class VI for "Tight shut off" (TSO) application shall be provided.	There are class IV valves in supplier's system (e.g. fuel gas flow control valves). But, in fuel gas system, fuel gas shut off valve with class V is used in addition to above fuel gas flow control valves.
46n		Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.8.3 Welding Procedures	Gas turbine interconnecting piping will be shop welded according to ASME B31.1 and all relating nondestructive testing will be performed according ASME B31.1. Test documentation related to these welds shall be submitted to Owner.	Based on ASME B31.1, gas turbine interconnecting piping will be shop welded as much as possible. But, for the equipment (e.g. gas turbine) which has dimensional tolerance when it is installed on site, the interface between the piping and the equipment should be welded at site in order to adjust the misalignment. Regarding the documentation related to the welds to be discussed at detail design stage.
46o		Mech	Appendix A2-4_Rumah and Nairiyah IPP OTS Part B Rev 0	5.8.6 Inspection of Welding	After the appliance of the PWHT on the G91, G22 and G11 an hardness test must be performed as follows: -100 % on Base Metals, Weld and Heat Affected Zones(3 identations in each area)	When using G22 and G11 (G91 is not used by supplier) steel plate for welding, 100% hardness test does not implemented after the appliance of the PWHT. But, as noted in supplier's ITP, by conducting non-destructive examination and visual examination based on ASME B31.1, supplier confirms the quality of supplier's equipment.

6. DESIGN AND OPERATIONAL REQUIREMENTS

Each Plant shall have GNCPC of 1800(±5%) MW net on natural gas and when operated on back-up fuel the net power output shall follow Appendix C in the EPC Contract. -The Plant shall comprise of Power Blocks consisting of Gas turbines, HRSGs and Steam turbine. The additional power capacity achieved by SF must not exceed 10% of the Contracted Capacity.

Refer to section3 and 4 of OTS Part A.

6.1 Plant Configuration

The Plant shall be designed as a Combined Cycle Power Plant, consisting of but not limited to:

- Dual Fuel Gas Turbine (GTs)
- Dry low NOx combustors;
- Heat Recovery Steam Generators (HRSGs),
- By-pass stack including diverter dampers horizontal and vertical sealing guillotine damper.
- Condensing Steam Turbine (CST) with Air Cooled Condenser
- Synchronous Generators for Gas Turbines and Steam Turbine
- Generator Circuit Breakers for GTGs and STGs
- 2 winding Generator Step-up transformers;
- 2 or 3 winding Auxiliary transformers
- HV Circuit Breakers for GSU Transformers of GTGs and STG

The plant shall be designed preferably symmetrically; this means the Plant shall consist of equally designed Groups (same number and type of GTs/HRSGs/ST).

6.2 Design Conditions & Requirements

In order to achieve an optimum cost solution, the final configuration and number of power and steam generation units to be proposed shall be based on the contractor's own optimization by taking into consideration the type and size of gas turbines available in the world market for meeting the required power generation capacity.

6.2.1 Air Cool Condenser

The condenser shall be capable of condensing the whole of the steam exhaust from the steam turbine (including the bypass systems) under all conditions of load and operation. For this the EPC Contractor shall provide with his proposal supporting calculations. Additionally, supporting calculations demonstrating that the wind direction at the site will not impact the performance of the ACCs shall be provided. The ACC location shall be selected so that it will not be in the predominant wind direction.

The systems and components for the ACC shall be provided by the Contractor and shall include but not limited to the following:

- Condensate extraction pumps and strainers;
- Vacuum raising and maintaining system;
- Turbine blow off tank;
- Condensate, air extraction, recirculation, air, drain and atmospheric exhaust pipework and valves;
- Vent pipework routed to a safe location;
- Flash boxes (with integral connections to condenser) for feed heater and other drains;
- Heat exchange sections;
- Support structure and fan deck;
- Steam duct / distribution system;
- Condensate collection system;
- Condenser air removal system;
- Cooling air fans with automatic controls
- Condenser atmospheric relief devices;
- Holding down bolts, plates, etc;
- Local ladders, galleries and hand railing;
- All driving motors with base plates, or stools for all auxiliaries;
- Local instruments, gauges, control and panels;
- Additional tapping points, complete with isolation and all necessary instruments for performance testing;
- Any special lifting tackle, spanners and tools;
- All other equipment required for the safe and efficient operation of the plant.

The exhaust steam from the Steam Turbine exhaust shall be directed to the air cooled condenser and condensing system via a steam piping and ductwork system; the water will then be forwarded to the condensate system.

The condenser and the cooling system, at a minimum, shall be capable of:

- i. Condensing the total steam flow from its associated steam turbine operating at maximum flow and at maximum ambient temperature;
- ii. Accepting the total steam system drain flows;
- iii. Condensing the full capability of the steam turbine bypass system(s); and
- iv. simultaneously operating with steam flow through the steam turbine and the steam turbine bypass system.

The condensing plant shall be designed and guaranteed to condense the maximum exhaust steam flow from the Steam Turbine; to accept the relevant steam system drains; and to condense the dump steam flows from

the IP and LP bypass systems under conditions of turbine start-up, rapid load reduction, partial or full load rejection and turbine trip.

The ACC shall be provided with N+1 fans where N refers to the number of fans required for the maximum load condition. There is no requirement to provide the ACC equipped with elevator.

The condenser vacuum shall be optimised by the EPC Contractor.

The steam turbine outlet connection pieces shall be designed for smooth transition with minimal obstructions from the LP turbine exhaust to the condenser to assure negligible pressure drop.

The turbine bypass dumps shall be designed with dispersion headers internal to the condenser to adequately disperse the steam safely into the condenser for all possible extreme transient scenarios.

The Air-Cooled condenser materials shall be selected to provide the necessary reliability at an economical cost taking into account that ACC is being operated with Demineralized water with proper water chemistry required by OEM guidelines. ACC OEM Design will take into consideration of proper material selection as per Demin water quality..

Vacuum raising and maintaining plant may be based on either steam ejectors or mechanical vacuum pumps. As a minimum, one start-up (hogging) duty and one running duty device shall be provided for each condenser.

Steam for air ejectors, if supplied, shall be taken from the main steam supply. A silencer shall be provided on each ejector outlet.

The hotwell tank and condensate system shall be sized in accordance with HEI standards.

The condensate system shall be equipped with instrumentation that will assist in trouble shooting of any condenser tube leakage.

6.2.2 Fuel Supply

The gas turbines of the Plant shall be designed for 100% dual fuel capability with Natural Gas as the primary fuel and liquid fuel (diesel) as back-up fuel and without any limitation on the duration of operation on either fuel. Plant fuel systems and control systems shall be optimized and have sufficient redundancy to ensure the maximum availability of the gas turbines including during a changeover between fuel types.

6.2.2.1 Natural Gas

The Fuel Connection Facility for gas will be located so that the area is directly accessible by Saudi Aramco from the outside without using the main gate of the Plant.

The EPC Contractor shall supply all civil works required for gas station according to the requirements for Rumah and Nairyah. This gas station shall be of sufficient size for any receiving/reducing/booster pumping installations along with the required metering necessities, I&C and electrical requirements.

The design for the gas installations shall be carried out in close coordination with SPPC and Saudi Aramco

6.2.2.2 Back-up Fuel (If Applicable)

The EPC Contractor shall connect the Back-up Fuel system of the respective plant of Rumah or Nairyah IPPs or both to the Back-up Fuel tank farm.

Each of the Back-up Fuel oil tanks shall be placed on reinforced concrete ring foundations and be located inside an oil and watertight tank basin of reinforced concrete slabs and surrounding walls with respective oil proof coating.

In addition, the EPC Contractor shall supply all civil works required for Back-up Fuel delivery by road tankers through an unloading station located adjacent to the Back-up Fuel storage tanks within the plant boundaries.

6.2.3 Plant Performance

The Power Plant shall be designed and sized for a total net capacity of 1800±5% MW at reference site conditions on natural gas and when operated on back-up fuel the net power output shall follow Appendix C in the EPC Contract. However, the capacity of each gas turbine shall not be limited below maximum gas

turbine output at full opening of GT compressor inlet guide vane during power block operation with single gas turbine, to compensate for GT degradation and during simple cycle operation.

The configurations as proposed by Contractor shall be based on reaching the lowest Guaranteed Contracted Net Heat Rate (GNCHR) while maintaining maximum possible GNCPC to be demonstrated at COD in conditions as defined in EPC Contract.

In order to increase the competitiveness of the proposal, Contractor shall consider:

- To limit its contractor margins and uncertainties to be applied for the purpose of establishing its GNCPC and GNCHR at Early Power Unit Commercial Operation Dates and PCOD;
- the application of triple pressure reheat HRSG with/without supplementary firing;
- the application of evaporative/fogging/wet compression system in GT;
- optimizing the HRSG exhaust gas temperature and gas side pressure drop, while considering proper means to reduce the risk of excessive corrosion;
- optimizing the feed water system heat recovery performances such as steam & water side pressure drops, pinch point, approach point;
- consider minimum Blow-down (0%) for the purpose of defining the GNCPC ;
- optimize the step-up transformers as to limit efficiency losses;
- minimize the auxiliary power consumption by;
 - use of best estimate auxiliary power consumption for setting the Contractor GNCPC.
 - Considering variable speed/hydraulic coupling for feed water pumps, and gas compressors (if applicable);
 - Minimizing building free volumes to avoid excessive HVAC
 - Use of Photo Voltaic panels for plant energy consumption.
 - Use of the best estimate for the equipment and Plant degradation.
 - Optimizing the margins on the design and sizing.
- Minimizing heat losses over the bypass stacks by providing air seal cooling ;
- means to increase the overall efficiency of the steam cycle;

6.2.4 HRSG design and output degradation

The HRSG shall be designed and sized based on the Design Point Performance (to be chosen at a later stage) operating on natural gas specified on OTS Part A Annex 4.1-1 Fuel Gas Specification.

The HRSG shall be equipped with cold end condensate recirculation system and condensate bypass arrangement. The HRSG shall achieve the desired plant performance at all the operating points.

In case of cold casing design, the HRSG design shall incorporate adequate hot & cold end corrosion protections. Hot and Cold end corrosion protective measures shall include Protective coatings on the casing and protective layers (Teflon or other suitable corrosion protection layers) to minimise the undesirable effects of sulphuric acid corrosion. Alternatively, hot casing design for HRSG shall be provided for hot and cold end corrosion protection. The corrosion protection methods employed shall be having proven commercial operational experience and shall be approved by the OWNER.

To satisfy the operational requirement as mentioned in OTS Part A. In order to model the degradation due to aging, the following information shall be supplied for HRSG for years following post COD starting from guaranteed values, but applying best estimate degradation in addition to submitting the long-term best estimate degradation curve or factors:

- Enthalpy of the exhaust gas
- Mass flow of the exhaust gas
- Temperature of the exhaust gas

- Exhaust Gas composition
- The pressure, temperature and steam flow of each pressure level

HP steam, after expansion in the HP ST, can be equally split over 2 or 3 HRSGs for reheating, without creating unbalances in the respective HRSGs (if reheat system is applied).

Condensate return temperature shall be above dew point temperature, to minimize corrosion risk.

6.2.5 HP/IP-LP design

HP/IP-LP bypass shall be 100% capacity.

- To cover the loss of ST with GT at full load (transient situation without switch over to open cycle mode)

6.2.6 Other

Essential and non-essential HVAC shall be separate systems. Essential HVAC shall be maintained during blackout start and safe shutdown conditions.

6.3 Emissions & Environmental Requirements

Emissions control system to meet the requirements of IFC guidelines, World bank, Equator Principles III (2013) and Saudi Arabia requirements. The gaseous emissions from the plant shall not exceed the limits stated in section 4.1.9 of the OTS Part A. The EPC Contractor shall submit all calculations and design data as appropriate to demonstrate full compliance with the Royal Commission requirements.

6.3.1 General

The Contractor will be required to design and construct the Plant in accordance with the applicable environmental laws and standards. The Contractor is also responsible for obtaining and maintaining all the necessary Permits for the construction of the Plant.

An Environmental Impact Assessment Study (EIA) will be completed by Owner and copy of the same will be made available to Contractor. A Preliminary Environmental Assessment ("PEA") or Environmental and Social Impact Assessment ("ESIA") was conducted by the Offtaker and is provided in OTS Part A, Annex 3.3-1 of Part III (Drawings and Diagrams).

It is anticipated that the Project's development will necessitate the involvement of international lenders for financing. Given this, the Project's assessment will need to account for the requirements of international lenders in addition to the local regulations. With the specific international lenders currently unknown, the key requirements, which have been considered within this Scoping report, comprise the Equator Principles III (2013) and associated IFC Performance Standards (2012). The Project will be expected to comply with the most stringent standards should national/IFC requirements differ.

The Owner will progress the EIA Study with the intention that the Environmental Permit is available soon after the time of Contract Award. Contractor shall be responsible for providing all the necessary technical and environmental inputs to the extent required for the EIA study. After Contract Award any amendment to the EIA Study and the Environmental Permit will be the responsibility of the Owner and the EPC Contractor.

Waste management during construction will be in accordance with the Waste Management Plan developed specifically for this project and in accordance with Government and Company approved plan.

Hazardous Area Classification drawings shall be developed in accordance with the latest revision of ANSI/API RP-505 and Company Standard SAES-B-068 during the PP Phase. The Area Classification drawings shall be prepared from the project plot plans and the process schedule of hazardous materials to determine the potential sources of hazard.

All areas of the where concentrations of gas may occur due to normal leakage or process fault or ventilation system failure shall be classed as hazardous locations. The Contractor shall calculate and produce area classification drawings and documentation for the whole site area. The classified area shall be as defined within the relevant regulations and standards. All equipment within each area shall be uniquely identified and be fully compliant with the standards and regulations.

In addition, all relevant national environmental laws and regulations of Saudi Arabia are to be complied with.

The Project designs shall ensure compliance with the most stringent source emission control parameters.

6.3.2 Air Pollution

The Project designs shall ensure compliance with the most stringent source emission control parameters. The maximum permissible air emission levels, which are to be met within the specified temperature range between minimum stable load (which is expected at about 30% of plant load) and maximum continuous rating of each gas turbine are shown in the following table:

Emission	Unit	Gas	Backup Fuel
NO _x (as NO ₂)	mg/Nm ³ (dry, 15% O ₂)	18 ^{2) 3)}	18 ^{1) 2) 4)}
CO	mg/Nm ³ (dry, 15% O ₂)	50	50
SO ₂	mg/Nm ³ (dry, 15% O ₂)	(According to Sulphur content in the fuel)	
Smoke Density	Bacharach	2	2
Reference - O ₂	Vol.-%	15	15

Notes:

- 1) NO_x emission for Back-up Fuel is based on maximum fuel bound nitrogen content of 0.015% by weight.
- 2) This value of 18 mg/Nm³ (or 9 ppmv) applies for combined cycle operation.
- 3) In case open cycle is implemented, this value may be exceeded up to a value of maximum 50 mg/Nm³ subject to the approval by the competent authorities
- 4) In case open cycle is implemented, this value may be exceeded up to a value of maximum 150 mg/Nm³ subject to the approval by the competent authorities

Emission guarantees are given for a daily average and for steady state operation and normal loading and deloading transients. Emissions guarantees will not apply to grid code compliance or frequency following transients.

The values in above table refer to the flue (exhaust) gas in dry condition, 0 °C temperature and 1013 mbar atmospheric pressure.

Emissions control system to meet the requirements of IFC guidelines, World Bank, Equator Principles III (2013) and Royal Commission requirements.

The gaseous emissions from the plant shall not exceed the limits as specified in the above guidelines and as per Environmental and Social Impact Assessment (ESIA) study report.

The emission guarantee of gas turbine is suggested for purpose of SCR design:

<Natural Gas firing>

-Ambient temperature is more than 15 degC: from 60% to 100 of the gas turbine load.

-Ambient temperature is less than 15 degC: from 80% to 100 of the gas turbine load.

<Back up Fuel firing>

-Ambient temperature is more than 15 degC: from 60% to 100 of the gas turbine load.

-Ambient temperature is less than 15 degC: from 75% to 100 of the gas turbine load.

6.3.3 Noise Levels

All plant shall run without undue vibration and with the least practicable amount of noise.

The Plant shall be designed to achieve all the relevant noise standards and guidelines. As a minimum the Plant shall be designed to meet the requirements as per OTS part A section 4.1.9.5.

The noise levels shall be measured according to ISO standards.

To meet the noise level requirements the Plant is required to be equipped with standard noise attenuation features. All machinery which exceeds the permissible noise limits shall be equipped with silencers or with lagging or specially designed acoustic enclosures.

6.3.4 Water Pollution

The Contractor shall take account of applicable environmental standards and regulations for wastewater re-use and discharge. Effluent discharge limits applicable to the Plant shall be as per EIA recommendations.

Permissible effluent emission limits for aqueous discharges into the soil are shown below:

Constituents	Unit	Allowable effluent Level
Physio-chemical pollutants Floatable	-	None
pH	-	6-8.4 pH units
Total suspended solids (TSS)	mg/l	35
Turbidity		max.5 NTU
Bio-chemical oxygen demand	mg/l	25
Chemical oxygen demand	mg/l	150
Total organic carbon(TOC)	mg/l	50
Total kjeldahi nitrogen(TKN)	mg/l	5
Total chlorinated hydrocarbons	mg/l	0.1
Oil and grease	mg/l	0
Phenols	mg/l	0.002

Non-organic pollutants	Units	Allowable effluent level 30 day average (mg/l)
Ammonia as nitrogen	mg/l	1.0
Arsenic(As)	mg/l	0.1
cadmium(Cd)	mg/l	0.1

Chlorine(residual)	mg/l	0.5
Chromium, total (Cr)	mg/l	0.1
Copper (Cu)	mg/l	0.4
Cyanide (eN)	mg/l	0.05
Lead (pb)	mg/l	0.1
Mercury (Hg)	mg/l	0.001
Nickel(Ni)	mg/l	0.2
Phosphate (total as Phosphorous)	mg/l	1.0
Zinc(Zn)	mg/l	4.0
	mg/l	
Biological pollutants		Allowable effluent level 30 – day average
Total coliform		2000 MPN per 100ml

Emission limits for aqueous discharges shall be stated by the contractor and must be agreed with the RC. This shall be in line with the environmental and social impact assessment and RC permitting. Emission limits for aqueous discharges shall be stated by the contractor and must be agreed with the RC. This shall be in line with the environmental and social impact assessment and RC permitting. In order to achieve the above mentioned permissible effluent limits wastewater treatment systems are required, which shall include an oil drain, a chemical drain and a sewage treatment plant. These wastewater treatment facilities shall be designed to cater for the entire wastewater of the Plant.

The sewage treatment plant and its ancillary systems shall fulfil the requirements to collect, treat/clarify and discharge the domestic and sanitary waste waters streams. The industrial wastewater shall be treated separately in adequate system. Both, domestic and industrial wastes consist of many compounds which could contaminate surface water) groundwater and soil. Furthermore, especially in the case of discharging untreated sanitary wastewater, the bacteriological contamination can result in the occurrence and spread of water borne diseases like cholera, dysentery, typhoid, etc. Therefore, special attention has to be paid by the Contractor during its investigation of the wastewater collection area to ensure that all waste water streams are considered and will be segregated, collected and discharged in a wastewater collecting system. This also includes all the sanitary and other domestic wastewater out of all permanent and temporary site facilities and the industrial wastewater out of the process like backwash water, rinse water, chemical and acid cleaning wastes and other process wastewater which cannot be used in other ways (blow-down water, etc.).

The wastewater shall be delivered to the wastewater treatment plant in sewers or pipes made of customary and proven materials in adequate diameters according to the amount of wastewater conducted in the pipes. In the case of geodetic constraints, pumping stations must be installed in adequate numbers. In the case of using cesspools, intermediate storage is allowed. The cesspools must be emptied by vacuum loader(s), which are also part of the scope of supply.

The collected wastewater should be treated in accordance with the above effluent limits.

In order to meet the required quality of clarification, sanitary and domestic wastewater streams should at least be treated by screening and primary sedimentation, biological clarification in activated sludge basins, sedimentation for removing the activated sludge and as a last stage an effluent disinfection for discharging

the wastewater into the sea. The treated sanitary wastewater should also be used for irrigation purposes if required.

All wastewater streams discharged into the domestic waste water treatment plant shall be free of toxic and inhibitory compounds concerning the process of biological clarification or compounds that can influence the further use of the sewage sludge.

The sewage sludge should either be treated or dried for use as fertilizer or disposed of in a suitably designed landfill according to US EPA/EU specification.

In order to meet the required quality values, industrial wastewater should be treated separately with special regard to the parameters listed in Table above.

Special regard is to be paid by the Contractor to the contamination of the several streams and the adequacy and suitability of treatment process wastewater streams with similar contamination can be summarized and mixed before treatment. Oil contaminated water have to be treated in oil separators.

All chemical polluted wastewater (coming from storage areas, chemical cleaning, acid cleaning) has at least to be neutralized and detoxicated.

6.3.5 Solid and Liquid Waste Management

The Project shall adhere to local solid waste management guidelines and implement the requirements of WBG for solid wastes and liquid wastes. Oil can become contaminated by oil and chemical spillage during construction, operation and maintenance of the Plant. The entire Plant shall be designed, constructed, operated and maintained in such a way that any soil contamination will be prevented.

The WBG standards liquid effluents originating from thermal power plants are summarised in Table below.

Effluents from Thermal Power Plants (IFC, Thermal Power Guidelines 2008)

Parameter	Maximum Value (Unit)
pH	6-9 (pH)
TSS	50 (mg/L)
Oil and Grease	10 (mg/L)
Total Residual Chlorine	0.2 (mg/L)
Chromium (Total)	0.5 (mg/L)
Copper	0.5 (mg/L)
Iron	1.0 (mg/L)
Zinc	1.0 (mg/L)
Lead	0.5 (mg/L)
Cadmium (Cd)	0.1 (mg/L)
Mercury (Hg)	0.005 (mg/L)
Arsenic (As)	0.5 (mg/L)

The plant shall be designed, operated, and maintained in such a way to prevent any soil contamination or impact on the terrestrial ecosystem. This shall be in line with the environmental and social impact assessment and RC permitting.

6.4 RAM (Reliability, Availability, Maintainability)

The Plant design and configuration shall provide suitable equipment size and number to ensure a reliable supply of power.

6.4.1 Introduction

This section describes the Owner technical specification for the RAM (Reliability, Availability, and Maintainability) modelling of the plant's equipment and systems. A set of RAM models are required to support and validate process decisions and equipment selection.

6.4.2 Objectives

The RAM modelling effort has several objectives based on the activity's timing and the level of detail incorporated in the model.

The main goals of the RAM assessment shall be:

- To test the capacity assumptions used in the initial design documents, at the unit and plant level, and identified equipment, systems or other factors contributing to the generation losses
- To calculate the design inherent reliability of the plant represented by electrical power continuity of supply reliability, based on the final design of the plant, including a sensitivity analysis in order to determine the effect on reliability output changes in the different input parameters
- Definition of conditions or assumptions under which that calculated reliability can be maintained as representative of the "reliability of the plant operation". Especially the related considerations of aspects, additional to design, that affect to the final achievable reliability like those which would be in the scope in the Operation & Maintenance of the plant stage.

6.4.3 Model Scope

The RAM assessment shall consider all systems, packages, sub-systems and equipment associated with all generating units of the plant and their utilities/ancillary systems/equipment required to maintain their operational status. The assessment will be divided in each unit and facility.

Finally, results of each specific unit and facility will be combined to assess overall reliability of the Plant. In the assessment, all relevant aspects that affect the expected reliability of the Plant shall be considered.

All systems that affect to electrical generation of the plant should be considered in the RAM assessment (to be discussed between the parties).

6.4.4 Study execution

Each unit model shall be developed using the following work process:

1. Overall Scope Definition
 - draft study execution plan
 - define equipment configurations
 - define production basis for units and overall generation
 - define the level of detail for each phase
 - Identify probable sensitivity cases for comparison with base model
2. Develop Unit models
 - Build Reliability Block Diagrams for each unit (RBDs)
 - Populate model with equipment failure data
 - Populate model with planned events (shutdowns, planned maintenance, etc.)
 - Run/debug unit model
 - Verify and validate the model with the project team

3. Analyse and Review

- Identify bottlenecks and constraints within the unit model
- Define and run comparison cases
- Analyse base case and comparison case results
- Define final model configuration and recommendations

6.4.5 Technical requirements

The RAM assessment to be developed by the EPC Contractor and shall include the following:

- The RAM software to be employed to perform calculations shall be based on Monte Carlo methodology and performed using Isograph.
- International standard EN-60812 or similar shall be considered.
- Establish a Base Case Model with all assumptions, including equipment reliability data used in the assessment.
- Prepare Reliability Block Diagrams (RBDs), including but not limited to configuration, sparing, and sizing arrangements. The RBD shall accurately model the system failure logic and include all redundancy in each specific system.
- Prepare a list of equipment or systems considered in the model with the Failure Mode and Failure Rates included in the models (MTTF, MTTR).
- Prepare a Base Case Model analysis, providing recommendations for the improvement of overall reliability. Perform sensitivity analyses, for the recommended parameters or equipment or for the client chosen parameters after analysing the first base case draft.
- Model uncertainty related to equipment failure, to prevent potential effects on the process and identify appropriate actions that will lead to reduced consequences
- RAM analysis shall be developed 'top-down' and expanded to lower levels.
- Expected average Plant availability and reliability
- Identification of all potential critical components and process bottlenecks
- Identification of failure modes and their consequences
- Decision support to determine the most cost-optimal solution to improve reliability results with respect to:
 - Process design
 - Spare part philosophy
 - Mobilization times and contracts
 - Preventive maintenance strategy
 - Identification of cost-effective measures
 - Knowledge of industry performance methodology

6.4.6 Facilitator Qualifications

The EPC Contractor shall resource the activities with a qualified facilitator and model operator. The RAM facilitator shall:

- Have a relevant reliability certificate such as CRE (certified reliability engineer by ASQ)
- domain industry experience (relevant experience in power plants operation and maintenance)
- hands-on experience in using the simulation solution platform.

6.4.7 Deliverables

- During the Design Stage, the model files will be retained by the EPC Contractor and updated as needed. Following Design Stage completion, model files shall be made available to Owner for future use.
- The RAM study deliverable for Design Stage consists of a report with attachments documenting the cases run, results obtained, and recommendations made.
- List of conditions or assumptions under which that calculated reliability can be maintained as representative of the "reliability of the plant operation", especially the related considerations of aspects additional to design that affect the final achievable reliability, like those which would be in the scope in the Operation & Maintenance stage of the plant.

EPC shall also provide the information of all installed equipment at the plant according to the templates available in Schedule E Equipment Data Templates.

6.5 Redundancy Concept

An appropriate level of redundancy shall be included in each auxiliary system. The Plant shall be designed so that no single auxiliary equipment failure or trip (except main equipment) shall result in the reduction of power capacity. In this context, it is required that all critical auxiliary equipment shall be provided with N+1 configuration. Station (common) auxiliaries shall include similar redundancy such that no single equipment failure shall result in any loss of Plant generating capacity. Main equipment refers to all GTs, HRSGs, STs, GRMS, condensers, generators, main transformers, aux. transformers, major electrical equipment, cables, pre-treatment. Redundant backup equipment shall start instantaneously to enable continued functioning of the process without consequent malfunction.

6.5.1 Mechanical equipment

All pumps and compressors shall be redundant with on-line switchover capability with following redundancy:

- Pumps, blowers, compressors, fans : N+1
- Lube oil pump for HP Feedwater Pump : 2x100% (for each FW pump)
- Lube oil heater for HP Feedwater Pump : 2x100% (for each FW pump)
- Power Plant Jockey pumps : 2x100%
- Heat exchangers (not applicable to fin fan cooler or to air cooled condenser). Redundancy should be considered to eliminate single point of failure that will result in reduced load.
- Cooling tower/fin fan cooler cells : N+1
- Tanks (if not mentioned hereunder) : more than 1
- CCW Expansion Tanks : 1 x 100%
- Atmospheric drain tank : 1 x 100%
- Vacuum Flash Tank : 1 x 100%
- Condensate Storage Tank : 2x50% or 1x100%
- Service air receiver tank : 2 x 100%
- Instrument air receiver tank : 2 x 100%
- H2 Tanks : 2 x 50%
- Diesel storage tank : 2 x 50%
- Gas Drain Collection Tank (GT Area) : 1 x 100%
- GT Fuel Gas Last Chance Filter (per GT) : 2 x 100% Chemical tanks
- DM plant : N+1 stream

- Air conditioning : N+1 (except for non-plant buildings, such as: Administration building/ workshop/warehouse/Gate guard house)
- Ventilation Fans : N+1 (except for non-plant buildings, such as: Administration building/ workshop/warehouse/Gate guard house))
- Gas reducing station : 3 x 50%
- Feed water control stations : 2 x 100%
- Fuel Gas Boosting Compressors: : N+1, where, N=Number of Gas Turbines
- Air compressors with after coolers (after coolers if required, will be confirmed during detail engineering) : 3x50%
- Air filtering skids : 2 x 100%
- Air dryers : 2 x 100%
- After Filters : 2 x 100%
- CO2 rack & manifold station for each GT : 1 x 100%
- CO2 rack & manifold station for each ST : 1 x 100%
- N2 rack & manifold station : 1 x 100%
- H2 generator : 2 x 100%
- Drain and vent valves : two valves in series (If operating pressure is greater than 4 MPa) otherwise single valve
- Level control valve for HRSG Drums and Feedwater tank : N+1 (not applicable for startup line, as applicable)

Provision of bypass valves for control valves shall apply for those systems/facilities which can also be reasonably controlled manually.

6.5.2 Metering System

- Power: Main and Back-Up Power Metering Equipment (Main and Back up Tariff Metering shall be at the HV side of the Transformer (Delivery Point) and meet the requirements of OTS Part A. GTG OEM shall provide the performance metering system.
- Gas: Main and Back-Up Gas Metering Equipment

6.5.3 Electrical System

- No single main equipment (GTU, STU, GSUT and HRSG) failure shall result in a Forced Outage of the entire Plant.
- No single auxiliary equipment (all other equipment than main equipment as specified above) failure shall result in the Forced Derating of Plant. The compliance with the above criterion requires to a large extent redundancy of the entire auxiliary equipment. Following examples shall illustrate this requirement:
- The Plant shall have a redundant power supply concept for the auxiliary services (redundant UATs, MV - distributions, LV- distributions, DC-system, UPS- system)
- Electronic devices of the protection systems must be redundant in case process redundancy is not available, while the requirement doesn't apply to MV&LV protection relay.
- Redundant active equipment shall be powered from separate buses and physically segregated where feasible. The system design shall be such that the standby equipment shall be activated automatically when the control system detects a failure of the running equipment to perform its intended function.
- GT MCC shall have the redundant / double incoming CB, and divided to two buses, normal bus and essential bus bar, with bus-coupler.

- The capacity of MV, LV bus sections and the sources feeding the bus sections should be adequate to take the load of the other bus section when the bus coupler closes in case of supply failure to other bus section.
- The n-1 redundancy criteria shall be maintained for each switchboard/MCC.
- Redundancy shall be provided for cooling fans such that with one group of fan or oil pump out of service the transformer would deliver the rated output within the specified temperature rise.
- Battery and battery chargers shall be provided with N+ 1 redundancy. DC System contains 2 x 100 % Battery and 2 x 100 % Battery Chargers to meet the redundancy requirements.
- UPS shall be provided with N+ 1 redundancy.
- Each LV system switchboard shall consist of two bus sections with two incomers and one bus- coupler. Each incomer shall be supplied by 1x100% capacity transformer.
- Diesel Generator shall be provided with N+1 criteria, where EDG cum BSDG concept to be adopted. The DG sizing shall consider all possible scenarios, including emergency shutdown during black start operation. Emergency cum BSDG DG sets shall have the capability of testing even if respective block is in normal operation OR when the respective block switchgear bus have no voltage. 1 x 100% battery starting system will be applied for BSDG.
- All electrical system switchboards shall be labelled for Hazard Risk category (HRC) as per results of ARC Flash assessment study using NFPA 70 or relevant IEC standards (as applicable).
- For Gas turbine package, redundancy concept of LV & DC distribution for equipment supplied by OEM would be of OEM's standard design.

6.5.4 Control instrumentation system

Redundancy for control & instrumentation shall be as given below:

- All redundant instrumentation shall have the hot swap capability. Critical Redundant equipment shall be change over automatically in case of failure one equipment / Instrument. (For Gas turbine package, the hot swap capability is not applicable due to safety reason. Especially, it is not allowed to enter GT enclosure during operation, hence any person cannot access the instruments in GT enclosure.)
- Electronic modules in GT control system shall not be removed while the gas turbine is in operation.
- In general, devices shall be preferred that contain intrinsic safety, to avoid expensive supervision/monitoring installations, possibly subject to failure (In Gas turbine package, explosion-proof type instruments will be used in the hazardous area.)
- Where redundant equipment is installed, it shall be possible to perform maintenance and repair works on one equipment while the others are in service.
- Hardware part of controls & instrumentation:
 - Turbine GT/ST protection system (fail-safe and dual redundant system), For Gas turbine package GT Protection Panel which will consist of Tricon CX triple modular redundant safety PLC system by Schneider Electric Systems as per Supplier's standard design is considered.
 - HRSG-protection (fail-safe and triple modular redundant for BMS)
 - Triple redundant transmitters for Plant protection signals such as steam drum and main steam parameters.
 - Critical measurements, which are supplied with redundant instrumentation shall be connected to different I/O cards.
- Software Protections:

SIL study shall be performed by the EPC in order to consider the voting logic function but not limited to the following:

- All protection signals leading to a trip of the Plant or to a load reduction shall use 2 out of 3 voting logic. For Generator Protection System, signal shall be achieved through 1 out of 2 with two multi-function protective relays shall be provided for generator.
- Other controls can be achieved through 2oo2 or 1oo2 voting as per OEM's recommendations.
- Protection for people and environment shall be considered in SIL study according to IEC 61508 and IEC 61511. Allocation of the safety related function (the SIL loop) and its SIL level will be determined according to result of the SIL study.
- All control panels, control systems shall be provided with redundant control supplies. Control and Protection Panels shall be fed from either two redundant DC Source or redundant AC Source (UPS). So, in case of failure one source, the second standby source will take over automatically and will not have any power interruption. (For Gas turbine package, the power supply of GT control system will be fed from essential AC and DC system as per Supplier's standards and power supply for the Gas turbine equipment and CCR will be fed from UPS.)
- Unit protection shall preferably be independently implemented in separate hardware to the control system(s) to avoid common mode failure and with appropriate redundancy to ensure security of operation.
- For redundancy provisions of equipment, no single fault within the control system shall cause the failure of the duty equipment and at the same time cause the standby equipment to be unavailable.
- Redundancy in the DCS shall include
 - Processor redundancy
 - Power supply redundancy
 - Network redundancy
 - Server redundancy
 - I/O card segregation
 - I/O card redundancy for critical signals (each of the 1oo2 or 2oo3 redundant signals will be hardwired to different I/O cards, with each cabinet having redundant power supplies)
 - In the case of redundant signals are connected to the different cards, redundant I/O cards are not required.

6.5.5 Redundancy Concept of GT OEM

Gas turbine

System	Equipment	Number (/GT)	Category of concept "N"	Remarks
Lube Oil System	Lube Oil Pump	100%x2 (AC) 100%x1 (DC)	- ②	DC type: Used for only emergency period
	Lube Oil Cooler	100%x2	-	
	Lube Oil Mist Separator	100%x1	①	With separator bypass line
	LO Vapor Extractor	100%x2	-	
Fuel Gas System	Fuel Gas Flow Meter	100%x1	③	Composed as Coriolis type which is high reliability
	Fuel Gas Heater	100%x1 (Shell & Tube type) 100%x1 (Radiator type)	③ ① and ②	Shell & Tube type: Based on OEM experienced standard Radiator type: If one fan stops, GT runs back. And used only for simple cycle operation.
	Fuel Gas Last Chance Filter	100%x2	-	
	Sweep Air Cooler	50%x2	③	Fintubed-pipes with high reliability
	Ignition System	100%x1	①	Used for only shutdown period
	Fuel Gas Unit	100%x1	③	Based on OEM experienced standard

	Fuel Gas Calorie Meter & Pressure Reduction Unit	100%x1	①	This equipment failure does not cause the loss of power production
Fuel Oil System	Main Fuel Oil Pump	50%x2	②	Used for only back up fuel operation
	Fuel Oil Drain Tank & Pump	100%x1	②	Ditto
	Fuel Oil Unit	100%x1	③	Based on OEM experienced standard
	Flow Divider Unit	100%x1	③	Based on OEM experienced standard
Water Injection System	Water Injection Pump	100%x1	②	Ditto
Purge Air System	Purge Air Compressor	100%x1	②	Used for only period from shut down after FO firing to re-start
	Purge Air Receiver	100%x1	②	Ditto
	Purge Air Cooler	100%x1	③	Based on OEM experienced standard
Air & Flue gas system	GT Cooling Air Cooler	100%x1 (Shell & Tube type) 100%x1 (Radiator type)	③ ① and ②	Shell & Tube type: Based on OEM experienced standard Radiator type: If one fan stops, GT runs back. And used for only simple cycle operation.
	GT Cooling Air Filter	100%x1	③	Its composed as inertia type filter which has no cogging risk
	2C Cooler	100%x1 (Shell & Tube type) /GT	③	
	4S Cooler	100%x1 (Radiator type) /GT	①	If one fan stops, GT runs back
	4S Cooling Air Filter	100%x1	③	Its composed as inertia type filter which has no cogging risk
GT Casing Cooling Air system	Casing Cooling Fan	100%x1	②	Used for only shutdown period
GT Protection System/GT Air Intake System	GT package Ventilation fan	50%x3	-	
	Turning Device	100%x1	②	Used for only turning period
	Air intake filter	With multiple filter elements	-	
	Evaporative Cooler System	100%x1 /GT	②	Operated only for low RH condition
	Wet Compression System	100%x1 /GT	①	Operated only for power augment at OP1
Enhanced Cooling Air System	Enhanced Cooling Air Cooler	100%x1 (Radiator type)	①	Radiator type: If one fan stops, GT runs back
	Enhanced Cooling Air Drain Mist Separator	100%x1	③	Its composed as inertia type filter which has no cogging risk
	Enhanced Cooling Air Compressor	100%x2	-	
	Enhanced Cooling Air Filter	100%x1	②	This is simple Y type strainer, which is high reliability item. Additionally, major filtering function is provided by Drain Mist Separator and this filter is only for emergency dust
Clean Agent FF system	Clean Agent FF system	100%x2 (including online reserve)	-	
Instruments	Instruments	100%x2 or 3 (control or inter lock)	-✕	Minor detection-use which does not cause the reduction of power
		100%x1 (detection-use)	①	
Valves, Piping, and others not specified in above	Valves, Piping, and others not specified in above	100%x1	③	Based on OEM experienced standard
AC Distribution System for GT Auxiliary	GT MCC	Normal Bus 100% x 1 Essential Bus 100% x 1	③	Tie CB between Normal Bus and Essential Bus is provided. Thus, the equipment has high reliability.
	GT AC DB	Normal 100% x 1 Essential 100% x 1	③	DB consists of simple components such as Busbar

				and MCBs. Thus, the equipment has high reliability.
DC Distribution System for GT Auxiliary	Battery & Charger	100%x2	-	
	DC DB	100%x1	③	DB consists of simple components such as Busbar and MCBs. Thus, the equipment has high reliability
	DC Motor Starter Panel		②	Used for only emergency period

Gas turbine generator

System	Equipment	Number (/GT)	Category of concept "N"	Remarks
GTG (H2&CO2 gas and seal oil systems)	Generator H2 gas dryer blower	100%x1	①	If the H2 gas dryer blower fails, generator can be continuous operation. It can also be replaced while the generator is running.
	seal oil pump	100%x2(AC) + Emergency Backup (DC)	②	The seal oil system is equipped with two AC seal oil pumps (which one pump is reserved as standby use) and a DC emergency seal oil pump. DC emergency seal oil pump starts automatically in case the AC seal pump fails due to any reason. The AC seal oil pumps can be changed as normal use and standby use at any time. This change is carried out by manual operating.
	Seal oil vacuum pump	100%x1	①	The proposed seal oil system is vacuum treatment type. If the vacuum pump fails, the purity and pressure of the hydrogen gas in the generator are gradually reduced. But it does not immediately affect normal operation. In addition, by manually adding hydrogen gas, the purity and pressure of the hydrogen gas can be maintained at almost normal conditions, generator can be continuous operation.
	Generator vapor extractor	100%x1	①	If the generator vapor extractor fails, generator can be continuous operation. It can also be replaced while the generator is running.

6.6 Plant Performance and Schedule

The Contractor shall provide the following as a part of Plant Performances and Schedule:

a. Correction curves

Performance correction curves and formulae for the correction of measured performance data at conditions different from the Reference Conditions shall be provided by the contractor for both the primary and the backup fuel:

- Gas Turbine Standard correction curves
- Gas Turbine generator capability curve
- Steam Turbine generator capability curve

- CC Start-up curves
- CC Shutdown curve
- Capacity of GT system vs ambient temperature
- Gas Turbine GSUT capacity curve
- Response capability curve (frequency response chart)
- Emissions table
- Steam turbine load changes
- Power output vs gas pressure
- CC Correction curves (to be provided for):
 - Evaporative Cooler ON (Only for Base Load),
 - Wet Compression On (only after Evaporative Cooler is ON)
 - Evaporative Cooler OFF (Base Load and partial loads)
 - Duct Firing On (if applicable)
- The corrections of the net capacity and net heat rate of the total Plant in combined cycle configuration shall be provided for:
 - ambient air temperature
 - ambient air humidity
 - ambient barometric pressure
 - Fuel Gas Lower Heating value (LHV)
 - Fuel Gas C:H ratio
 - Inlet Air Condition Difference Between Cooling Tower/ACC and Gas Turbine
 - Grid Frequency
 - Power Factor
 - Cycle Unaccounted Losses
 - Natural Gas Pressure at the Terminal Point
 - Natural Gas Temperature at the Terminal Point
 - Partial load (only for EVC OFF case)
 - Degradation with GT fired hours
 - condensate return flow and temperature

b. Outage modes

The important and different outage modes encountered during the operation shall be furnished by the Contractor. Corresponding H&M balance drawings are to be supplied by the Contractor.

The outage durations of GT, HRSG and ST shall be defined.

The outage frequency shall be defined for HRSG, ST and GT (in line with the typical LTSA assumptions).

6.6.1 Operating Points

The following reference operating points are assumed for the Rumah 1 IPP and Rumah 2 IPP or Group dispatch:

Key Item	Unit	RSC	OP1	OP2	OP3	OP4	OP5
Fuel			Gas	Gas	Gas	Gas	Gas
Air Temperature (dry bulb)	°C	48	48	40	35	25	15
Relative humidity	%	20	20	30	60	70	70
Atmospheric pressure	mbar	955	955	955	955	955	955
Net Dependable Capacity (Export)	%		100	90	75	60	30
Annual operation hours	h/year		100	3400	3300	1150	810

The following reference operating points are assumed for the Nairyah 1 IPP and Nairyah 2 IPP or Group dispatch:

Key Item	Unit	RSC	OP1	OP2	OP3	OP4	OP5
Fuel			Gas	Gas	Gas	Gas	Gas
Air Temperature (dry bulb)	°C	48	48	40	35	25	15
Relative humidity	%	30	30	30	60	70	70
Atmospheric pressure	mbar	1005	1005	1005	1005	1005	1005
Net Dependable Capacity (Export)	%		100	90	75	60	30
Annual operation hours	h/year		100	3400	3300	1150	810

7. CIVIL STRUCTURAL AND ARCHITECTURAL WORKS

7.1 Scope of Civil Works

The civil engineering and building works to be provided by the Contractor shall comprise the full detailed design of all permanent and temporary works, including detailed assessment of existing site conditions, structures and any required clearance tasks, supplementary site investigation studies, detailing and preparation of design and construction drawings, construction as built drawings, and completion and maintenance of all the various foundations (including transmission towers), structures, plant related buildings, support services buildings, installations and services etc. required for the installation, commissioning, operation and maintenance of the equipment and plant for the Facility.

In general terms, a common and consistent set of standards shall be used throughout the facility for Civil, structural and Architectural work. If any requirements mentioned in Section 7 of OTS with respect to any codes and standards, doesn't exactly match with the requirements mentioned in the code and standard EPC Contractor wants to follow within the Codes and standards envelopes mentioned in OTS, the equivalent or more stringent requirement shall be followed.

The clear Headroom of for the ceilings shall be minimum 3.0m, OEM recommendation, as per relevant applicable code and Operational requirement whichever is stringent.

Following scope of work shall be included for Civil/Structural and Architectural works;

Studies and Surveys

- Site survey;
- Topographical survey;
- Bathymetric Survey if required;
- Investigation of onshore and offshore subsoil conditions and study of geological formation and seismic conditions of the area;
- Geotechnical and Geophysical Survey work;
- Hydrology and Flood risks;
- Meteorological studies;
- Hydrographic studies;
- Meteorology Survey;
- Air recirculation study;
- Hydraulic calculations - steady state and dynamic;
- Logistic Survey for O&M purpose
- Security Risk assessment Study as per local regulations

Preparatory Works

- Due diligence of all applicable aspects:
 - a. Investigation of site physical conditions
 - b. Investigation of shipping & transportation logistics
 - c. Site Survey, including property lines, setting of permanent site benchmarks
 - d. Soil survey including borings
 - e. Hydrological survey/investigation
 - f. Determination of seismic requirements
 - g. Determination of flood potentiality
 - h. Gather, verify and present all input data to be applied to the design of a technically and economically optimal power plant.
 - i. The EPC Contactor has to investigate the local conditions with respect to planning and applicable laws.
 - j. Technical requirements from involved authorities.
- Provide Plant data regarding interface points;
- Collection of meteorological data;
- Removal of existing temporary facilities, site offices if any;
- Relocation of any buildings, cables and pipe crossing the site, including replacement material;
- Site preparation including compaction of soil, filling of low areas with imported fill and grading of the entire area of the site to the required lines levels and slopes, as required;
- Soil improvement;
- Provision of temporary lay-down areas, warehouses, workshops, vehicles, equipment etc. all as necessary for the construction phase;
- Provision of precast yard as necessary;
- Earthwork cutting, Backfilling, levelling, grading as required;
- Soil improvement as required;
- Provision of temporary firefighting and alarm system;
- Provision of temporary site drainage including storm water and sanitary drainage including for the site, site facilities, temporary lay-down areas, warehouses, workshops, housing compounds and camp facilities etc., disposal of sewage as necessary;
- Provision of temporary housing compound and camp facilities;
- Provision of temporary roads and temporary fencing;
- Provision of temporary site fencing including gates, as required;
- Provision of first aid, site safety and security system for the construction phase;
- Provision for site office for the Client and Client's representatives
- Provision of temporary offices and facilities on the Plant Site for the Owner, the Engineer

- (Owner's Engineer) and Owner's Operator;
- Provision of temporary offices and all facilities for the EPC Contractor; and its sub-Contractors staff.
- Site services as required for the construction and Commissioning of the Plant such as:
- electricity supply during construction;
- potable water, demineralized water, etc.,
- instrument and service air,
- fuel supply during construction,
- telecommunication,

Other Civil Works

- All the Foundation works;
- All Civil-Structural and Architectural works including and up to the Electrical, Fuel supply, water supply, access road terminal points.
- All Civil-Structural and Architectural works for Renewable Energy generation facilities if any;
- All Civil-structural works for carbon reduction and abatement as required;
- Boreholes for monitoring of soil contamination during O&M stage as recommended by ESIA report;
- Back up Fuel unloading station;
- Service water Unloading station;
- Service water underground buffer tank, pumping station and storage tanks;
- Fire Brigade building (if required)
- All types of foundations including pile
- Civil works for Storm water and wastewater drainage system;
- Roads, Paving and Fencing works (Recommendations of NFPA 850 shall be applied);
- Erosion Protection works;
- Landscaping as per ESIA requirement;
- Restoring damages to the existing properties if any caused due to the EPC design, Construction and Testing-commissioning activities;
- Civil works related to Intake and Outfall facilities
- Modification for the existing facilities as required;
- Access to the interface facilities for the O&M period;
- All the Civil works for and up to the Civil, electrical and Mechanical Interfaces;

7.2 Design Requirements

7.2.1 General

The Contractor is responsible for all aspects of civil engineering design. This shall include architectural, civil, structural and building services design for all temporary and permanent works. The civil works shall be designed to have a minimum working life of fifty (50) years before any significant repair or replacement of the main structural elements and secondary structural elements/cladding. This section identifies the Owner's minimum design requirements and parameters that are to be provided by the Contractor.

The Contractor shall provide all building services (electricity, telephone, computer, water, HVAC, fire protection etc.) and ensure that they are readily accessible for maintenance and repair and adequate clearance is provided from main structural members and foundations.

The choice of external building materials shall be durable against sandstorm erosion, saline and corrosive environment.

Any pipe or underground cables crossings outside the plant boundary under Roads and Railroads shall be designed and constructed such that, proper access for maintenance of the damaged part can be possible during O&M period without disturbing the traffics under Roads and Railroads.

7.2.2 Ground and Soil Conditions and Environmental requirements

The Contractor, pursuant the EPC Contract, shall satisfy himself that the findings of the site investigations and surveys referenced as following adequately reflect the geotechnical, topographical and environmental

conditions at the Site. The Contractor shall carry out additional investigations and surveys at his own expense to satisfy himself as to the geotechnical, topographical and environmental conditions at the site and to establish any information/data necessary prior to Contract and designing the Works. The Contractor shall prepare both factual and interpretative reports for any such studies.

Further soil investigations shall be carried out by the Contractor as required for justification of the selected design parameters.

- Geotechnical Survey including seismic study
- Topographic Survey
- General Survey work
- Any other survey as required for the implementation of the project and indicated in OTS.

7.2.3 Earthworks

Backfill placed for embankments, non-supportive and unpaved areas shall be compacted to at least 75% relative density (ASTM D 4253 and ASTM D 4254) for free draining soils containing less than 15% by weight finer than 75 micron sieve non plastic material or 90% of the maximum density as determined by ASTM D 1557 for soils containing more than 15% material passing the 75 micron sieve.

Backfill placed below all the supports, roads and paved areas shall be minimum compacted to 85% relative density (ASTM D 4253 and ASTM D 4254) for free draining soils containing less than 15% by weight finer than 75 micron sieve non plastic material or 95% of the maximum density as determined by ASTM D 1557 for soils containing more than 15% material passing the 75 micron sieve.

A cap layer shall be placed to provide wind erosion protection in all the outdoor exposed areas. The cap layer shall be a minimum of 150 mm thick. The contractor to use soil-cement asphalt paving or sprayed-on polymer-based erosion inhibitors on that capping layer. Gravel layer is acceptable, as long as no erosion possibility is there due to water and wind. Gravel layer thickness 100 mm can be considered. However, this is subject to the Geotechnical recommendation.

For the design purpose, Design Ground water level shall be considered as per Geotechnical information, however seasonal variation of groundwater and capillary rise of groundwater shall be taken into the consideration for design and construction work and shall be substantiated before the start of design and construction work.

No foundations shall be supported on backfilled soil. Only less important and lighter foundation can be rested on backfilled soil subject to detail engineering and Owner's approval fulfilling the settlement requirement.

A geotextile layer shall be put between the filling layer and the soil after removal of top layer if recommended by EPC Geotechnical study.

Selected fill

Selected fill shall have the following properties:

Well graded having uniformity index of not less than 5. Selected fill shall be non-cohesive, free of plastic clays and having silt content not greater than 5%, salt content less than 3% and organic matter not greater than 2%. Decomposing or compressible materials shall not be acceptable.

The material shall be of such nature and character that it can be compacted to the specified densities. It shall be free of all materials subject to decay, decomposition or dissolution, and of cinders or other material that will corrode piping or other material.

The intention is to use selected fill below structures, below pipelines, backfill to structures, roads, parking areas and hard standings etc.

Sub-base material

Sub-base shall be gravel or crushed rock of suitable maximum size and approved grading that can be used as sub-base material for roads, tanks and switchyard areas.

Rip-rap and rock fill

Rip-rap and rock fill shall be of weather and waterproof stone that shall not suffer from any ill effects through the action of ground, river/canal water or brine water. Selected fill, riprap or rock fill of suitable size and grading may be used below selected fill of foundations or for stabilization of sub-grade under roads.

7.2.3.1 Earthwork Quality Control and Testing

Selected fill

As a minimum the Quality Control shall be as follows:

- Unit dry weight
- Heavy dynamic sounding, to DIN 4094 or equivalent standards: 20 blows per 10 cm penetration or alternatively for Dutch cone test, DIN 4094 or equivalent standards, 200 bar point resistance
- Plate load test:
 - E_{v2} not less than $1200 \text{ kg/cm}^2 = 120 \text{ MN/m}^2$
 - E_{v2}/E_{v1} smaller than 2.2

Where:

- E_{v1} being the modules of deformation of the first loading process and
- E_{v2} being the modules of deformation of the second loading process (reloading after relieving once)
- Tests shall be performed for each 500m^3 of fill.

7.2.4 Erosion Protection Works

If the surface soils is susceptible to wind and water erosion these shall be stabilized with Cement. The soil and cement should be well mixed, moisture conditioned to within 1 per cent of optimum moisture and compacted to at least 98% of the maximum dry density. The optimum moisture content and maximum dry density of the soil-cement mixture should be determined by the ASTM D-558 test method. Traffic, except as required to maintain the surface of the soil cement in a damp condition, should not be permitted for at least 24 hours after final compaction.

For slope protection against erosion from wind, stabilization of surface soil with Type V Portland Cement as per ASTM C 150 may be adopted. For slope protection against erosion from water, slowing down of surface water is needed. To accomplish this, erosion control blankets, fibre rolls may be used. Erosion control blankets are usually woven type from a chosen material, with lots of ridges and obstructions. These materials are either synthetic or natural. Fibber rolls are also made of similar material as used in erosion control blankets but are rolled into large diameter logs. Coir type logs/ Logs made of rice/wheat straw also may be used. These logs are nailed into the slope to slow down the surface water. Tensar type geo-grids also may be effective. The source/ availability of the appropriate material is essential in selection of the technique, and it shall be duly approved by the Owner's engineer.

Gravel layer is acceptable, as long as no erosion possibility is there due to water and wind. Gravel layer thickness 100 mm can be considered.

7.2.5 Buildings, Structures and Foundation Design Requirements

On the basis of a site-specific geotechnical investigation, the Contractor shall determine the foundation requirements for the Site. The Contractor shall verify the accuracy and completeness of all/any information provided in this regard and shall take full responsibility for all foundations works. All foundations shall be capable of withstanding all dead, live, site, climatic, meteorological and other environmental loadings.

Piles shall be designed to resist all combinations of compressive, tensile, lateral and transient loads from installation, plant/building dead and live loads and environmental loading including seismic loads.

The safe working capacity of the pile(s) shall be determined by calculation using recognized soil mechanic principles and equations. The Contractor shall carry out a programme of preliminary test piling at site to verify his designs prior to commencing the installation of permanent 'working' piles. The maximum load applied during the preliminary pile testing programme shall be 200% of pile working load. The settlement limit in this case shall follow relevant applicable International/local standards. Upon completion of testing preliminary piles shall be cut down and abandoned, such piles may not be incorporated into the permanent works construction.

For the structural design of the piles the recommendations in ACI 318M-08/318RM-08: "Metric Building Code Requirements for Structural Concrete and Commentary" and ACI 543R-00 Design, Manufacture and Installation of Concrete Piles and the requirements of this OTS shall be followed.

Permissible pile settlements at under load for compression, tension and lateral load tests shall not exceed settlement criteria imposed by plant manufacturers, local regulations or shall conform with the BS 8004, whichever is the most stringent. With $\gamma_s, \gamma_{sl} \geq 1.0$, the settlement of an individual pile shall be limited to less than 3% of its diameter, D; with 1.2, the settlement shall be limited to less than 1.5% D complying BS 8004 requirements.

If piles are required to resist dynamic loads they shall not be loaded to more than 50% and 85% of their safe working load during normal (dead, live and other sustain load) and abnormal operating conditions (wind, seismic and other transient load) respectively.

The design of the foundations for plant subject to dynamic loads shall be in accordance with DIN 4024 : Part 1 (1988) "Machine Foundations" and CP 2012 : Part 1 "Foundations for Reciprocating Machines". If manufacturer's design guide is not coincided with above mentioned code, then manufacturer's guide shall be considered preferentially

The bearing pressure under foundations subject to dynamic loads shall not exceed 60% and 90% of the safe ground bearing pressure during normal and abnormal operating conditions respectively.

The overall and differential settlement of foundations for plant shall strictly adhere to the plant manufacturer's requirements or accepted standards which is more stringent.

The allowable overall settlement for the isolated foundations shall not exceed 25 mm. The allowable overall settlement for the raft foundations shall not exceed 50 mm. The allowable differential settlement for the tanks ring beam foundations shall be designed in accordance with API 650. The allowable overall settlement after the hydrotest for the tank foundations supported on ring beams shall not be more than 50mm.

The Contractors design(s) for concrete elements shall comply with the crack width limitations noted in this section. The Contractors design(s) shall incorporate an assessment of the anticipated concrete crack widths for the reinforced concrete structural elements in their design reports to cover both early thermal and tensile/flexural cracking.

The Contractor shall carry out reinforced concrete crack width calculations to satisfy the limits under service loads listed in the following table or as per local regulations whichever is more stringent:

Exposure Conditions	Crack width maximum limit (mm)	Compliance
Air , soil	0.3 mm	ACI 224 R

Seawater and sea spray, wetting and drying Concrete piles	0.15 mm	ACI 224 R
The crack width requirement shall follow the local regulations, standards. For the water retaining structure, the crack width can be 0.2mm as per BS 8007/BS EN 1992-3. Leak proof/tight tests before backfilling shall be carried out as per relevant standards. All other requirements as per the same code shall be considered.		

Tanks that are supported on flexible foundations shall have a perimeter reinforced concrete ring beam designed in accordance with API 650 to contain the enclosed fill. The fill supporting the tank shall be capped with asphalt concrete surfacing minimum 50 mm thick. If cathodic protection is used asphalt concrete surfacing won't be required.

All the structures below ground level shall be designed to take account of their exposure to groundwater. In some cases, this shall mean protecting the concrete against the corrosive effects of sulphate/chloride attack only. In other cases, it shall mean protecting the concrete against sulphate/chloride attack and attaining the required degree of watertightness.

Following design criteria, parameters and philosophy shall be considered by the Contractor so as to attain the required degree of water tightness in concrete structures below ground.

The Contractor shall be responsible for the design of all water-tight (i.e. water retaining and water excluding) concrete structures.

The Contractor shall adopt the following design philosophy for the design of watertight concrete:

- All water retaining/water excluding structures and cable tunnels shall be designed and detailed in accordance with BS 8007 (BS 8102, Type B construction) to achieve a performance level as defined in BS8102 for a Grade 2 structure. Additionally, all cable tunnels that are founded below a level determined as High Ground Water Level plus 1.5 metres (HGWL + 1.5 m) shall be tanked. Where it is determined that tanking shall be required, it shall be applied as a minimum to the underside of the base and to the walls up to the HGWL + 1.5 m level or shall totally enclose the tunnel if the top of roof slab level is below HGWL + 1.5 m level.
- Where the design deems that building basements are necessary, whether above or below ground water level, then these shall be tanked to achieve a performance level as defined in BS 8102 for Grade 2 structure. The tanking shall be applied to the basement structure and extend up the walls to ground level;
- To reduce thermal and shrinkage cracking, joints shall be provided at centres generally not exceeding 5.0 m (or in accordance with BS 8007 if more onerous); joint spacing greater than 5m can be adopted with supporting calculation of rebar ratio for crack control as per BS 8007.
- Concrete floors, walls and slabs shall be provided with all necessary contraction, expansion and control joints. These joints shall be provided by discontinuity joints, saw cuts and weakened planes. Joints shall be located so as not to weaken the structural integrity of the building or structure. Joints shall be filled and sealed with suitable materials as appropriate;
- All joints in the concrete, including expansion, contraction, partial contraction and construction joints shall include a proprietary water-stop system;
- Joint details shall comply with the requirements of ACI 504R-90, Guide to Sealing Joints in Concrete Structures.

The design of slabs on grade shall be in accordance with ACI 360R

7.2.6 Turbine Building (s)

The Steam turbine buildings shall provide protection of the equipment, electrical enclosed installation and controls from the environmental conditions. The buildings shall be designed for noise attenuation and be of non-combustible construction. The buildings shall be dust proof type and ventilated so that the building internal Design Temperature should not be greater than 5°C above maximum ambient temperature. The entrances shall be sufficiently sized to provide access for the largest item to be located within respective Building.

The building shall be of sufficient size to facilitate the installation operation and maintenance of all components and equipment housed therein. Requisite lay-down areas shall be provided for O&M purpose. Entrances shall be suitably sized and located to facilitate maintenance requirements.

Proper handling arrangement shall be provided for the equipment for O&M period.

Sufficient capacities of cranes shall be erected for normal maintenance of the gas turbine auxiliaries, including lifting heavy parts such as the casing and rotor.

Suitable access to the turbine building shall be provided to allow access for single complete rotors. A rotor unstacking area shall be provided with a suitable foundation to enable unstacking of a single rotor in the upright position and all necessary equipment shall be provided to manoeuvre and support the rotor. The turbine laydown/maintenance area may also function as the rotor unstacking area; however the Contractor shall take due care of the space usage and functionality.

Suitable paved areas outside the Turbine Buildings to put the mobile crane and a low bed trailer, for transportation of the disassembled parts, in a stable way.

The separate turbine enclosures will be foreseen with easily removable panel(s) that can be lifted by the crane and can be put aside and foresee walkways with railings, sufficient lighting, around the roof and around the roof opening(s) that, with a connection in between the walkways.

The separate turbine enclosure shall be foreseen with removable beams to connect hoists in strategic locations, such as above the combustion chambers.

The floor slab within the building, which incorporates pipe and cable trenches shall be constructed of reinforced concrete laid flat and with local dishing incorporated adjacent to floor gully inlets. Floor drains shall discharge via a piped drainage network into oily water separators. Trench covers shall be designed to accept the appropriate floor loading according to their location and shall be manually removable.

Emergency exits fitted with panic escape bars shall be provided to all areas of the turbine buildings and annexes and boiler enclosures. Steel doors and frames should comply with the fire rating requirements. These openings must be kept clear of all obstructions.

7.2.7 Turbine supporting structures

The GT and ST foundations shall be a monolithic raft supported by piles if required as per the Geotechnical report. The exhaust stack may be supported by a separate foundation or incorporated within the foundation if required by the Contractor.

The contractor shall ensure that the following points are included in the GT and ST foundation works:

- a. A minimum rebar content of 80 kg/m³ of concrete shall be provided.
- b. The design of the GT and ST foundation shall satisfy, in addition to an appropriate internationally recognized design code on machine foundations, the following codes: ACI 351, CP 2012 and DIN 4024. CP 2012 requires an internal reinforcement cage system which is essential for the foundation to act as a homogenous mass and provide a second line of defence to the machine anchoring system with the holding down bolts. DIN 4024 provides guidelines for carrying out dynamic analysis.

- c. The GT and ST foundation shall be isolated from the rest of the works by floor joints.
- d. The ratio of mass of the foundation to mass of the GT and ST plant shall be at least 2.
- e. To reduce the risk of dynamic compaction the maximum load on any pile supporting a GT and ST foundation shall not exceed 60 per cent of its working capacity during the normal operating conditions and shall not exceed 90 per cent in extreme conditions. For a foundation supported by soil the same limits apply to the allowable bearing capacity of the soil.
- f. The design of the GT and ST foundations shall include the thermal effects on the foundation and its ability to withstand the extreme differential temperatures likely to be experienced by the foundation, during the construction as well as throughout its design life. Protective measures during construction shall be part of the design.
- g. The section of the GT and ST foundation below ground shall be protected with a coating system to resist attack from the sulphate content in the soils and chlorides from the ground water. Further protection against corrosion of the reinforcement (which leads to expansion of the concrete and cracking) due to the severe chloride ion content of the soils and groundwater shall be provided as covered by the concrete section of this Specification.
- h. The GT and ST foundation shall be poured in one operation and construction joints are not permitted unless in exceptional circumstances and with the approval of the Owner/Owner's Engineer.
- i. The design, detailing and construction of the GT and ST foundations shall allow fully for the "buildability" of the foundations, particularly where congested areas of reinforcement are required.
- j. In addition to the static structural design the contractor shall prove the design of the turbine supporting structure dynamically by modelling the bearing points, the foundation and the soil body together, taking the full consideration of the coupling effect and interaction between the three systems.
- k. The design of turbine supporting structures, if applicable, shall be same as that supporting the gas turbines except in some designs a table top structure and columns are erected over a mat foundation. The supporting structure if cast monolithically shall be analysed with due consideration of the flexibility of each structural component. If the table top structure is supported by springs the substructure may be designed as a static structure if its superstructure proves a sufficient isolation achieved by the springs.

The Contractor shall submit thermal curing procedures for the steam turbine and gas turbine foundations to the Owner/Owner's Engineer for approval prior to commencing construction of these works.

7.2.8 Building Roof Access

Areas of roofs requiring access shall be provided with safety rails.

7.2.9 HRSG Foundations

A reinforced concrete slab shall be provided under the HRSG and under the feed pump enclosure, which shall incorporate bunding and falls to drains. All drains from all buildings in the area shall be contained and routed properly so that they are discharged to the wastewater system. Bunding shall be adequate to prevent any overspill of blowdown water under any conditions.

Any floor mounted main equipment within the Plant buildings shall be mounted on a 200mm high concrete plinth.

7.2.10 N.A.

7.2.11 Piling and isolated footings

The Contractor shall pay especial attention for the unknown amount of cavities, voids and interconnecting discontinuities and shall ensure that the locations of all significant plant and building foundations are thoroughly probed for voids and discontinuities as part of his supplementary ground investigation studies.

To mitigating the settlement risk the cavities and voids may be grouted injected if they are found below the major foundations. Alternately the foundations may be founded on piles.

The design of piled foundations for major structures and buildings shall conform to the settlement criteria stated earlier of the specification.

Following activities are mandatory for the Foundation design and construction:

- The depth of the Borehole shall be considered based on following criteria as a minimum: British Standard: Code of practice for site investigations BS 5930:1999 Section_“12.7.2 Foundations for structures
- Geophysical tests at the key foundations (to be agreed with Owner) locations to be carried out before the construction/erection of foundations
- If cavities are found, required soil improvement/ foundation design criteria to be carried out subject to OE approval.
- Dewatering methodology will have to be demonstrated is fit for purpose according to soil conditions and submitted for Owner approval. Particular attention to be paid to those soils that can be degraded and/or settled due to dewatering process.
- Dewatering shall ensure to restrict the movement of fine grain material below the foundation level.
- The dewatering work shall be performed with reputed local Specialize contractor. The Contractor for dewatering and the Dewatering process shall be approved by the Owner.

Dewatering design shall be submitted for owner's approval.

Following criteria shall be given special consideration while designing the foundation:

- Effect of surrounding foundations shall be considered when pressure bulbs supposed to be overlapping. For key foundations raft foundations will be prefer design criteria.
- During construction EPC will take measurement of settlement and send a result that differential settlement within the limit complying with OEM requirement is achieved.
-
- Detail geotechnical analysis will have to be performed by EPC contractor by a first class engineer for the key foundations as a detail input of the main foundation design.
- Main foundations design basis will be subject to Owner approval.
- Foundation design will obviously consider all the applicable loads including backfilling, foundation weights, dead and live load, equipment dead load and dynamic loads, hydrostatic load, environmental load and any other load as applicable.
- Maximum Differential settlement value for the foundations shall be submitted for Owners approval and as a general criteria 1/500 rad. or as per Equipment manufacturer's recommendation whichever is stringent shall be followed.
- Foundation designs will have to comply with applicable local regulations including local, municipality and country applicable regulations.
- Settlement monitoring system shall be considered for the key foundations.
-

Following foundations shall be considered as Key foundation:

- Steam and Gas Turbines foundations

- Steam and Gas turbines Building foundations if applicable
- HRSG Foundations
- Air cooled condenser
- GT and UAT foundations
- Deaerator and Feed water Structure foundations and Tank/Pump foundations
- GT and UAT foundations

Following sensitive structures and foundations shall be designed by the reputed Consultant subject to Owner's recommendation and approval.

- Steam and Gas Turbines foundations
- Steam and Gas turbines Building foundations if applicable
- Main Building foundations as control and electrical buildings
- Deaerator and Feed water Structure foundations and Tank/Pump foundations
- Main and Bypass Stack foundation
- Key dynamic equipment foundations
- Air Cooled Condenser Foundations
- GT and UAT foundations

Earthwork and levelling shall also be designed and performed by the reputed Contractor and Consultant as per Owner's recommendation and approval

7.2.12 Road Layout and Design

Design and construction standards and criteria for roads shall be in accordance with the local regulations and AASHTO requirements.

The final road network design shall incorporate two major considerations:

- Full accessibility to the offshore facilities and on-shore plant ensuring that all operational and maintenance requirements can be fulfilled in an efficient manner including interfaces with the existing road network;
- Security, protection and safety of the employees and the Plant

A road system shall be provided to serve all buildings on the Site.

The roads shall be constructed between raised kerbs and graded to falls leading surface water into gullies or catch pits that will connect into the surface water drainage system. Gullies shall be provided with grating to prevent the ingress of debris into the drains.

Road pavements are to be designed to for a 25 years design period.

The roads shall be constructed in reinforced concrete or in flexible bituminous concrete (asphalt minimum of two layers) construction, depending upon the location. Concrete roads shall be used in areas of heavy and abnormal loading, where oil spillage may occur and where a durable and maintenance free construction is essential to the movement of plant and equipment. All other roads shall be of flexible construction with a bituminous concrete (asphalt) finish.

Sidewalk shall be provided for all the roads where it is necessary to satisfy person's walking requirement.

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Main and Secondary Access Gate and Main and Minor Gate house shall be considered for entry/exit the plant.

7.2.13 Fencing

Internal fences for the safety of any particular installation within the Plant Site shall be erected completely with all necessary doors and gates. The American standards may be used in addition to the local code.

Each type of internal fencing shall suit the level of security required by each installation. In general, each level of security shall be agreed with the Owner, however a 3 level master key security system shall be provided as a minimum.

Straining posts shall be provided at all ends and corners, at changes in direction or acute variations in level and at intervals of not more than 60m in straight lengths of fence. Struts shall be fitted to all straining posts, in the direction of each line of fencing. Steel struts and posts shall be bolted together. Intermediate posts shall be provided at intervals, measured centre-to-centre, of not more than 3m.

All building doors, equipment room or container doors and fence gate lock keys shall be operable by Grand Master key and to the approval of the Owner.

As per NFPA regulations, transformer shall be enclosed by fire walls. The entrance side shall be provided with a removable security internal fence fitted with a lockable personnel access gate compatible with Master key system.

The plant Area and sub-station area shall be closed by fence or wall meeting the national and local guidelines. The plant fence or walls shall be maintained throughout the construction phase and at no time security of any of the plant sites shall be compromised. The fencing shall be HCIS Category -1.

The design of the fences, gates , boundary walls and all other security installations shall follow the local requirements. Internal fences for the safety of any particular installation within the Plant Site shall be erected completely with all necessary doors and gates. The American standards may be used in addition to the local code.

Internal fencing shall suit the level of security required by each installation. In general each level of security shall be agreed with the Owner, however a 3 level master key security system shall be provided as a minimum.

Mechanized Vehicular gates shall be double opening. Gates shall be furnished with hinges of a type which prevent the easy removal of gates from the gate posts. Gate hinges shall allow the gate to swing through 180°. The gates shall use the same type of wire mesh as used for the fencing. Emergency opening mechanical push bars shall be provided to all personnel gates. Locking bars shall be provided to all vehicle gates and provision made for the attachment of a padlock which shall be accessible from both sides of the gate.

Hazards areas like transformer, fuel areas shall be protected with the internal fence and gate.

All building doors, equipment room or container doors and fence gate lock keys shall be operable by Grand Master key and to the approval of the Owner.

Sand preventing arrangement/sand fence shall be provided as required for the smooth operation of the plant to prevent the ingress of sand. The structural design of the fence shall consider loads if any from the deposited sand.

7.2.14 Permanent access road and entrance

To provide completely separated and independent access to the Site new access roads shall be constructed by the Contractor. Details about the routing shall be agreed with Owner, relevant local authorities, the Municipality and the Public Road Department. These roads shall be constructed with complete road drainage and lighting system with street lighting poles every 25 m as a minimum. The completion shall be envisaged at the end of the construction/erection period.

For the entry and exit purpose of the plant at least two (2) entrances and exits shall be constructed fulfilling the HCIS and OTS part A requirement..

Access to all external and Internal Interface points shall, be provided by the Contractor.

7.2.15 Surfacing

All buildings in the Plant shall be paved with minimum 150 mm concrete extending from the edge of the building for a distance of 1.2m with proper slope to drainage Surface Water Drainage System and additionally asphalted paving shall be extending from 1.2 m up to a distance of 5.0m with proper slope to drainage Surface Water Drainage System if the soil is erosion prone due to water and wind.

A surface water drainage system shall be provided for the Plant. The design shall allow for the separate collection, treatment and disposal of surface water over the whole of the Site (arising from run-off of natural water from buildings, roads, ground surface, paved areas, etc.). All buildings shall be provided with roof drainage for disposal of surface water, and be connected to the surface water drainage system. The surface water drainage system shall be arranged and designed to effectively manage all run-off and discharge arising from the Site.

The Storm water Drainage system shall be designed considering 100 years rainfall return period and rainfall intensity equal to the retention period.

Basements, cable and pipe trenches and tunnels shall be constructed with adequate provision for drainage. Floors to basements, trenches and tunnels shall be laid to falls and/or drainage channels provided therein. Any water entering the basements trenches or tunnels shall be routed to sumps and connected to the surface water drainage system. Where connection to the surface water drainage system is not practical, then provision to pump-out water from the trench or tunnel drainage sump, by permanent and/or portable sump pumps shall be made. Fire deluge water shall be gathered into storm water catch pits and discharged via main storm water system and prevented from entering tunnels.

Primary site drainage and sub-soil drainage shall be provided, where necessary, on external areas generally to protect the site from run off into building areas etc.

Storm water drainage sumps, pumps and electrical equipment shall be located to avoid hazardous areas. If unavoidable then the pumps and electrical equipment shall conform to the requirements in accordance with API and NFPA for operation in hazardous environments. Pumps shall be able to be withdrawn from the sump at surface level for maintenance and repair. The sump and valve chamber shall be equipped with access covers and ladders for maintenance purposes. The sump chamber shall be designed hydraulically so as not to impair pump performance.

Additionally, ESIA requirement shall be completely followed for storm water design including for the requirement of Storm water basin.

7.2.16 Plant Wastewater Drainage System

The Contractor shall be responsible for design of wastewater drainage system complying with the Environmental Regulations and other applicable regulations with respect to wastewater. The Contractor shall provide the following to comply with local environmental protection requirements.

The Plant shall be designed with Zero Liquid Discharge (ZLD) which means that no emissions of liquid effluents other than uncontaminated storm water will be discharged to the environment. Hence, all wastewaters will be directed to an evaporation pond where the accumulating sludge will regularly be disposed.

The Contractor shall present a concept for the wastewater treatment plants and demonstrate the major waste water treatment steps including the corresponding maximum and normal flows.

Treated waste waters from operations that are known to produce effluents will be directed to a dedicated sump where they can be neutralised before being pumped to an evaporation pond. Transformer compounds: provide storage contamination and separation facilities for oil spillage;

- Main building plant drainage: oil spillage, effluent from HRSG cleaning, HRSG blowdown, air heater wash and other non-oily contaminated sources, provide oil separators, chemical treatment and sedimentation facilities/evaporation pond.

The treated effluent shall be discharged by pumping or gravity, ultimately to the final discharge point.

Effluent discharged from station processes shall be treated to comply with local environmental protection requirements. Effluent discharged from station processes into the chemical drains, dirty drains and clean drains shall be routed directly, after treatment, to the final discharge point.

7.2.17 Sanitary Wastewater and Sewage Drainage System

A sanitary wastewater and sewage drainage system shall be provided to allow for the separate collection, treatment and disposal of foul water (arising from toilets, urinals, sinks, etc.).

The sanitary wastewater and sewage drainage system shall be laid in straight lines between manholes. The systems shall be self-cleansing and function with the minimum of maintenance, and access shall be provided so that it is possible to rod the whole of the drainage system. The effluent shall be directed to a self-contained package type sewage treatment plant. The quality of all temporary and permanent drainage discharges from the Site shall comply with the local authority requirements.

Sanitary wastewater shall, to the extent possible, be plant internally re-used after its treatment for irrigation purposes. Its sludge shall be used, if applicable, as fertilizer.

For this re-use, the PME's/NCEC's General Environmental Regulations and Rules for Implementation apply. The sewage system shall consist of a piping system with inspection manholes every 60 m and at each change of direction

The piping and collection system shall provide holding capacity adequate for a 72 hour failure of the treatment system without overflow.

7.2.18 Underground Tanks

Providing of any underground storage tank of any size designed to contain petroleum products or Hazardous Substances should be avoided. If underground installations is deemed necessary, all such tanks shall:

- Be provided in an impervious liners or structures (e.g. concrete vault) under or around tanks and lines that direct any leaked product to monitoring ports at the lowest point of the liner or structure or constructed with double walls; and
- Be provisioned with gauges to measure the material volume inside the tank for reconciling tank content, leak detection, automatic fill shutoff valves and level overflow alarms that are transmitted to a continuously attended workstation (e.g. from the Control room).

Impervious liners or structures shall be fabricated of material appropriate for the substance being contained in the underground storage tanks. If a double walled tank is provided, means shall be provided to detect leaks between the walls.

Contractor shall report to the Owner why an underground tank is proposed and how its design meets Owner's minimum requirements for underground tank installation and operation.

7.2.19 Unloading and Handling areas

All areas used to unload/load transportation vehicles (e.g. tank, trucks) of liquid hazardous substances or petroleum products shall be provisioned with impervious secondary spill containment provisioned with a means to drain accumulated rain water in a controlled fashion (i.e. using a lockable drain valve). The capacity of the impervious spill containment system must be capable of containing one hundred ten percent (110%) of the largest spill (e.g., truck capacity) and a free board of 250mm. Check valve and leak-proof piping connections shall be used for the transfer pipe system to prevent a release in the event of a system failure.

Back-up Fuel oil tanks shall be placed on reinforced concrete ring foundations and be located inside a oil and water tight tank basin of reinforced concrete slabs and surrounding walls with respective oil proof coating.

7.2.20 FRP Pipeline

In addition to the requirements provided hereunder, the EPC Contractor shall follow ACWA's FRP Management system document.

Proper Geotextile layer shall be provided for the FRP pipe trenches to ensure no loss of fine particles are possible from pipe surrounding and pipe bed material. Sufficient gap shall be kept between the pipe edge and trench to allow proper compaction.

When a valve is connected to a FRP pipeline, expansion bellows are to be considered helps in valve replacement and maintenance without cutting the pipelines. The final locations of bellows shall be decided during detail engineering stage subject to Owner's approval.

AWWA M45 shall be followed for the installation of FRP pipes. All fibre glass pipe joints shall be laminated, complying with the manufacturer's recommendation and the relevant internationally applicable and accepted codes and standards.

The laying and installation of FRP pipes shall be as per following requirement:

- The final locations of bellows shall be decided during detail engineering stage subject to Owner's approval.
- The stiffness value has to be ascertained carefully depending on the route and overburden loading.
- Heavy Vehicle routes to be marked clearly with labels and signage.
- A specialized independent third party shall be appointed by the Contractor with the approval of the Owner for design review and Supervision of in-situ installation and testing.
- Following requirement shall be considered as mandatory.
- The specialized independent third party to review and approve the type of joints for the FRP pipe.
- The specialized independent third party to supervise of laying of FRP onshore pipeline.
- The specialized independent third party to review and approve the design for FRP pipeline installation and backfilling, MoS for the laying of FRP pipeline including the backfilling and bedding material quality, FRP Pipe trench sizing and type of FRP pipe joints.
- The specialized independent third party to involve in measurement of deflection and gap for the onshore pipeline during project execution stage.
- The specialized independent third party shall be subject to the approval of the Owner
- Proper marking on roadways for the movement of heavy equipment on FRP pipeline routing.
- Where applicable, thrust blocks, which are concrete pipe thrust restraints or anchors placed up against underground pipe fittings, should be used to prevent movement of underground pipes from thrust forces in water mains.
- Also, and where applicable, permanent markers for underground utilities like cable duct banks, direct buried cables, pipes etc. shall be specified.
- When encasing FRP lines in concrete, we should use a Neoprene pad to protect the pipe, or if the pipe crosses a road, we could install it within a trench or sleeve, which is a good solution because there will be no weight transfer to the pipe. The bottom line is that the concrete must not come into direct contact with the FRP pipe.

In addition to the above requirements, the Document Management system for the Fibre Glass Pipeline, shall be included as Schedule R of this Specification and shall be followed by the Contractor. This management system shall include the requirement of Design, material selection, supplying, trenching, laying, jointing, workmanship, testing of Fiberglass pipeline to be followed by the EPC Contractor.

7.2.21 Architectural Requirements

The Contractor shall develop an architectural design philosophy to provide a consistent design solution throughout the Site. Unifying elements of materials, colour and detailing appropriate to the form and function of the buildings and their location, and for the associated plant enclosures and exposed plant shall be selected to develop an appropriate theme for the development at the specified site. The Contractor's philosophy and theme etc. shall be subject to approval by the Owner.

The Contractor shall be responsible for developing the overall architectural concept for the Plant. The Contractor's tender shall provide proposals for the site layout and the architectural treatment for main and ancillary buildings, taking account of the visual impact and function of the buildings, and their proposed relationship to each other on site.

All buildings shall be designed to ensure that internal noise, vibration, temperature and dust levels are kept within acceptable limits to provide proper operating conditions for the plant which is to be installed and a comfortable working environment for the Owner's operations and maintenance staff throughout the year. Noise limitations have been specified around the turbine housings and the design and construction shall fully take this into account.

Roofs – typically flat roofs with single ply solar reflective waterproofing membrane (with protection) over rigid insulation over concrete slabs, however the material and construction are the Companies design. However, maximum U-value shall of 0.19 W/m²K shall apply to habitable or air-conditioned buildings. The above requirements or local requirement and standards which one is stringent shall be followed.

Walls – concrete structures with concrete masonry units as infill with insulation and a decorative render finish however the material and construction are the Companies design. However, maximum U-value shall of 0.2 W/m²K shall apply to habitable or air-conditioned buildings. The above requirements or local requirement and standards which one is stringent shall be followed.

Rock wool (Type A) insulation shall be used for the insulated cladding.

The Contractor shall make provision in the designs for the maintenance of all high-level buildings, both internally and externally.

The architectural concept for the Plant buildings, including colour schemes and finishes schedules, shall be finalized and agreed with the Owner/Owner's Engineer at an early stage in the design process.

The dimensions of buildings shall provide adequate space for the safe installation and operation of all plant and equipment, its replacement, and for maintenance procedures.

7.2.22 Buildings, Structures and Foundation Design Loads

The loads identified below shall be considered in design; where relevant, the appropriate codes and basic design parameters are also given.

- Dead – All dead loading shall be in accordance with ANSI/ASCE 7-16 "Minimum Design Loads for Buildings and Other Structures " and more specifically BS 648. The self-weight of structure, floors, roofing, side cladding and internal partition walls shall be treated as dead loads. Fixed plant loadings shall be treated as dead loads.
- Live – Live loading shall be in accordance with ANSI/ASCE 7-22 "Minimum Design Loads for Buildings and Other Structures". The Contractor shall determine the correct superimposed loading for all floor areas based on design criteria. ASCE 7-22, IBC 2024 and supplier's input details.

In addition to all plant loadings, the minimum live loads to be considered in the HRSG structure, turbine housings and annexes shall not be less than:

- | | |
|---|-----------------------|
| • Turbine House suspended concrete floors live load | 10 kN/m ² |
| • Firing aisles on burner Platforms in the boiler structure | 7.5 kN/m ² |
| • Turbine House open grid floors live load | 7.5 kN/m ² |
| • Auxiliary cable floors live load | 7.5 kN/m ² |
| • Switchgear and control room floors live load | 7.5 kN/m ² |

7.3 Materials and Workmanship

7.3.1 General

All materials shall conform to the latest edition of the standards cited within this OTS and within the design standards referred to therein. Any materials not fully specified and for which there is not a relevant standard shall be the best of their kind.

Samples of all materials for the Works may be called for at any time by the Owner/Owner's Engineer and shall be furnished by the Contractor free of charge and without delay. The cost of all routine tests required by the relevant standards shall be borne by the Contractor.

All the civil-structural and architectural works shall be designed for a design and service life of 50 years.

7.3.2 Site Clearance, Earthworks, Excavation and Filling

The Contractor shall grub up and remove vegetation of any kind which is established on the site of the new works. The removal and reinstallation of the existing facilities shall be within EPC Contractor's scope. The modification and adjustment of the existing facilities if required shall be within the scope of the Contractor.

The modification, removal, reinstallation work shall be carried out as per Owner and Offtaker approval.

The Contractor shall carry out excavation in any type of soil found. Stripping off of topsoil as required shall be carried out for both the plant areas. Any obstructions encountered in the excavations such as old foundations, abandoned drainage systems, etc. shall be broken out and removed in their entirety. Any excavations so formed shall be filled with well compacted granular fill to the approval of the Owner.

The gradation of granular fill materials used in the works shall produce a stable compacted fill.

Satisfactory soils shall be identified by the Contractor, from his geotechnical investigations and written recommendations. Satisfactory soils shall be capable of being compacted to meet the requirements of the Specification whilst also fulfilling the requirements of the Contractors detailed design of the works, including loading and safe bearing pressure, settlement, stability, drainage, buildability, etc. They include ASTM D 2487 and ASTM D 3282 Soil Classification Groups: GW, GP, GM, SW, SP, and SM, or a combination of these groups; AASHTO M 145 Soil Classification Groups A-1, Plasticity Index less than 6, free of rock or gravel larger than 75 mm in any dimension, debris, waste, vegetation, and other deleterious matter. In addition, the satisfactory soils will be applied to the structural area such as equipment foundation and buildings and may not be applied to other non-structural areas.

Unsatisfactory soils shall be identified by the Contractor from his geotechnical investigations and written recommendations. They include ASTM D 2487 Soil Classification Groups: GC, SC, ML, MH, CL, CH, OL, OH, and PT, or a combination of these groups. Also, AASHTO M 145 Soil Classification Groups A-2-6, A-2-7, A-4, A-5, A-6, and A-7. The Contractor shall notify the Owner/Owner's Engineer regarding the discovery and extent of any unsatisfactory material with recommendations for remediation. Unsatisfactory materials shall not be mixed or stockpiled with satisfactory materials.

7.3.3 Concrete Requirements

To achieve the service life specified, a high quality, durable concrete shall be provided to protect reinforcement, embedded metals and concrete against attack from aggressive chemicals such as chlorides, sulphates and other agents.

The Contractor shall take the following key exposure categories into account whilst preparing the design mix(s) requirements. Design mix(s) shall be subject to the Owner's/Owner's Engineer's approval.

- Concrete below ground level.
- Internal concrete.

All structural concrete shall develop a minimum compressive cylinder strength (f'_c) of 35 N/mm² at 28 days. (This shall be equivalent to a cube strength of 42 N/mm²). However, for the superstructure located in ambient conditions, cylindrical strength of 31 N/mm² at 28 days will be acceptable if proven for required durability and Local regulations.

Blinding concrete shall have a minimum thickness of 75mm and minimum 28 day cube strength of 20.0 N/mm². Also, the blinding concrete in contact with Sea water shall have a minimum 28 day cube strength of 20.0 N/mm², unless more stringent strength requirement is there as per local regulations. Non-structural mass filling concrete can be considered to have a minimum 28 day cube strength 15 N/mm². The minimum thickness of lean concrete shall be 75mm.

In the event that slip formed construction is offered for any part of the works then a special concrete design shall be developed by the Contractor and offered to the Owner/Owner's Engineer for approval.

The outer surface of the concrete in contact with ground and ground water has to be protected with a waterproof bituminous membrane. Minimum one additional corrosion protection method such as use of corrosion inhibitor, cathodic protection or coated reinforcement.

For all reinforced concrete, the recommendation for Geotechnical Investigation report and CIRIA guidelines shall be followed.

Admixtures shall be selected by the Contractor to accommodate his requirements for placing of fresh concrete.

The Contractor shall demonstrate to the satisfaction of the Owner/Owner's Engineer that the intended durability of the concrete mix for the required design life can be satisfied for the particular exposure environment.

The Contractor shall ensure that the Concrete is not adversely affected by the climate factors, namely, higher air temperature, low relative humidity and wind velocity. Air and concrete thermometers shall be kept on the site. Concrete shall not be placed if the shade temperature exceeds 40°C or the concrete temperature exceeds 30°C. Temperature thermocouples shall be placed in the mass concrete foundations to monitor the heat of hydration for concrete. The differential temperature between core and surface shall not be more than 20°C.

The Contractor shall be responsible for ensuring that all constituent materials used for the concrete works (e.g. cementitious, aggregate, reinforcement, water, admixtures etc) comply with recognized international material standards and methods of testing and meet the requirements of this Specification. The use of proposed constituent materials shall be agreed with the Owner/Owner's Engineer prior to their use on the Project.

The following values of minimum reinforcement cover shall be provided:

Precast Concrete (onshore/offshore)	25/50 mm
For all external works at faces cast in contact with the ground water, capillary zone of groundwater or seawater (Onshore)	75 mm
Reinforcement in slabs, columns and beams (onshore)	
- exposed to ambient weather	50 mm
- exposed to ground and not in contact with groundwater	75 mm
Reinforcement in slabs, columns and beams (offshore):	
- not permanently/intermittently submerged in sea water or exposed to sea water splashing,	75 mm or as per applicable code
- permanently/intermittently submerged in sea water or exposed to sea water splashing	75 mm or as per applicable code

Cover for the concrete in contact with seawater and splash zone shall follow the relevant codes.

In addition, the Contractor shall provide sufficient concrete cover and overall cross-sectional dimensions to ensure the correct fire protection to the various elements of structure (where fire protection is required) are achieved. This shall be in accordance with ACI-216 1-97/TMS 0216.1-97 "Standard Method of determining Fire Resistance of Concrete and Masonry Construction Assemblies.

For below ground concrete protective measures shall, as a minimum be provided to meet the requirements of and as specified below. The contractor shall assess sulphate, chloride and pH levels existing in representative soil and groundwater samples as part of his programme of supplementary geotechnical investigation of the site. The concrete protective measures set out below shall be upgraded where necessary based on these.

Location	Sulphate levels below 6.0 g/l (groundwater) 6.7 g/l (soil).	Sulphate levels in excess of 6.0 g/l (groundwater) 6.7 g/l (soil).
Parts of the structure within the capillary rise zone or below ground water table	Bitumen emulsion paint and minimum cover to reinforcement shall be as per ACI standard and local regulations.	Bitumen emulsion paint and minimum cover to reinforcement shall be as per ACI standard and local regulations.
Precast concrete piles	Steel moulds.	Steel moulds

Note that the depth of the capillary rise zone shall be taken as 1.5 m above the highest ground water level (considering seasonal variation) for underground facility design and construction. Sulphate and chloride levels shall be determined by the Contractor's soil investigation.

Bitumastic paint shall be applied on top of all blinding concrete if recommended by Geotech report except where horizontal shear resistance is required.

Embedded materials shall be accurately fabricated and assembled to suit the construction interface required.

All steel embedment in concrete shall be accurately positioned and securely anchored either directly to the formwork or by templates prior to pouring the concrete. Pockets for later insertion of assemblies generally will be allowed.

Embedment shall be clean both before they are installed and after placement of the concrete.

Where embedment are required for major equipment items e.g. Turbine Generator, Pump Foundations, etc, then anchor bolts, attachments and embedment shall be located and secured prior to concrete placement with accurately made steel templates. In the above cases tolerances shall be as specified by the equipment manufacturer. Templates shall be interconnected and braced with steel members that maintain the anchor bolt alignment and position.

However, penetrating type of bolts are accepted if recommended by OEM.

Welding of embedment, to the reinforcement cage to secure their location, will not be accepted.

Corrosion protection shall be provided to all embedment, suitable to the environment in which they are cast. Generally, the following exposure conditions shall be considered as a minimum:

- Dry internal environments;
- Wet internal environments;
- External environments not exposed to seawater or seawater spray;
- External environments exposed to seawater spray/splash zone;

The materials that will be used for embedded items shall be of stainless steel or carbon steel (e.g. sub-plate) as per proven experience and aggressiveness of the environment. The selection of material will be based on strength and durability requirements. Additionally, the choice of material may be dictated by interfacing issues with plant requirements.

The necessity of welding plant fixings etc. to embedment exposed at the concrete surface in exposed environments shall be considered at design stage and wherever possible alternative fixing details shall be provided in order to preserve the original corrosion protective coatings.

Where this detail cannot be designed out then to protect the embedded item(s) from corrosion, for the life of the station, the following minimum material requirements and corrosion protection criteria shall be provided.

- All anchor bolts shall have a concrete cover of at least 100 mm.

The corrosion protection system provided to concrete embedment shall also consider the corrosion protection system(s) provided to adjacent plant and the dual requirements shall be considered collectively, where applicable, to select the most appropriate solution.

Concrete construction tolerances shall comply with ACI 117.

The minimum lateral distance between adjacent individual spread footings bearing at the same grade on soil shall be $B/2$, where B is the footing width or diameter of the larger footing. For Square or rectangular footings, of width B, the minimum lateral separation distance should be no less than B. Where the lateral spacing is less than recommended, the stress distribution between the adjacent footings should be analysed.

Net design bearing capacities and settlement estimates for tanks shall be based on the assumption that the foundation soils underlying the tank are not significantly affected by adjacent tanks. The lateral distance between adjacent tanks should be limited to a distance B, where B is the diameter of the smaller of two adjacent tanks. If the tanks are spaced at a distance less than B, the differential settlement between the centre and edge could be about 15 percent larger.

Furthermore, the landscaping shall be at least 3 m away from the proposed structure perimeters. A 1.2-m wide concrete or asphalt apron could be constructed adjacent to buildings around the perimeters to protect the foundation soils from surface water and sloped away from the structures to facilitate drainage. Water lines should, if possible, be kept away from the foundations. In addition, consideration should be given to the use of an impermeable membrane placed around building perimeters to a distance of approximately 1.5 m beyond the edge of the foundations and sloped away from structures to provide drainage.

7.3.4 Structural Steelwork

Steel sections shall be new and shapes shall conform to American, British, European, Standards or equivalent international standards subject to approval by the Owner during detailed Engineering stage. All structural steel sections shall be hot rolled with a minimum grade A572 in accordance with ASTM standards and the AISC manual for steel construction. Contractor can use built up section higher than 400mm as per the design requirement subject to the QA/QC of the built up section is approved by the Owner. In this case the utilization ratio for built up members shall not be more than 90% for design. Other structural shapes for minor secondary structures (ladders, handrails, platforms, pipe supports, etc.) shall conform to ASTM A36 or equivalent S275 if allowed by local regulations and HSE standards.

Pre-Fab/Pre-Engineered building for shelters, Workshop-Warehouse can be accepted subject to OE's approval for functional and local requirements.

- a. The relevant standards as per ANSI & EN codes
- b. Proper corrosion protection is considered as per OTS.
- c. HVAC design and Fire protection design for PEB buildings shall meet DM regulations.
- d. The design life shall be 50 years.
- e. Metal insulated cladding shall be provided as per OTS.
- f. Relevant certification shall be provided as per the industry standard, if applicable and available
- g. Proper mitigation against water leakage.

- h. Acoustic requirements are achieved.

Hollow sections are allowed provided proper corrosion protection is considered

All bolts in elevated steelwork connections shall be minimum grade ASTM A325 or A490 high strength bolts to suit the required design. All holding down bolts shall be minimum grade ASTM A307. Finishes etc. shall be selected such, to eliminate galvanic corrosion. Bolt finishes shall conform with the finishes applied to the steelwork elements which they connect. Welded connections will comply with the AWS D1.1 of the American Welding Society.

Non Destructive Inspection shall fulfill the requirements of AWS D1.1 of the American Welding Society.

The Contractor shall select coating systems on consideration of climatic conditions prevailing at the site. Such systems shall have a design life to first maintenance of at least 25 years. The contractor's particular attention is drawn to the maritime environment and high levels of windblown salts prevailing at this site. This shall consist of shop applied coatings and site applied finishing coat(s) all in accordance with BS EN ISO 12944. The final site coat(s) shall be applied after steelwork erection/alignment and bolt tightening activities have been completed, unless otherwise agreed with the Owner.

Fire protection to the structural steelworks/ steel structures which are exposed to lube oil fire, shall be intumescent paint system in exposed solutions, and fire-resistant board system or vermiculite or spray system in concealed areas to give 2 hour fire protection when tested in accordance with NFPA.

When required the paint shall be sprayed or applied in accordance with the manufacturer's approved methods onto previously primed structural steel. Suitable mesh 'reinforcement' shall be incorporated to prevent loss insulation when subject to physical damage.

7.4 Electrical Buildings Facilities and Services

7.4.1 Scope of works

The scope of works for the Electrical Building Services shall include the following:

- Lighting systems;
- Small power installation including cables/wiring and distribution systems;
- Lightning protection system.
- Control wiring, etc.

The Contractor shall be responsible for the complete design, detailed design calculations, equipment selection, installation, testing and commissioning and testing of the complete electrical building services systems.

Design calculations, system diagrams and construction drawings shall be submitted for approval. The Contractor shall ensure that the equipment is provided suitable for the location it is to be installed in taking into account temperature and environmental conditions expected on site.

The Contractor shall include for providing all as constructed drawings, which shall be prepared as the works proceed. Completed sets of Operational and Maintenance Instructions including all test, commissioning and any other documentation required to maintain the works.

7.4.2 Lightning Protection System

The Contractor shall provide and install a lightning protection system to provide the necessary protection to each building. Each building shall have its own air terminal network, down coming tapes and earth points.

The Contractor shall connect together with a perimeter conductor all earth rods of each individual lightning system. From the nearest point of the perimeter conductor the Contractor shall provide a link to the Plant earthing system.

The building lightning protection system is to be executed to:

- IEC 62305-1:2010

- IEC 62305-3:2010 and
- IEC 62305-4:2010

And alongside protection of human life, shall prevent damage not only to buildings, but also to electrical and electronic installations.

7.5 Mechanical Building Facilities and Services

7.5.1 Scope of works

For piped services, the Service Water and Compressed Air Installations shall be extended into each building as required to provide wash down/maintenance facilities. The site Potable Water distribution system shall be extended into each building as required for domestic hot and cold-water supplies. The domestic hot and cold-water services shall comply with the requirements of BS 6700:2009 – “Design, installation, testing and maintenance of services supplying water for domestic use within buildings and their curtilages”. Plant areas and emergency washing/shower facilities are also to be supplied with Potable Water.

The heating, ventilation and air-conditioning (HVAC) systems shall be installed to provide the required inside conditions as indicated in the Specification below. Design, installation, testing and commissioning of systems shall comply with the latest applicable Codes and Standards, e.g. ASHRAE, NFPA, SMACNA, ARI.

The Contractor shall be responsible for the complete design, detailed calculations, equipment selection, construction, controls, testing and commissioning of the whole of the Mechanical Building Services, subject to approval. The above systems shall comply with the relevant sections of the OTS. These shall include, but not be limited to, the following:

- Potable water;
- HVAC
- Service water;
- Service (compressed) air;
- General mechanical plant;
- Thermal insulation;
- Fire detection;
- Fire suppression;
- Testing and commissioning;
- Handover, O&M manuals, record drawings.

7.5.2 Design Information

Design calculations, system diagrams and construction drawings for all services and systems shall be submitted for approval, before procurement/construction is commenced. Design information shall be submitted with descriptions and calculations under the following headings.

- Objective;
- Concept of system;
- Method of calculation;
- Design criteria;
- Calculations;
- Appendices.

7.5.3 Quality control

All equipment for the Mechanical Building Services shall be obtained from member firms of the relevant Trade Association(s).

The Contractor shall submit full details for approval, including samples where necessary, of the following components that he proposes to use:

- Refrigeration plant and accessories;
- Air handling units;
- Fans (all types);
- Air filters;
- Grilles and diffusers;
- Fire/Smoke dampers;
- Air cooling coils;
- Air heating coils;
- Control systems and BMS;
- Water storage tanks;
- Pipes and fittings for all systems.

7.5.4 Domestic Hot and Cold Water Services

The domestic hot and cold-water services within the buildings shall comply with the requirements of BS6700: 2009 – “Design, installation, testing and maintenance of services supplying water for domestic use within buildings and their curtilages”

When the direct potable water supply is not applicable, the Administration and Control Building, Social Building, Workshop and General Store and Gatehouse shall be provided with a cold-water storage tank of sufficient capacity to meet one day's consumption for the building or area served. The tanks shall be of GRP insulated construction, shall be located so as to prevent water damage or consequential losses in the event of a tank leakage for any reason and shall feature condensation drip trays to prevent nuisance damage. All tanks which are located above water sensitive areas should be surrounded by a bund wall with adequate evacuation ducts.

The hot water systems shall be served from local electrical water heaters, with central storage systems being provided only for large buildings with heavy demand for hot water.

The Contractor shall also include for Potable Water supplies to emergency wash-down and shower facilities in battery rooms and chemical areas, and to HVAC humidifiers.

The Contractor shall include for all necessary inspection, testing and commissioning, including disinfection, of the complete hot and cold-water services installations.

7.5.5 HVAC Systems and Design Conditions

HVAC systems shall be designed, installed and commissioned in accordance with the latest applicable codes of the following:

- ASHRAE American Society of Heating, Refrigerating and Air Conditioning Engineers.
- SMACNA Sheet Metal and Air Conditioning Contractors National Association.
- NFPA National Fire Protection Association.
- ARI American Refrigeration Institute.
- ANSI American National Standards Institute.
- AABC Associated Air Balance Council.

All systems shall be designed and installed to provide acceptable environmental conditions for each area. The conditions of temperature, humidity, air movement and air filtration shall be controlled as required in the respective areas in summer and winter.

HVAC systems serving critical areas such as Control Rooms shall be provided with duty/standby air handling and mechanical cooling plant to ensure continuity of operation.

HVAC systems shall be arranged to shut down in the event of a fire alarm, with installation of the fire dampers for each individual room, seal supply and extract ducts as required for the application of clean agent fire suppression systems and start up smoke control pressurization systems. Provision shall be made in all plant areas for the venting and clearance of smoke and fumes in the event of fire, by means of roof vents or extract fans.

All systems and components shall be suitable for the design life of the Plant, and be designed for minimum life cycle cost and complexity, consistent with functionality, ease of maintenance and reliability.

For all general cases the HVAC systems the ambient design conditions shall be as ASHRAE recommendations to suite the site conditions. The HVAC systems shall be designed to obtain the internal conditions and system redundancy as indicated in the OTS. Central air-conditioning systems having humidity control may be provided with air handling units for fresh air supply. These units shall be provided with cooling coils and steam humidifiers to handle the full dehumidification/humidification load for the central system, with dew point control. Fresh air-cooling coils shall have 10 per cent margin on heat transfer surface to cater for periods of abnormally high humidity. Face air velocity shall not exceed 2.0 m/sec.

Humidification shall be by electric powered steam generators with proportional control and steam lance injection.

The sound levels in plant and occupied areas shall not exceed the Appendix C of the EPC contract performance guaranties and the noise limits for HVAC systems should be in line with latest ASHRE guidelines.:

Sound attenuators shall be provided in duct systems, as required, to achieve the above sound levels.

All HVAC units and systems shall be provided with local control panels for local manual/automatic control. The operation status such as alarms, humidity, temperature, fault indication, status indication of each air conditioning system shall be indicated and available at the local control panel and any individual fault signal results in a group signal, which shall be transmitted to the central control room.

In addition, the following criteria shall be considered in the design of the HVAC system:

- (i) A general lighting level dissipation of 33 watts/m²
- (ii) Sensible and latent heat gains for occupancy should be allowed on the basis of one person per 10m².
- (iii) A minimum quantity of fresh air should be provided at the rate of 0.00125m³/s per m² of floor area or 12.5 l/s/person, whichever is the greater and this should be shown in the calculations. All equipment gains within each room should be allowed for in determining Plant capacities.
- (iv) Re-circulated air should be filtered. Air from areas where pungent fumes are likely to be present should not be re-circulated.

Battery rooms should be air conditioned by self-contained air conditioning units. The design of the air conditioning system shall avoid recirculation of air back into the battery room. These units should be of the non recirculative type and sized to provide the internal conditions previously specified with a minimum of five air changes per hour. Duplicate extract fans should be provided, mounted at high level and positioned such that accumulations of hydrogen cannot occur. The fan units, ducts and dampers should be suitable for long life in the acidic environment.

The WTP chemical area should be provided with acid vapor preservation systems, otherwise these areas should be isolated as individual rooms with HVAC materials properly selected for acidic environment.

7.5.6 Fresh Air Requirements

Fresh air shall be supplied for the following purposes:

- Fresh air for occupied areas, minimum 12.5 litres/second/per person;
- Free cooling during periods of low ambient temperatures;

- Make-up air for exhaust ventilation systems;
- Building pressurization to exclude dust. Minimum fresh air supply shall not be less than 5 per cent of supply air volume.

Fresh air supplies shall be obtained from areas not subject to contamination from fumes. Air intakes shall be located at maximum available height, to reduce dust load.

Fresh air intakes on large systems shall have inertial sand filters with bleed fans and access sections. Small systems with fixed amounts of fresh air shall be provided with sand louvres.

All supply systems shall have two-stage air filtration, comprising washable primary panel filters and secondary bag filters, efficiency as required for the areas served.

7.5.7 Mechanical Cooling

Complete cooling systems shall be provided. The units and complete installation shall comply with ASHRAE 15 – Safety Code for Mechanical Refrigeration – and ASHRAE Guideline 3 for Refrigerant Leaks, Recovery, Handling and Storage Requirements.

Critical air-conditioning systems shall be provided with 100 per cent standby plant. Critical air conditioning systems are those which are essential for the continued operation of the buildings and allow continued occupancy of essential working areas.

HCFC refrigerants shall not be used.

7.5.8 Ductwork

Ductwork shall be fabricated, installed and tested in accordance with the latest SMACNA Standards. General supply and extract system ductwork shall be of galvanized sheet steel construction. Irrespective of pressure/velocity classification, ductwork at low level in plant rooms and in areas subject to mechanical damage shall be constructed (as a minimum) to medium pressure standard.

All duct systems shall be complete with dampers to facilitate balancing and fire dampers. Splitter dampers will not be accepted. Access doors shall be provided as required for inspection/maintenance of dampers, fire dampers, instruments etc. Final connections to ceiling-mounted supply and extract terminals may be carried out in insulated flexible ductwork.

Extract system ductwork from battery rooms and chemical areas shall be of PVC construction.

High-pressure ductwork shall be pressure/leak tested. Low and medium pressure ductwork need not be tested, except where located outdoors.

The following ductwork shall be insulated:

- Fresh air ductwork
- Low Temperature supply and extract ductwork.
- Medium temperature supply ductwork.
- Ductwork in Electrical rooms, where subject to condensation.

Fire dampers to UL Standards and NFPA-90A shall be provided on ductwork systems passing through firewalls and floors, and as required for compartmentation of gas fire suppression systems.

Dampers shall be operated by fusible link and (re-settable) damper motor, suitable for actuation through the fire detection system and fire damper control panel. Smoke dampers for clean agent fire suppression systems shall be motor operated/solenoid actuated for closing and opening purposes. Damper status shall be indicated on the control panel mounted adjacent to the Fire Control panel.

7.5.9 Air Filtration

All fresh air intakes on large central systems shall be provided with inertial type sand filters, with access sections behind. Small systems employing fixed volumes of fresh air shall be provided with sand louvres.

Primary and fresh air filters on all systems shall be permanent metal panel washable type, dust spot efficiency 20 per cent. A complete set of spare cells shall be provided for cleaning, along with coating fluids. Washing facilities shall be provided, with service water being used for cleaning.

Secondary air filters on medium temperature systems serving plant areas shall be replaceable bag filters, dust spot efficiency 50 per cent. Secondary air filters on low temperature systems serving offices/control rooms/electronic areas shall be replaceable bag filters, dust spot efficiency 90 per cent.

All bag filters shall be supported in proprietary mounting frames for front or side withdrawal. The Contractor shall supply a sufficient number of replacement cells to last during the test periods and during the Defects Liability Period, with one complete set being provided at that the end of the Defects Liability Period.

7.5.10 Power Supplies

The Contractor shall include for all power and control cabling and containment, MCC and controls for the complete HVAC systems.

Dual redundant power supplies (normal ac and safe ac) shall be provided for systems requiring 100 per cent standby plant, including mechanical cooling.

7.6 Miscellaneous Buildings Facilities and Services

7.6.1 Administration Building

- 4 No. Executive Office (Owner Representative) - Single occupancy office with sofa, coffee table and meeting table and 6 chairs
- 7 No. Manager Offices; - single occupancy office with small meeting table and 4 chair.
- 1 No. Administration Office – 4-person occupancy
- 3 No. General Offices (Trading, Engineering and Chemist); – 8-person occupancy
- 2 No. Meeting Room (20 people);
- 1 No. Training Room (20 people)
- 1 No. Prayer Room.
- 1 No. Data/Filing Room;
- 1 No. Kitchen and Messing Room.
- Male and Female Toilets.
- First Aid Room.
- Chemical Laboratory.
- 1 No. Storeroom (10 m²);
- Shaded Car Park – 50 parking area.
- One designated smoking area.
- Office for the External guests (one with 4-person occupancy and one with 8-person occupancy)

The final size shall be decided during detail engineering stage subject to Owner's approval.

7.6.2 Electrical and Control Building

- 1 No. Control Room (at ground level or first level);
- 1 No. Server Room.
- 1 No. Data/Filing Room.
- Male and Female Toilets.
- 1 No. LOTO/PTW Room.

- 1 no. of Locker Room (10 lockers).
- Pantry/Dining area.
- Meeting Room

7.6.3 Workshop and Stores

The building shall house at least the following rooms and facilities:

- 1 No. Workshop Area;
- Mechanical and maintenance workshop;
- Electrical maintenance workshop;
- I&C maintenance workshop;
- Store room for big tools and spare parts;
- Store room for small tools and spare parts;
- Store rooms for chemicals;
- Domestic storage room;
- 1 No. General Stores Area;
- Offices; (5 offices with 4-person occupancy)
- 1 no. of locker room (20 lockers)
- Janitor's rooms;
- HVAC room.
- Adequate no. of Toilets and Showers; (considering 60% Male and 40% Female occupancy)
- 1 No. Environmentally Conditioned Store Area;
- Overhead EOT crane(s) as required for mechanical, electrical equipment and sized for the heaviest equipment + safety margin as per local law and regulation but not less than 10%
- Adequate entry and exit plan for the Trailers shall be considered from two different ends of the Warehouse.

The final size shall be decided during detail engineering stage subject to Owner's approval.

7.6.4 Hazardous and chemical storage area

- 1 No. Hazardous Equipment and Gases Store Area;
- 1 No. Hazardous Materials Store Area;
- 1 No. Hazardous Chemical Storage Area;
- Washing facilities;
- Eye Wash Shower;

The final size shall be decided during detail engineering stage subject to Owner's approval.

This facility shall be considered additionally at a separate location and away from Plant main workshop, store buildings and offices.

The facility shall be fenced and shall have access control (key or badge access). The facility shall be cover against sun, rain and dust-storm.

7.6.5 Social Building

- 1 No. Locker Room (50 lockers);
- Toilet Rooms, Wash Basin, shower Room for Male and female.

- Canteen.
- Kitchen and Dining Hall.
- One Designated smoking area (to be arranged outside the building);
- One Social Room for the plant O&M staff.
- Prayer room.
- Shaded Car Park – 50 parking area.

The final size shall be decided during detail engineering stage subject to Owner's approval.

The administration building shall be combined with Social building.

The Contractor shall provide all necessary expertise, labour, construction plant and equipment, and materials for the satisfactory construction of the Works.

On the basis of the approved design calculations and design drawings, the Contractor shall prepare all necessary construction drawings and design works associated with these drawings including, but not limited to, architectural construction and detail drawings, building services construction and detail drawings, layout and placing drawings for reinforcing steel, reinforcing bar detailing and bar schedules/drawings, concrete placement drawings, formwork and shuttering drawings, design of connections between structural steelwork members, preparation of detailed shop structural steelwork fabrication and erection drawings, material lists and weight schedules etc.

The scope of supply shall cover, in addition to any other necessary components which may be required for the proper and continuous commercial operation of the complete installation including any and all auxiliary systems, the complete construction of all components specified in this OTS (and including but not limited to all labour and material procurement, delivery, transport, supplementation, maintenance, consumables, dismantling of all auxiliary and ancillary work and furthermore, all insurance, handing over of all work in satisfactory condition, maintaining during the guarantee period until the final acceptance of the entire complete works).

The Contractor's scope of work shall include all activities necessary to properly design, supply and execute all works for the facilities, infrastructure, substructures and superstructures of all the above items, as detailed in this OTS. Any works that are not specifically mentioned but are evidently needed for the proper supply and execution of the Work, shall also be included and shall be carried out in accordance with the relevant standards of this OTS.

The EPC shall repair all facilities Owned by other agencies, public and/or private facilities damaged through his use up to the satisfaction of the Owner.

7.6.6 N.A.

8. PLANT CONTROL AND OPERATIONAL SYSTEM

The plant shall be provided with a process control system based on redundant configuration for safe and reliable operations with high availability. The design, control philosophy, redundancy concept, grade of integration, etc. will be subject to review, comment and approval by the Offtaker. The scope of supply shall include - but not be limited to - the following systems and components:

- Primary sensors, transmitters, actuators minimum equipment required shall be according to relevant IEC code or International Standards
- Plant control and monitoring system (DCS – Digital Control System) for CCR & Supervisory Control and Data Acquisition (SCADA) for the Substation and other Plant packaged systems, including all necessary software licenses for the Plant with redundant operator stations and additional one operator / engineering station.
- The plant and control and monitoring system shall be designed to fulfill all Cyber Security regulations including HCIS requirements and international standards such as ISO 27000 series (especially ISO 27001, ISO 27002 and ISO 27019)

- The network architecture shall be designed using quadrant segregation and will fulfill all requirements as per ACWA cyber security policy.
- Long-term data server via RAID station (Redundant Arrays of Independent Disks) for central back management and archiving.
- Network with routers and switches
- Copper/FO-converters where applicable
- Plant monitoring system and performance calculation (at all design conditions).
- Settlement metering system including check meter shall be provided in line with the latest version of the Grid Code at the date of the Bid submittal.
- Hardware and software for Plant Accounting Settlement System (PASS)) including and other requirements (data transmission, Remote control from LDC, cybersecurity, AMI integration) as defined by Offtaker.
- Internal communication systems (telephone, public address system, LAN/ WAN system)
- GPS-based clock systems
- All relevant BOP system like HVAC, Fire-fighting and Fire detection systems, CCTV and surveillance systems in accordance with the country local requirements i.e. HICS KSA .
- Ergonomic designed control room
- Field testing equipment
- Instrument workshop equipment.

8.1 Operational Philosophy

The plant control and operation system shall meet all OTS requirement. Irrespective of contractor using any configuration of an individual and specialized DCS system in conjunction with black box systems supplied by the subcontractors, from an operator's perspective the Plant DCS and field equipment shall provide for the control of fuel system, Gas Turbines, HRSGs, Steam system, Steam Turbines and required ancillary plant. Therefore, all the required control systems shall be fully compatible with each other and shall be available on a common operator interface for the entire plant.

The system shall be designed to have a functionally and geographically distributed architecture utilizing a number of independent outstations. The equipment located in the DCS cubicles shall include all necessary control processors, input/output (I/O) cards, power supplies, data highway interfaces, marshalling and termination facilities,

The microprocessor-based distributed control system shall comprise of multifunctional modules operating over a network. The operators have access to all significant process information and all control. The DCS will have redundancy throughout and will have no single point of failure that will result in reduced load,

All necessary hardware for marshalling and terminating incoming and outgoing plant cabling shall be provided. This shall include gland plates, glands, terminal blocks, intrinsic safety barriers or explosion proof instrument, isolation devices, labelling and wiring. Equipment shall be designed to operate in hazardous areas where necessary,

Analogue and Digital signal will have separate cables. Analogue cables shall be twisted pair and screen protected.

All Processes tags available at Operator workstations (up to 1 sec. frequency) and CMS tags (up to frequency required by CMS provider) shall be made available to the Owner at OPC server level.

The DCS shall be designed to ensure maximum availability by the inclusion of built-in redundancy for both hardware and software. This shall include duplicated control processors, data highway and power supplies

with automatic changeover to the standby unit upon detection of a fault or failure of the operating unit. Where appropriate, I/O cards shall also be duplicated. The failure of any single element shall not affect the operation of any item of operating Plant. In particular, the duplicated control processors shall be capable of fail over without change to Plant status, sequence status or control actions, at any stage of Plant operation.

The DCS shall be supplied with all hardware, system and applications software (VDU displays, alarms, etc.) fully developed, tested, debugged and installed for all analogue and digital I/Os. An overall 10 percent spare capacity shall be provided to all necessary hardware including I/O units and associated wiring, supplied and tested.

- The plant control system shall have an OPC (read/write) DA server and all the tags including Substation, GTs, HRSGs, STs & BOP (including Meteorological station & Fire detection systems) should be mapped in the OPC Server. The OPC tag list should be provided in csv format with Tag Name, Measuring unit, Range, High and low limits.
- The Site historian (ODBC) shall be provided to archive the data with 1 sec interval and third party system connectivity shall be enabled for fetching the data
- Above mentioned OPC DA server shall be supplied with necessary client license for third party system connectivity for the data access.

Above OPC system network architecture shall be subject to ACWA cybersecurity policy and shall include the necessary firewalls, IPNs etc..

The Contractor has to ensure that the Plant DCS and Plant Substation Automation System are interconnected so that the substation can also be controlled from DCS system and remotely through OPC connectivity.

The Plant should have full remote monitoring and control capability and the Contractor should demonstrate such capability during commissioning and at the time of handover. Any required licenses for such purposes should be part of the Contractor's scope.

If communication between plant control room and Offtaker dispatch centre (Hotline) is installed, a duplicate line shall be provided to ACWA / NOMAC Monitoring and Prediction Centre remotely either at Jeddah or Dubai.

The Instrumentation & Control (I&C) system shall support an efficient and reliable operation, control, and supervision of the whole plant by a minimal operational staff. Start-up, synchronization and loading of the GT units, the HRSGs and the ST unit from the Central Control Room (CCR), shall be possible automatically. through pressing one pushbutton after the completion of manual preparation for BOP facilities by the operator. Plant load control shall be possible from a remote Load Dispatch Centre (LDC), automatically via appropriate interface lines, if required.

The supervisory section incorporates operator stations, an engineering station archiving and communications and the control section comprises control outstations (automation stations) supplied by the DCS vendor and/or programmable logic controllers (PLCs) supplied by packaged plant vendors. CCR should be provided with fire and gas display panel, communications and security television screens etc. in addition to the operator and engineering stations

If Plant Billing System will be provided by others, the interfaces to the Plant Billing System and other operator's computer systems are to be included. Plant billing system shall be available with dedicated servers and shall not depend on Plant DCS servers, except for data retrieval. Main and Back-Up Metering Equipment for water and fuel shall be provided by the EPC Contractor. The Main and Back-up Metering Equipment for Electrical Energy shall be provided by EPC Contractor in compliance with Metering Code of Grid Code.

The contractor shall ensure that the control system provided for plant will have at minimum the following basic functionalities:

Upward Compatibility with newer versions of control system, to avoid control system being obsolete during project life.

Online modification should be possible (for control logics and HMI graphics) without the need to take the equipment out of service and the control modules should be hot swappable.

Control and Operating system should be the most updated and latest version and all control system software and operating systems will be updated with latest versions at PCOD.

A centralised facility for management of patches (including software updates and revisions) shall be provided. The solution shall incorporate all supervisory layer Servers, Operator stations, Workstations, etc (including Plant DCS, Steam turbine DCS, CGT DCS, etc).

A centralised system for backup and restore shall be provided. Such solution should also enable a complete hard-drive image backup for all workstations (including Engineering workstation), servers and control logics, which are part of the control system. The back-up and restore shall be capable of being performed to a network server and to removable storage media. This functionality should be provided using standard, commercially available back-up and restore software, which should be approved by the Client

Availability and redundancy

The control systems shall have an availability of at least 99.98%. To achieve this, critical parts of the control systems shall be arranged in a redundant configuration, including the Rumah and Nairyah plants control and monitoring system servers, OPC servers, the Rumah and Nairyah plants control and monitoring system controllers and the network communication equipment. For all control systems, critical communication, critical security systems and electrical protections, the data highways and data servers shall be fully redundant. .

The design shall be such that the impact of a failure of any single piece of auxiliary equipment on the output of the Rumah and Nairyah plants is minimal and shall not cause a failure to the entire Rumah and Nairyah Plant and shall not result in the loss of generation. Any system, sub-system, component, instrument that, in case of failure, might cause the outage of the complete Rumah and Nairyah Plants shall be designed for high reliability and maintainability. Redundancy criteria shall be adopted to avoid single point of failure. N+1 redundancy shall be used for critical control loops, for instruments with Dual/Triple redundancy shall not require additional N+1.

The Rumah and Nairyah Plants shall be monitored and controlled by the DCS Control system which shall be interconnected, through redundant communication data link, with central control system.

Any system that can cause a complete outage if one (1) component fails shall be designed as redundant equipment and for high reliability, ease of maintenance and quick system backup support i.e., no single point of failure. If a failure in an instrument or in a control component can directly or indirectly cause the failure of the whole system, this component shall have a redundancy factor. The trip or outage of any single item of equipment or any single item of auxiliary equipment shall not affect the operation of the remaining equipment; in other words, priority shall be given to stability of the electrical grid system.

Reliable power supply and remote control of common facilities shall be maintained in case of outage of one single component.

Critical equipment shall be powered by two sources with a changeover arrangement with sufficient speed to prevent disruption in the equipment operation.

If a failure of the duty equipment is detected, the system shall automatically transfer to the backup equipment. Failure of backup equipment shall be detected as soon as it occurs rather than when it is required to operate.

Redundant AC, Safe AC and DC auxiliary energy back-up systems shall be provided. Each battery shall be able to supply the 100% loads for one (1) hour in case of a total supply voltage failure. DC and UPS system batteries shall be Ni-Cd type (For Gas turbine package, vented lead acid type batteries with 1 hour discharge period) and shall be sized for a discharge period of one (1) hour at 100% load. Valve Regulated Lead Acid (VRLA) type batteries shall not be considered. UPS and DC system to share the common battery bank is acceptable to provide complete static and dynamic load.

The entire plant shall be designed to minimize common mode failure. A unitization philosophy shall be adopted for all systems and devices to ensure that the risk of unit trip due to common mode failure does not occur.

Every safety-related loop must be verified according to the standard IEC 61508, IEC 61511 to ensure that the requested and necessary Safety Integrity Level (SIL) is fulfilled for the individual loop. The contractor must assess all safety-related loops to ensure compliance with the related requirements and must provide a protocol about these studies. Moreover, I&C equipment located in the hazardous/ flammable zones are to be certified (e.g. ATEX) for their respective zone classifications.

The instrumentation and control equipment shall have high electro-magnetic and radio frequency interference immunity and shall not be affected by portable radio transmitters operated in the vicinity of the equipment. Any limitations shall be stated.

All I&C equipment shall have enclosure classification not less than IP 54 according to IEC 60529 when mounted in an enclosed building and IP 65 for mounting outdoors. Control cubicles installed in air-conditioned rooms shall be at least IP 41.

Sunshades shall be provided for all cubicles located outdoors and all cubicles shall be adequately ventilated or air-conditioned, if necessary, for operability.

The DCS shall be fully integrated with the control, instrumentation and alarm equipment of the plant being provided, in order to maximise automation of the Plant and reduce the operator manning levels.

Annunciation will be predominantly in the DCS. Major packaged sub systems will have a local alarm system, with a single trouble alarm to the control room. Major alarms and protections will be time tagged. A dedicated annunciator graphic display page will be provided in the DCS for these miscellaneous alarms.

The manufacturer's standard packaged control systems may be provided for such plant as Steam Turbine, generators, steam generation, BOP, water treatment plant, in addition to the DCS, for local control and monitoring facilities, but these control systems shall not be specifically designed for this application. The standard packaged control systems may include control, instrumentation and alarm equipment, and they shall be integrated with the DCS to allow a high level of centralised control and monitoring.

Centralised control and monitoring shall be provided from the CCR via the DCS. The DCS shall obtain plant data from a set of outstations. After suitable processing, the data shall be transmitted in digital form over the data highway to the CCR where the data will be presented to the operator(s) on VDUs. The data shall also be logged for analysis at a later date. The control outputs generated for each mode, for example start-stop and set-point raise-lower instructions shall be transmitted to the outstations over the data highway. The system design shall be such that failure of the DCS shall not affect the safe operation of the Plant. Control processors, operator and engineering stations with 16 bit microprocessors are not acceptable.

As an example: Control of individual drives of the package auxiliary systems will be possible only from the operator interface of the package control system. From the DCS it will be possible to perform only commands related to the package intended as a function group. The required signals for the DCS will allow monitoring of the package system.

The Steam Turbine shall be controlled by the DCS via hardwired connection and be monitored in the DCS by serial Modbus interface. However, if ST control system is the same type as the main DCS supplier it will be monitored and control via the data highway.

Serial interfacing with other Computer or Programmable Logic Controllers (PLC) is allowed for monitoring purposes only. This means that signal interfacing with serial link is not allowed for controls, and control signal interfacing will be hardwired to packages from DCS.

8.2 I & C Concept

These objectives will be implemented by a supervisory system that is based on a functionally and geographically distributed microprocessor-based plant level controllers using redundant architecture techniques. The redundancy shall be complete including the processors, interface main communication module, power supply modules, station bus. The man machine Interface (MMI) in the control room shall include all necessary visual display units, keyboards and other devices required to operate, configure and

monitor the Facility. The number of hard-wired and/or conventional instruments shall be minimised commensurate with safety and legal requirements; in particular, the Contractor shall provide hard-wired pushbutton switches for emergency shutdown of each gas turbine and each steam turbine. Moreover, where equipment presents a danger to people (e.g. rotary screens), the Contractor shall provide local hard-wired pushbutton switches in general in accordance with any legal requirements.

A consistent instrumentation and control philosophy shall apply throughout the plant and shall be implemented in terms of a range of equipment exhibiting a minimum diversity of type and manufacturer. The objective shall be to standardize measurement and control equipment throughout the plant in order to rationalize operation, maintenance and reduce spares holding.

The main items of plant such as the GTs, steam turbines, water treatment etc. may be supplied with proprietary, purpose-designed, systems for control, monitoring and protection. Such systems shall be designed to function independently of each other but shall be interfaced to the supervisory level to allow co-ordinated operation of the Facility. The system shall have a high level of availability. As the C&I system employs duplication of facilities at the supervisory and MMI levels and they must be unitised at the automation (plant) level, no single C&I failure shall:

- Cause danger to personnel, environment and plant;
- Invalidate protection by inhibiting a trip or interlock (except on minor plant items);
- Shut down more than one major plant item (equipment or machine);
- Simultaneously trip a plant item and invalidate its auto-change-over;
- Cause a control failure and simultaneously inhibit protection;
- Cause a control failure and simultaneously inhibit the alarm indicating the consequent condition of the plant;
- Affect more than one control area.

There shall be no single common cause failure possibilities (e.g. air supplies, static switches, fuel supply etc.). The automation level outstations shall be able to continue to operate safely in the event of communication failure and/or failure of the supervisory level. Safety critical inter-trips between plant sections shall be hard wired in addition to any protection relying on data communication (and must meet the single failure criteria above). This shall be achieved by the use of redundant design techniques and equipment.

The interface point is the interface panel(s) provided as part of the Interface Room in Grid Station and power plant Interface Room:

- Transfer of signals between the Grid Substation and the Plant (in both directions); and
- Data exchange between the Plant and the control centers of the TSP

Critical plant control/protection (i.e. potentially causing danger to personnel or damage to plant e.g. overspeed, drum level, BMS etc.) may be achieved by separate dedicated systems or vested in the DCMS. Precautions shall be taken to ensure a single failure cannot invalidate protection (e.g. separate input cards, two-out-of-three (2 out of 3) failsafe outputs on separate cards dual processors etc.). The protection of plant items vested in the DCMS shall use (2 out of 2 or 2 out of 3) voting logic, limited to major plant components. Control cabinets and MMI equipment located on the Site for the critical plant control shall be designed based on DC power and AC (emergency AC) power for each control system and UPS power for MMI equipment and printers.

For Gas turbine package, triple redundant sensors with 2 out of 3 voting logic for GT trip function and dual redundant sensors for modulating control loops will be considered.

At each measuring point, sensor will be common for both control and monitoring function. In case that redundant analog sensors are provided for protection, they will be also used for both control and monitoring function.

Redundant analog and digital I/O signals will be assigned to different I/O modules.

A single sensor will be applied to the following cases:

- Sensor used for process status monitoring
- Sensor used for status monitoring of auxiliaries
- Sensor used for any correction of modulating control loops and process
- Sensors used during start-up period
- Redundant sensors cannot be provided due to physical limitation at a measuring point
- Sensor used for modulating control and auxiliary controls for back-up fuel (fuel oil and water injection system)

Machines over 500 kW shall be equipped with vibration transmitters (4-20 MA directly connected to DCS and installed on the pump side bearings and vibration protection system (interlock). Temperature monitoring shall also be provided where technically feasible. Machines over 2500kW shall be equipped with vibration monitoring system.

Local starting and protection of diesel fire-pump(s) and emergency generators shall not depend on the DCMS, as it may not be available when these are required. They shall be monitored by the DCMS. In case DCMS remote starting is provided, provision for local starting shall also be made available for emergency situation.

Except where protected by covered cable trays, protective conduits or other protective cable routing materials, I&C cables shall comply with IEC standards.

8.2.1 Operating Systems

Any system(s) version – operating system(s), software(s), DCS/ICS/SCADA system(s), firmware(s), equipment(s), firewall(s), switch(s), etc. installed within the network shall be the latest version existing on or around the PCOD and should stand valid for support for at least 5 years after the PCOD. If the EPC Contractor fails to comply with the aforementioned requirements, the EPC Contractor shall remedy such non-compliance and bear all costs in doing so including but not limited to the costs in connection with upgrading the relevant system(s) version – operating system(s), software(s), DCS/ICS/SCADA system(s), firmware(s), equipment(s), firewall(s), switch(s), etc. installed within the network.

8.2.2 Instrumentation

The following basic requirements must be observed:

- Selection of instrumentation shall consider high long term stability/accuracy under the prevailing environmental conditions, high reliability, low power requirements and low maintenance
- All instruments shall be of high standard industrial type and be as uniform as possible
- All instruments shall be solidly built using well proven high quality components of the latest up-to-date technology
- All parts of the instruments in contact with the process fluids shall be fully compatible with those fluids and shall not deteriorate under operational environment.
- Particular attention shall be paid to the ease of access to all instruments. The instruments shall be suitable for mounting in visible positions with easy access for adjustments.
- Measuring errors and response time shall be as low as possible and technical reasonable.
- For connection of the process values to the control system electronic type instrumentation shall be used. Primary instrumentation shall be transmitters having rough and reliable measuring cells with microprocessor-based electronics protected in a tight housing. Transmitters of the SMART type with incorporated local digital displays shall be used. Where instruments are located in inaccessible locations, local indicators shall be provided. (In Gas turbine package, HART type pressure, differential pressure and level transmitters with a built-in digital indicator will be used. However, the transmitters located in GT enclosure, fuel gas unit and fuel oil unit will not have the digital indicator because it is not allowed to enter the areas during GT operation due to safety reason.)

- All instruments shall have terminal stripes (no flying leads for external termination).
- The field instrumentation shall be weatherproof and rated at least to IP 64 according to EN60529 when mounted in an enclosed building and IP65 for mounting outdoors.
- Vibration sensors for the gas turbine do not have IP degree, and GT combustor pressure fluctuation sensors have IP degree less than IP 52 as per Supplier's standards.
- Manufactured, assembled and tested in the factory in conformity with certified international quality assurance and quality control standards (ISO).
- The instruments shall guarantee the accuracy values specified in this section. Accuracy will be expressed in percent of adjusted span as far as not specified in percent of the measured value.

The given %-values represent each instrument (including entire loop).

For Gas turbine package, the instruments of the generator and its auxiliaries are designed according to GT OEM standard in the following systems. .

- Generator Cooling Water System
- Generator Seal Oil System
- Generator H₂ & CO₂ Gas System
- Generator H₂ & CO₂ Gas Supply System

Instrument description	Unit of Measure	Accuracy	
		% of span	% of measured value
Transmitter (transmitter and converter)			
- pressure	Bar	±0.25	
- diff. pressure (Barton type)	Bar	±0.50	
- flow (billing)	m ³ /hr		±0.50
- flow (other types)	m ³ /hr		±1.00
- temperature	°C	±0.50	
- level	m (mm)	±1.00	
Switches			
- pressure		±1.00 ¹	
- diff. pressure		±1.00	
- level ¹⁾		±1.00	
- others		±1.00	
Direct Type Indicators			
- pressure (Bourdon type)	bar	1.0	
- diff. pressure (Barton type) ²⁾	bar	1.0	
- flow (standard)	m ³ /hr	1.0	
- flow (billing)		0.2	
- temperature	°C	1.0	
- level	m (mm)	1.0	
- others		1.0	

Note 1. For Gas turbine package, GT OEM shall try to meet the required accuracy level. In case, GT OEM supplier are unable to meet the requirement, then GT OEM will provide the most accurate switch available.
Note 2. For Gas turbine package, +/- 1.6 % or better in general.

The instruments shall guarantee the accuracy values specified in this section. Accuracy will be expressed in percent of adjusted span as far as not specified in percent of the measured value. The given %-values represent each instrument (including entire loop).

All Instruments shall be mounted in such a position as to guarantee the specified accuracy of measurement, easy access for maintenance, good visibility of indicating devices and standardization of installation.

Note: the accuracy class of billing instruments shall be coordinated and approved from the project company.

8.2.2.1 Control and Instrumentation Signal Levels

The following signal levels shall be used for instrumentation and control valve actuating:

- i. Electronic signals 4 to 20 mA at 24 V dc
- ii. Conditioned signals 1 to 5 V dc
- iii. Solenoid at relay drive 24 V dc at 500 mA maximum
- iv. Alarm contact drive 48 V dc maximum
- v. Alarm contact potential free
- vi. Pneumatic 0.2 to 1.0 bar

General for all DCS interfacing with:

- a) Packages
- b) Electrical units like MCC, Breakers Transformers and OLTC etc.
- c) Field instrumentation direct connected from the field.
- d) Computers or Programmable Logic Controllers (PLC)
- e) Junction Boxes and terminal points

Following are all signals read as I/O from or to the DCS system except for serial interfaces and for temperature measuring points. No other signal Inputs or Outputs are allowed from or to the DCS system.

AI (Analogue Inputs) 4-20 mA

1. Smart transmitters 2 wire system 24 VDC / 4-20 mA DCS Active
2. Optical or galvanic separated by the source 24 VDC/4-20 mA DCS Passive 1)
3. Smart position transmitter 2 wire system 24 VDC / 4-20 mA DCS Active

AO (Analogue Outputs) 4-20 mA

1. Smart Valve positioners 2 wire system 24 VDC / 4-20 mA DCS Active
2. Optical or galvanic separated by the source 24 VDC / 4-20 mA DCS Active 2)

PI (Pulse inputs) 24 VDC

1. Metering device MW Potential free contact 24 VDC fast pulse
2. Metering device for Flow 24 VDC fast pulse

DI (Digital Inputs) 24 VDC

1. Potential free contact in the field 24 VDC NO or NC
2. Potential free contact from a package 24 VDC NO or NC

DO (Digital Outputs) 24 VDC

¹ 2% hysteresis for all switches

- | | |
|-------------------------------------|--------------------------------------|
| 1. 24VDC source from DCS to Package | 24VDC Interp. relays in package/ MCC |
| 2. Potential free contact from DCS | Voltage coming from a Package system |

Notes:

- 1) Signals coming from packages and powered from a source other than DCS
- 2) Signals going to packages which are powered from a source other than DCS

8.2.2.2 Field Mounted Instruments

Unless otherwise specified or approved by the Owner in accordance with the Contract, all parameters (e.g. flow, level, pressure, temperature) for indication on local control panels or control panels remote from the point of measurement shall utilise transmitters which convert the measured parameter into a standard dc electrical signal. For specific requirements, RTD/TC can directly be connected to DCS after discussing with OE. Steam Turbine/Generator set OEM standard and general industrial practice are acceptable.

Separate transmitters shall be provided for critical control and indication purposes. Indication shall include alarms. If there is no control function, indication and alarms shall have separate transmitters in case hat switches are not used separately for alarm and interlock. A single transmitter shall be supplied for non-critical and performance monitoring.

All outdoor field mounted electrical items shall be weatherproof to IP 65 (IEC 60529) or equivalent and suitable for the local ambient weather conditions. Pneumatic instruments shall be weather-proof to IP 54 or equivalent.

Each transmitter, which is used to transmit a signal to the DCS, shall be powered from 24 V dc supply and utilise a two-wire transmission system.

All electrical and pneumatic transmitters shall be of the indicating type or equipped with separately mounted local indicator. All transmitters shall be provided with test connections and additionally pressure and differential pressure transmitters shall be provided with instrument isolating valve manifolds. (In Gas turbine package, all pressure and differential pressure transmitters will be provided with test connections and instrument isolating valve manifolds)

Switches for pressure, flow, temperature and level monitoring shall be of the heavy-duty type with double pole changeover contacts. Contacts shall be rated to suit their required duty. Mercury switches shall not be used.

8.2.2.3 FLOW INSTRUMENTS

Flow measurement devices operating on differential pressure principles shall be designed in accordance with the requirements of BS 1042 or ASME. Design the orifice plates and nozzles shall be according to ISO 5167.

For high-pressure applications, the carrier for the primary device shall be of similar material to the pipeline in which it is to be mounted and shall be connected into the pipeline by butt-welding. Other applications shall use flanged connections wherever possible. For line sizes below 50 mm, a precision pipe assembly shall be utilised with a minimum line size of 50 mm.

All differential pressure devices shall be provided complete with tapping point isolation valves, 3 valve manifolds and drain valves or 5-valve manifolds, as required.

Orifice type flow meters shall be used for gas flow measurements. (In Gas turbine package, coriolis type flow meter for GT fuel gas flow measurement is provided) The flow meters shall in all cases be suitable for the process conditions applicable to the measurement location and the environment in which it is mounted. Meters used for tariff metering purposes shall be supplied, complete with adjacent pipe sections where necessary, with calibration certification from an approved laboratory. Displacement flow meters shall be capable of accuracy to within ± 0.25 percent.

Where density, temperature or pressure corrections to tariff metered signals are required then such measurements shall be made close to the flow meter location.

Flow meters employing alternative measurement principles may be utilised for appropriate applications subject to the approval of the Owner in accordance with the Contract.

Flow integrating motors shall provide output in the form of pulses from volt -free contacts to the DCS.

8.2.2.4 LEVEL INSTRUMENTS

Level measuring devices may be of the direct measurement, differential pressure or electrical/electronic type as appropriate to the application.

Reflex-glass or magnetic level gauges will be used. Gauge glasses used in conjunction with level instruments will, as a minimum, cover a range in excess of that required for alarms and trips. Level gauges will be selected so that the normal vessel level is approximately at gauge centre, where practical.

Level indication for corrosive service will use devices other than reflex-glass gauges. Level gauges for high pressure service will have suitable personnel protection.

Each measuring instrument shall be capable of being removed without the vessel or other instruments being taken out of service.

Differential pressure transmitting devices shall preferably be used on all services other than slurry.

Displacement type instruments and switches shall be mounted in external cages with flanged connections, rating same as the vessel. This type of instrument shall not be used for applications involving viscous, corrosive or flashing liquids.

Direct measurement of level by means of internally mounted floats, etc, shall only be used when the switch point is either well defined in advance or is adjustable in service, and the vessel can be emptied and/or de-pressurised for the removal of the switch without effect to the normal operation of the Plant, or where choking of extended connections is likely to occur.

Float switches shall be glandless with magnetic coupling.

In Gas turbine package, guided wave radar type and displacement type level transmitters will also be used as per Supplier's standard design.

8.2.2.5 PRESSURE INSTRUMENTS

Low pressure gauges may employ manometer or barometer devices to indicate small gauge, differential pressures or vacuums.

Sensing element materials shall be selected to resist corrosion by the process medium.

When gauges or transmitters are to be used for duties where the process medium temperature exceeds 70°C, siphon loops shall be utilised. (Siphon loops will not be utilized in gas turbines, gas turbine generators and their associated auxiliaries. Instead, impulse piping for pressure gauge or transmitters is considered in the GT package.)

Gauges measuring low ranges that can be exceeded by a higher containment pressure shall be fitted with over-range protection. Gauges shall be fitted with vent/test valves. Blow out devices shall be provided for high-pressure applications.

Pressure switches shall be such that under normal process operating conditions, the contact in the switch shall be closed unless otherwise approved by the Owner in accordance with the Contract. Switches shall have adjustable set points and adjustable hysteresis, as required. (In Gas turbine package, hysteresis of pressure switches will depend on manufacturer's standard specification.)

8.2.2.6 TEMPERATURE INSTRUMENTS

The method of temperature measurement to be employed shall be selected for the particular application bearing in mind requirements for accuracy and reliability.

Sensing elements for air, inert gas and radiant temperature measurements shall be complete with suitable protective sheath. For steam, water and hazardous applications, Thermowells shall be utilised.

For Turbine/Generator set OEM standard and general industrial practice are acceptable.

Resistance temperature detectors (RTD's) shall be to IEC 60751 Grade II, resistance of 100 Ω at 0°C, using 3-wire connections and applications up to 600°C. On differential temperature measurements RTD's to IEC 60751 Grade I will be used. RTD's shall be used only where they can be readily replaced without loss of Plant capability.

Where thermocouples are employed then cold junction compensation shall be provided.

Where an external junction box is provided it shall be located in a position so as not to be in excess of 50°C or a lower temperature commensurate with the insulation material of the compensating leads.

In the case of direct immersion metal couples of the mineral insulated type, the couple up to the associated junction box shall be capable of easy replacement. A flexing coil having a minimum diameter of 250 mm shall be provided in the couple between the junction box and the carrier.

Where thermocouple assemblies are supplied, a disconnection head shall be arranged clear of the pocket and associated lagging and shall be sufficiently long as to prevent excessive temperature of the head assembly.

Local temperature indicators shall be supplied with a thermowell and shall preferably be of the bimetallic type and of the multi-angle variety.

No temperature measuring system shall use mercury as its sensing medium.

In general, temperature instruments will have scales with temperature units in degrees Celsius. Dial thermometers will have 100 mm or larger dials and white faces with black scale markings and will be every-angle type and bimetal actuated. Dial thermometers will generally be visible 3 m from an operator's normal stance at floor level, platform, or stairs (viewing area) and will be resistant to plant atmospheres.

Temperature elements and dial thermometers will be protected by thermowells except when measuring gas or air temperatures at atmospheric pressure and for Steam Turbine/Generator set OEM standard and general industrial practice are acceptable. Temperature test points will have thermowells and caps or plugs.

RTDs will be either 100 ohm platinum or 10 ohm copper, ungrounded, three-wire circuits (R100/R0 = 1.385). The element will be spring loaded, mounted in a thermowell, and connected to a cast iron head assembly.

Thermocouples (if used) will be dual-element, grounded, spring-loaded, Chromel-Constantan (ANSI Type E) for general service. Thermocouple heads will be the cast type with an internal grounding screw. If a thermocouple is inaccessible, the leads will be brought to a junction box in an accessible location. Where thermocouples are employed then cold junction compensation may be provided as per manufacturer's standards practice. For Turbine/Generator set OEM standard and general industrial practice are acceptable.

Where ASME Performance Test Codes (e.g., PTC-6 Steam Turbines) are applicable to power cycle piping, they will be used as the criteria for determining well lengths.

Inside GT package, standard specification of thermocouples/RTD's is as follows;

- Thermocouple;

Type E, manufacturer's standard element diameter, ungrounded, double element, 316 stainless steel sheath, manufacturer's standard terminal head with screw type terminal block inside and terminal head cap with chain

Accuracy (tolerance) of thermocouples will conform to ASTM E230 special.

- RTD;

Pt 100 ohm, 3-wire, ungrounded, double element, 316 stainless steel sheath, manufacturer's standard terminal head with screw type terminal block inside and terminal head cap with chain, mounting on thermowell or protection tube

Accuracy (tolerance) of RTD's will conform to ASTM E1137 Class A.

Temperature measuring devices will be installed in thermowells or protection tubes except for such as devices mounted on the turbine and metal temperature measurement.

8.2.2.7 POSITION INSTRUMENTS

Position measuring devices employing magnetic, proximity, electrical (e.g. potentiometers and limit switches) or optical principles may be utilised as appropriate to the application.

Mounting locations shall, wherever possible, be remote from sources of heating or leakage of process, gases or liquids. Where vibration-free locations are not possible, the mounting arrangement shall incorporate flexibility in the coupling between the device and the point of measurement and the transducer mountings.

For position instruments in high temperature applications such as steam valves, etc, the calibration of position transducers and the setting of limit switches shall be unaffected by the hot/cold operating temperatures of start-up and shutdown.

8.2.2.8 ANALYTICAL INSTRUMENTS

Analytical instruments monitoring chemical or physical properties of process fluids and substances may be installed either directly in the process line or vessel or at a distance and connected by means of sampling systems.

Analytical instruments shall be selected to operate continuously, infrequently or on a continuously interrupted cycle as most suited to the measurement and the intended application.

Sensors shall be of ruggedized construction and shall not require frequent maintenance or recalibration. Where sensors require the use of calibration solutions or gases then a minimum of one year's supply at normal usage shall be supplied. Where sensors require replacement or removal for recalibration or rejuvenation at regular intervals, then an adequate number of spare sensors shall be supplied for one year's normal usage.

Self-cleaning pH and conductivity probes are preferred.

8.2.3 Vibration Monitoring

Vibration monitors will be provided on all major high-speed rotating equipment greater than 500 kW, such as steam turbines and boiler feed pumps.

As a minimum, the following functions shall be provided:

- a) Bearing pedestal vibration measured by seismic mass type velocity transducers or piezoelectric accelerometer, if required by the Owner.
- b) Axial displacement of thrust collars of Steam Turbine monitored by means of a non-contact proximity probe. If it is not feasible to measure axial displacement of the Steam Turbine, other means of high-speed protection in the event of axial displacement occurring shall be provided.
- c) A phase marker facility and associated non-contact proximity probe.
- d) Relative shaft vibrations shall be measured by two non-contact proximity probes fitted into each journal housing 90° apart.
- e) High vibration alarm and trip facilities.
- f) Display vibration levels on DCS in CCR.

For Gas turbine package, the vibration monitoring system (vibration probes in x-y direction and monitoring rack) of GT relative shaft vibration, GEN relative shaft vibration, GT absolute vibration, GEN absolute vibration and pinion shaft vibration sensors (in x-y direction, 4-20 mA directory connected to GT control

system) of GT Enhanced Cooling Air Compressor are provided. Vibration monitoring system will not be provided for any other rotating equipment in the Gas turbine package.

GT absolute vibration measurements will be used not for trip and alarm functions but for monitoring only. Axial position and shaft eccentricity measurements will not be provided for gas turbine and gas turbine generator as per GT OEM standard design.

8.2.4 Distributed Control System

A fully integrated, DCS shall be provided to monitor, control, display, alarm and record selected physical and electrical parameters associated with all plant areas.

The Plant (each Unit) shall be operated, monitored and controlled from a Control Room (CR).

- The minimum requirements of the Plant's DCS shall be, but not limited to:
- Allow for automatic control of all the systems of the Plant from the CR,
- Ability to collect signals and alarms from all the necessary systems.
- Critical signals shall have a minimum level of redundancy to guarantee reliability and availability of the Plant in any circumstance.
- Provision of a remote monitoring system and software with supervisory role and access to historic values.
- This shall include 2 web based monitoring for Offtake connection along with any license required.
- Remote access shall be allowed for maintenance and backup purpose.

The system shall be designed to have a functionally and geographically distributed architecture utilising a number of independent outstations. The equipment located in the DCS cubicles shall include all necessary control processors, input/output (I/O) cards, power supplies, data highway interfaces and termination facilities.

The microprocessor-based distributed control system shall comprise of multifunctional modules operating over a network. The operators have access to all significant process information and all controls Analogue and Digital signal will have separate cables. Analogue cables are twisted pair and screen protected.

Outstations shall be capable of autonomous operation and perform data acquisition, calculation, digital (open loop) and analogue (closed-loop) control functions.

Operator's workstations shall allow monitoring, control and adjustment of Plant operation conditions via high-resolution "windows" style graphics displays. All VDUs, keyboards, printers and peripheral devices and their associated control desks shall be supplied.

One 72" large screen graphics wall monitor for the plant shall be provided in the CCR which will primarily be used for:

- I. Display of important data and graphics from the DCS, video, etc.;
- II. To provide real time clear luminous information to operators and visitors;
- III. The whole screen of the large screen graphics wall should behave as a single logical screen and the control of the same shall be from a dedicated graphical control unit having a keyboard and mouse. However, as an option it should be possible to control the Graphics Wall from a PC on LAN having Windows NT/2000 OS;

The complete system shall be interfaced on 10/100 MBPS LAN as a minimum. Provision for additional VDU shall be provided for future additions. The man machine interface shall be to the review and approval of the Owner in accordance with the Contract. The display screens shall be developed in conjunction with the Owner and shall be to the review and approval of the Owner in accordance with the Contract.

Communications between the outstations located in different plant areas and the Operator's and Engineer's workstations located in the CCR shall be performed over a redundant high-speed serial data highway. The

data highway shall be designed to perform at the speed necessary to ensure that all variables are updated and control commands are issued without loss of system performance. The data highway shall include all control equipment, error detection and correction facilities and cabling.

The redundant data highway will have different routing.

The DCS shall be designed to ensure maximum availability by the inclusion of built-in redundancy for both hardware and software. This shall include duplicated control processors, data highway and power supplies with automatic changeover to the standby unit upon detection of a fault or failure of the operating unit. Where appropriate, I/O cards shall also be duplicated. The failure of any single element shall not affect the operation of any item of operating plant. In particular, the duplicated control processors shall be capable of fail over without change to Plant status, sequence status or control actions, at any stage of Plant operation.

Comprehensive system diagnostic facilities shall be incorporated to assist in maintenance and troubleshooting. These shall include memory, control processors, I/O cards, data highway, storage devices, as a minimum.

The DCS shall have a master clock of GPS satellite controlled clock for time synchronisation with package computer systems. To allow the DCS equipment to be synchronised irrespective of the time shifting in summer.

The Contractor shall ensure that DCS system is provided with sufficient storage capability to perform manual and automatic back-ups.

8.2.4.1 System Response Times

The response times for the Operator's workstations under normal conditions shall be as follows:

The DCS, as finally configured and installed, shall exhibit response times which do not adversely impact the operation of plant equipment nor impede the operator's response to events. To this end the maximum response times defined within this specification shall not be exceeded.

- a) Control response time is defined as the elapsed time between the moment a signal change is originated at the input card of a processing unit until a controller action takes place in another processing station, assuming the worst communication situation possible, including maximum data highway traffic level, longest cable length and largest access time to the data highway. This time shall not exceed the following values:
 - i. Slow Loops 1000 millisecond (such as temperatures and large tank levels)
 - ii. Fast Loops 500 millisecond (such as most flows and some pressures)
 - iii. Discrete/Logic Control 500 millisecond (motor control/interlocks)
- b) Monitoring function response time is defined as the time between initiation of an action or event and the time at which the operator is advised of the event through a LCD display or a printer output. These times include the elements of the control response times, including consideration of worst case communications, as well as the additional delays associated with operator station equipment such as LCD buffers, display drivers, and memory access. These times shall not exceed the following values:
 - i. Operator Commands to Equipment 1000 millisecond (exclusive of external equipment reaction times)
 - ii. Alarms/Annunciation 1000 millisecond (including pre-processing for alarm suppression)
- c) The time between selection and display of a VDU, fully updated, from the database shall not exceed 2 seconds with a typical workstation loading of 4 graphic displays, 2 trends and 1 alarm display in 'windows' on the VDU.
- d) The time between the occurrence and change of a signal at the origin and the change of state, value or alarm showing on the VDU shall not exceed 2 seconds.

8.2.4.2 Spare Capacity

The DCS shall be supplied with all hardware, system and applications software (VDU displays, alarms, etc) fully developed, tested, debugged and installed for all analogue and digital I/Os. An overall 10 percent spare capacity shall be provided to all necessary hardware including I/O units and associated wiring, supplied and tested.

8.2.4.3 Software

The latest version of the manufacture's system software shall be provided fully developed, tested and installed.

The application software and associated displays, reports, databases, etc. shall be pre-formatted when installed at Site. The creation or modification of displays, reports, databases or control strategies shall be possible with the system on-line by the Owner's engineering personnel without affecting the Plant operation. Modifications to the software shall be done in straightforward and logical steps using, wherever possible, graphic displays. It shall not be necessary to have specialist knowledge of the underlying software in order to carry out modifications.

All necessary software licences shall be included. This shall lead to unrestricted use of all specified software packages by the Owner. Access to software and application programmes at various levels shall be possible by operating or engineering personnel using passwords. This shall protect the software from unauthorised access.

8.2.4.4 OPERATOR INTERFACE SOFTWARE

The operator interface software shall enable the Operator to carry out the necessary actions in a safe and efficient manner. In particular the following features shall be included, as a minimum, but not be limited to:

- a. Software shall be structured in such a way so as to provide a hierarchy of control from automatic sequential Plant start-up through to manual control of an individual item.
- b. During manual intervention by the Operator (sequence initiation and individual device control under normal and failure conditions), all software derived, hardwired permissive and overrides shall be visible to the Operator.
- c. Where process measurements have been duplicated into the DCS for improved reliability, the signals shall be averaged for control purposes and quality monitoring shall be employed to automatically switch to the good signal in the event of an interruption. The Operator should have, for the purposes of maintenance, the on screen facility to select either signal or the average, as the measured variable for control.

A philosophy shall be developed to define the action of the sequence logic in the event of a step failure, Operator intervention or the initiation of a sequence with not all Plant conditions satisfied. It shall be possible to bypass a required plant condition but this section shall be alarmed and logged and only valid for one attempt for that step. This philosophy of operation shall be to the Owner's approval in accordance with the Contract.

8.2.4.5 REPORTS

The DCS shall be capable of presenting to the Operator, pre-defined reports. The information on the reports may be either in tabular form or in a pre-formatted report form and may be automatic at specific times or on request. Reports shall include periodic logs, daily logs and group logs, balance of energy and efficiency log. In addition, reports shall be produced to meet the requirements of the Environment Agency, and all report formats shall be to the approval of the Owner in accordance with the Contract.

8.2.4.6 ALARM MANAGEMENT

All plant and system generated alarms shall be prioritised, logged and displayed by the DCS. There shall be a dedicated alarm display which shall show all the Plant alarms or groups of alarms. The Operator's workstations shall have a visual and audible warning which will raise the attention of the Operator upon the receipt of any alarm. Alarms shall also be printed on the alarm/event printer for DCS, ST and GT.

The Operator shall have the facility to select different alarm displays, summary of alarms, historical alarms on global or point based, unacknowledged alarms, alarms by priority, alarms by pre-defined system groups.

There shall be a facility to suppress an alarm or groups of alarms as a result of a particular alarm or other Plant condition in order to minimise the number of alarms under transient conditions.

All alarms generated from analogue I/O shall be configured with a dead band of approximately one percent (1%). This shall be an adjustable value, which forms part of the configuration software. All alarms generated from digital I/O shall have the facility to be supported with an adjustable time based dead band, which will defer the reset by the allocated time. This value of time shall form part of the configuration software.

Plant level Alarm System shall be engineered /designed to minimise number of alarms in the plant level to meet the latest revision of ANSI standards (Number of alarms for Operator) - ISA-18.2 or IEC 62682.

Alarm management shall be based on Alarm EEMUA 191 guidelines.

8.2.4.7 HISTORICAL DATA STORAGE AND RETRIEVAL

The DCS shall include a mass storage memory device to provide storage of operating data for the plant. Data storage shall be non-volatile. Hard disks and optical disk drives are the preferred media for mass storage of data.

A magnetic tape drive, magnetic cartridges, optical laser disk type device shall be provided for permanent long term storage of data. The system shall automatically dump accumulated data to tape or cartridge when the storage media nears capacity, or at user defined intervals, whichever occurs first. The system shall generate an alarm if the drive is not in the ready condition to receive data. Sufficient time shall be provided to allow operator intervention before loss of data.

It shall be possible to retrieve data from the hard disk or permanent storage media from any operator station to display time based trends of stored data. The hard disk mass storage media shall have a capacity of 2 Gigabyte or of a larger storage capacity as required by the system design.

In addition to process data, the system shall record manipulations to the database to include, but not be limited to:

- a. Adding/removing points from scan
- b. Adding/removing points from alarm
- c. Changing alarm limits
- d. Algorithm tuning changes
- e. Force/clear of digital point status
- f. Processing unit/operator station configuration change
- g. Force of analog signal value

Each operator station shall be capable of retrieving and trending stored data. Operator-adjustable time frames of 10 minutes (with 1 second resolution) to 24 hours shall be provided as a minimum.

As a minimum, the system shall be capable of archiving data from all system points.

8.2.4.8 DATA LOGGING AND TRENDS

The DCS shall incorporate long term data logging facilities for all analogue, digital, serial data, alarms, condition monitoring and other internally generated points including the overall extension capacity specified. All signals shall either be scanned periodically or reported by exception, and updated values or digital status changes shall be stored. Data shall be stored for up to 3 months on a suitable storage medium. An alarm shall be given to the Operator when the storage medium approaches its maximum capacity. All data shall be automatically archived and retained for future reference in either optical or magnetic medium.

The operator shall have the facility to present real time data, recall data held in the data logger memory or from archives over a specified period of time. The requested information may be presented on the Operator's workstations either in tabular form or as selected variables on a trend display. It shall be

possible to compare on the same display, multiple trends of real-time data or historical data from the data logger or archives.

The format and content of the reports for data retrieval from the logging facility shall be subject to approval in accordance with the Contract.

8.2.4.9 SEQUENCE OF EVENTS

A sequence of events (SOE) recording system shall be supplied to allow analysis of the causes of trips or Plant disturbances. The SOE recorder may be either integrated within the DCS (preferred method) or a stand-alone item of equipment. The SOE recorder shall scan all designated binary inputs continuously with a resolution of 1 millisecond.

The status of each point shall be time tagged and stored in a database together with other relevant information for a specific time period. Data will be continuously deleted from the database after the specified time period has elapsed. Facilities shall be provided for data to be archived for future reference.

On the occurrence of a nominated event (e.g. Steam Turbine, generator or pumps trip) or manual initiation from the DCS, all data for the time period before the event shall be retained and continue to be recorded for a period after the event.

The SOE recorder shall automatically reset after the elapsed time period and continue to monitor the Plant in the normal way.

SOE reports shall be generated and printed using the stored data and shall be used for post trip review and analysis. The report shall be presented in a clear and logical format.

8.2.4.10 SAFETY

All operator control command functions (both modulating and discrete) shall include a tag-out feature which permits the control function to be placed into a secured mode (e.g. stopped, off, minimum and output) with normal operator commands inhibited. Software logic shall inhibit any device from being placed in the tag-out condition until operator action has placed the field device in the safe and secured mode (e.g. motor stopped and valve closed). Any function placed in the tag-out mode shall be identified as such on all displays of the function or controlled equipment. Operator action shall be required to set or remove the device from the tag-out state. Means shall be provided for logging and printing tagged-out mode points and functions. This tag-out function should be a standard feature of the DCS not requiring additional memory or engineering to implement.

8.2.4.11 Emergency Shutdown System

An emergency shutdown system shall be provided to automatically shut down discrete equipment, a system, or the Plant, in a safe and controlled manner, in the event of abnormal operating conditions which could cause permanent damage to the plant endangering human safety or exceed operating limits to the detriment of Plant life.

The Contractor shall be responsible for the co-ordination and provision of an integrated tripping system, which will monitor, initiate and interface with other systems which, will initiate the protection requirements of the Plant.

The Contractor shall develop in conjunction with all his sub-suppliers a Plant inter trips philosophy document/drawing, which will be used to control and implement the protection requirements of the Plant.

Emergency push button and programmed logic shutdown system shall be provided for the safety shutdown of the plant. The Emergency Push Button in the main control room and at locally shall be hardwired to control system (directly to the output card) for the plant critical process /equipment. Meanwhile programmed logic trip system could also be applicable as per manufacturer standard control system.

The system offered shall have a successful record of proven usage in the same or similar duty. Each field input shall be separately protected against possible faults or induced voltage pickup and line monitoring shall be provided. A simple arrangement for each I/O to indicate the presence or absence of I/O shall be provided, and thus allow problems to be quickly identified.

The design of the protection system shall be failsafe that is to fail to the condition, which would normally protect the Plant.

All critical inputs which, if present will cause loss of Plant performance, shall be engineered and included, but not be limited to, the following:

- Devices leading to Steam Turbine shutdown.
- Facilities shall be provided to allow routine testing of trip systems without causing Plant shutdown, by inhibiting the trip systems, while tests are being carried out. The inhibits should be designed in such a way as to provide restricted use and application of such inhibits shall be alarmed to the Operator.

The emergency shutdown system providing automatic trips shall be sufficiently reliable or contain sufficient redundancy to produce the required level of security.

It shall be possible to initiate from the workstations in the CCR, the shutdown of complete Plant. These features shall be provided by hardwiring from the control area, direct to the emergency shutdown systems.

The emergency shutdown system shall be operational at all times, irrespective of the equipment/package control mode.

Remote and local start-up trip override and trip reset facilities shall be provided to assist in commissioning and the auto sequence control of the Plant by the DCS. The shutdown system either in part or as a whole shall be capable of event analysis either by discrete data logging or transfer of information to the DCS.

Where 2 out of 3 trip logic has been incorporated, then 1 out of 3 discrepancy alarms will be generated for the attention of the operator, via the DCS. All trip initiations from the emergency shutdown systems will be alarmed to the DCS for event logging.

8.2.5 Performance Monitoring Package

As optional, (in case of awarding this scope should be treated as optional), the Contractor shall offer an online performance monitoring package commercially available in the market for monitoring net output of the plant exported to the grid with correction applied as required under the Power Purchase Agreement for site ambient conditions and the net plant heat rate without applying any corrections as well as monitoring output and heat rate applying the corrections agreed in the Contract and applicable PTCs. The package shall provide instantaneous and historical trends of corrected output, auxiliary loads, and heat rate and efficiency of each generator and shall have diagnostic features for investigating the causes of any performance degradation, benchmarking against performance as of new conditions, and finding solution for that. The package shall use the historical data from the DCS for long term trending.

8.2.6 Telephone Interconnection and Private Automatic Branch Exchange

Integrated services digital network (ISDN) type PABX, approved for connection to the public switched telephone network (PSTN) and situated within the Plant Administration Building shall be provided.

The installed capacity shall be 16-exchange lines, 16-tie lines and 50-extension lines. A personal computer (PC) based telecommunication console shall be provided together with a VDU, keyboard and headset.

The Contractor shall provide necessary equipment for VHF/UHF/microwave wireless connection between the nearest public telephone exchange and install tower on the site for connecting of the PABX with the public telephone network system (this connection could also be provided by means of optical fibre). The wireless/OF equipment shall be fully compatible with the KSA Telephone Exchange system. The Contractor shall be responsible for taking approval from KSA Telephone Exchange for such connection and the PABX and making payment of all fees and charges for installation of the system.

All other internal telephone connections within the Plant boundary, including cabling to and within the administration, control and turbine area, etc, shall be the responsibility of the Contractor.

Facilities Required

The following basic facilities shall be included in the PABX design:

- i. Extension to extension calls shall be made by direct dialling.
- ii. Extension to exchange lines for outgoing calls shall be made by dialling a single access digit.

- iii. Operator access from extension by dialling a single digit is required.
- iv. Trunk barring on outgoing calls shall be provided as required on selected extensions.
- v. Provision for limited barring on outgoing calls from selected extensions shall be provided.
- vi. Incoming calls from the public exchange shall normally be routed to the operator. However, the exchange shall be suitable for direct dialling-in. The ringing period for such calls shall be limited to 30 seconds after which the calls shall be diverted to the operator.
- vii. Operators recall from an extension engaged with an incoming or outgoing call shall effected by operation of a recall button followed by dialling the operator access digit.
- viii. Call transfer between extensions on incoming and outgoing exchange calls shall be provided.
- ix. Operator intrusion into an established call-in order to offer an exchange call or message shall be provided. An intrusion tone shall be introduced.
- x. Night service facilities shall be provided by extension of exchange lines to selected extensions.
- xi. PABX alarms shall be displayed on the telecommunication console.
- xii. It is desirable that a limited number of extensions (not exceeding 6 in the first instance) shall have a priority facility to intrude into extension to extension calls. An intrusion tone shall be introduced.
- xiii. Telephone handsets shall be provided with the facility for 'hands free' operation, and these shall be suitable for desk and wall mounting.
- xiv. Industrial type wall mounted telephone handsets shall be provided. These handsets shall be of rugged design for use within a Plant and equipped with a large visual indicator suitable for mounting separate from the handset, for use in noisy environments.
- xv. Acoustic hoods shall be provided.

8.2.7 Leased Lines/Direct Ring Telephones

The direct ring telephones on the leased lines shall enable the CCR to call the National Grid Control Centre. When the telephone handset in either the CCR or the Grid Control Centre is lifted the telephone in the other location shall ring.

SYSTEM FUNCTIONS

The system shall have the following functions, as a minimum, but not limited to:

- a. Automatic calling upon lifting handset at "calling" end.
- b. Operation by the PABX battery supplies.

8.2.8 Emergency warning system

The power plant will be provided with an emergency warning system in conformity with the international standard and shall include necessary siren and emergency warning lights. This system will be activated in the control room of the power plant in case of emergency and will warn by a sound transmitted to all the power plant components through the above paging system together with sparkling alarm lamps in the noisy areas.

The emergency warning system will be supplied with an UPS, battery and standby power supply in case of interruption of the main power supply.

8.2.9 Power Supplies

The following electrical power supplies should be made available for the C&I equipment:

- A. 480 V or 400 V ac (3 phases + neutral).
- B. 277 V or 230 V ac (1 phase + neutral + protective earth).
- C. 227 V or 230 V ac (1 phase uninterruptible power supply).

8.2.10 Electrical

In general, the power supplies available in electrical installations can be subjected to voltage and frequency variations, harmonic distortions, interruptions, and transient voltage peaks. Equipment shall be designed and proved by test, to the following unless otherwise specified.

Short time duration peak voltages can occur on both ac and dc supplies, typically due to switching operations, and are unpredictable in magnitude. For example, on AC systems, the peak may be up to 5 times the rated rms voltage, and on dc systems, the peak to rated ratio may be even higher. Sensitive equipment shall therefore be protected against such transients at its supply terminals.

Equipment shall ride through interruptions up to 100 milliseconds duration. (GT control system can ride through interruptions up to 40 milliseconds duration, since redundant power supply is provided)

8.2.11 Power Supply Switching and Fusing

Connections to the external power sources shall normally be made via switches and fuses or via miniature circuit breakers.

8.2.12 Insulation and Isolation

All circuits intended to be electrically isolated from external circuits shall be designed to withstand the following voltage levels:

All circuits requiring isolation (including contacts of switches, relays or contactors) shall have an insulation resistance across the isolating barrier of not less than 20 mega ohms when measured at 500 V dc.

A circuit intended for connection to 110 V ac or 125 V dc and above shall withstand 2 kV 50 Hz rms between itself and all other points for one minute.

8.2.13 Interference

All control and instrumentation equipment shall comply with the requirements specified in IEC 61000-4 series of standards for electromagnetic interference and immunity.

Equipment proposed shall not generate electrical or electromagnetic interference at a level detrimental to the performance of the equipment itself, other equipment or to the hazard or discomfort of personnel.

Equipment proposed shall not be susceptible to the effects of radio frequency interference which could emanate typically from a 1 watt ERP mobile radio telephone operating at VHF or UHF at a distance of 1 metre from the equipment.

8.2.14 Earthing

All non-current carrying metallic parts of plant, equipment, components, enclosures, protective screens, etc, shall be provided with earthing facilities for personnel safety reasons and for obtaining the maximum possible rejection of electrical interference.

Earthing should be generally designed and installed in accordance with IEC, IEEE 80 unless otherwise approved.

Protective screens around insulated cables shall be insulated from field instruments, connected through any field junction boxes, and be continuous up to the earthing point in the destination cubicle or panel.

Cable armouring shall not be used as a protective screen. Earthing of cable armour shall be arranged for electrical safety purposes and not for noise reduction. Cable armouring shall be insulated from any cable protective screening.

Where the Contractor's design requires a separate and dedicated earthing arrangement for example, a "clean" earth for electronic equipment then the Contractor shall provide such earthing facilities as required. All instrument clean earth bars shall be clearly labelled as such.

Earth bars inside cabinets of GT control system will not have any label. Cables connected to the earth bars will have a wire marker label such as PE (Protective Earth), FE (Functional Earth).

Where the system design requires a single connection from the C&I equipment to the main Plant earth bus, then a removable link shall be provided to connect the equipment earth to the main Plant earth bus for testing purposes. This removable link shall be clearly labelled with a warning indication.

Where a cubicle, frame or enclosure is earthed via the above facilities then it should be insulated from structural steel works or any other plant earth.

8.2.15 Testing

In addition to the manufacturer's production tests, the Contractor shall carry out routine tests on completed equipment. The Owner reserves the right to witness such tests.

Routine tests on each piece of equipment shall include at least the following:

- a) Visual inspection.
- b) Performance test.
- c) Control loop testing
- d) Voltage withstands test and insulation test.

Where a number of items of equipment are intended to operate together, the complete assembly shall be tested to demonstrate that the interconnected items are compatible. Design consideration should be given to the testing requirements of the Plant as defined in the OTS.

The Site acceptance tests shall include, but not be limited, to the following:

- e) All outstations are tested without connection to workstations in the CCR.
- f) Data communications to workstations are established for each outstation.
- g) Commissioning tests including, in-situ instrument calibration check to DCS control loops, DCS device drivers, protection and interlock logic, sequences, station master load control, alarms, and all other equipment supplied (s), firmware(s), equipment(s), firewall(s), switch(s), etc. installed within the network.

8.2.16 Identification

Each item of equipment shall be durably and legibly labelled as required by the Owner, indicating the purpose, plant identification number and where necessary any operating position functions.

The material of all labels and the dimensions, legend and method of printing shall be to approval in accordance with the Contract. Surfaces of labels for cubicles and control equipment shall have a matt or satin finish to avoid dazzle. Colours shall be permanent and free from fading. Danger labels shall have red lettering on a white background.

Labels shall be provided on front and rear access doors of all cubicles. Labels shall also be provided inside cubicles to assist the identification of apparatus and terminals.

All actuated valves shall be provided with labels, which shall include the valve and actuator reference number.

8.2.17 Fuses and Links

Fuses shall be of the high breaking capacity cartridge type of approved manufacture and shall comply with IEC 60269.

Fuses, link carriers and bases shall be of approved manufacture and shall be of good quality moulded insulating material. Control circuit fuse carriers and bases shall be black and for links they shall be white.

The arrangement of the fuses and the design shall be such that fuses, when operating, will not cause an arc to form between phases or to earth. Fuses and links shall be grouped and spaced according to their function

in order to facilitate identification. Fuse carriers shall be capable of taking any size of fuse, up to the marked full rating of the carriers.

All incoming circuits in which the voltage exceeds 110 volts shall be fed through insulated fuses and/or links, the supplies being connected to the bottom terminal, which shall be shrouded. The contacts of the fixed portion of the fuses and links shall be shrouded so that accidental contact with live metal cannot be made when the moving portion is withdrawn.

Isolating links shall be provided on each panel to facilitate isolation of the dc circuit (where used) to permit testing or other work on individual equipment on the panel without danger to other circuits, and such links shall be provided with labels, indicating their function and polarity.

Distribution of 220VDC power and power feeds to power units (24VDC output) inside cabinets of GT control system can be isolated by circuit protector.

8.2.18 Documentation

In addition to the general requirements for Operations and Maintenance Manuals detailed elsewhere in the OTS, the Contractor shall provide the following specific requirements for computer-based systems.

Operator assistance documentation comprising:

- i. Operation and control philosophy.
- ii. Menu guides to DCS facilities.
- iii. Description of all sequences in structured step-by-step formats.
- iv. Printed error message guide.
- v. Alarm message guide.
- vi. Brief description of any hardware back-up features.
- vii. Process parameter values under normal operating regimes.
- viii. Hardware trouble-shooting guide.
- ix. Instructions for changeover from DCS control to local and vice-versa.
- x. Operator Graphics, and Trend Display and Report Layouts.
- xi. System Functional Design Specification
- xii. Software documentation including, as appropriate:
- xiii. Plant configuration and database listing.
- xiv. Plant operating system listing.
- xv. User-configurable software listing.
- xvi. System error messages listing.
- xvii. Trouble-shooting guidelines.
- xviii. On and off-line diagnostic software.
- xix. Communications network management software.
- xx. Hardware documentation including, as appropriate:
- xxi. Hardware list and system diagrams.
- xxii. Start-up instructions for all hardware.
- xxiii. Listing of system error messages.
- xxiv. Description of links or interface hardware between Equipment Supplier control packages and the DCS.
- xxv. I/O card listing identifying location and function.

xxvi. Recommended spare part List

8.3 Master Clock System

A master-clock system shall be provided and interconnected with the GPS, which shall also form part of the scope of the Contractor's supply.

8.4 Earthing

The whole C&I and communication systems shall have carefully designed earthing system(s) for protection, to minimise noise pick-up and ensure Electromagnetic Compatibility.

8.5 N.A.

8.6 Functional Requirements

Manual operations may be used to prepare the plant when cold, but, having completed these, only a single operation shall be required to start major sections of plant. At appropriate points (e.g. synchronisation, etc.) the sequences shall pause and ask the operator for permission to proceed.

The plant shall be equipped with an Automatic Generation Control (AGC) system through which the each generation unit can be remotely controlled from the Load Despatch Centre (LDC). In principle, the AGC system will enable the output of each generating unit on the system to be controlled directly from the LDC (either automatically or manually) to match the system demand. The AGC logic shall be designed to prevent the application of inappropriate or undeliverable remote demands and to ensure that remote control signals are consistent with the technical and commercial provisions of the Offtaker.

8.6.1 Gas Receiving Station and Treatment

The Gas receiving and treatment shall be in accordance with the OTS.

EPC shall design & supply the metering facilities at the existing Gas Connection Facility, and hand over the same to the Owner. After handing over, the Gas Delivery Point at the downstream of the Seller's Metering Facility will be the operational interface point between the Seller and the Owner during the operational period.

The gas system is a potential single failure point so special attention must be given to meeting the criteria of OTS. Every effort shall be made to maximise reliability and safety. Common reduction/compressor stations (if applicable) shall meet the requirements of OTS under all circumstances. Main and Back-Up Metering Equipment for gas shall be provided by the Contractor. Instruments for Gas metering shall have an accuracy of 0.5% or better and as per Client approved Standards and Specifications.

An on-line calorimeter and spectrometer shall be provided and shall also be interfaced to both the DCMS and flow computers.

8.6.2 Gas and Steam Turbines

Governors and AVR's shall meet the requirements of OTS. They shall be proprietary items and interfaced to the DCS so the supervisory functions are available via the DCS MMI. A separate MMI may be supplied for intimate plant level control for commissioning/fault finding/repair/special monitoring purposes if these are not available through the DCS. If over-speed protection is fully two (2) out of three (3), fully fail-safe and self-checking, the same transducers may be used for speed control and over-speed protection. Otherwise, separate equipment is required.

It shall be possible to start the gas turbine by a single operation in the CCR and steam turbine and its control system shall be suitable for all modes and conditions of operation.

The governors shall be capable of withstanding full load rejection without tripping, operation in island mode, at fixed power, and for the gas turbines, frequency support (droop) mode and in accordance with the requirements of OTS.

The HRSG stacks and bypass stacks shall be equipped to online monitoring system and record NO_x, SO₂, CO and O₂ emissions. At any time either one or the stack – either HRSG stack or bypass stack will be in service while Gas Turbine in operation.

8.6.3 HRSG and Steam Cycle

Control will be by the DCS automation outstations, with additional protection. Less critical protection, such as maintenance of desuperheater spray water, can be vested in the DCS outstation, however it shall not be possible to override this protection in normal operation (e.g. by not permitting manual operation of a valve under certain circumstances).

The Burner Management System (BMS) in case supplementary firing is provided, shall be proven, be fully failsafe and meet the requirements of OTS. The International Standard used for BMS and boiler controls shall be stated. The BMS shall be separate from all other outstations (in accordance with NFPA 8502, Section 4-3.2.3.1.)

Turbine and superheater water ingress protection shall meet or exceed ANSI/ASME standard TDP-1 Recommended Practices for the Prevention of Water Damage to Steam Turbines used for Electric Power Generation.

The steam and feedwater systems shall be fully automatic in normal operation and automatic sequences shall be provided for all start-up and shut-down operations (eg. BMS sequences, floating together steam systems etc.). The sequences shall pause for operator permissive where required, but emergency actions (such as dump condenser, bypasses etc.) shall operate immediately.

8.6.4 Electrical System Monitoring

The HV and MV electrical systems shall be monitored and controlled by the DCS. For operator actions, the interlocks provided by the electrical system shall be exactly duplicated in the DCS so the operator knows of any missing permissive.

The DCS shall also monitor the DC and UPS systems.

Generator parameters such as vibration, temperature, coolant flows etc. shall be monitored but protection is provided by the generator protection system. Transformers shall be similarly monitored.

An active generator capability diagram shall be provided.

The fault monitoring system shall be interfaced with the DCS with a view to collection of post-incident data for archiving purposes (see earlier).

Time stamping shall be coordinated with the DCS. It is not intended that relay settings can be changed through the DCS.

Currents of all MV motors shall be monitored and running hours shall be recorded. Currents and running hours of other motors shall be monitored and recorded where necessary.

The DCS shall be interfaced to Generator and Transformer AVR's to provide manual raise/lower facilities (see earlier).

Automatic synchronisation can be initiated by the DCS (generally after release by an operator).

All metering shall be connected to the settlement system and also totalised by the DCS.

8.6.5 Gas Detection and Fire Detection/Fighting

The main requirements are stated elsewhere in this OTS Part B.

The fire detection system shall cover the entire new Plant and gas detection shall cover all necessary areas within the new Plant. An interface with the DCS shall be provided for information and protection of the operators. The system shall interface with the other fire detection systems, if any, to inform the control room operators of fire elsewhere on the site. The master panel shall be mounted in the control room with repeat indication in the Gate House. Repeat panels shall also be provided at the entrance of other buildings where information is required for protection and/or safety.

8.6.6 Common Services

In all common services (air, electrical, cooling systems water treatment etc), special attention to the design is required to ensure the common fault criteria are met.

8.6.7 HVAC

The control of the HVAC shall be independent of the DCS, but alarm(s) is/are required to alert the process operator investigation is required.

8.6.8 Meteorological Station

A meteorological station shall be provided to measure and record with required precision:

- Atmospheric pressure;
- Wind speed;
- Temperature;
- Rainfall;
- Relative humidity;

Data shall be repeated to the DCS.

8.6.9 Performance Monitoring System

The Contractor shall install systems for continuous monitoring of performance of the Plant such that information required by the Owner for the production of invoices meets the requirements of Offtaker. All data required to produce invoices shall be automatically forwarded to Offtaker's electricity control centres.

The performance monitoring systems shall be integrated with all the correction curves and sufficient enough to generate daily performance reports and support NDCT reporting.

8.6.10 Communication and Security

The system will include and an OPC server for access from external networks to the Control System, with consideration of cyber security. OPC server shall include all parameters and signals provided from DCS/SCADA system.

The plant control systems are critical for plant safety and operation. If degraded or sabotaged they put people, environment and assets at risk. The requirements for security are therefore high.

To ensure availability, confidentiality and integrity, at least the following points shall be taken into account:

- Be designed with security measures, segregation and availability to reflect the plant requirements;
- Have measures to protect itself from unauthorized access from external systems including the corporate office data network;
- Have measures that enable secure remote operation according to process requirements;
- Have measures for disaster recovery;
- Operate safely also if communication to external systems fails; and
- Be documented and maintained in a manner that reflects the requirements of the connected systems.

The measures shall be based on international standards such as ISO 27001, IEC 62443 and NERC 1300.

For more detail refer to Schedule C Network Security to this OTS.

All external communication and data links shall have all necessary protections in order to achieve a maximum level of cyber security. The system shall be designed with security measures, segregation and availability to reflect plant requirements and have measures for disaster recovery. The measures shall be based on international standards like ISO 27001, IEC 62443

and ISA S99.

A CCTV system shall be supplied to cover the entire Plant including stacks, GT&ST turbines, control room entrance, etc.

Anti-virus & Malware Protection: The Contractor shall provide the most recent versions of anti-virus and malware protection software at the end of warranty period or Defect Liability Period (DLP). It is recommended that antivirus software should be in line with ACWA Power ICS guidelines.

8.7 Cyber Security Requirements

The cyber security measures implemented at the Plant shall at all times comply with the internationally accepted cyber security requirements.

The Contractor shall follow the same cyber security measures as per international standards (like ISO 27001, IEC 62443 and ISA S99), section 4.2.13.7 of Part A and as well as Schedule C or any other specific Local requirement from Client when extending and/or integrating Client systems (SCMS, LDC SCADA, Telecom, etc.). The system will include a Web server and an OPC server (UA type which has capability to store historical) for access from external networks to the Control System, with consideration of cyber security. OPC server shall include all parameters and signals provided from DCS/ SCADA system.

8.7.1 General Requirement

All external communication and data links shall have all necessary protections in order to achieve a maximum level of cybersecurity. The system shall be designed with security measures, segregation and availability to reflect plant requirements and have measures for disaster recovery.

8.7.2 Specific Cyber Security Requirements

8.7.2.1 Introduction

This section describes the technical requirement for the cyber security measures necessary for the safe and efficient operation of the Plant. Such requirements are to be implemented by the Contractor in the phases described below in this section. Each phase can be completed consecutively or concurrently.

8.7.2.2 Phase 1: Approval of cyber security compliant network based on quadrant strategy

The Contractor shall propose and submit to the Owner for approval the network architecture for the binding of 4 quadrant subnetwork(s) comprising of SCADA/DCS, metering, historian, LDC, remove support connectivity in Quadrant 4, iPAS, PI or Plant performance monitoring system, Offtaker and business users connectivity in Quadrant 3, O&M IT network in Quadrant 2 and Project Company IT network in Quadrant 1.

Prior to submission, the Contractor will have discussed with the Owner, all of the necessary equipment (e.g. firewalls, AV, Endpoint protections, P2P connections etc.) that are needed in order to ensure the overall Plant architecture is cyber security compliant.

8.7.2.3 Phase 2: Implementation of Data Diode

The Contractor shall implement a data diode device between Quadrant 4 and Quadrant 3. The data diode device shall enforce unidirectional data transfer from the source to the destination.

The Contractor shall use the OPDS-100 data diode device (including the relevant support service(s)) produced by OWL Technologies (as set out below) and shall coordinate with the Owner who shall provide further details in this respect.

8.7.2.4 Phase 3: Testing and Implementation of Disaster Recovery Procedures

The Contractor shall propose a disaster recovery plan for the complete network for approval by the Owner.

In addition, the Contractor shall ensure unitary testing along with end-to-end testing is in place before handover to the O&M Contractor.

The Contractor shall furnish the Owner with the relevant testing schedules and logs on a regular basis.

8.7.2.5 Phase 4: Implementation of Monitoring and Protection Service

The Contractor shall implement a continuous monitoring and protection service for the monitoring of the Plant network.

The Contractor shall use FireEye, Inc. services and hardware to implement the monitoring and protection service and shall coordinate with the Owner who shall provide further details in this respect.

8.7.2.6 Phase 5: External and Internal Penetration Testing

The Contractor shall arrange for a third-party audit of the cyber security measures to ensure compliance with the specifications set out in each of phases 1-4 above (subject to that, Each Phase must pass such audit before handover to the O&M Contractor).

The Contractor shall use THALES Holding Maroc SAS to perform the audit and shall coordinate with the Owner who shall provide further details in this respect.

Any non-conformances identified by the auditor shall be notified to the Contractor. The Contractor shall have 30 days (unless the parties agree otherwise) to implement such comments and make the necessary changes to the cyber security measures at the Plant. The audit process shall be repeated in respect of the relevant Phase(s) until the auditor declares a satisfactory result is achieved. All costs related to additional audits shall be borne by the Contractor.

8.7.2.7 Regulatory Compliance

The cyber security measures implemented at the Plant shall at all times comply with:

- i. local laws, ordinances, regulatory requirements, codes and standards of Saudi Arabia (if any);
- ii. Schedule C; ACWA – Requirements – 001 ICS Network Security– v2.0 policy document & Document Technologies Unification.
- iii. Essential Cybersecurity Controls – ECC-1: 2018 requirements developed by Kingdom of Saudi Arabian Cybersecurity Authorities for KSA Project or any specific requirement of Country where the Project is located.
- iv. provisions of the Project agreements (including the Power Purchase Agreement); and
- v. Requirements set out in phases 1-4 above.

Any inconsistency or conflict between the provisions shall be resolved with priority being given as follows: (i), (iii), (ii) and (iv).

8.8 Supervisory System

8.8.1 General

The common Plant system shall be DCS-based, with incorporated display hierarchy and display selection concept as follows:

- | | |
|--------------------|---------------------------|
| • Power Block | Overview level, |
| • Functional area | Area level control, |
| • Mechanical units | Group/Unit level control. |
| • Components | Item level control |

Item level control may be by means of pop-ups.

Operator communication and monitoring shall be carried out via monitors providing the window(s) to the process.

The display layout of an operator station shall be as follows:

- A frame with header and footer is always visible. It contains selection fields for displays and additional information as well as collective displays, which shall also indicate faults in displays not currently visible,
- Basis diagrams occupy the rest of the screen. They provide information on the process in the form of plant diagrams, curves, characteristics, etc.
- Windows are selected in addition to the basic diagrams. Normally the window is allocated to a particular object, e.g. pumps, and provides detailed additional information in graphic or alphanumeric representation.

- The control room operator shall be presented with displays for information and control functions, which are especially designated for monitoring, intervention and clarification at the process, plant or control system level. The displays shall present process sequences and statuses in a functionally related manner. This shall be achieved by means of three types of displays with different information content, such as:
 - Plant displays which show the physical items in the plant such as pumps, fans, valves, etc. and their state by the means of the symbol, colour, blinking, (in acc. to DIN 2481) to facilitate operator intervention,
 - Representations of physical and technical relationships in form of curves, characteristics, bar graphs, digital displays, to provide a view the process. They provide the operator with historical, current and forecast states of process.
 - Control system displays showing function diagrams for instrumentation and control as well as device configuration diagrams.
 - Step-by-step indication of sequences' progress and permissive awaited.

The above shall be available whether a DCMS vendor's outstation or packaged plant vendor's PLC provides control.

These different types shall mutually supplement each other and can be called up by the plant personnel in accordance with their access authority. Mixed displays shall be possible.

8.8.2 Alarm System

The DCMS shall incorporate an alarm system with the main tasks of providing a comprehensive and clearly arranged representation of faults in the power plant, facilitating quick localisation of the plant sections affected by faults.

There shall be a Common Alarm Display facilitating blinking marks in diagrams, menus and the overview single lines permanently displayed. Signal disturbances from the complete plant, regardless of the current display and operator control situation, shall be indicated at the operator's workstation.

It shall be possible to identify every measurement in alarm whether displayed in a table or on a mimic, and the state of its acknowledgement. (e.g. by blinking or different colours.). Sequence time-outs shall be treated as alarms. The event sequence display additionally shall list events, which do not need to be acknowledged, if this cannot be accommodated by the "sorting" process.

For alarms at the operator stations limited consequential alarm suppression shall be provided but not on printed alarm logs (see below). The extent of suppression shall be agreed in engineering phase. In addition to display on the screen, audible signalling of faults shall also be possible.

The Alarm Sequence Display / Event Sequence Display shall be facilitated by listings in tabular form of currently pending, non-acknowledged alarms / faults in chronological order and in plain text. Even if ASD has not been selected, the alarms shall be stored in the background and new incoming alarms shall be indicated in the overview line. It shall be possible to designate one monitor exclusively for ASD operation, however, additionally it shall be possible call up the ASD immediately on any other monitor.

The event sequence display additionally shall list the event alarms, which do not need to be acknowledged. The Contractor shall supply a separate alarm list with initiation points and all corresponding operator actions.

The alarm sequence display shall be stated according to:

- Alarm class; and
- Functional area.

All the non-acknowledged alarms are written into new pages whereas all alarms acknowledged, but still pending, shall be shifted to the old pages and marked with blinking display. Additionally, it shall be possible to print out all alarms (alarm log).

In addition to display on the screen audible signalling of faults shall also be possible.

Alarm management shall be based on Alarm EEMUA 191 guidelines.

8.8.3 Logs/Plots

A log shall be the filtered output of data from the short- and long-term archives supplemented by titles and explanations.

The following logs shall be available:

- OM-configurable logs (to be freely programmable);
- Status logs;
- Sequence logs;
- ES-configurable logs.

Logs of this nature shall be available as status logs, sequence logs and incident review logs.

On-line logs are “open” sequence logs, which are continuously routed to a printer starting at the time of their activation and shall show:

- Trends (with history)
- Characteristic curves (X-Y plots).
- All alarm/event logs

The print of the above shall be visible as soon as it is printed.

All signal changes occurring within recent hours shall be archived in the short-term process data archive located in the processing units, where a replica of the process signals shall be maintained.

8.8.4 Archiving

The long-term process data archive shall be located in server units. For this purpose, all changes of process signals stored in the short-term archive are transferred to the long-term archive where they are archived in chronological sequence.

The storage on the archive server shall be reasonably compatible with writing to an optical disc and when data occupies more than 80% of the hard disk drives, the data shall automatically written to an optical disk, the drive being integrated in the server unit.

8.8.5 Hardware/MMI

The hardware shall comprise, as a minimum:

- operator workstations, each with 21” monitors, dedicated keys, QWERTY keyboard, mouse, etc.
- Supervisor’s workstation, similar to operator stations;
- Engineering workstation, for configuration of both supervisory and automation level equipment;
- Printers for alarms and logs (printing of alarms, events, Sequence-of-Event (SOE) and Post Incident Recorder (PIR) to be immediately visible);
- Colour screen dump writer (incl. facility to suppress background colour);
- Servers and associated accessories as required to connect LANs, for archiving, performance monitoring, condition monitoring, etc.;
- Hot spares, with test facilities to connect “Smart” instruments (if any);
- Facilities to connect other (future) computers via the LAN or servers;

A dual redundant GPS master clock system (possibly shared with other systems).

HMI shall be possible to display the process with corresponding graphic displays, control loop displays, electrical single line schematics, group displays, etc., all with dynamically updated process parameters on the video display units (“VDU”). The system shall be able to create and to show displays like the following:

- Overview display;

- Process graphic display;
- Alarm summary display;
- Loop display;
- Sequence control status display;
- Trend display (real time and historical trends); and
- Logging reports.

The human machine interface ("HMI") system shall be capable of processing all inputs in such a manner that all displayed data shall be updated within typical one (1) second for analogue and digital inputs. The system must be capable of accepting spontaneous events from the process with a time resolution of 100 ms (1ms for sequence of event recording ("SER" function), independent from the above mentioned scanning cycle times.

Abnormal operating conditions and events in the Plant have to be annunciate. Under operation control, it shall be possible to print all or selected alarms on demand.

I&C systems shall be designed in such a way that the Plant can be remotely controlled from the central control room ("CCR") during start up, normal operation and shut down. Local activities shall be not necessary, except after long term outages.

The operation concept shall consider in the CCR at least four (4) operator stations for each power generation control desk (one per Plant Unit) as well as large screen projection and another four (4) Operator stations for the common and electrical services control desk. In addition, each one (1) operator station with access to supervise all Plant Units and systems shall be foreseen for the shift leader and in the superintendent's office. Each operator station shall be equipped with two (2) VDUs.

The CCR shall be designed (interior architecture, lighting, colour, material and furniture) according to the state-of-the-art of the latest technology in CCR design worldwide and shall comply with the international ergonomic standards.

8.9 Plant Automation System

Plant level automation outstations may be proprietary controllers supplied as part of packaged plant (e.g. Water Treatment) or manufactured by the main DCS vendor (e.g. BOP). Fuel system, governors, HRSG, and steam cycle controllers shall be a minimum of dual processor (and preferably triplicated) because protection is vested in them, but all outstations shall meet the operability and reliability requirements stated in OTS. It shall not be possible to invalidate protection provided by dedicated protection systems.

GT Control Panel will be dual redundant microprocessor based system as per GT OEM standard design.

8.10 Safety and Protection System

The CCR shall include a dedicated site emergency panel with all functions hard wired from the field and these signals shall also be duplicated on the control system and shall include (without limitation) the following:

- Gas and steam turbine emergency trip;
- Auxiliary boiler emergency trip;
- Generating unit trip button and ESD trips;
- Firefighting system;
- Gas pipeline shutoff valve.

Components being of high importance for safe operation like burner management system or general boiler/plant protection are to be realized with fail-safe technique. Single fault tolerant design shall be provided to avoid tripping of a boiler or turbine as a result of malfunction of the protection equipment.

Permissive interlocks shall be active at all levels to prevent incorrect operation of the and to ensure a high level of integrity of operation. Interlocks shall be capable of being tested with the equipment in operation. The protection and interlocks systems shall incorporate comprehensive diagnostics to input level and shall

be visible during the start-up and shut-down sequences via the control system displays for troubleshooting purposes.

As a general rule, measuring points and measuring equipment for interlocking and protection purposes shall be separated and not combined with measuring equipment for monitoring or automatic control equipment. For flow measurements one common primary element can be used.

Components being of high importance for safe operation like burner management system or general boiler/Plant protection shall be realized with fail-safe technique. Single fault tolerant design shall be provided to avoid tripping of a boiler or turbine as a result of malfunction of the protection equipment.

The process requires an evaluation to determine the appropriate independent protection layers ("IPL") to control hazards. Safety instrumented systems ("SIS") are intended to provide protection for people, property and the environment against deviations in the processes. IEC 61508 and IEC 61511 standards shall be followed to determine, design and build independent protection layers and safety instrumented systems. Every safety related loop has to be verified according to the standards IEC 61508 and IEC 61511 that the requested and necessary safety integrated level ("SIL") is fulfilled for the individual loop. The EPC has to assess all safety related loops to ensure compliance with the related requirements and has to make a protocol about these studies.

8.11 Settlement Metering

Settlement metering system including check meter shall be provided in line with the latest version of the Saudi Arabia Grid Code Chapter 7 "Metering Code".

All check meters and associated equipment and facilities including CTs and VTs shall be in accordance with the Saudi Arabia Grid Code and approved by the relevant authority.

For the metering system, the transmission protocol IEC 62056 is to be followed. The final amount and type of signals to be exchanged have to be agreed with SEC during engineering and detail design stage.

8.11.1 Plant Settlement

Software shall be developed and installed by the Owner in accordance with Good Utility practice to calculate payments due from Offtaker in accordance with the terms of the payments, such software to include the Fuel Demand Model. The Plant Settlement System hardware supplied by the EPC contractor, the Plant Settlement System shall be capable and be designed to interface with Offtaker Central Settlement System. All necessary integration with DCs, control centres, interface panels in GIS and Plant shall be done by the EPC Contractor.

A ready configured operator station (incl. computer, monitor, etc.), on which a certified copy of the settlement software is installed, shall be provided to Offtaker by Owner. The equipment to be provided to Offtaker shall contain also all future updates and training in its use and licenses for appropriate use of the software. The settlement software to be provided for Offtaker shall be able to run on a virtual server on Windows basis as well.

8.11.2 Telemetry

It is required that all main Plant electrical variables are transmitted to the Load Dispatch Centre (LDC) via SCADA links. The SCADA equipment will be provided by others; however, the Contractor shall make all necessary provisions to ensure that the required signals are made available at the telemetry interface point located in the HV substation building. This shall include fully redundant data network links or hardwired connections as applicable.

All primary sensors such as transducers, contacts, adapters etc. shall be provided along with all cabling and communications equipment. All circuits shall be brought to a telemetry interface point, which shall be installed in a communications room within the GIS substation. Connection from the telemetry interface point to the SCADA equipment will be by others. Details of the telemetry interface point shall be agreed with Offtaker's LDC during the details design phase.

The exact signalling requirements shall be determined during the design phase through coordination with Offtaker's LDC, however they are expected to include the following as a minimum:

- Active / Reactive power / energy measurements for each generator;

- Active / Reactive power / energy measurements for each Transmission Line;
- Measurements of temperature and humidity of the site;
- Status indication of all 380 kV circuit breakers, isolators and earthing switches;
- Main Power Plant alarms;
- Protection trips for all generators, generator transformers, unit transformers, station transformers and HV Cables;
- Auxiliary power consumption of each unit;
- Total generated active/reactive power;
- Incoming signals from NCC;
- Firefighting alarms; and
- Disturbance data pertaining to generator units.

A number of signals will be transmitted from Offtaker' LDC to the Plant. It is expected that these will initially be of an advisory nature, although provision is required for future remote control of the Plant from Offtaker' LDC. A suitable interface shall therefore be provided at the DCS to receive the necessary signals, equipped with all necessary software/hardware modules. Connections between the DCS and the telemetering as well as all other metering equipment of the project shall be made as part of this Contract.

IEC 61724 must be considered as a guideline for measurement, data exchange and analysis.

IEC 61446 must be considered as a minimum requirement for system documentation, commissioning tests and inspection.

IEC 61850-7-420 communication standard for distributed energy resources shall be considered.

Due to the requirements of availability, reliability and function, the control and monitoring tasks of the Plant processes shall be performed by a state-of-the-art control system.

8.12 Fire Detection and Alarm System

The firefighting systems and equipment shall be designed in accordance with the NFPA recommendations and with any additional requirements of local authorities and shall comply with the local regulations in respect of supplier, installer and system approval.

The local regulation and standards shall have the highest priority in case any conflict with the international standards i.e. NFPA etc.

The firefighting system shall also cover the fire protection system and the fire and gas detection system and shall be one unified system of the same make and type for the entire plants. Approval shall be received also from authorized insurance company.

For fire protection and detection purposes the overall Plant site shall be divided into separate fire zones and the alarms and control requirements for each zone shall be actioned through a local control and alarm panel, including alarm signal to the central alarm panel of the fire brigade. All penetrations between two different fire zones shall be sealed with fireproof sealing material of listed type. Warning, Safety, Exit, No Smoking Signs shall be provided. The labels shall be weatherproof and UV resistance.

An automatic fire detection and alarm system shall be provided, with detection devices and with manual control switches selected to suit particular risks, and with a control system designed to provide operating and fire brigade staff with sufficient information to identify and respond to any fire detected.

The firefighting water system shall be designed as continuously pressurized ring system to serve outdoor hydrants at strategic locations and to supply water to indoor hydrants, hose reels, sprinkler and spray water systems and shall be equipped with sectionalizing valves so that a failure in any part of the system can be isolated while allowing the remainder of the system to function properly.

Generally, it shall be considered that the following firefighting installations shall be foreseen:

- Oil tanks, lube oil tanks of turbines, Gas generators, other related facilities: Foam/water spray systems;

- Oil filled transformers: Automatic fixed spray water systems;
- cable tunnels, cable floors: Automatic sprinkler systems; Cable room shall be provided with fire detection and suppression system as per NFPA 850, Cl:7.8.2.1, The Cable spreading rooms and Cable tunnels should be protected with automatic sprinkler, water spray, water mist or automatic gaseous extinguishing system
- Ventilation systems: Portable extinguisher
- Switchgear facilities, electronic and control rooms and buildings: carbon dioxide extinguishing systems and/or mobile extinguishers; and
- Other buildings and open air facilities: Outdoor/indoor hydrants, hose cabinets.

The firefighting water source shall be pre-treated water from the services/firefighting water tanks. Two (2) 100% firefighting water pumps shall be foreseen, one (1) equipped with electric motor drive and the other with diesel engine drive. Redundant jockey pumps (2 x 100%) shall be considered to maintain pressure in ring main. The firefighting water pumps shall be installed in a separate building and the Diesel Pump and Diesel tank shall be installed in separated rooms at the building at the water storage tank area.

A central fire brigade station shall be built for the Plant which shall house the mobile fire equipment (vehicles) and a command room, the main alarm panel, offices, staff rooms, a fully equipped medical station with a first aid room, workshops, stores, etc. and refilling units for portable fire extinguishers and breathing apparatus, as well as any other equipment required for firefighting and rescue.

For fire protection and detection purposes the overall Plant site shall be divided into separate fire zones and the alarms and control requirements for each zone shall be activated through a local control and alarm panel, including alarm signal to the existing central alarm panel and mimic board at the fire brigade station. A fire and gas alarm panel shall be provided in the central control room of the Plant.

An automatic fire and gas detection and alarm system shall be provided, with detection devices selected to suit particular risks, and with a control system designed to provide operating and fire brigade staff with sufficient information to identify and respond correctly to any fire detected.

The contractor shall provide temporary firefighting and alarm services during construction according to own assessment and requirements.

8.13 Security Access and CCTV System

The Communication system for plant shall be in accordance with the OTS. Plant communications shall include a telephone system (up to PABX terminal box and suitable for connection to the public system) with co-ordinated pagers, public address covering the site areas, fax and email facilities as required by the Owner, connected to the SCADA system.

A CCTV system shall be supplied to cover the entire Plant including stacks, GT&ST turbines, control room entrance, all others, etc, the CCTV system shall also be supplied, covering the Plant, boundary fence and Gatehouse for security. Number and exact location of cameras and monitors shall be fixed during the detailed design stage.

Monitors of process related CCTV shall be placed at least one in the central control room.

Digital colour CCD cameras with motorized zoom objective, placed on suitable masts in environmental protection housing and equipped with pan and tilt drives shall be used.

External drive shall be foreseen to archive the records.

Both systems shall have repeat monitors in the Gatehouse.

8.14 Additional C&I Works

The requirements for the Load Dispatch Centre are defined Grid Code for the KSA issued by the regulators.

8.14.1 Plant Management System

A Plant Management System (PMS) shall be provided which is the highest level in the control hierarchy and shall be passive in nature.

High end quality hardware and software tools should be used for PMS. The PMS server shall be connected on-line to the DCS and all other control systems. The metering data shall be handled by the metering system but also shown at PMS. Data exchange to an external server shall be foreseen to provide information for the office LAN. The PMS shall provide the following functions: operator guidance, optimal load distribution, a visual display of the efficiency deviations, mass and energy balances, performance optimization of the major equipment.

8.14.2 SCADA/Telemetry Communications

All communications necessary for the operation of the Plant and the LDC shall be supplied, including those listed in OTS and the latest version of Saudi Arabian Grid Code, . All transmission media, termination equipment and servers/firewalls shall also be supplied.

Communication is required between the Plant and the LDC as described in OTS.

8.14.3 Implementation of LDC interface

It is required that the Plant is equipped with an Automatic Generation Control (AGC) system through which the plant can be remotely controlled from the Load Control Centre (LDC) via the AGC.

The requirements in connection with AGC are specified in OTS.

8.14.4 Telecommunication System

Telecommunication systems shall be:

- Designed to prevent unwanted operations such as those caused by equipment or personnel,
- Powered by the sub-station's batteries or other sources independent from the power system, and monitored in order to assess equipment and channel readiness;
- Protection relay settings shall be coordinated by the Transmission System Operator (TSO) and IPP across the connection point. In case of a dispute between any parties on such issues, the TSO shall be the final deciding authority;
- The components and software used in all protection systems shall be of proven quality for effective utility application and follow good utility practice;

8.14.5 Telephone system

A telephone system shall be provided by the EPC. Beside wall mounted telephones in the Plant and desk telephones in offices/control rooms, mobile telephones (DECT phones) shall be provided for manager, engineers, operation shift leaders and maintenance shifts.

The system shall be modular in design, enabling flexibility in re-configuring, building up and expanding the system when required. The system shall be connected to the public telephone system.

Communication systems for process plants shall be fed by the most reliable power source of the plants such as UPS.

Radio Paging System

Following should be provided as mobile radio telephone system for general plant communications.

- At least fifteen (15) sets of Battery operated handheld transceivers for general use in the plant and suitable number for plant.
- At least ten (10) sets of walkie-talkies of different frequencies are to be provided in the plant and suitable number for plant.

8.14.6 Public Address System

A public address system (loudspeaker system) shall be provided to cover entire plant for the broadcast of announcements throughout the power station. Loudspeakers shall be connected on a zonal basis with the call station and zone selection facilities located in the CCR. The complete PA system shall conform to the latest state-of-the-art. The system shall be complete including central amplifier station, master call station, remote call stations, loudspeakers, interfaces, cables, etc.

The loudspeaker lines shall be organized in different zones.

Different tone signals shall be provided for at least the following purposes:

- Call announcement;
- Operational alarm;
- Fire alarm; and
- All-clear signal (for resetting of alarm).

The sound output level of the PA system shall be designed to achieve a sound pressure level 10 dB higher than the ambient noise in the area to ensure speech intelligibility.

The PA system shall incorporate a self-surveillance facility for monitoring the status of the PA equipment including loudspeakers. Appropriate alarms shall be raised automatically to alert the operator when a fault is detected in the system.

The Contractor shall be responsible for ensuring that the loudspeakers supplied are suitable for the environmental conditions and that the loudspeakers are positioned and oriented to give the required sound coverage.

Power amplifiers shall be sized to provide sufficient power to drive all the loudspeakers under normal operating conditions plus 20 percent overload margin.

The public address system shall be interconnected with the PABX system for access from each extension.

9. ELECTRICAL SYSTEMS

9.1 Electrical Connection Equipment

The generator transformers shall be connected to the National Grid Saudi Arabia (NGSA) through a new 380 kV GIS Substation to be constructed by NGSA. (As per RFP / OTS Part A requirement).

The Interface Point / Point of Common Coupling (PCC) with Offtaker Grid shall be in compliance to RFP / PPA and the Saudi Arabian Grid Code (The interface point for the electricity connection is SF6/XPLE (cable termination) of the 380 kV GIS bus duct connecting, cables of the generator step-up transformers to 380 kV GIS at the Grid Substation located directly outside the 380 kV GIS hall of the substation.

The EPC shall include provision of all signals for signal exchange with LDC (Load Dispatch Centre) via interface panel in compliance to RFP / PAA and Country Grid Code or as per connection agreement of the Project.

The Power System Automation system shall also communicate with the Offtaker. Interface point will be in the Plant's control room.

The generator transformer cables shall be solidly earthed in the GIS and sheath voltage limiters shall be installed at the transformer end of the cable. It is the contractor's responsibility to determine the suitability of the sheath voltage limiters.

2x100% Redundant Numerical protection for HV cable between GSUT and the GIS or AIS shall be provided.

The generator transformers are connected to 380 kV GIS by 380 kV Cables with proper termination at both ends (from respective OEM) in accordance with Offtaker's termination practices and relevant IEC standards.

A 380 kV circuit breaker, equipped with protection and control, shall be available for each generator transformer circuit belonging to the incoming units from the plant under SPPC scope.

Details of the sealing and earthing shall be agreed with the Offtaker, the Offtaker's GIS substation contractor, the EPC and the EPC's cable contractor.

Interface panel will be provided by the EPC in the GIS building / control room for interfacing with various services such as control, metering, monitoring, protection, cabling with the Plant.

The 380 kV generator transformer cables shall be solidly earthed in the GIS and sheath voltage limiters shall be installed at the transformer end of the cable. It is the Contractor's responsibility to determine the suitability of the sheath voltage limiters.

380 kV GIS substation for the project shall be provided by the Offtaker based on the RFP (OTS Part A) requirement.

The protection of the GIS Substation shall be provided by the Offtaker, the EPC shall advise the Offtaker on the technical requirements to ensure compatibility of the protection schemes. The EPC shall provide all the necessary cabling of the protection schemes and inter-tripping from the Plant to the Offtaker interface cubicles in the GIS Substation. Redundant protection for HV cable between GSUT and the GIS shall be provided by the EPC in coordination with Offtaker.

Plant normal and abnormal operating parameters i.e. Voltage, Frequency and Power Factor at Electrical Delivery Point shall be in accordance to the Saudi Arabian Grid Code. or Interconnection conditions to be agreed with Offtaker.

9.2 Network Connection Conditions (Local Grid Code / Regulation requirements)

In general, the network connection has to be designed in compliance to the Saudi Arabian Grid Code for Connection of Electric Power Producers (if any) or as per agreed with Offtaker.

The delivery point is defined at the high voltage side of the step-up transformer or as per PPA and Grid Connection Agreement with Offtaker.

The metering point will be on the high voltage side of each unit step up transformer / delivery point. Metering facilities at the delivery point shall meet all the requirements specified in Section 9.2.4 of this OTS and comply with the Grid Code or as per agreed with Offtaker.

9.2.1 SCADA Signals

As a minimum the SCADA signals requirement shall be in accordance with the Grid Connection Agreement and as agreed with Offtaker.

9.2.2 Protection and Alarms

As a minimum the Protection and Alarm requirement shall be in accordance with the Grid Connection Agreement and as agreed with Offtaker.

9.2.3 Telecommunication

As a minimum Telecommunication System Technical specification requirement Grid Connection Agreement and as agreed with Offtaker.

9.2.4 Metering Facilities

As a minimum the Measuring, Metering & Protection requirement shall be in accordance with the Grid Connection Agreement and as agreed with Offtaker

In general the metering facilities shall be installed by the HV Switchyard Contractor at delivery point and consist of a Main and Back up metering system and a back-up metering system with an accuracy of at least 0.2%.

The meters shall separately record the input and output Active and Reactive Energy. The Reactive Energy metering shall provide separate measurements for each quadrant. The output from two or more instrument transformers or meters may be combined into one integrating Recorder provided all the requirements of this Section are met.

The outputs that need to be integrated in one recorder are:

- (a) Active demand incoming and outgoing in the transmission system; and
- (b) Reactive demand incoming and outgoing in the transmission system

Provisions shall be made to permit on-site as well as remote interrogation of the Recorder. Both, main and back-up revenue meters must have the following features and otherwise fully comply with the Grid Code and agreed with Offtaker:

- The meters shall measure, record and locally display at least kW, kWh, kVAR, kVARh and cumulative demand, with the additional features such as time-of-use, maintenance records, power quality

monitoring and pulse output. The pulse output of for remote metering must be for all major measurement features.

- Nonvolatile fail-save memory storage for program and register data.
- Communication interface for meter programming via PC.
- Security codes and switches to prevent unauthorized reset or reprogramming.
- The ability to reset the meter registers with high security level by software only.
- All the functions are fully programmed from the software.
- The ability of partial programming without resetting registers.
- The insulation test voltage shall be 1000 VAC, 50 Hz and applied for one minute.
- Real time synchronizing.

9.2.5 Substation Control and Monitoring System (SCMS) (under SPPC scope)

As a minimum technical requirement for Switchyard Control and Monitoring System (SCMS) shall be in accordance to the Grid Code for Connection of Electric Power Producers (if any) or as per agreed with Offtaker and Grid Connection Agreement with Offtaker.

For control and monitoring of the Switchyard a digital Switchyard Control and Monitoring System (SCMS) with related HMI systems shall be provided in the "switchgear and control building". For future remote control from the LDC the required signal exchange via the telecommunication equipment and the OPWG of the OHL's shall be provided.

Specific requirements for installed measuring equipment, measuring accuracy and master data must be made available by Offtaker for a CCP plant to be connected to transmission network.

The EPC or HV Switchyard Contractor shall coordinate with Offtaker in order to obtain the corresponding information and requirements. The EPC shall ensure continuity in sending active power, reactive power, voltage and frequency to the network operator as appropriate.

9.2.6 Technical interfaces

The interfaces are defined according to the limits of supply for the OHL or Underground Cable transmission line and the telecommunication system (SCADA) Interfaces with Offtaker and National Control Centre;

In order to link the CCP Plant to the National Control Centre (NCC), the EPC or HV Switchyard Contractor shall make available all analogue and digital signals as requested by the NCC at an interface cubicle (provided and owned by EPC or HV Switchyard Contractor and located in an interface room in the control building of the Power station).

EPC will provide and own the interface cubicle, cable protection cubicles shall be provided but located in a separate room in the control building of the power station, for HV-cable protection.

Fiber optic cable or Optical Ground Wire terminations at the Optical Distribution Frame ("ODF") are to be provided and owned by the EPC or HV Switchyard Contractor and located in an interface room in the control building of the Switchyard. In the ODF all interface signals from/to the NCC shall be provided and data transmission is facilitated by the telecom equipment to be supplied under the scope of work. All interface equipment (hardware and software) together with the cables connecting them to the ODF shall be included in the scope of supply of the HV Switchyard Contractor.

The preliminary list of signals which shall be considered as a minimum scope shall be provided by EPC or HV Switchyard Contractor based on the requirements of the Network Connection Agreement with Offtaker.

The following signals shall be transmitted to Offtaker Control Centres upon agreement:

- All signals transmitted to the NCC; or Offtaker Control Centres
- All settlement metering and back-up metering signals;

- All further signals from the Central Control Room of the CCP Plant as required by Offtaker.

9.2.7 Grid Connection

The interface for the primary circuits is the termination of the conductors at the 380 kV GIS or as agreed with Offtaker. For telecommunication and SCADA the interface are the OFC or OPGW connection box.

However regarding the interfaces the manufacturer and type of equipment has to be agreed with Offtaker.

9.3 Signal Exchange between Load Dispatch Centres

Signal exchange between the Plant and the Offtaker's electricity control centres shall be via communication protocol of IEC 101 interface located at the GIS Substation. The interface shall be provided by the Offtaker and the EPC shall cable to the interface from the Plant. The EPC shall comply with the Offtaker's ICS specifications for all data communications between the Plant and Offtaker's electricity control centres. All measuring devices, auxiliary contacts and interposing relays and selector switches together with wiring and associated facilities for the Plant are to be supplied by the EPC.

- The EPC shall develop and agree with the Offtaker schematic screens of the sections of the Plant which are connected to either side of the generator transformers. These shall be compatible with the Offtaker existing schematic screens at the Offtaker's electricity control centres and be provided to the Offtaker for integration into the existing schematic screens.
- The EPC shall undertake end-to-end testing for all communications between the Plant and Offtaker's electricity and water system control centres including tests associated with the Fuel Demand Model.
- Overall policies for Plant monitoring and governor control shall be developed by the EPC in consultation and agreed with the Offtaker.
- Integration of the required telephone shall be carried out in line with the Offtaker's telephony plan.
- A Supervisory Control and Data Acquisition (SCADA) facility list is envisaged which shall be required to provide status and alarm signals and metering signals for the following;
 - Generator circuit breakers and related equipment;
 - Generator transformers;
 - Communications;
 - Generator and generator transformer protection;
 - Metering of Electrical Energy Delivered and Electrical Energy imported;
 - Metering of Natural Gas received

9.4 General Electrical and Control requirements

The design of the electrical system shall take into consideration reliability, availability, interchangeability and maintenance aspects as well as overall economics.

All electrical equipment shall be suitable for operation satisfactorily under the site ambient conditions / environment conditions.

Ambient temperature for outdoor installations (in the shade): 55°C (as per site ambient conditions)

Design temperature of electrical equipment (outdoor without the shade): 55°C

Design temperature of electrical equipment (outdoor under the shade): 50°C

Design temperature of electrical equipment (indoor without HVAC): 50°C

Design temperature of electrical equipment (indoor with HVAC): 40°C

All the selected material and equipment will be suitable for carrying continuously the specified currents and loads without undue heating and will be designed to ensure satisfactory operation under sudden

variations of load and voltage as may be encountered during normal / abnormal operating conditions of the system

Electrical equipment shall be designed to ensure continuity of operation of the main Plant under all working conditions. The Plant shall also be capable of meeting all transient, abnormal and load shedding scenarios.

All reasonable precautions shall be taken in the design of equipment and of works to ensure the safety of personnel concerned with the operation and maintenance of works. Safety, isolation, locking and interlocking facilities shall comply with international standards and practice.

Electrical equipment shall be constructed to withstand the specified maximum short circuit currents and duration without the temperature exceeding the value permitted for the related insulation class. The equipment shall be considered as being operated at steady state maximum permitted current prior to the inception of short circuit current.

The final temperature attained as a result of the passage of short circuit current shall not cause permanent damage, deformation or deterioration sufficient to reduce the normal operating characteristics below those specified.

Insulation materials shall be suitably finished so as to prevent deterioration of their qualities under the specified working conditions.

Anti-condensation heaters shall be provided for switchgear and control panels in areas without redundant air conditioning. The heaters shall be permanently connected without thermostats and energised via a local isolator supplied from a local distribution board. Anti-condensation heaters for motors shall be interlocked with the motor supply.

Electrical Panels shall be designed to facilitate thermography inspections of panel by provision of IR windows/ suitable design. (For Gas turbine package, IR windows are not applicable.)

Auxiliary power systems shall be arranged to minimise operational constraint and be capable of maintaining or rapidly restoring the full continuous power output in the event of a single component or circuit failure.

Generator and Generator Electrical Electric Auxiliary System shall in complete and include:

- Generators and Generators Neutral Grounding
- Generator Step up Transformers
- Generator Busbar and Generator Breakers;
- Generator Excitation and Synchronization
- Unit Auxiliary Power Transformers;
- MV / LV Distribution Transformers;
- 13.8 kV or 11kV or 6.6 kV (as per the RFP requirement of voltage level) MV Power Distribution;
- 400/230 V LV Power Distribution;
- Rectifier / Battery Chargers, Batteries and Inverters;
- Emergency Diesel Generator;
- IPB bus duct & MV/LV cables;
- Electrical Protection;
- Electrical Control;
- Cable Systems and Tray, Conduits, Trenches, etc.;
- Earthing and Lightning Protection as per earthing and lightning study and calculation results. (International standards like NFPA or IEC standards or IEEE recommendations)
- Cathodic Protection System;
- Intercom. loudspeaker and time distribution System; and

- Lighting and small power installations.
- Emergency Diesel Engine Generator

Electrical Auxiliaries:

For the design of the electrical auxiliaries system, all requirements specified under the Mechanical requirements shall be followed accordingly e.g. electrical redundancies shall not be less than mechanical ones, reliable power supply and remote control of facilities shall be maintained in case of outage of one single component. An appropriate level of redundancy shall be included in each auxiliary system.

- Area Electrical sockets shall be of industrial type. (CEE norm, with cover, minimum with IP 54). Classified area shall be provided with hazardous protection as per area classification
- Ground fault circuit interrupter (GFCI) shall be provided for all electrical circuits where human safety is required (electrical LV circuits)

The design consideration of Electrical System shall be in compliance to the OTS. The following general electrical requirements shall apply:

- Unit auxiliary transformers and switchgear shall be designed to continue normal operation with one unit auxiliary transformer out of service. A change-over scheme shall allow transfer, without loss of output, between normal system operation and outage condition;
- The mechanical protection class of all MV switchgears shall be minimum IP41 and all & LV switchgears shall be minimum IP41. DC rectifiers and inverters (UPS) shall be minimum IP31 (Installed Indoor Air Conditioned Room).
- Junction Boxes, Control Instruments, Power Receptacles installed Outdoor shall be minimum IP 54W / IP 55 and Junction Boxes, Control Instruments, Power Receptacles installed indoor shall be IP 54.
- All AC Electric Motors shall be of IP 54 and DC motor IP 23. Motors installed outdoor shall be of IP55W.
- Electronic cubicles located within buildings without air conditioning shall be IP41 and located in air conditioning rooms shall be IP 32 (IP 41 for GT Control cubicle).
- Electrical equipment installed outdoors shall be protected by sunshades where necessary, while sunshades are not required when the equipment meets OTS design temperature and met the service life requirement. .
- All electrical equipment related to units like MV, LV switchgear, DC & SAFE AC system, diesel generators and electronic cabinets shall be located in separate fully air-conditioned electrical room or in separate fully air conditioned electrical container for each Plant Unit near the respective units with at least n - 1 A/C redundancy.
- For the design of the electrical and instrumentation and control facilities all requirements specified under Section (Operational requirements) and the mechanical particular requirements shall be followed accordingly e.g. electrical and I & C redundancies shall not be less than mechanical ones, reliable power supply and remote control of common facilities shall be maintained in case of outage of one single component
- At least all equipment which when failure could cause an outage of a total Plant Unit and/or the total Plant, shall be provided with 100% redundant electrical supply equipment.
- The safe shutdown scheme shall include duplication and auto-changeover facilities for the essential auxiliary supply section, such that the UPS equipment will only switch-over to any alternative supply, or the diesel generators automatically start-up on, a complete loss of supplies to the site.

The design consideration of Instrumentation and Control System shall be in accordance to the OTS. Information, Control and Communication Systems shall in Complete and including:

The control system shall consist of block-wise functionally distributed control system (DCS) for overall control & monitoring of the Power Facility from the central control room - including direct DCS control of HRSG equipment and majority of the balance of plant (BOP) equipment. This control system shall be integrated with /aided by GTG control system (gas turbine & its auxiliary controls), STG control system (steam turbine & its auxiliary controls) and local proprietary/ autonomous (PLC or DCS based) control

system for each stream of the SWRO plant and some of the BOP packages such as compressed air system, HVAC system, fire and gas detection system, etc

- DCS and Information storage systems;
- Plant Settlement System;
- Time Synchronising System: Master clock system with GPS (Global Positioning System) & external time synchronisation;
- CCTV systems;
- Site alarm system and public address system;
- Environmental monitoring system;
- Field testing equipment;
- Instrument workshop equipment;
- External and Internal Telecommunication systems;
- Interfaces to third parties as necessary;
- Central Control Room inventory;
- Plant operator working stations (e.g. thin clients) with Human Machine Interfaces;
- Power diagnostic systems as necessary; and
- Maintenance Management Systems.

All electrical equipment shall be routine and type tested in the manufacturer facility or type tested in third party facility. Type tests shall not be repeated if type test certificates of the same equipment (same design and related technical parameters) are available. For the avoidance of doubt, the type test shall be delivered to the Owner.

9.5 Generators and Exciters

9.5.1 Generator

Gas Turbine Generator and Steam Turbine Generator (STG) shall generate power at a terminal voltage of 15.75 ~ 22 kV or any other voltage level according to OEM standard of the generator, at the power frequency of 60 Hz. The generator shall be provided with all associated auxiliaries such as cooling system, and all other associated instrumentation and auxiliaries of proven type with suitable redundancy in equipment.

The generators shall comply with the requirements of the relevant parts of IEC 60034 and shall be rated to match the turbine output over the full range of ambient temperature under all operating conditions. Each generator shall be subject to factory tests as required by those Standards. The generator shall be of an existing design which has a proven record of reliability for similar applications. Prototype designs are not acceptable.

Due to reasons of transmission operation needs the generator sub – transient reactance (saturated) shall be minimum 12%.

Turbine generators and related cooling systems shall be designed to class F and operated to maintain internal heating within insulation class B limits according to IEC 60034-1 throughout the entire operational range and throughout the ranges of ambient temperature.

Generator design consideration shall be in accordance to OTS. Generator performance requirement, voltage variation and frequency variation shall be in accordance with the Grid Code. Stable Generator Unit operation shall be possible at any power factor within the capability limits of the generator performance chart. Each generating unit must be capable of meeting any Network Connection Conditions for the 380 kV Grid and compliance to the Saudi Arabian Grid Code latest version.

The generator shall be either air or hydrogen cooled. For generators with a hydrogen cooling system, it is required that there shall be local hydrogen production and that provision shall be made for any hydrogen

leakage into the cooling water system in service to be detected, alarmed and the rate of leakage measured. Similarly, all appropriate generator protection systems and related instrumentation including partial discharge detection and fire protection systems shall be provided and comply with the relevant NFPA standards.

In case of closed air cooling system, the generator air coolers shall be sized so, that maximum rated generator output can be maintained with 15% of the tubes blocked and with 15% of the tubes blocked and one cooler section out of service (cleaning) with temperature rise according class F. The cooler and isolating valves shall be arranged to permit isolation and cleaning of individual sections whilst the generator is on load.

In case of closed hydrogen cooling system, the hydrogen shall be circulated by fans mounted on the generator shaft in a closed circuit system incorporating H₂ to water coolers. The generator hydrogen cooler etc. shall be sized so, that maximum rated generator output can be maintained with 15% of the tubes blocked and with 15% of the tubes blocked and one cooler section out of service (cleaning) with temperature rise according class F. The cooler and isolating valves shall be arranged to permit isolation and cleaning of individual sections whilst the generator is on load.

Continuously acting rotor winding earth fault monitoring, Generator online partial discharge monitoring system and flux probe for each generator.

Generators shall be provided with Online Partial discharge monitoring device with facility to monitor PD remotely in DCS/Control Room utilizing bus couples monitoring system in accordance to IEC 60270 for permanent installation on site and a Vibration monitoring system. Generator shaft voltage monitoring and Flux probe monitoring shall also be provided.

Generator associated component i.e. Collector Ring, Shaft Grounding, Exciter Fan , Brush Holders shall be of proven design and similar design machines are being operated for at least since 10 years (a confirmatory letter for the same to be provided by OEM).

Generator shall be provided with controlled space heaters working according to design control set to avoid any condensation and controlling humidity (as applicable for Air Cooled Generator).

Continuously acting rotor winding earth fault monitoring, Generator online partial discharge monitoring system and flux probe for each generator with a common portable type online monitor (not continuous monitoring) as optional shall be provided with each GT and ST Generator.

Generators shall be provided with Online Partial discharge monitoring device with facility to monitor at remote DCS. Generator shaft voltage monitoring and Flux probe monitoring shall also be provided.

PDM (Partial Discharge Monitor) in the Gas turbine package

PDM(partial discharge monitor) will be provided as the generator on-line condition monitoring system with microstrip antenna type sensor manufactured by MELCO (MELCO PDM),

MELCO PDM has a lot of proven track record.

For large capacity turbine generators, antenna type sensor is recommended because the sensor is not installed on to the high voltage winding, and they are not subject to any high electrical stresses.

- On-line partial discharge monitoring system with microstrip antenna for turbine generator

MELCO PDM (and also other vender's PDM) do not comply with IEC 60270 but comply with IEC 60034-27-2 for the following reasons.

(1) IEC 60270 is the standard for high-voltage test techniques of PD measurement at low frequency below 400Hz, not the standard for on-line PD measurement.

(2) IEC 60034-27-2 is the standard for on-line PD measurement on generator stator winding insulation.

Generator associated component i.e. Collector Ring, Shaft Grounding , Exciter Fan , Brush Holders shall be of proven design and similar design machines are being operated for at least since 10 years (a confirmatory letter for the same to be provided by OEM).

For Shaft grounding copper braids shall be preferred over the graphite brushes for better performance and reliability.

NOTE: Generator Mechanical design shall ensure that all assembled parts nuts & bolts with lock washers.

Hydrogen seal oil units shall be complied to requirements of NFPA850, Cl:7.7.4.4 (i.e. automatic sprinkler system or foam system).

All six terminals shall be brought out of the machine and shall be available for external connection to the main isolated phase bus ducts (IPBs) and bus ducts to the neutral earthing transformer.

The turbine generators (GTGs and STGs) shall be designed in compliance with the latest edition of IEC 60034, except where more stringent rules are stated.

The requirements of the Grid code shall apply.

The STG shall have sufficient capacity such that it does not limit the capability of the steam turbine over the range of ambient conditions.

The EPC is requested to design the components (turbine, generator, generator transformer) so, that the following conditions will be met within the complete range of defined range of maximum and minimum design-temperatures_ PT (Transformer power (MVA) at nominal power-factor and tap changer in middle position "0") $\geq$ PSG (Generator output power (MVA) at nominal power factor as input power to the generator transformer) $\geq$ PST (Turbine power (MVA) at nominal power factor and consideration of the generator efficiency)

A graph shall be submitted with the bid showing in one diagram the power outputs PT, PSG, PST and PWT (Turbine (active) power (MW) as input power to generator) versus minimum to maximum designed ambient air temperature (+4°C to +50°C).

Turbine generators and related cooling systems shall be designed to class F and operated to maintain internal heating within insulation class B limits throughout the entire operational range and throughout the ranges of ambient temperatures and cooling temperatures.

Capacity curves, V%/f% curves, capability diagram and also permissible single phase and unbalanced 3-phase loading plotted against time shall be submitted with the Bid. All diagrams shall be valid for operation under ambient air conditions from +4°C to +50°C.

The bearings shall be of the metal sleeve and self-alignment type and designed so as to be readily accessible and replaceable without the removal of the rotor. (For Gas turbine generators, tilting pad bearings are applied for better rotor shaft vibration characteristics.)

Air Cooled or Hydrogen cooled generators shall be provided. However, hydrogen cooled generators with stator winding direct-cooling clean-water cooling system will also be accepted, if of manufacturer standard design.

If hydrogen cooling equipment is applicable, the cooling system shall comply with the requirements of IEC 60034-3 for hydrogen cooling equipment. The cooling classification shall be stator/rotor IC7 (8) H1W7 and if applicable for stator-winding IC9W7W7.

The generator coolers etc. shall be sized so, that rated generator output at class B operation can be maintained with 5% of the tubes blocked

Additionally, the numbers of coolers (preferably 4 coolers) are to be chosen so, that with one cooler section out of service (cleaning), a maximum generator output can be maintained. The requirements of IEC 60034-3, Para 4.21 are to be fulfilled; (The hydrogen gas coolers of the proposed generators comply with the requirements in IEC 60034-3:2020, Para 4.22.)

The cooler and isolating valves shall be arranged to permit isolation and cleaning of individual sections, preferably whilst the generator is on load. (For Gas turbine package, the hydrogen gas cooler cannot be cleaned during generator operating condition considering safety concern.

A complete set of brushless or static excitation system including power system stabilizer (PSS) and excitation limiting devices.

A numeric type automatic voltage controller (AVR) with power factor and reactive power control

The excitation system shall be capable of producing a steady state short circuit current to ensure that the main and back-up protection are sensing and clearing the fault on the generator terminals.

A block diagram of the complete excitation system shall be provided with the Tender, together with the time-constants of the transfer functions for the control and stabilizing circuits.

Provision shall be made for testing the excitation equipment without the operation of the main generator.

Rotating exciters (RE) shall be of the enclosed, ventilated type, IP 54. They shall be directly driven by the generator. Ventilating air for rotating AC exciters shall be closed circuit cooling, incorporating an air-to-water heat exchanger, and shaft mounted fans.

For Static excitation (SE) the thyristor converter shall incorporate a number of three-phase full-wave bridges and shall be arranged so that a minimum of 20% spare capacity is incorporated. The converter should be of modular construction, so that a minimum of one bridge may be removed for maintenance, whilst maintaining unrestricted operation of the main generator. This shall apply also for the field flashing facilities. The SE system shall be equipped with forced cooling in 2 x 100% design with automatic switchover and monitoring from the CCR. The cooling medium for slip rings and commutators shall be separated from the exciter- and generator cooling circuits. This cooling circuit shall be provided with filters (easily maintainable) against carbon dust from the brush-gear.

In Gas turbine package, thyristor rectifier cooling fans are provided for each thyristor element in GT OEM standard system.

The redundancy is N+1 and that is as same as thyristor rectifier. Even if the fans are out of service in the one parallel of thyristor rectifier, excitation cubicle maintain generator normal operation.

The fan is applied DC fan which has features of long life and easy of maintenance and etc. Cooling fans power source are supplied by Distribution Panel.

A minimum excitation limitation device shall be incorporated, to prevent the AVR reducing the generator excitation below a value, which might endanger power system stability limits. The characteristic of the var-limiter shall be adjustable (also remotely) and facilities shall be provided so, that the var-limiter operating characteristic can be adjusted to any setting within the specified range.

For each generator a state of the art generator on-line condition monitoring system shall be implemented to show the momentary condition of the rotating machine respectively as well as long term trends. The generator on-line condition monitoring (partial discharge monitoring) system shall consist of sub-systems for specific monitoring tasks and shall comprise at least the following main functions:

Electrical real-time partial discharge (PD) monitoring system in accordance with IEC 60270 for permanent installation on-site;

Vibration monitoring system shall be provided as detailed below covering the vibration and displacements measured at each main bearing housing (angle of 90 degrees between the two probes) and by direct measurement off the shaft adjacent to each main bearing plus axial position of the shaft also. Measurement and indication of shaft eccentricity shall also be provided. All measured values shall be readable in the control room.

9.5.2 Excitation Systems and Equipment

The excitation system for the generators may be brushless system or static excitation system. Two automatic independent digital AVR shall be provided along with a manual channel. The regulator will be equipped with the required limiters for limiting the stator current, load angle / under excitation limiter, power factor, v/f, VAR etc., as per modern engineering practices. Power system stabilizer shall be provided as required by local regulations. The excitation system design and equipment shall conform to the Grid code / or any Government Authorisations Regulation requirements and IEC / IEEE standards requirements or as per agreed with Client.

The excitation system and equipment shall be complete with fast response continuously acting dual channel automatic voltage regulators, (Dual AVR modules with automatic and manual mode), power system stabiliser, limiters and stabilisers, field suppression equipment and all necessary control, monitoring and protection equipment. The static excitation systems shall include redundancy for thyristor bridges and cooling fans. Brushless excitation systems shall include redundant AVR.

Must have provision of complete control and monitoring of excitation system (AVRs) in remote control room through DCS. (For ST OEM, control and monitoring of the excitation system of STG will be sent to DCS from STG control system via OPC server. However, complete control and monitoring requirement of this document should be fully complied.

The generator excitation systems of high response static type complying with the requirements of relevant IEC 60034 or IEEE standards and Grid Code shall be provided. Automatic excitation control system of a proven design shall be provided to control the generator voltage without hunting/instability over the entire operating range of the generator.

The excitation equipment shall operate reliably and shall not impose restrictions on the availability of the generators due to excitation equipment faults or malfunctions according to excitation architecture and relevant redundancy.

Each static excitation system shall utilise a three phase fully controlled thyristor bridge rectifier energised from the machine terminals or medium voltage busbar through a suitable transformer. The transformer shall comply with the requirements of IEC 60076-11 (dry type transformer) as applicable. Excitation transformer is connected to MVSWGR (medium voltage busbar) considering black out start.

Each excitation systems rated output voltage and current shall be at least 110 percent of the corresponding generator field (rotor) voltage and current for generator rated output overexcited.

In order to provide adequate safety margins, the diode and thyristor devices in the power rectifiers shall have current (IFAV) and voltage (VRRM) ratings, which are at least 1,6 times the rated values and 1,1 times the rated current, unless otherwise approved by the Owner in accordance with the Contract. The Contractor will need to demonstrate its design is in line with industry standards. It shall be possible to detect failure of a diode / thyristor.

Excitation Transformer shall be located indoor i.e. Turbine House building and shall have the Thermal Insulation class F but the Temperature rise class shall be Class B. Following shall be considered in Excitation Transformer:

- Temperature detectors in LV coils of each phase
- Temperature detector in the middle core shank
- Tapping links on the HV side which can be disconnected under de-energized condition.
- For outdoor installation a metal enclosure is provided with IP23 class protection. (For Gas turbine package, oil immersed type excitation transformer will be considered and located outdoor.)

In case of GTG / STG OEM offer Dry Type Excitation and SFC Transformer for the Generator Excitation System, following consideration are recommended for design consideration:

- a) Excitation Transformer manufacturer to provide detail design of electrostatic screen and its termination with ground. Electrostatic screen ends shall be labelled with bold tag with instruction to terminate it with ground. OEM documents and drawings shall be available with guideline of electrostatic screen connection with ground.
- b) EPC to ensure completion of all transformer commissioning works and confirm electrostatic screens are properly connected with ground

The excitation systems and equipment shall be complete with fast response continuously acting (AVR) automatic voltage regulators one in operation, one in standby and one manual regulator (MVR) in standby, power system stabiliser, limiters and stabilisers, quadrature droop compensation and cross circuit compounding, field suppression equipment and all control, monitoring and protection equipment to enable operation from the Central Control Room. The excitation system shall be provided with a fast-acting MVar limiter so as to prevent the generator output falling below its safe limit. Protection features as part of the system shall include over-voltage, overcurrent, VT fuse failure, diode failure, and AVR power supply failure.

A field shorting or discharge switch feature shall also be included in the system as protection against overstressing the generator insulation in the event of a fault.

Manual excitation control facilities shall be provided as a backup to the automatic control and shall have an adequate range to allow for control of excitation for testing purposes. A true null balance shall be provided to allow for smooth excitation transfer between manual and automatic control.

The system shall be one of high-initial response as defined in IEEE Standard 421.1. The continuous current capability of the excitation system at maximum ambient conditions shall be adequate for the operation of the generator at full load and rated power factor at 95% -105% terminal voltage, with the generator field at maximum temperature.

The excitation system shall be capable of supplying its ceiling voltage and the growing field current associated with the sudden application of ceiling voltage to the generator field for at least 5 seconds. The ceiling current shall be at least 2 times the rated field current. The transient thermal capabilities of the excitation system for periods of more than 10 seconds and up to 120 sec shall be higher than the short-time thermal capabilities required in Section 7.2 of ANSI Standard C50.13 or IEEE C50.13 section 4.2.1&2 for the generator field winding.

The excitation system shall meet the requirements of the interconnection system but in any case, the AVR shall be capable of maintaining the steady-state terminal voltage of the generator within +/- 0.2 % of the set point according IEEE 421.1 or 0.5% over the entire operating range of the synchronous machine.

Over excitation limit value setting shall be adjusted automatically according to rotor temperature (proportional to rotor cooling air outlet temperature).

An adjustable device shall be incorporated to limit the per unit ratio of generator voltage/frequency (over flux) to such a value that the maximum ratio shall not exceed 1.1 per unit.

Under excitation, over excitation, stator current and overfluxing limits shall also be incorporated to prevent excitation falling or rising above the values necessary to ensure steady state stability and rotor and stator thermal limitations are not exceeded. The regulator equipment shall be free from voltage drift and temperature errors and insensitive to frequency variations between - 6% and + 10% of nominal.

The excitation equipment shall operate reliably and shall not impose restrictions on the availability of the generators due to excitation equipment faults or malfunctions according to excitation architecture and relevant redundancy.

The static excitation system shall consist of:

- Excitation transformer;
- Solid state Thyristor based rectifiers;
- Field DC circuit breaker; and
- Excitation controls, Limiters and Protections.

In Gas turbine package, AC field circuit breaker (a 3-pole breaker on exciter AC side) system with discharge circuit composed by thyristor and discharge resistor without breaker on DC side will be provided as GT OEM standard.

The excitation control system shall be fully redundant as far as reasonably possible, such that failure of any one component does not cause any loss of functionality or degradation of performance. Thyristor rectifier cubicle (including redundancy to satisfy reliability requirement under this document)

The excitation transformer shall be oil filled or dry type. It shall be connected from the generator bus and it shall be capable of providing the secondary voltage for the high ceiling voltage required. Its rated capacity shall be not less than 1.1 times the field load at rated generator output and 5% over voltage. The generator field breaker shall be provided with a field discharge resistor or other means to dissipate the field energy when the breaker is opened.

The excitation system shall be provided with a field ground detection system that monitors the soundness of the field insulation. The detection shall be continuously acting during operation.

Exhaust Fan System for Excitation Transformer Cabinet to be designed with Auto and Manual Mode.

9.5.3 Temperature Detectors

Resistance temperature detectors shall be provided in the stator winding and in the cooling system. Resistance temperature detectors or thermocouple detectors shall also be provided to monitor the temperature of bearing metal. Bearing metal temperature monitoring shall follow manufacturer proven standard.

9.5.4 Generator Main Connections and Neutral grounding

- The generator main connections are to be designed as isolated - phase bus duct system (IPB).
- IPB shall be pressurized with instrument air (cleaned and dried air).
- Pressure relief devices shall be provided to prevent over pressurizing of the bus ducts.
- Where the bus ducting connects to the transformers at one end and to the generator at the other end, the connections shall be made with flexible copper

9.5.5 Generator Bus Duct and Generator Circuit Breaker

The generator circuit breaker has to fulfil the requirements according to IEEE (ANSI) C37.013 and IEC 62271 series of standards;

Exclusively circuit breakers of the SF6 type with 125 VDC motor-driven stored energy mechanisms are to be provided. All CBs shall be fitted with operational counters;

All current and voltage transformers are to be provided in accordance with IEC 61869 series of standards;

Motor-driven earthing switches of short-circuit-proof design are to be provided. The necessary interlocks with the 380 kV line isolator, generator circuit breaker etc. are to be provided

The generator shall be connected to the generator circuit breaker, excitation transformer, generator step-up-transformer and unit auxiliary transformer via Isolated Phase Bus bars (IPB), which shall be enclosed in an insulated aluminium bus duct slightly pressurized with clean cooled dry air, including the necessary tee-offs.

The design and mechanical construction of all bus bars, boards and links shall safely withstand the thermal and dynamic stresses under normal operation and fault current conditions. Provisions shall be made for free expansion of conductors and enclosures due to thermal stresses by means of expansion pieces. Transmission of vibrations occurring at the generator or transformers to the busbar systems shall be avoided by use of flexible and adequately dimensioned connection tapes.

The system shall contain in addition to the main circuit busducts:

- a. Terminal boxes for the generator enclosing the main terminals and current transformers around the terminal bushings;
- b. Voltage Transformer, Surge Arrester and Protective Capacitors should be located inside GCB cubicle at the unit-transformer side.
- c. Generator neutral busducts and neutral earthing equipment cubicle;
- d. Removable three phase short circuit bridge for testing purposes, rated according to the rated current of the system and located between the generator and generator circuit breaker; and
- e. Tee-offs for the transformers (unit, unit auxiliary and excitation).

The internals of the line side busducts shall be arranged with a positive pressure to keep the system clean and dry. The busducts are to be naturally cooled.

The connections to the generator, generator circuit breaker and transformers shall have flexible joints to prevent the transfer of mechanical loads and forces. The enclosure connections to transformers shall be by means of expansion bellows made of electrically isolating and easily removable material for access to the transformer terminals.

Generator circuit breaker needs to provide ease of testing during O&M phase for Electrical testing with ease of doing and removal of main enclosure / major fitment.

On the conductor, aluminium flexible laminations, with the same total cross sectional area of the conductor shall be welded between the two sections of the conductor, to avoid sliding or bolted contacts.

The generator line-end bushing cubicle, which encloses the line-end bushings and allows the connection between generator and bus duct, shall be metal-clad isolated phase type, the cubicle shall be made of aluminium. The terminal block for connection of secondary windings of CT's mounted on bushings shall be arranged in this cubicle. The star point cubicle shall be made up of a sturdy structure of aluminium fitted with all accessories necessary for connection to generator neutral grounding cubicle. The star-point cubicle shall enclose the generator star-point bushings and the relevant current transformers. For Steam Turbine/Generator set, OEM standard and general industrial practice are acceptable.

Temperature Limits rise for Conductor and enclosure of Isolated Phase Bus duct shall be as per ANSI/IEEE C37.23. The temperature of the conductor shall not exceed 90°C and for the enclosure the temperature shall be limited to 70°C.

The Generator circuit breaker and the bus duct shall be rated for the continuous operation of the maximum apparent power at minimum generator voltage.

The generator shall be provided with a (GCB) generator circuit breaker. The GCB shall comply with IEEE Standard C37.013. The generator shall be synchronized with the power system across the generator circuit breaker and the generator circuit breaker shall be suitable for this purpose in all respects.

The generator circuit breaker shall be sufficiently rated to break the maximum fault current under any fault condition without damage to itself or any other item of plant. The generator circuit breaker shall be of the SF6 type and arranged for assembly in conjunction with the busducts. The breaker shall be equipped with a series-disconnector on the transformer side and with earthing switches on both sides. The earthing switches shall have motor operated actuators. The breaker and disconnector shall normally be controlled remotely.

All the breaker unit functions shall be operable from the local control panel at the breaker unit.

Generator Bus Duct and Generator Circuit Breaker shall be subject to a type test, certified by a third party.

9.5.6 Generator Synchronization

Automatic and manual synchronising facilities backed up with check synchronising circuits shall be provided for the generator circuits. Synchronising equipment shall operate satisfactorily in conjunction with the automatic excitation control and turbine speed governing equipment.

Synchronizing of the generator by means of the generator circuit-breaker shall be carried out automatically from the generator control desk located in the CCR and automatically and manually (with synch. check) from the same location. (For Gas turbine package, manual synchronization device is not provided in CCR.) Synchronizing of the 380 kV feeder circuit breaker at grid substation (by the client). Under exceptional circumstances, at the sole discretion of the Client, the plant shall be permitted to open the Client's 380 kV Power Unit Cable Circuit Breaker(s) or due to transfer trip in case of cable protection / 380kV grid switching station bus bar protection operate. The Generator shall be able to open Client's 380 kV Power Unit Cable Circuit Breaker(s) only in an Emergency.

Each generator shall be provided with synchronizing relays which by means of a selector switch at the generator control desk shall provide the following facilities for the associated generator circuit-breaker:

Automatic: Where control of speed and excitation is carried out automatically and a suitable relay is arranged to automatically close the circuit-breaker when speed, voltage and phase relationship are within permissible limits.

On the turbine control desk at CCR indication shall be given that synchronizing control is available at DCS. The DCS will include the necessary equipment for initiating an automatic synchronizing procedure.

The scheme shall be arranged so that the circuit breaker will either close directly once synchronism is achieved or closing can be delayed pending further operator action. A "ready for synchronizing" indication shall be given.

Synch-check: Where speed and excitation are controlled manually and closing of the circuit breaker is also under manual control by means of the control switch on the generator control desk. In this mode, the synchronizing relay provides a check feature to ensure that closure of the circuit breaker is inhibited if speed, voltage or phase relationships are outside the permissible limits.

Manual: Closing of the circuit-breaker is under manual control the operator using the instruments to ensure that conditions are correct. The same shall be interlocked with Synch- check.

Selection of the method of synchronizing shall be by means of a key operated selector switch having the following positions AUTO/OFF/SYNCH-CHECK (lockable in all three positions).

Selection of the 'MANUAL' mode shall be made by putting the above selector switch in the 'SYNCH-CHECK' position.

A set of synchronizing instruments comprising incoming and running voltmeters and frequency meters, a synchronous scope shall be provided on each generator control desk at Central Control Room.

Each generator unit shall be provided with an automatic synchronizing relay and a separate check synchronizing relay and this could be through generator control system facility.

Synchronization scheme for Plant Intermediate substation(s) feeder bay(s) and Client 380 kV feeder bay(s) subject to approval of Client respective Transmission department.

The generator shall normally be synchronized with the grid by the generator circuit breaker (GCB). Provision has to be provided that, It may also be resynchronized with the grid by the 380 kV circuit breaker of Plant Intermediate substation(s) / Client 380 kV remote substation. As the distance between Offtaker's 380 kV breaker and the plant is long then the VT circuits cannot be extended directly to the plant. Therefore, the synchronising relay will need to be installed in the 380 kV Client's site and synchronizing with 380 KV networked through to the plant controller.

The generator shall have an automatic dual channel synchronizer. Synchronization shall be carried out by auto-synchronization mode and as well as by Manual synchronization mode. A synchronization checking device shall be provided to control and supervise the Auto / Manual synchronization operation.

9.5.7 Generator Construction

The entire generator shall be mounted on a single fabricated frame supporting, the stator frame and the pedestal bearings. The stator frame shall allow the radial expansion of the core and absorb the core vibration without undue transmission of noise to the outside. The frame shall provide enough strength against the torque caused by the electric load and short circuit currents. In case of faulty synchronization, the stresses in the materials shall not exceed 2/3 of the elastic limit. The stator frame shall allow adequate air circulation for cooling of the rotor and stator.

The stator magnetic circuit shall be built with high permeability; low-loss grain-oriented stampings stacked on the frame key bars and pressed to form a single continuous core, with spacers as required for allowing an effective circulation of cooling air.

As an alternative, it can be non-grain oriented stator core laminations.

The stator windings shall consist of single-turn copper bars with the conductors transposed in the slot portion by the Roebel method and joined at their extremes. Thermosetting insulation shall be used, with spring type fillers and ripple springs to ensure that the bars will remain tight into the slots. The slot portion shall be covered with a semi conducting varnish and the end-turn portion shall be provided with graded semi conducting varnish over the main insulation. The end-turns shall be adequately supported and secured against the dynamic forces arising from solid short circuit currents between the generator terminals.

The rotor body shall be a one-piece forging machined to receive the copper field windings. The field winding shall have an ample cross section for a high ceiling current.

All components of the rotor insulation system shall be to class "F" standards or better. Rotor earth fault protection shall be provided. A continuous monitoring electronic system for brushless electronic protection equipment is preferred for brushless excitation system.

The retaining rings shall be shrunk on the rotor to support the rotor field end turns. They shall be made from 18Cr-18Mn stainless steel alloy. The rotor shall be provided with radial fans at both ends for ventilation of the rotor and the stator. The rotor shall be dynamically balanced at the manufacturer's shop, and it shall be mounted on tilting-pad type bearings. The bearing pads shall be insulated to prevent the flow of shaft currents. The number of bearings shall provide for ease in removing and inserting the generator rotor and exciter.

Shaft voltage measurement facility shall be provided by suitable means. One end of the rotor shaft shall be earthed by means of a strap brush which shall be easily re-measurable for the purpose of insulation checks. Where the bearing insulation is not easily accessible test terminals shall be brought to convenient point outside the bearing

9.5.8 Accessories

The generator shall be provided with all required accessories for an efficient and continuous operation within its whole range of operation, including hydrogen gas to water heat exchangers inside the generator, bearing oil cooling system using oil to water coolers, lubrication oil supply, RTDs for thermal protection relays, indication, etc. A closed circuit water cooling system shall be provided for the cooling of the generator hydrogen gas coolers and bearing oil. Current transformers at the neutral and at the terminals shall be provided as needed for all the protection, metering and indication functions.

9.5.9 Generator Neutral Grounding

Generator neutral grounding equipment shall be provided for the generator. The grounding system shall consist of a single-phase, encapsulated dry-type distribution transformer with secondary loading resistor located in an outdoor compartment near the generator neutral terminal. The resistor and the distribution transformer shall provide high-resistance grounding to the generator to limit the magnitude of any ground fault current to approximately 10 A. The connection from the generator neutral to the grounding equipment cubicle shall be as short as practically possible if the grounding equipment cubicle is not connected directly to the generator enclosure.

The generator neutral connection shall be fully insulated from surrounding metallic enclosure and the connection between generator neutral terminals to earthing transformer shall be fully insulated and primary terminal of earthing transformer shall also be fully insulated.

9.6 Power Transformers

The transformers shall comply with the Saudi Arabian Grid Code especially with Section 2.5.5 of Chapter 2: Connection Code (refer to Section 4.1.11 of Part II of OTS Part A (Technical Specifications) of the RFP) and all NGS related standards.

Oil-immersed power transformers shall comply with the requirements of IEC 60076 and series of standards including IEC 60076-7 and shall be of the outdoor type. The selection of insulating oil shall take into consideration the most up to date protection against sulphur corrosion (according to IEC 62535) and without metal passivator (according to IEC 60666). The transformer oil must be pure virgin of naphthene base petroleum product, without any antioxidant (uninhibited) in accordance with latest edition of IEC 60296. Thermally upgraded paper to be considered for GSU/UAT transformers.

The generator step-up transformers and the isolated phase busducts shall be designed to be able to transmit the maximum output of the corresponding unit at all possible ambient temperatures. Contractor shall provide with its proposal a diagram showing the capacities of the turbine, generator and generator transformers versus all possible ambient temperatures.

The transformer shall be designed to operate without exceeding the temperature limits of IEC 60076-2 under the continuous maximum output conditions of the Power Plant and during all other operational sequences. All transformers shall be of the core type, Cu/Cu winding and shall comply with the latest edition of IEC 60076.

Cooling classification shall be ONAN/ONAF or ONAN/ONAF/OFAF or ONAN/ ONAF / ODAF, except for plant auxiliary distribution transformers as that may be of the ONAN cooling for oil type or AF cooling for

dry type, subject to provisions specified below. For oil immersed transformers, the radiators/coolers may be mounted separately from the tank or mounted on the transformer tank.

All transformers shall be equipped with tap changers that have local manual control as a minimum. The tap range, impedance and losses shall be selected to enable full output under normal conditions which allow for the highest and lowest system voltage operation and shall not be restricted over the specified ambient range.

The motor drive, plus all auxiliary equipment for operation of the tap changer, shall be incorporated in a rigid control cabinet, protection class IP56 and shall be mounted onto the transformer tank in a convenient floor height. The driving gear shall be of the belt-type or equivalent dry-type gear. Oil filled driving gears are not acceptable.

The generator unit main transformers and the unit auxiliary transformers shall be provided with diaphragm type of conservator and online transformer condition monitoring and diagnostic system including dissolved gas analysis (DGA). Online Dissolved Gas (Multi-gas) and Moisture Analyzer along with all required accessories shall be provided with each Unit Step transformer for measurement & analysis of dissolved gases and moisture in the oil. Interpretations shall be as per IEC 60599. These all DGA signals shall be wired to DCS for remote monitoring as well.

Each Generator Unit Step up main transformer shall also be equipped with in line partial discharge monitoring system, whose functioning is based on the simultaneous use of three unconventional methods of PD detection: high frequency (HF), and acoustic emission (AE), vendor proposal can be acceptable during detail engineering stage subject to in compliance with IEC standard. PD system shall be integrated according to the IEC 61850 standard with SCADA (Supervisory Control and Data Acquisition) system to registering voltage, active power, and oil temperature of the monitored unit.

For the oil type transformers, the permissible temperature rises will be according to the IEC-regulations (IEC 60076-2). The top oil temperature rise is 40 K, the average winding temperature rise is 45 K for cooling method ONAF. The winding hot spot will be also selected in accordance with the IEC loading guide (IEC 60076-7). Considering the temperature rises mentioned before the winding hot spot will be 58 K at design ambient temperature of 55°C (outdoor without the shade) + 5°C. The winding and the oil temperature rise shall match this requirement as required by the OTS Part A. For the liquid filled transformers shall be provided with embed temperature sensors in the critical points of the winding to ensure hot spot temperature measurements during the factory tests. For dry type transformers position in line with IEC 60076 for temperature measurement shall apply.

The step-up Transformers and the isolated phase bus ducts shall be designed to be able to transmit the maximum output of the corresponding unit at all possible ambient temperatures. the unit auxiliary Transformers should also be designed for the ambient conditions on site and a sufficient spare capacity of at least 10%.

All oil cooled transformers shall be complete with the following:

- oil conservator, oil level indication/alarm
- oil sampling device
- silica gel breather fitted on main transformer conservators. The silica gel material shall be of a type approved for use in the applicable country; in particular it shall not contain cobalt chloride
- winding temperature/alarm/tripping (local and remote monitoring)
- pressure relief device with alarm
- Buchholz gas relays with tripping/alarm
- Oil temperature indicator with alarm/tripping (local and remote monitoring).
- Surge Arrestors (In excitation and SFC transformer of Gas turbine package, not considered)
- Transformer control cubicle
- All Auxiliary Panels for control, tap changers as required for monitoring signal requirements shall comply with the specific Area classification requirements. To be protected against rain, mechanical

damage and oil spillage (In excitation and SFC transformer of Gas turbine package, tap changers are not considered)

Each transformer shall be fitted with a gas and surge actuated Buchholz relay of approved make and pattern, having alarm and tripping contacts which close simultaneously in response to both liquid surge and low oil level.

Power transformers shall be provided and installed outdoors in the main substation building. These transformers shall be designed in accordance with IEC 60076.

Approved facilities shall be provided for inspection, testing and maintenance access to the gas and oil-actuated relays and conservators associated with the transformers.

Transformers shall be located in suitable open concrete enclosures and including oil/water catchment sumps which discharge to the oily water separator drainage system. The durability of the protective paint system shall comply with class VH specified in ISO 12944-1 / C4 and the covering coat shall be RAL 7035.

All Auxiliary Panels for control, Tap changers / as required for monitoring signal requirements shall comply with the specific Area classification requirements. To be protected against rain, mechanical damage and oil spillage.

All links contained within the transformer tank for earthing and other purposes shall be readily accessible and the links and their fixings shall be of the captive type.

Transformers shall be provided with embed temperature sensors in the critical points of the winding to ensure hot spot temperature measurements during the factory tests.

Thermally upgraded paper shall be considered for GSU/UAT transformers, as applicable.

For cast resin/dry type indoor/outdoor transformers the respective maximum and average room, and enclosure temperature to be considered. The average winding temperature rise of the cast resin/dry type of transformers shall comply with IEC 60076- 11.

The tap changer setting range of the step-up transformer shall be minimum $\pm 12 \times 1.25\%$ or in compliance to Country Grid Code Voltage variation of rated grid voltage.

All transformers shall be equipped with the appropriate class of protection/metering Current Transformers (CTs), as required.

Vector Group of Power Transformers, earthing transformer and impedances shall be according to substation design. OLTC and AVR shall be properly designed to achieve best operation condition.

Approved facilities shall be provided for inspection, testing and maintenance access to the gas and oil-actuated relays and conservators associated with the transformers.

Transformers shall be located in suitable open concrete enclosures and including oil/water catchment sumps which discharge to the oily water separator drainage system. The durability of the protective paint system shall comply with class VH specified in ISO 12944-1 / C4 and the covering coat shall be RAL 7035.

All Auxiliary Panels for control, tap changers / as required for monitoring signal requirements shall comply with the specific Area classification requirements. To be protected against rain, mechanical damage and oil spillage.

All links contained within the transformer tank for earthing and other purposes shall be readily accessible and the links and their fixings shall be of the captive type.

Transformers shall be provided with embed temperature sensors in the critical points of the winding to ensure hot spot temperature measurements during the factory tests.

For cast resin/dry type indoor/Outdoor transformers the respective maximum and average room, and enclosure temperature to be considered. The average winding temperature rise of the cast resin/dry type of transformers shall comply with IEC 60076- 11.

All transformers shall be equipped with the appropriate class of protection/metering Current Transformers (CTs), as required.

Generator Step up Transformer and Power Unit Auxiliary Transformer shall have Automatic Fire Sprinkler System as per relevant NFPA standards and local regulation as may apply.

For all transformers, as a minimum all electrical tests reports will be provided after commissioning phase (and before energization) for Owner review. As a minimum insulation resistance, ratio, polarity, tap changer test results, tan delta and capacitance test of the HV bushings, SFRA, DFRA and oil sample reports. Where possible, RIP bushing should be used but in in case of HV Oil Bushing, tan delta, capacitance and bushing oil DGA at site shall be performed through OEM and report shall be submitted.

9.6.1 Generator Step Up Transformer (With On Load Tap Changer)

Each Generator shall be connected to a generator step-up transformer to connect it up to the 380 kV GIS or AIS substation bay. The Two winding type generator transformers suitably rated to match the output of the prime movers over the entire ambient temperature range of the site and in every mode of operation, shall be provided. They shall also have ONAF/ONAN rating.

The generator step-up transformers shall be of ONAN/ONAF or ONAN/ONAF/OFAF or ONAN/ONAF/ODAF type of cooling. The ONAN rating shall be at least 60% of the generator MVA output rating over the whole range of site temperatures up to the maximum site ambient temperatures. However, ODWF type cooling with 4 x 33% oil/water coolers will be accepted. The generator step-up transformers shall be procured from one of the manufacturers as identified in the Agreed Approved Vendor List. If alternative manufacturers are proposed, then those will be subject to Owner approval.

The transformers shall be designed so that the 380 kV connections can be made by means of cable sealing end fittings with protecting funnels, motor operated isolators switches and earthing switches and with surge arrestors (all SF6 gas insulated).

The transformers shall be located in an outdoor compound adjacent to the generators each having firewalls, fire water spray system, oil collecting pit and a common oil separator. The transformer shall have an on-load tap changer. The tapping range shall be selected to meet the voltage and reactive power control and operating requirements of the transmission system. The transformer LV terminals shall be connected by phase isolated bus bars to the generator. Generator step-up transformers shall have ONAN/ONAF/OFAF cooling.

The single phase fault contribution from each of the generator transformers to the 380 kV system shall be minimised by designing the transformers with the ratio X_0/X_1 not less than 0.95. The X_0/X_1 ratio of 0.95 shall be the absolute minimum and no tolerance below this figure shall be allowed. Such a design should ensure that there are no voltage regulation problems.

Each generator transformer shall be capable of continuously delivering the rated MVA of the generator over the Site ambient temperature range.

Neutral Point of each Generator step-up transformer (at 380 kV side) shall be solidly grounded as per detailed design engineering with compliance to requirement of the Grid Code. The star points are to be fitted with motor operated single-pole earthing switches and lightning arresters in parallel with the earth connections or grounded with 40 Ω impedance to limit the short circuit current.

Generator transformers shall be fitted with On Load Tap Changer with AVR. The numbers of taps and tapping range shall be in accordance to meet the voltage variations at 380 kV (Grid Voltage). The OLTC shall have provision for manual local control, both mechanically and electrically, as well as manual control from a remote control room. Moreover, for automatic voltage control by means of a suitable voltage regulator.

The generator transformer shall be fitted with on-line gas monitoring equipment. For the oil type transformers, the permissible temperature rises will be according to the IEC-regulations (IEC 60076-2). The top oil temperature rise is 40 K, the average winding temperature rise is 45 K for cooling method ONAF. The winding hot spot will be also selected in accordance with the IEC loading guide (IEC 60076-7). Considering the temperature rises mentioned before the winding hot spot will be 58 K at a design ambient temperature of 55°C (outdoor without the shade) + 5°C. The winding and the oil temperature rise shall match this requirement as required by the OTS Part A

Technical Specification of Generator Step up Transformer shall be as under:

Bushings, surge arresters and cable sealing ends utilized for outdoor connections shall have a minimum creepage distance of 31mm/kV based on maximum phase to phase voltage (SCD) The creepage distance value is to be finalized as per the severity of the Site pollution characterization qualitatively defined in IEC 60815-1. The most stringent requirement between these two shall apply. The Transformer shall have on-load tap changer with a tapping range of $\pm 12\%$ in steps, each step not greater than 1.25 % of nominal Voltage. However, this may be revised according to country Grid Code for voltage variation by respective Offtaker / Competent Authority requirements,

Each generator step-up transformer OLTC (On Load Tap Changer) shall be equipped with an automatic voltage regulator (AVR). The AVR shall be active only if the generator circuit breaker is in off position

Transformer shall be designed to deliver maximum power at the worst voltage and power factor conditions and maximum design temperature.

Transformers with forced cooling shall be capable of remaining in operation at full load for 15 minutes after the failure of the fans, without the rise of temperature above the calculated hotspot temperature.

All transformers equipped with radiator units shall be equipped with one unit more as required.

The transformer neutral (star) points on the high voltage windings are to be solidly earthed with the possibility of isolating them when necessary. Therefore the star points are to be fitted with motor operated single-pole earthing switches and lightning arresters in parallel with the earth connections.

The low voltage side is to be equipped with connection flanges and a connection frame to connect an isolated phase generator main connection.

The step-up transformer on-line condition monitoring system shall consist of sub-systems for specific monitoring tasks and shall comprise at least the following main functions:

- Electrical real-time partial discharge (PD) monitoring system in accordance to IEC 60270 for permanent installation on-site;
- Gas-in-oil detection (DGA) by monitoring dissolved hydrocarbon and combustible gases in oil, especially the significant gases for partial discharge activities;
- On-load tap changer (OLTC) monitoring system;
- Temperature supervision system.

The transformer oil must be non-corrosive, non-aging, hydro-treated, pure virgin of naphthane base petroleum product, without any anti oxidant (uninhibited) , free from additives, extremely good heat transfer characteristic and shall have properties such as to avoid formation of copper sulphide under continuous heavy loading conditions in accordance with latest edition of IEC 60296. The liquid shall not contain, nor be contaminated with, polychlorinated biphenyls (PCBs) and shall be of low sulphur corrosiveness and shall be in accordance with latest edition of IEC 60296: 2003. The transformer oil shall pass the corrosion tests according to ASTM D1275 method B, IEC 62535 and DIN 51353.

Each Unit Step up main transformer shall also be equipped with on line partial discharge monitoring system, whose functioning is based on the simultaneous use of three unconventional methods of PD detection: high-frequency (HF), ultra-high frequency (UHF), and acoustic emission (AE). PD system shall be integrated according to the IEC 61850 standard with SCADA (Supervisory Control and Data Acquisition) system registering voltage, active power and oil temperature of the monitored unit.

Generator step-up transformer impedance voltage to be minimum 15%, however shall be adjusted based on the fault level contribution limitation.

Bushings, surge arresters and cable sealing ends utilized for outdoor connections shall have a minimum creepage distance as per IEC 61643 for low voltage, and IEC 60815 & IEC 60601-1 ("Guide for selection of insulators with respect to polluted conditions") for high voltage equipment. Self-cleaning polymer bushings will be accepted with creepage distance of 42 mm/kV.

9.6.2 Power Unit Auxiliary Transformers

Unit Auxiliary transformer (UAT) shall be provided which will receive power from the tap off connection of the isolated phase bus duct connection between the generator and the generator transformer after the

generator circuit breaker. UATs shall be designed for Plant's maximum auxiliary power demand (in 3x50% configuration). The UAT will step down the voltage to the MV voltage for unit and station Auxiliary loads. The UAT shall be a 2 or 3 winding oil filled transformer with LV windings which will get connected to the MV bus to serve unit and station loads. The transformers shall be adequately rated with suitable design margins to provide the auxiliary power supply systems and to be provided with off-load tap-changers as well as with winding temperature indication and alarm devices and gas and oil actuated relays. The transformers shall comply with the requirements of IEC 60076 for temperature rise.

The Contractor shall suitably size these transformers taking into account power requirements of the Plant.

All transformers to be equipped with the appropriate class of protection/metering CTs, as required.

Oil-immersed unit auxiliary transformers shall have ONAN/ONAF cooling;

All off load / off circuit Tap changers must have interlock arrangement so to ensure its operation is only possible in de energised condition. It must be provided with lock and labelled clearly for Safety Signage "DO NOT Operate in Energised condition".

9.6.3 Auxiliary Distribution Transformers

The numbers and ratings of the various auxiliary transformers required around the Plant are to be determined by the Contractor based on load estimates for each service or building.

The LV auxiliary transformers shall be of two winding, outdoor oil filled type with ONAN cooling or cast resin dry type with AF cooling suitable for indoor location. Transformers with ratings less than 1 MVA may preferably be dry type.

Dry type transformers shall comply with the requirements of IEC 60076-11 and be mounted in protected enclosures installed indoor with minimum mechanical protection IP23. Heat dissipation must comply with requirements of IEC 60076-11 (Para 4.2.3). Design of enclosure to guarantee to confirm and design to ensure that adequate supply air (clean, cool and dry) shall be provided to dissipate total heat as per design calculation of transformer.

Climate, environment and fire class shall be E2, C2 and F1 as per IEC 60076. Insulation class shall be F or H.

The no load voltage ratio of the auxiliary transformers shall be stated in the Contractor's proposal.

MV/LV Auxiliary power transformers shall be provided with off-load tap changer with minimum $\pm 2 \times 2.5$ % steps or as per Grid Compliance study results. Generally, the transformers will be fed in MV via cables whereas the LV side will be connected to the MCCs through busbars.

The current transformers in all transformers, shall be wired to short-circuiting, screw type terminal blocks mounted in the transformer control panel.

9.6.4 Other Transformers

Dry type transformers shall be of cast-resin type designed for class F insulation and shall be operated with class B temperature rise. Temperature supervision for each winding shall be provided.

Cast resin power transformers shall be located in transformer housings with a degree of protection of min. IP23 and shall be located indoors in air-conditioned rooms.

The numbers and ratings of the various auxiliary transformers required around the Plant are to be determined by the Contractor based on load estimates for each service or building.

The LV auxiliary transformers shall be of two winding, outdoor oil filled type with ONAN cooling or cast resin dry type with AF cooling suitable for indoor location or AN cooling suitable for outdoor installation. Transformers with ratings less than 1 MVA may preferably be dry type.

Dry type transformers shall comply with the requirements of IEC 60076-11 and be mounted in protected enclosures installed indoor with minimum mechanical protection IP23 or outdoor with protection IP54. Heat dissipation must comply with requirements of IEC 60076-11 (Para 4.2.3). Design of enclosure to guarantee to confirm and design to ensure that adequate supply air (clean, cool and dry) shall be provided to dissipate total heat as per design calculation of transformer.

Climate, environment and fire class shall be E2, C2 and F1 as per IEC 60076. Insulation class shall be F or H.

The no load voltage ratio of the auxiliary transformers shall be stated in the Contractor's proposal.

9.7 Switchgear

Design temperature of electrical equipment installed indoor in non-air conditioned rooms shall be based on 50°C and Design temperature of electrical equipment installed indoor in air conditioned rooms with HVAC n-1 redundancy shall be based on +5°C to 40°C and the relative humidity 10% to 100%.

MV switchgear bus of each GT shall each fed from UATs of the respective GTGs. And each section shall be connected to the MV switchgear of next GT with High-speed automatic transfer switch in order to achieve the full fed redundancy of each section to avoid a single point of failure concept.

The EPC has to provide type test certificates according IEC62271-200 for all switchgear in his scope of supply. Type test certificates shall also prove that the switchgear supplied is internal arc classified IAC = AFLR (arc duration 1 sec) according to IEC 62271 Part 200.

The switchgear shall be of metal-clad air insulated compartmented type tested design for indoor installation and shall comply with IEC 62271-200, IEC 62271-1.

The MV switchgear shall withstand a short-circuit current of minimum 10% higher than required by calculations for a duration of 3 seconds.

As basis for short-circuit calculation IEC60909-0 is suitable.

The cubicles are to be provided with single copper busbar system and draw-out type circuit breaker.

All electrical system switchboards shall be labelled for Hazard Risk category (HRC) as per results of ARC Flash assessment study using NFPA 70 E

The MV and LV electrical system will be fitted with the necessary electrical and mechanical safety interlocking devices. Electrical interlocking will be in place to ensure that all closing and tripping functions (incl. in between, upstream, downstream systems) take place in a particular sequence (including busbar earthing switches). The interlocking scheme will be presented to the Owner for approval. The switchgear apparatus concerned by the interlocks will be equipped during the manufacturing with the appropriate keys and locking devices (type Castle Key or equivalent)

Arc flash study shall be conducted in accordance with the relevant categories are defined as per NFPA 70E (Test shall not be repeated if Certificate shall be provided by OEM). MV Switchgear comply with IEC 62271-200 and LV switchgears shall comply with IEC 61641 for the internal arc flash resistance and testing. A label with the arc flash level and the PPE required to operate the equipment shall be provided in all MV cabinets/circuits. The label shall be of weather/UV resistant and posted in front of the equipment.)

The current transformers in all transformers, shall be wired to short-circuiting, screw type terminal blocks mounted in the transformer control panel.

Earth fault supervision/monitoring must be provided for all medium voltage and low voltage networks.

Panels to be considered with inspection windows so to facilitate IR Inspection

Cable termination boots to be provided at cable chamber of Form 3b type switchgear assemblies.

HSBT systems to be added for Unit and common MV SWGRs

Block interconnection: In order to avoid import power from Grid for auxiliaries for stand by units / block, there should be a provision to interconnect and receive power from operational units / block.

Minimum spare breakers and panel to be provided at MV/LV switchgears with fully refurbished items.

Supports of busbars shall be of porcelain or of cast resin material. All bus joints shall have bolts with vibration proof lock washers.

For control, signalling and measuring systems two (2) DC external power supplies shall be provided from separate sources, the OEM will provide in the switchgear the required diodes and independent MCB's for each auxiliary supply circuit.

All incoming and outgoing feeders including motor feeders with vacuum circuit breakers with motor-driven stored energy mechanism. The circuit breaker shall have a defined test position. In this position, the main inserts of the breaker shall be isolated from the busbar and the cables and the shutters shall be closed.

The bare copper conductors of the busbars shall be provided with temperature indicators.

Any internal fault shall be cleared quickly by detectors sensitive to light (internal arc), pressure, heat or by the differential bus bar protection system. LV switchgears are not required differential bus bar protection and arc detection.

For earthing and short-circuiting the busbars, fault making earthing switches are to be provided on each section. The earthing switch shall be interlocked with the circuit breakers of all feeders connected to the relevant switchgear section.

The individual cubicles are to be bulkheaded from one another by double sheet metal partitions or approved equivalent system. The busbar compartments must be bulkheaded from panel to panel by arc barriers. Within the cubicles, the compartments for circuit breakers, busbars, instrument transformers and cable connections as well as the relay recesses must be bulkheaded from one another.

Sufficient pressure relief must be provided in the case of fault arcs. For each bulkheaded compartment a separate pressure relief must be provided. Care must be taken that in the event of arcing no hot gases escape to the front of the cubicle (operational side)

A high-speed transfer system from one incoming feeder to the other incoming feeder shall be provided for MV-switchgears which are required for power auxiliary supply of the plant which when failure could cause an outage of a total Plant Unit and/or the total Plant

High-speed transfer systems shall be provided between the power sources with the required interlocking to avoid overloading of the related unit auxiliary transformer. Each of the systems shall be provided with facilities for selection of the changeover sequence/priorities.

The high-speed transfer system shall be started as follows:

- manually (local or remote through the DCS)
- automatic by protection relays

Each high-speed transfer system shall comprise following:

Logical analyser for all required control, interlocking and monitoring functions, Phase comparison device for continuously monitoring the phase angle, frequency and voltage, Test facilities for testing the complete transfer system up to the closing and trip coil of the circuit breakers whilst the system is ready for operation.

All main switchboards, MCC, relay panels, control equipment etc. shall be provided with duplicate power supplies for control and alarm purposes.

The LV (30 or 400 V) switchgears will have two sections with a bus coupler. Each of the bus section will be fed power supply from respective sections of the 13.8 kV Block switchgear through a 13.8 kV /380 or 400 V. The bus coupler air circuit Breaker (ACB) will be interlocked with the incoming ACBs of the two sections.

380 or 400 V normal cum emergency auxiliary bus shall supply normal auxiliaries and essential auxiliaries of the Plant. This bus shall also receive power supply from the emergency supply bus. The essential bus provides power supply to chargers of the battery system, 380 or 400 V/220 or 230 V UPS systems, AC lube oil pumps, lighting system, etc.

The 380 or 400 V switchgear system main incomers, interconnectors and bus-section circuits shall exclude check-synchronising facilities but shall be electrically interlocked to prevent the paralleling of two incoming supplies to a switchboard.

Each switchboard shall have the degrees of protection IP40 specified for the enclosures and partition. The switch room floor shall not form part of the enclosures.

Circuit breakers supplied shall be identical to those used for the type tests for which certificates are submitted.

MV switchgear and all main LV switchgear shall be of withdrawable type. Fault levels and thermal withstand time of these switchboards shall be in accordance with the technical requirements of the appropriate standards.

The Contractor shall perform a short-circuit calculation, in accordance with standards IEC 60909 and IEC 60865, in order to define the fault levels at all the different points of the Plant electrical network.

A two position (Local/Remote i.e. Switchgear/CCR) selector switch shall be provided at the switchgear for all switchgear incomer, bus-section, feeder and interconnector circuits respectively for switchgear emergency control and isolated testing/CCR remote control.

The Plant shall be provided with earthing and lightning protection system, illumination system. The Plant illumination system shall include emergency lighting system.

The Plant cabling system and cable support system shall follow Good Practice, local regulations. A fault monitoring and disturbance recorder system shall be provided to cover the entire electrical system including incoming / outgoing feeders and Bus tie / coupling breakers.

The MV and LV electrical system will be fitted with the necessary electrical and mechanical safety interlocking devices. Electrical interlocking will be in place to ensure that all closing and tripping functions (incl. in between, upstream, downstream systems) take place in a particular sequence (including busbar earthing switches). The interlocking scheme will be presented to the Owner for approval. The switchgear apparatus concerned by the interlocks will be equipped during the manufacturing with the appropriate keys and locking devices (type Castle Key).

All Switchgears shall have a design lifetime of at least 25 years.

Cable termination boots to be provided at cable chamber of Form 3b type switchgear assemblies.

HSBT systems to be added for Unit and common MV SWGRs.

Block interconnection shall be provided in order to avoid import power from Grid for auxiliaries for standby units, there should be a provision to interconnect and receive power from operational units.

Minimum spare breakers and panel to be provided at MV/LV switchgears with fully refurbished items.

Supports of busbars shall be of porcelain or of cast resin material. All bus joints shall have bolts with vibration proof lock washers.

For control, signalling and measuring systems two (2) DC external power supplies shall be provided from separate sources, the OEM will provide in the switchgear the required diodes and independent MCB's for each auxiliary supply circuit.

For Gas turbine package, Generator grounding cubicles and SFC start-up switching cubicle are acceptable as OEM standard design.

9.7.1.1 MV Switchgear

The MV switchgear shall be of metal-clad air insulated compartmented type tested design for indoor installation and shall comply with IEC 62271-200, IEC 62271-1.

As basis for short-circuit calculation IEC60909-0 to be considered.

The MV switchgear shall withstand a short-circuit current of minimum 40 kA or subject to the confirmation of electrical studies calculations for a duration of 3 seconds..

The cubicles are to be provided with single copper busbar system and draw-out type circuit breaker. EPC to consider the 20 % spare feeders in each bus section shall be provided.

Withdraw able three pole circuit breakers, using SF6 interrupters or vacuum interrupters, shall be used, unless otherwise approved.

The cubicles are to be provided with single copper busbar system and draw-out type circuit breaker.

All incoming and outgoing feeders including motor feeders with vacuum or SF6 circuit breakers with motor-driven stored energy mechanism. The circuit breaker shall have a defined test position. In this

position, the main inserts of the breaker shall be isolated from the busbar and the cables and the shutters shall be closed.

The bare copper conductors of the busbars shall be provided with temperature indicators.

The MV switchgears shall be designed for a nominal voltage, type tested and internal arc certified IAC=AFRL (arc duration 1 se c) according to IEC 62271-200.

The switchgear shall be of metal-clad air insulated compartmented type tested design for indoor installation and shall comply with IEC 62271-200, IEC 62271-1.

The MV switchgear shall withstand a short-circuit current of minimum 10% higher than required by calculations for a duration of 3 seconds.

The cubicles are to be provided with single copper busbar system and draw-out type circuit breaker. For earthing and short-circuiting the busbars, fault making earthing switches are to be provided on each section. The earthing switch shall be interlocked with the circuit breakers of all feeders connected to the relevant switchgear section.

The individual cubicles are to be bulk headed from one another by double sheet metal partitions or approved equivalent system. The busbar compartments must be bulk headed from panel to panel by arc barriers. Within the cubicles, the compartments for circuit breakers, busbars, instrument transformers and cable connections as well as the relay recesses must be bulk headed from one another.

Sufficient pressure relief must be provided in the case of fault arcs. For each bulk headed compartment a separate pressure relief must be provided. Care must be taken that in the event of arcing no hot gases escape to the front of the cubicle (operational side).

Where operation of a switching device can give rise to surge voltages, which would be detrimental to the insulation of the equipment controlled, surge suppressers shall be installed in the switchgear to control these voltages to a safe level. Details of the maximum surge voltages which can arise shall be provided by the Contractor and shall state the criteria on which the provision of surge suppressers is based.

Feeder protection shall be by a digital multifunction relay. The MV switchgear shall be designed for remote control via the DCMS.

Low voltage devices and protection equipment shall be mounted in the door or on a mounting plate inside the low voltage compartment of the switchgear panel.

Switchgear design with two level arrangements of circuit breakers shall be available.

The earthing switch shall be interlocked (Electrically and mechanical with castle key or suitable arrangements).

Interlocks shall be provided to prevent the door opening (including the back panel), when the circuit breaker / contactor is in service position. With the door closed, the indication of position and the status should be Indicated.

Status of the earth switch and Switchgear shall be available on SCADA to ensure safe remote operation.

Medium Voltage Circuit Breakers will have a trip circuit supervision. This shall be provided to continuously supervise the integrity of the circuit breaker tripping circuits and give an alarm in the event of the following fault conditions:

- Loss of dc tripping supplies, e.g. MCB tripped;
- The alarm shall be time-delayed preventing it operating during momentary dips in the DC supply.

MV switchgear shall be provided with circuit labels on the front and rear of the panels. All enclosures and apparatus shall be clearly labelled, on both the front and back, indicating and at a minimum the full equipment identification number for the device being controlled as per equipment coding system of project. The labels shall be corrosion resistant material, fade resistant and securely fastened.

9.7.1.2 LV Switchgear

Low voltage switchgear shall comprise IEC 61439 and IEC 60947 series type tested switchgear and control gear assemblies, factory assembled with all the internal mechanical connections and components fitted.

The design shall be according to IEC 61439 – 1 form 4a for incoming feeders, form 3b for outgoing feeders and shall have a short circuit rating of 50 kA for 1 second or according to requirements of short circuit level defined through plant short circuit studies (whichever is higher). Busbars shall be air insulated.

Incoming circuits and tie feeder circuits shall be equipped with withdrawable air circuit breakers. The LV-incomers and feeders will be protected by the internal protection of the MCB's, MCCB's and circuit breakers. Outgoing feeders shall use MCCBs and be installed on modular drawers assembled to form a combined power distribution board and motor control centre.

Motors will be protected by comprehensive digital protection relays or MCCBs with special motor protection functions, subject to design requirements.

Incomers and bus-sectionalisers shall be equipped for remote control via the Facility DCMS. Other feeders shall be suitable for remote control operation as required.

LV distribution boards not controlled via the DCMS shall be equipped with indication lamps on the front door to assist personnel in the safe operation of the equipment and plant.

Switchgear design with two level arrangements of circuit breakers is available.

Motor control centre is available to be erected back-to-back.

There are no local control cabinets for all motors operated by motor control station (MV/LV) or process operation station in the central control room.

9.7.1.3 VARIABLE FREQUENCY DRIVES

Medium Voltage & Low Voltage Variable Speed Drives shall be provided wherever the process demands variable speed.

VFDs shall comply with IEC 61800-2, IEC 60146-1-3 and other relevant parts.

The converter shall comply with IEC 61800-3 with respect to EMC.

The converter equipment shall generally comply with IEC 60146 series of standards where appropriate.

Harmonic distortion shall not exceed specified limits as per IEC 61000/IEEE standard 519. All VSDs shall be located indoor in the Electrical buildings. For Medium voltage VSD 18-24 pulse type shall be used, whereas for Low voltage VSD active front end type shall be preferable.

All VFD's shall be suitable for control and indication through a serial link to the DCS process control system.

VFD for GT auxiliaries will have hardwired connection to GT control system without serial link. The converter unit shall be stand alone for medium voltage drives. They should be self-contained in a sheet steel enclosure with IP31 degree of protection, with inlet and outlet filters where necessary, and shall be located indoors. The cubicle and equipment therein shall comply with the appropriate clauses of this Specification.

VFD design shall be reviewed and get confirmation from OEM that how installed design is protected against high voltage peak stresses (dv/dt) associated with High frequency pulse width modulation (PWM). It is also one of the factors which deteriorate cable insulation.

VFD shall be housed inside MCC for low voltage drives.

9.8 Main Connections and Generator Circuit Breakers (GCB)

The main connections between the generator, the GCB and generator transformer, as appropriate, together with the associated tee-offs to the unit transformers shall be made using isolated phase busduct (IPB). Busducts shall be complete with short circuit testing facilities and shall be pressurized with their own clean and dry air which may be supplied via a dedicated instrument air line from the instrument air receiver tank.

The busducts shall be of the self-supporting rigid type and shall be suitably rated up to the upper ambient temperature limit specified in this OTS and shaded (via sunshade or enclosure) against the direct rays of the sun. The conductors shall be of high conductivity aluminium and all joints shall generally be fully welded. Some joints of the conductors may be fitted with flexible / expansion conductors and connectors to prevent stress from affecting the conductors and the thermal insulation of equipment. The busducts shall

have continuous and short-circuit ratings appropriate to the connected equipment for all site ambient temperatures in the design range.

As required, aluminium terminal enclosures shall be provided for the connected equipment and these shall incorporate expansion compensators, which electrically insulate the busbar enclosure from the equipment and provide a flexible connection to allow for thermal expansion and vibration.

As appropriate, conductors shall be connected via flexible joints. In the event of a unit auxiliary transformer failure requiring prolonged outage, it shall be possible to disconnect the UAT from the main busduct run to allow generator unit operation to continue.

The busduct shall be complete with sealed and fully insulated wall plates, fixings, supports and steelwork. Enclosures shall be pressurised from the instrument air system to prevent the ingress of dust and dampness. Earth bars and flexible earthing connectors shall be provided between individual phases and equipment as appropriate.

Temperature Limits rise for Conductor and enclosure of Isolated Phase Bus duct shall be as per ANSI/IEEE C37.23.

Generator circuit breakers shall be provided for all gas turbine generator units. Generator circuit breakers shall be designed manufactured and type tested for generator switching duty in accordance with the requirements of IEEE/ANSI C37.013. The inherent design of these circuit breakers shall be such that they produce over voltages (less than 2.2 per unit) on switching conditions including inductive and capacitive switching to IEC62271-110.

Depending on the Contractors' proposed arrangement, the circuit breaker shall be located within the isolated phase busduct between the generator and generator transformer and the installation shall include an earthing switch on the generator side and voltage transformers.

Generator circuit breaker needs to provide ease of testing during O&M phase for Electrical testing with ease of doing and removal of main enclosure / major fitments.

9.9 System Earthing/Earthing Grid

System earthing shall be as follows:

- Generator transformer HV winding :with 40 Ω impedance
- Generator neutral :High impedance (resistance loaded VT)
- MV systems :Low impedance (current limiting to 600A)
- LV systems :Direct

The LV network shall be either of TN-S (separate PE and N conductors) or of the TN-C-S type with combined PE&N conductor (TN-C) from transformer to the main distribution, but separate PE and N conductors in the distribution and all connected consumers (TN-S).

DC and UPS system earthing shall be supplied as appropriate to the proposed equipment.

All non-current carrying metallic parts of plant, equipment, components, enclosures, protective screens, etc, shall be provided with earthing facilities for personnel safety reasons and for obtaining the maximum possible rejection of electrical interference.

Earthing should be generally designed and installed in accordance with IEC, IEEE 80 unless otherwise approved.

Protective screens around insulated cables shall be insulated from field instruments, connected through any field junction boxes, and be continuous up to the earthing point in the destination cubicle or panel.

Cable armouring shall not be used as a protective screen. Earthing of cable armour shall be arranged for electrical safety purposes and not for noise reduction. Cable armouring shall be insulated from any cable protective screening.

Where the Contractor's design requires a separate and dedicated earthing arrangement for example, a "clean" earth for electronic equipment then the Contractor shall provide such earthing facilities as required. All instrument clean earth bars shall be clearly labelled as such.

Where the system design requires a single connection from the C&I equipment to the main Plant earth bus, then a removable link shall be provided to connect the equipment earth to the main Plant earth bus for testing purposes. This removable link shall be clearly labelled with a warning indication.

Where a cubicle, frame or enclosure is earthed via the above facilities then it should be insulated from structural steel works or any other plant earth.

9.10 DC/UPS Systems

Redundant (2 x 100%) direct Current (DC) and redundant (2 x 100%) with Nickle Cadmium batteries rated for standby period of at least 1 hour at 100 % continuous load of its associated distribution board (2x100%). (For Gas turbine package, vented lead acid type with 1 hour is considered as OEM standard). UPS and DC to share the common battery bank is acceptable to provide complete static and dynamic load.

Each charger shall be rated to supply 100% of the normally running DC system load and the required float charge of the connected batteries. The Contractor shall also provide five (5) additional battery cells per DC/UPS system as part of its spare parts provision. (Except Gas turbine package).

DC systems shall be provided for instrumentation purposes and to power motor drives of switchgear and other consumers, as required for the safe shut down of the Plant.

UPS inverter systems supplied from LV AC or DC systems shall be provided for loads which cannot tolerate an interruption in supplies. These supplies or the loads shall be fully redundant.

Switchgear for DC and essential AC supplies may be of the fixed pattern.

The DC and UPS system for GTG will comply with GTG OEM's standard. If the OEM standards are based on approved international standards and codes than they shall apply.

DC/UPS system should feed to 2 x 100 % distribution board.

Chargers should be available with auto restart facility at restoration of main AC power supply.

Control DC power supply for plant critical circuits like protection relays, Control DC supplies of critical switchboards shall be provided with redundant dc sources fed from 02 different DC Banks.

Online boost charging provisions to be added for charger along with dropper circuit, required blocking diodes to be added to prevent circulating currents between the banks.

9.11 LV & MV Motors

All motors shall be designed and manufactured in accordance with IEC 60034 and other related standards. Motors in hazardous areas shall be suitably certified.

All motors with rated power of 200 kW and above must be designed as HV motors, connected to the MV switchgear and shall be fed via a circuit breaker.

The rating of each motor shall be adequate to meet the requirements of its associated load which is designed for the whole lifetime of the plant, without infringing upon the service factor. The service factor is defined as the ratio of the installed motor output to the required power at the shaft of the driven machine at its expected maximum power demand including specified 10% excess capacity pump. Service factor employed shall be 1.1.

Electric motors shall be designed for class F insulation. During operation at rated power of the driven machine, the motor insulation must only be stressed in accordance with the requirements of class B insulation.

Anti-condensation heating shall be switched into operation when the motor feed circuit is in the 'off' position. Heater supply power 'on' indicating lamp shall be provided. Condensation outlets shall be provided at the lowest point in the housing.

Motor starting current must not exceed 5.5 times rated current for MV Motor DOL and 7 times for LV Motors DOL of rated current (referred to motor rating at 40°C ambient temperature)

The voltage drop at direct-on-line (DOL) starting must not exceed 10% at the motor terminals for MV and 15% at the motor terminal for LV.

Air cooling for the motors is the preferred method; where water-cooling is applied, conditioned, treated water is to be used. Cooler construction shall be per cooling water used. Material for cooler, cooler tubes, etc shall be as per water used.

The running quality must be within the classification of "Class III, Zone A" according to the ISO 20816 or equivalent, i.e. the vibration velocity must be not greater than 1.8 mm/s (rms)

Design and construction of the motors shall comply with IEC 60034-1 and 60034-5.

Protective features

- High voltage motors must be fitted with measuring points for the determination of slot and bearing temperatures
- The winding temperature has to be detected with RTD's at high temperature and alarm to be produced, at extra high temperature the motor to be tripped (In Gas turbine package, RTD is not applied to the LV motor (<100 kW) of the generator's auxiliary systems as per Supplier's standard design.)
- For motors ≥ 1000 kW winding and bearing temperature readings shall be provided in the DCMS
- Motors above 1000 kW shall be additionally protected by a high impedance differential protection relay

Motors of the same type must be fully interchangeable and shall comply as far as applicable with IEC standard motor dimensions. AC motors shall be of the squirrel cage induction type provided with either deep slots or double squirrel cages.

Motors shall have Class F insulation but the temperature rise shall not exceed Class B. However, for some fractional horsepower motors, such as air-conditioning motors, it may be Class E or, as per manufacturer's standard.

DC motors and starters shall be designed for high reliability with direct-on-line starting using a permanently connected series resistance in the armature.

Emergency stop buttons of the stay-put key-reset red button mushroom head type shall be provided adjacent to all MV motors and all lubricating oil pumps (except for gas turbine package). These buttons shall directly trip the motors, prevent a start until reset and alarm their operation to the Central Control Room. The buttons shall all use a common reset key.

For motors rated above 55kW, anti-condensation heating shall be switched into operation when the motor feed circuit is in the 'off' position. Heater supply power 'on' indicating lamp shall be provided. Condensation outlets shall be provided at the lowest point in the housing.

All motors shall normally be of the totally enclosed fan-cooled type or water-cooled type, of protection class IP55 according to IEC recommendations 60034 and 60529. Where motors are installed outdoors a weatherproof design IP55W shall be chosen according to IEC recommendation 60034 and 60529. Motorised valves and submersible pump motors shall be to protection class IP65.

LV-motors with ratings of 55kW and above shall be equipped with anti-condensing heater.

All motors shall be designed as 3-phase squirrel-cage motors suitable for direct online starting

The motors must be capable of being switched on to a large and stable network and with phases in opposition.

LV-motors with ratings of 55kW and above shall be protected against overcurrent conditions by PTC-thermistors.

Motor rating to be at least 110% of the maximum required electric power consumption of the driven machine.

Continuous delivery of rated power at voltages of 90 - 110% of the rated voltage at rated frequency within the limits of thermal class F

Rated power output at ambient temperatures existing at the particular site of installation and under continuous running conditions.

For motors rated above 11 kW, anti-condensation heating shall be switched into operation when the motor feed circuit is in the 'off' position. Heater supply power 'on' indicating lamp shall be provided. Condensation outlets shall be provided at the lowest point in the housing.

The service voltages of the motors will be:

- 13.8 or 11 or 6.6 kV, 3-phase AC, 60 Hz, for motors above 200 kW (On Line Starting).
- 480/277, or 400/230 V, 3-phase AC, 60 Hz, for motors up to 300 kW (VFD Starting).
- 480/277, or 400/230 V 3-phase AC, 60 Hz, for motors 200 kW and below (On Line Starting) .
- Calculations of the voltage drop at start-up shall be made for the maximum capacity motor to decide if the motor can be directly fed from a 400 V switchboard or through a starting device.
- Motors above 1,000 kW shall be additionally protected by a high impedance differential protection relay.

The rating of each motor shall be adequate to meet the requirements of its associated load, without infringing upon the service factor as per relevant IEC or NEMA and/or ANSI standards.

Except for low-speed motors (below 1000 rpm) three-phase AC motors of 500 kW and above shall have power factors not below 0.9 at rated power. Motors of a power lower than 500 kW shall be designed and manufactured in accordance with IEC60034.

Motors shall be capable of operating continuously under rated output conditions at any voltage deviation between 90% and 110% of the nominal voltage however their performances may be altered during such events. A transient overvoltage of 130% of the nominal voltage shall also be sustainable. Further, the motors shall be capable of delivering the maximum torque as required by the driven machine when running at 70% nominal voltage for a period of 10 seconds without damaging, overheating or slipping.

All motor driven auxiliaries shall be designed for 20 percent over speed for two minutes.

The pull-out torque for continuously loaded motors shall be at least 160% of the rated motor torque and for intermittently loaded motors 200% of the rated motor torque.

In case of Motor design and construction as per IEC standards with Voltage and frequency characteristics of motors are as per IEC 60034-1, Motor will start with allowable voltage fluctuation range, max 10 % drop as per IEC 60034-1.

Fixed speed AC motors shall be designed for direct on-line starting.

With voltage and frequency within permissible variation band, all motors shall be capable of two starts from cold in quick succession with a third start five (5) minutes thereafter all with full load. The motors shall have a hot restart capability of one immediate restart followed by two (2) starts at fifteen minute intervals, all fully loaded. Apart from this, the motors shall also be capable of three equally spread starts per hour. At the end of the three (3) starts the temperature of the insulated windings shall not exceed the maximum permitted temperature for class B insulation, and the cage windings shall not suffer damaging temperature rise.

In addition, each 13.8 or 11 or 6.6 kV motor shall be capable of enduring two (2) successive starts with the motor initially at operating temperature and the temperature of the winding at the end of start shall not exceed the Class B limit. Each 400 V motor shall be capable of withstanding two successive starts under the same conditions or once every twenty minutes without damaging temperature rise.

Vertical motors shall have combined thrust and guide bearings. Vertical motors above 37kW and all other motors above 200kW shall have roller type bearings.

All motors utilizing maintenance-free bearings must be clearly and permanently indicated as full. All other bearings must be provided with a lubricator that can be used while the machine is running. Over-lubrication must be avoided by means of a lubricant controlling device.

Motors of the same type must be fully interchangeable and shall comply as far as applicable with IEC standard motor dimensions. AC motors shall be of the squirrel cage induction type provided with either deep slots or double squirrel cages.

All motors shall normally be of the totally enclosed fan-cooled type or water-cooled type, of protection class IP54/IP54w according to IEC recommendations 60034 and 60529. Where motors are installed outdoors a weatherproof design (minimum IP54w / 55) shall be chosen according to IEC recommendation 60034 and 60529. Motorised valves and submersible pump motors shall be to protection class IP65.

Motors shall have Class F insulation but the temperature rise shall not exceed Class B. However, for some fractional horsepower motors, such as air-conditioning motors, it may be Class E or, as per manufacturer's standard.

DC motors and starters shall be designed for high reliability with direct-on-line starting using a permanently connected series resistance in the armature.

Emergency stop buttons of the stay-put key-reset red button mushroom head type shall be provided adjacent to all MV motors and all lubricating oil pumps (except for gas turbine package). These buttons shall directly trip the motors, prevent a start until reset and alarm their operation to the Central Control Room. The buttons shall all use a common reset key.

For motors rated above 55kW, anti-condensation heating shall be switched into operation when the motor feed circuit is in the 'off' position. Heater supply power 'on' indicating lamp shall be provided. Condensation outlets shall be provided at the lowest point in the housing.

9.12 Cabling and Cable Installation

The Contractor shall undertake all relevant design and engineering of the cable systems for equipment in its scope of supply, and shall prepare the cable connection schematic diagrams, terminal diagrams, cable lists, cable schedule and the cable route plans. All cables shall comply with the relevant IEC standards.

Power cable, Control, Instrumentation cables and thermocouple extension wire) for GT package including GT enclosure, fuel gas unit, around gas turbine exhaust duct, and so on. "Non-armoured type ETFE (Ethylene Tetrafluoroethylene Copolymer) insulated and sheathed" cables complies with the IEC 60332-3 Cat.C for suitable high temperature condition in GT package.

All power cables shall be of cross-linked polyethylene (XLPE) insulated type and shall be capable of continuous operation at the highest system voltage specified with a maximum conductor temperature of 90°C, under rated conditions and a maximum temperature under fault conditions of not more than 250°C.

For Gas turbine package, armoured type ETFE (Ethylene Tetrafluoroethylene Copolymer) insulated and ETFE sheathed cables (lighting power cables, control cables, instrumentation cables and thermocouple extension cables) for inside GT package including GT enclosure, Fuel gas unit, Fuel oil unit and exhaust gas duct area will be considered.

Low smoke: IEC 61034 and Zero Halogen: IEC 60754-1/2 cannot follow due to the special cable insulation and sheath required to fit environmental conditions.

All power, control and instrumentation cables shall have an over sheath of the flame retardant or reduced fire propagation type, or of the low smoke, zero halogen type complying with relevant IEC standards.

Cables for analogue signals shall have a common screen of metal tape, cores shall be twisted by pairs, for process computers, each pair shall be separately screened.

Every cable end of power cables shall be sealed and protected in an approved manner consistent with the voltage levels, cable type, and place of installation.

"Plug-in" cable terminations (if supplied) shall be in accordance with the characteristics of the switchgear.

Special care to be given to critical equipment like lubrication of the turbine and power for hydrogen seal-oil pumps for power and control cables. Like,

- a. Route the cables to minimize exposure to potential oil fires
- b. Locate control and instrumentation cables for individual units so as to minimize the chance a fire will result in an extended outage to multiple units
- c. Provide separate routing of cables for primary (AC) and backup (DC) lube-oil pumps

- d. Cable for operation of lube oil pumps should be protected from the fire exposures as per the NFPA850. Protection can consist of separation of cable for AC and DC oil pumps and minimum for 1-hour fire resistive coating and aligned with local standards.

Fire barriers separating fire areas should be a minimal of 2-hour fire resistant rating as per guidelines of NFPA 850 for separation distance.

MV, LV , DC, and control cables shall be to be laid on cable racks or trays within plant areas and in cable trenches on racks/trays or buried cables ducts (outside buildings). Any cables direct buried shall be armoured or mechanically protected. No cables shall be installed in tunnels.

Cables laid on trays shall be of adequate strength and size to carry in one layer (for power cables) or in several layers (I&C cables and small power cables), the weight, number and configuration required for each tray plus additional cables, not less than 25% by weight and dimensions, in excess of those installed. Cable tunnels shall not be used.

In case of buried cables or to be laid on trays / conduits installed in cable ducts / trenches / tunnels, anti-rodent and anti-termite protection shall be provided. Armour cables or cables installed in conduits do not require anti-termite protection.

Segregation of the various services and voltage levels shall be achieved by use of separate trays and marshalling boxes for each voltage grade of cable used. The highest voltage cables shall be laid on the bottom rack with lower voltage cables next. Control and protection cables shall be laid on the highest levels.

Cable termination boots to be provided at cable chamber of Form 3b type switchgear assemblies.

For the data highway communication link, spatially separated routing with the cables running in galvanised steel conduits of approved type shall be supplied and installed.

Communication Cable (FO) shall be routed on covered cable trays, protective conduits or other protective cable routing materials or galvanised steel conduits under concrete area or crossing roads.

If cables of separate units are included in the same cable route, they shall be segregated by the use of barriers.

Cables shall be suitably derated as appropriate to the requirements of the installation. Unnecessary cable joints shall be avoided. Cables shall be permanently identified with steel or plastic cable markers. Cables laid direct in ground shall be armoured.

Multicore cable cores shall be terminated using crimps or other suitable termination. Bare cores will not be accepted. Cores shall be provided with some permanent means of indelible identification. This may include colour coding or identification ferrules.

Fire barriers shall be installed on all cables at penetrations through building walls and other building structures.

The following raceway service levels shall be adopted:

Level	Service
A1	AC Power cables above 125V
A2	Unit Auxiliary transformer cables and Incoming cables to switchgear- above 1000V
B1	DC Power cables and AC power cables below 1000V
B2	Auxiliary transformer cables and Incoming cables to switchgear- below 1000V
B3	Control circuits - 125V DC

C1	Measuring & control cables for 60V & above
C2	Instrumentation & control cables for below 60V

Note: 1.) Cables in category B1 to B3 can occupy the same raceway provided they satisfy the requirements of Code NFPA 70-2008 Art. 300-3-C(1).

2.) Cables in category C1 to C2 can occupy the same raceway provided they satisfy the requirements of IEC/IEE/Local Wiring Regulations, BS 7671, Article 528-01-

3) In Gas turbine package, following raceway service levels for inside GT package including the GT enclosure, Fuel gas unit, Fuel oil unit and exhaust gas duct area will be considered.

- LV Power cables
- Control cables
- Instrument cables and thermocouple extension cables

Following minimum separations shall be observed between above defined levels for assurance against electromagnetic interference:

Level	Separation
(A1, A2) and (B1, B2, B3)	300 mm
(B1, B2, B3) and (C1, C2)	300 mm
(A1, A2) and (C1, C2)	600 mm

All holes in ceilings, floors and wall made for cable routing must be sealed fire-resistant, when the cables have been laid. This also applies to switchgear panels and cubicles, passages between cable ducts, at vertical risers etc. As there is an increased danger of fire during the construction period the holes shall be fire-resistant sealed at that stage already.

The laying of cables outside of buildings should be preferably on cable bridges where the cables are to be naturally ventilated. In areas where a risk of mechanical damage exists the cables shall be protected by suitable metal covers.

Cable risers near gangways or in electrical rooms which are exposed to possible mechanical damage shall be protected up to 1.5 meters from ground by suitable metal cover.

If the cables are laid in the ground, after excavating the trench a layer of sand must be introduced. On the sand the cables are laid separately from each other according to voltage systems. The cables must be laid at a depth of 0.8 m at least. The trench is filled with sand and covered with concrete covers or bricks. The thermal resistance of the backfilling shall not exceed 150 Kcm/W. A ground temperature of 40°C has to be considered.

Every cable end of power cables shall be sealed and protected in an approved manner consistent with the voltage levels, cable type, and place of installation.

"Plug-in" cable terminations (if supplied) shall be in accordance with the characteristics of the switchgear.

MV, LV, DC, and control cables shall be to be laid on cable racks or trays within plant areas and in cable trenches on racks/trays or buried cables ducts (outside buildings). Any cables direct buried shall be armoured or mechanically protected. No cables shall be installed in tunnels.

Cables laid on trays shall be of adequate strength and size to carry in one layer (for power cables) or in several layers (I&C cables and small power cables), the weight, number and configuration required for each tray plus additional cables, not less than 25% by weight and dimensions, in excess of those installed. Cable tunnels shall not be used.

Segregation of the various services and voltage levels shall be achieved by use of separate trays and marshalling boxes for each voltage grade of cable used. The highest voltage cables shall be laid on the bottom rack with lower voltage cables next. Control and protection cables shall be laid on the highest levels.

For the data highway communication link, spatially separated routing with the cables running in galvanised steel conduits of approved type shall be supplied and installed.

If cables of separate units are included in the same cable route, they shall be segregated by the use of barriers.

Cables shall be suitably derated as appropriate to the requirements of the installation. Unnecessary cable joints shall be avoided. Cables shall be permanently identified with steel or plastic cable markers. Cables laid direct in ground shall be armoured.

Multicore cable cores shall be terminated using crimps or other suitable termination. Bare cores will not be accepted. Cores shall be provided with some permanent means of indelible identification. This may include colour coding or identification ferrules.

Fire barriers shall be installed on all cables at penetrations through building walls and other building structures.

The medium and low voltage cables shall be plastic-insulated copper cables. For cables that are exposed to ambient temperatures above 60°C Teflon or silicone cables are to be provided.

The cables must be suitable for laying indoors, in the open (direct or indirect sunlight), in ducts, on trays, in the ground and in water.

Fire-resistant cables shall be provided for the feeders:

- of auxiliaries required for safe shut down of unit including motors for lubricating oil pump, turning gear, jacking oil pump, air pre-heater, fuel line control valve and any other critical equipment for safe shut down not mentioned herewith;
- for control of unit circuit breaker and turbine lockout;
- between emergency diesel generator and associated switchgear;
- Fire-detecting and fire-fighting systems.

The ratings for continuous loading of the cables shall be calculated according to the relevant IEC regulations.

LV & MV cable sizing & ampacity calculation shall include but not limited to: cable voltage drop, cable short circuit capability, cable damage curve, grouping factor, insulation power factor, load factor, % fill (cable in trays), motor service factor, temperature rise derating factors, etc.

The voltage drop on various feeders shall not exceed the following specified percentage limits as referred to nominal voltages:

- | | |
|--|-----|
| - Main feeder cables to switchboards at rated current | 1% |
| - At consumer terminals at rated current. | 5% |
| - At motor terminals during DOL starting | 10% |
| - At motor terminals during DOL starting for LV motors | 10% |
| - Supply cables for lighting and fail-safe equipment | 2% |

As all cable terminations of Form 3b switchgear assemblies will be in same cable chamber, it is advisable to provide insulating boots on terminals to protect them from falling metallic objects or tools while working on other isolated feeders.

9.13 VFD Cable Design:

Shielded cabling to be used for all VFD installations as per EMC emission standards IEC 61800-3 and following shall be considered for the Cable Insulation Design used in VFD starting Motors / Consumers:

A. Outer PVC Sheath: PVC sheath material shall be suitable for its application, withstand maximum operating temperature and environment it is installed in. Enhanced outer sheath material shall be considered if the project installation environment requires the same.

B. XLPE Insulation: Thickness of main conductor and reduced conductor shall be designed to provide lower capacitance for cable length and higher temperature stability.

Cable sizing and overall design must be finalised with recommendations of VFD manufacturer.

Cable design to cover details showing clearly that how each requirement of VFD manufacturer is completed for its sizing (main conductor, PE, copper shield, other layers), type (symmetrical), laying arrangement, cable length, termination of cable shield (360 degree connection at both ends) and grounding.

An overall inspection (VFD, Cable laid, grounding and bonding) shall be arranged by EPC for joint review and approved by owner Engineer - OEM VFD prior energisation of drives at site. Inspection to confirm that approved design is completed in all respects and is ready for energisation.

Cable calculation and sizing report shall be completed and based on actual site condition mainly as following:

1. Cable Technical Data sheet: Cable resistance, maximum short circuit withstand, others
2. Cable laying arrangement
 - a. If laid in air, site maximum air temperature expected shall be used.
 - b. If laid in ground- actual factors shall be used for maximum ground temperature of site, depth of laying as planned, soil thermal resistivity
 - c. To mention how cable current capacity is calculated if cables are laid in multiple laying design (free in air / pipes, underground direct / in ducts, any other)
3. Cable length shall be optimised and to be kept as short as possible
4. Design to show details how impedance match is achieved among motor and VFD Cable.

9.14 VFD Cable Laying and Installation:

Cable run shall be designed in manner to avoid crossings with process piping as minimum as possible.

Option of cable run in overhead trays (properly designed) shall be considered. This is to minimise cable interface and crossings with process piping and avoid any probable leak.

Cable trays shall be grounded and tray to tray connections must ensure continuity of ground connection. Cable trays shall be inspected if there is any corrosion or non-bimetallic connections. Access to this area to be barricaded since drive current measurement may not also be accurate.

Cable laying shall be done as per best industry practices and project specifications.

Spacing of cable trays, spacing of cable among different voltage levels must be ensured at site during SAT. Cable bends shall not exceed the permissible bending radius.

Design of cable crossings in road and piping (if any) shall be completed with best engineering practices and project specifications with provision of future maintenance. Sufficient spare sleeves shall also be provided to accommodate cable repair / replacement.

Cable trenches and pits must be constructed to protect against any water ingress from process, rainwater and water accumulation. Auto dewatering of cable trench shall be considered.

9.15 Lighting & Small Power, Earthing & Lightning Protection

Lighting systems shall be designed such that failure of the auxiliary supply system will not prevent safe escape from Plant areas in darkness. This philosophy shall be achieved using a combination of self-maintained AC fittings, DC supplies, essential AC supplies and normal AC supplies as appropriate. Levels of illumination during normal operation and under emergency lighting conditions shall be appropriate to the area function. Energy efficient lighting to be provided. Normal AC lighting voltage shall be 220V or 230V single phase.

Low voltage power supplies shall be 400/230V or 380 / 220 kV three phase, 60 Hz, four wire (TN-C-S).

A completely integrated earthing scheme shall be provided in accordance with the requirements of "IEEE Guide for safety in Substation Grounding", IEEE Std. 80-1986. The earthing scheme shall include all structures on the Plant. A lightning protection scheme shall also be provided.

Earthing & Lightning Protection shall be as per earthing and lightning study and calculation results according to International standards like NFPA 780.

For special application like sockets, lighting and heaters, earth leakage circuit breakers with residual current rating of maximum 30mA shall be installed.

Storage batteries used as source of power of emergency lights, shall be of suitable rating and capacity to supply and maintain the total load for a minimum period of 1½ hours without the voltage applied to the load falling below 87.5% of normal as per NFPA70 article 700.

9.16 Electrical Protection and Synchronising Facilities

9.16.1 Generator , Transformer, HV Cable, Switchgear and Motor Protection

All protection setting files as per approved design duly verified, software program for relays, connecting cables to interface with relays shall be provided.

Protection system shall be designed to isolate a faulty section of electrical power system from rest of the live system so that rest of the portion can function satisfactorily without any damage due to fault current. Main and backup or dual protection schemes shall be installed using modern numerical protection relays to protect the generating plant. The design of the systems, selection and setting adopted for protection devices ensure that, in case of a failure, only the protection device of the direct consumer will operate to isolate the fault. However, on non-clearance due to any reason (protective device / switchgear, etc) the back-up protection will clear the fault with selectivity as required.

The electrical protection system shall comply with the requirements of the Grid code and regulations.

Only multifunctional digital/numerical type, self-monitoring protection relays (MFR) fitted with test facilities are to be used throughout.

Generator and Transformer protection shall also be in accordance to the IEEE standards C37.102-1995 and IEEE Guide for AC Generator Protection

The numerical multifunctional relays (MFR) shall have fault/event recording functions. The MFR shall and shall be integrated into the plant DCMS.

All protection and control systems shall be 100% redundant including all Unit protection systems. The details shall be described in detail engineering stage with Owner's Approval.

Multifunctional digital protection equipment shall be provided for:

- Redundant generator protection;
- Transformer protection;
- MV switchgear and feeder protection;
- LV switchgear protection (air circuit-breaker incomers and couplers); and
- MV motors.
- All High Speed Transfer Systems (HTS)
- All Automatic Transfer Switches (ATS)

For transformer protection proper rated CTs shall be provided on the neutral points/bushings of the transformers for earth fault and restricted earth fault protection.

Separate protection panels (sheet-steel) to be provided for:

- generator
- each generator step-up transformer

- each unit and common auxiliary transformer.
- 380 kV cable

Generator and Transformer protection shall also be in accordance to the IEEE standards C37.102-1995 and IEEE Guide for AC Generator Protection.

The protection relays shall be of the numerical type with a 16-bit microprocessor design incorporating the following features: -

- Each relay shall be completely autonomous containing DC/DC converter for power supply, potential and current transformers, A/D converter, computer chips, watchdog circuit, alarm and trip relays;
- Fully digital measured value processing and control; extending from data acquisition and digitalization of measured values up to the trip decision for the circuit breakers;
- Single processor design, providing self monitoring and annunciation of malfunctions;
- In the event of faults that activate a protective device, the current and voltage values shall be recorded and stored in the device to facilitate trouble shooting;
- All operational alarms, global alarms as well as settings, self monitoring alarms shall be stored;
- Insensitive to VT and CT errors, transient conditions and interference;
- The trip relays and the trip matrix (by software) are included in the new multi-function protection;
- The protection cubicles shall be supplied from a common power supply fed by two coupling diodes connected to two independent DC sources OR redundant DC powered with a fast acting relay for break free changeover, thus providing 2 x 100% redundancy.

A. Generator Protection System

A minimum of two multi-function relays shall be provided for each generator, fed from a separate set of current & voltage transformers in order to provide redundant protection from sensing devices to tripping mechanism. The multi-function relays shall include the following protective functions:

- i. Generator differential relay
- ii. Generator loss of excitation relay / Field failure relay
- iii. Generator rotor ground fault (64F)
- iv. Stator earth fault protection (Stator earth fault protection 95% and 100% shall be separate protection)
- v. Volts/Frequency relay
- vi. Reverse power relay
- vii. Minimum impedance
- viii. Synchronism check relay
- ix. Overall differential protection
- x. Negative phase sequence relay
- xi. Under frequency relay
- xii. Generator overvoltage relay
- xiii. Voltage balance relay
- xiv. Unit differential protection
- xv. Lockout Relays (to trip the switchyard circuit breakers and mechanical equipment (such as turbine inlet valves); and
- xvi. Exciter Field Earth.
- xvii. Island Protection / Network Relay Protection

A.1 Excitation Transformer Protection

- xviii. Definite time overcurrent protection 50/51 (Redundant in both group of Generator Protection).
- xix. Earth fault on LV side of transformer 59N or 64F (In One group of Generator Protection)

B. Transformer Protection System

A minimum of two multi-function relays shall be provided for each Main Power Transformer and Unit Auxiliary Transformer fed from a separate set of current transformers to provide redundant protection. The multi-function relays shall include the following protective functions:

- i. Transformer Differential
- ii. Unit Overall Differential;
- iii. Over fluxing protection
- iv. Earth fault protection (for generator transformer neutral)
- v. Phase Overcurrent
- vi. Oil Level (Alarm Only)
- vii. Oil Temperature (Alarm and Trip)
- viii. Buchholz relay (Alarm and Trip)
- ix. Winding Temperature (Alarm and Trip) and
- x. Overpressure (Alarm Only).
- xi. Over fluxing protection to be included for GSU transformer

Redundant Transformer differential protection and separate redundant overall differential protection shall be provided.

C. H.V. Cable / Overhead Line Protection

Numerical / digital Main Protection and Backup protection shall be provided as per the following protective functions:

- i. Cable differential Protection (Redundant) as Main
- ii. Over Current & Earth Fault Protection as Back up
- iii. Distance Protection as Backup
- iv. Short Circuit Protection (Instantaneous)

D. Switchgear and Motor Protection System

The microprocessor based multi-function relays shall include, as a minimum, the following protective functions. The relays shall be mounted on the related compartments of the respective switchgear.

- i. Generator Circuit Breaker
 - a. Circuit breaker failure
- ii. Medium-Voltage Switchgear
 - a. Bus Under & Over voltage
 - b. Bus Earth Fault
 - c. Incoming Feeders Phase Overcurrent and Earth Fault and
 - d. Outgoing Feeders Phase Overcurrent and Earth Fault.
- iii. Low-Voltage Switchgear
 - a. Bus Under voltage and
 - b. Incoming Feeders Phase Overcurrent and Earth Fault.

- iv. ARC Flash protection based on current and light detection principle so to reduce ARC Flash levels of SWGR

E. Medium-Voltage Motors:

The microprocessor based multi-function relays shall include, as a minimum, the following protective functions. The relays shall be mounted on the related motor feeder compartments of the respective switchgear.

- i. Phase overcurrent;
- ii. Earth overcurrent;
- iii. Phase current unbalance;
- iv. Locked Rotor;
- v. Number of Starts;
- vi. Bearing high temperature;
- vii. Stator high temperature; and
- viii. High-impedance differential protection (For Motor above 2.5 MW).
- ix. Lock rotor
- x. Number of starts

All protection setting files as per approved design duly verified, software program for relays, connecting cables to interface with relays shall be provided.

9.16.2 Protection Relays and Digital Fault Recording

Local access to the protection relays shall be possible through notebook computers.

Moreover, the protection relays shall serve as fault recorder, with the stored data of the same resolution as applied for the protection functions. Additionally, to the storage of analogue data, logic data like protection trips shall be stores to provide valuable information for fault analysis.

A. Protective Relays

All relays and protection shall comply with the requirements of the appropriate parts of the IEC 60255.

All relay cases shall prevent the ingress of dust and shall meet the requirement of IEC 60529, classification IP 50, category 2.

Electronic protection relays, preferably of the programmable microprocessor type incorporating fault and transient measurement memories, matrices for programming tripping sequences and auxiliary relays shall be provided.

B. Digital Fault Monitoring and Disturbance Recorder

Fault Monitoring and Disturbance Recording System (FMS) shall be provided to cover selected measurements, trip- and status signals within the turbine/generator/transformer/MV switchgear system. The FMS shall assist in the fault analysis of major disturbances of the generator/grid interconnection and shall cover:

- I. All Main Plant Processes
- II. Generators, step-up transformers and unit auxiliary transformers
- III. All MV switchgear feeders except motor feeders (motor feeders shall be monitored via the DCMS);
- IV. Incomers and couplers of all main low voltage (LV) switchgears.
- V. All generating unit breakers and switches of Plant HV switching Station.
- VI. FMS should accommodate the fault signals of 380 kV GIS or AIS substation bays of respective units.

The FMS shall be housed in a steel sheet panel and shall be located in the electrical rooms close to unit protection and synchronizing panels. The system features shall be:

- Modular, initially provided with 48 analogue and 128 binary signals;
- Multifunctional (Fault Recorder and Sequence of Event Recorder);
- Include a remote terminal unit to connect to the DCMS via standardized protocol.

The FMS shall be time synchronized with the master-clock system DCMS

Other required performances:

- Accuracy of the acquisition: better than 1 % on any type of analogue channel
- Min. 12 bit data acquisition at 10 kHz
- Time resolution: record tagged to 1 ms
- Analogue inputs: 140 and 300V, current shunts up to 50A
- Trigger channels: level, rate of change
- Power supply: redundant
- Common mode insulation EN 60255-5: 2.5 kV RMS
- Storage capacity: minimum 200 events

9.16.3 Synchronisation

Synchronization of each Power Unit to the Grid is the responsibility of the EPC and normally shall occur through the Plant generator synchronizing breakers (GCB). The generator circuit breaker (GCB) installed between generator and generator transformer will be used for synchronizing the generator with the Grid through the Generator transformer. The Plant generator circuit breaker units shall have integral isolating facilities downstream of the breaker.

In the 380 kV GIS Substation, the 380 kV generator transformer circuit breakers shall be configured for operation from the Plant and also from GIS Substation control room and the electricity system Control Centre, with proper interlocks and control. The only permitted operation of the 380 kV generator transformer circuit breakers from the Plant shall be to open these in the case of an emergency.

Automatic synchronisation units of the numerical type shall be provided to synchronise the generators via the generator breaker (GCB) or the HV circuit breaker in the 380 kV GIS Substation.

If the circuit breaker of the connected 380 kV GIS or AIS Grid substation is not able to switch the steam turbine generators (ST-generators), all ST-generators shall be equipped with generator circuit breakers. It is the responsibility of the bidder to clarify this technical detail with the Grid substation supplier and Transmission.

The synchronisation process shall be initiated by the start-up program of the DCS or by manually given operator command after the desired synchronisation point (i.e. HV CB or Generator CB) has been selected.

The synchronisation unit shall automatically check whether the necessary conditions are being met, i.e. matching phase angle, voltage and frequency on both sides of the open breaker.

Manual control facilities in the local synchronisation panel shall also be provided for synchronising the units to allow operators to start or stop the synchronisation process manually. Synchro check relays shall be provided for use in manual synchronisation of the turbo sets. The voltages, frequencies, phase angles, and the status of the synchronising unit shall be indicated on the synchronisation panel.

In general, all necessary frequency, voltage and phase synchronization equipment shall be provided by the EPC allowing the resynchronization of the with the grid as per the latest version of Saudi Arabian Grid Code.

9.16.4 Auto Transfer Scheme

Where loss of MV supply would cause a shutdown of generators or water production, fast automatic transfer schemes shall be provided. Where loss of LV supply would cause a shutdown of generators or water production, slow automatic transfer schemes shall be provided. Such schemes shall not operate in the event of a fault such as a busbar fault.

Such schemes shall have an automatic facility and a manual facility for maintenance purposes whereby the operator selects an operation and initiates it by a single action.

9.16.5 Tariff & Performance Metering

The contractor must propose in its Proposal a scope of work for installing a set of instruments and meters for metering and quality monitoring of potable water, demin water, boiler feedwater, LP steam, HP steam, power, fuel gas, electricity, Pentane and steam condensate outflows and inflows. The metering systems shall consist of settlement meters, back-up meters and quality measurements as described in OTS.

Within the Offtaker's GIS or AIS substation two sets of approved tariff quality metering equipment shall be supplied and installed by the Offtaker to measure the Electrical Energy Delivered at and to the Electrical Delivery Point. These shall be owned and maintained by the Offtaker. The metering equipment shall give, for each direction of flow, the following quantities:

- a continuous automatic record of both kilowatts and kilo vars;
- an automatic record of the kilowatt hours and kilo var hours for each sixty (60) minute interval beginning or ending on an hour; and
- a continuous integrating record of the kilowatt-hours and kilo var hours.

One set of the meters shall be used for the Electrical Energy export and import by the Plant to the GIS substation. The second set of the meters shall be used for verification purposes. Each set of meters shall be connected to a separate set of current transformers.

Measuring points and measuring equipment for interlocking and protection purposes shall be separate and not combined with measuring equipment for monitoring or automatic control equipment.

Power metering is in scope of Offtaker, however EPC to implement integration of data between the Offtaker's power tariff metering system, the Plant SCADA, and with control centres' including Offtaker's LDC/others' panels. It will be EPC's responsibility during the project implementation to obtain required information from Offtaker and ensure integration of metering system with the Power Facility.

For the measuring of generator voltage, current, frequency, active and reactive power, VTs, CTs and measuring transducers (4 to 20 mA) shall be provided.

One combined meter, for two directional flows of active and reactive power, shall be provided for each generator for performance monitoring.

All SWGRs & critical MCC's (MCC's changeover require unit shutdown) should have make before break changeover schemes for planned changeovers.

9.16.6 Electrical System Control, Indications and Alarms

All gas turbine generator controllers shall be equipped with automatic generation control (AGC) capabilities, for automatic regulation of the total power output of the Plant within predetermined limits. The set point for such regulation shall be through communication links between NCC and the Power Block equipped for such control.

EPC shall provide instrumentation for alarm and trip signals to and from the Plant central control room as may be necessary.

The Contractor shall be responsible for the design, manufacture, supply, installation, testing and commissioning of control, monitoring and protection systems associated with its electrical installation. Control and alarm facilities for the Plant electrical MV, LV and DC systems shall be provided.

The scope of supply shall comprise all necessary control, protection, monitoring and supervisory equipment, cabling and wiring, regardless of whether it is specifically mentioned or not, for safe and efficient operation of the Plant. The control and monitoring of the electrical system and other associated common systems shall include at least the following:

- Control, monitoring and protection of MV system;
- Control, monitoring and protection of auxiliary power distribution (LV, DC, UPS);
- Energy meter cubicles;
- Alarm cubicles;
- Auxiliary and marshalling cubicles.

MV and LV AC electrical systems (excluding LV sub-distribution systems) shall normally be monitored and controlled from the Central Control Room including for all transformers, switchgear incomers, bus-section, feeders and interconnector circuits via the plant DCS. The voltage of each switchboard, and single-phase current of all main circuits (incomers/outgoing, bus-sections and interconnectors) shall be displayed locally at each switchgear, and for motors of 55 kW and above repeated in the Central Control Room via the plant DCS.

Motor circuits shall normally be controlled and monitored from the Central Control Room via the plant DCS system, except for station minor plant needing only local control. For motors of 20 kW and above the phase currents of each motor circuit shall be displayed locally at each switchboard.

Alternative locations of control of switchgear and motors shall be at the switchgear (under Permit to Work for operation or testing) or plant (for operation or maintenance).

LV sub-distribution systems and the DC systems shall be locally controlled only, but alarms shall be transmitted to the Central Control Room via the plant DCS.

The switchgear shall include all related circuit prominent coloured LED type status and alarm lamps. The associated protection relays shall also include LED status/alarm indicators or flags.

9.17 Black Start / Emergency Supplies

9.17.1 Emergency Supply

The Plant shall be provided with emergency power supplies through diesel generating units to ensure the safe and controlled shutdown of all Units in the event of loss of all electrical supplies without damage to any equipment.

Emergency diesel generators for safe shutdown of the plant including supply of the essential HVAC load, where the essential HVAC load list to be submitted for Owner's approval during detailed design stage. It is suggested to adopt the concept of Two (2) emergency diesel generator (EDG), one for Power Block loads and 2nd one for Common loads (including BOP building HVAC requirements) and for safe shutdown loads of the unit such as Emergency drives, UPS, DC system and Emergency lighting etc. These EDGs will supply backup power to respective Normal cum emergency switchgears of the plant, in case of black-out.

Control Circuit of EDG shall be designed in manner to provide high availability in plant black out conditions. For example, supply of dc control supply for EDG Control circuit shall be provided from two separate DC sources.

Fuel level of Diesel tanks shall be monitored in control room.

The Emergency Diesel Generator Set shall be installed outdoor in the IP 54 Protection Class Canopy / Soundproof Enclosure / Containerized Solution in accordance to the IEC 60529.

Emergency diesel generators shall be designed with a class F insulation and shall be operated with class B temperature rise.

The EPC shall provide the sizing calculation for Emergency and Black Start Generators. Further requirements are set out in following sections.

Emergency diesel generator including MV switchgear, distribution cables, MV/LV transformers and emergency power distribution for all units, auxiliary boiler and utilities.

The Plant shall be provided with emergency power supplies through diesel generating units to ensure the safe and controlled shutdown of all Units in the event of loss of all electrical supplies without damage to any equipment.

In minimum each Group shall be provided with 1 x 100% emergency diesel generator which shall cover the essential loads of the whole Group and the related common essential loads. The design shall make provision for 1x (100% +10% margin) emergency diesel generator. The diesel generating unit shall be equipped with switchgear, control, protection and synchronising equipment to enable all operation to take place automatically following detection of a loss of AC supply. Manual facilities shall be provided to allow units to be regularly synchronized with the normal AC system for load testing purposes.

Control Circuit of EDG shall be designed in manner to provide high availability in plant black out conditions. For example, supply of dc control supply for EDG Control circuit shall be provided from 02 separate DC sources.

Fuel level of Diesel tanks shall be monitored in control room.

The emergency diesel generator shall be provided with all necessary auxiliary systems, but not limited to the following items:

- Container for outdoor service with adequate thermal and acoustic insulation. The diesel generator units and corresponding auxiliary equipment shall be factory tested inside the container.
- Common base frame for the engine and generator. The base shall provide both structural integrity and vibration isolation (by means of spring elements). All components that physically connect to the diesel generator set must be flexible in order to absorb the vibration movement without damage.
- Complete compressed air starting system or battery type engine starting system for five (5) consecutive start-ups.
- Complete engine cooling system.
- Lubricating oil system.
- Fuel storage tank
- Combustion air intake suction system.
- Gas exhaust system.
- Ventilation and cooling air intake
- HVAC system for control room of the container.
- Fire protection and detection systems of the containers
- Safety protections for personnel.
- All the necessary auxiliary systems for lighting and ventilation inside the container.
- Auto & Manuel Synchronizing devices to allow on load test during operation.

The DG Units shall be designed, manufactured and tested in compliance with the latest versions of the following standards:

- IEC 60034 Rotating Electrical Machines
- IEC 60085 Thermal Evaluation and Classification of Electrical Insulation
- IEC 60529 Degrees of Protection provided by Enclosures (IP Code)
- ISO 8528 Reciprocating internal combustion engine driven alternating current generating sets
- ISO 20816 Mechanical Vibration
- ISO 3046 Specification for Reciprocating Internal Combustion Engines
- ISO 9000 Recommendations for Diesel Engine Acceptance Tests Quality assurance

9.17.2 Black start generators (if applicable for the project)

Black start generator set(s) with N+1 redundancy shall be installed to start one gas turbine. The black start generators shall be fired with Back-up Fuel. Number of black start generators required shall be determined by the EPC considering the load required to start one gas turbine comprising GT starting load, GT associated auxiliary system load and related plant BOP necessary load. The black start system shall be configured to select and start any GT.

The black start generator shall have the same features stated for Emergency Diesel Generator.

The Black start generators shall have the feature to start and synchronise with mains to share load in parallel.

Black start Diesel Generator set shall be able to be tested on any GT of a block, while the other GTs and the ST of the same block are still in operation. During the test the BSDG set shall serve start-up the selected GT which shall be completely isolated from the neighbouring groups and from the Grid. The test is considered successful once the GT, that was started with the BSDG, set is successfully synchronized with the Grid and the BSDG is load is transferred to the normal power supply, without tripping the unit.

The Facility shall be capable of restarting from complete blackout conditions. This could either be achieved by a suitably sized emergency blackout diesel generator or a gas turbine generator and which shall be capable of starting any of the Power Blocks capable of starting the Plant from blackout conditions.

EPC shall install Black Start Facilities and all necessary auxiliaries of sufficient capacity to start any one of the gas turbines, connect to de-energized transformer circuit breakers located in the Offtaker's 380 kV GIS Substation, energize the dead 380 kV bus bars, energize the outgoing transformer feeders and energize the inter bus transformers and reloaded to its maximum capacity under the PPA or PWPA. The Black Start Facilities shall have 100% capacity to start each one of the gas turbines even with the loss of one diesel generator.

The Black Start Facilities should be capable of operation if all other sources of electrical power either within or outside the Plant are not available. The Black Start Facilities shall use liquid fuel and be provided with sufficient fuel storage capacity for 6 hours operation without replenishment of fuel supplies.

The Black Start Facilities shall be capable of being tested whilst the Plant is in operation and without shutting down or interrupting or affecting the operation of other parts of the plant. Provision shall be made available for the functional test of the BS facilities to be conducted to carry out the black start of any GT without shutting down or affecting the operation of any other parts/system of the plant."

It is to be provided with feature to start and synchronize with mains to share load in parallel.

Non-availability of the Black Start Facilities due to failure shall be immediately advised to the Offtaker. All planned maintenance shall be agreed in advance with the Offtaker.

EPC shall undertake a detailed study into the operation of the Black Start Facilities and gas turbine and generator control systems design such that a gas turbine can be started, be connected to the Grid and be loaded in the event of a complete system failure. The results of the Study shall be presented to the Offtaker.

The Black Start Facilities shall be commissioned and tested prior to the Scheduled Early Power Commercial Operation Date (as per the PPA).

9.18 Construction Power Supply

Electricity required during construction shall be provided by the EPC contractor.

9.19 Mechanical Protection

Depending on the installation and application, the following degrees of protection for enclosures, in line with IEC 60529, are required for the various electrical equipment.

Equipment	Internal location		External location	Below ground	Submerged
	Dry	Damp			
MV switchgears	IP 41 ¹⁾	---	---	---	---
Dry type transformers in steel enclosures	IP 21 ¹⁾	---	---	---	---
DC rectifiers, inverters (UPS)	IP 31 ¹⁾	---	---	---	---
LV switchboards, distribution boards, local control panels	IP 41 ¹⁾ IP 52 ²⁾	IP 54	IP 65W	---	---
MV motors and generators	---	IP 55	IP 55W	---	IP 65
MV motor/ generator terminal box	---	IP 56	IP 56W	---	IP 65
LV motors and generators	IP 54	IP 55	IP 55W	IP 58	IP 65
LV motor/ generator terminal box	IP 54	IP 56	IP 54W	IP 55	IP 65
Motor operated valve actuators	---	IP 57 IP55 (Control valve inside GT Enclosure)	IP 57W	IP 57	---

¹⁾in air-conditioned rooms

²⁾ in all other dry rooms

9.20 Connection to Generator

The line side connection to the generator shall be designed as a single phase encapsulated IPB. The IPB must be sealed air-tight to the generator bushings to impede ingress of hydrogen into the IPB for the case of hydrogen leakage at generator terminals. A forced ventilation system, installed in each phase of IPB enclosure near to the generator bushings shall prevent a concentration of hydrogen by means of provided outlets towards turbine hall. The connection between IPB conductors and generator terminals shall be realized by flexible copper links with silver-plated contact surfaces. The conductor connection shall be provided as such that a removal of generator bushings is possible without any cutting and re-welding on the IPB enclosure. The generator and the IPB shall be designed in a way that current transformers of ring core type can be installed at line side within the IPB enclosure or at the generator terminals.

The built-up of neutral shall be realized within a special generator neutral box. A forced ventilation system, installed for each of neutral bushing shall prevent a concentration of hydrogen by means of provided outlets towards turbine hall. The generator and the IPB shall be designed that way that current transformers can be installed within the box or as per Contractor's standards. The generator neutral shall

be connected via a busduct or a cable with the grounding transformer inside the neutral grounding cabinet. For Turbine/Generator set, OEM standard and general industrial practice are acceptable.

In Gas turbine package, forced ventilation system is not applied to the neutral cover of the proposed generator. Instead, ventilation holes in the cover is considered to enhance air in-flow through it.

9.21 Transformer Connection

The connection of IPB to transformers is designed as single-phase encapsulated solution. The mechanical connection between the IPB enclosure flange and the transformer flange shall be carried out insulated. The flange connection shall be sealed by a gasket. The contact surface of connection between IPB conductor and transformer terminal shall be provided by means of silver-plated flexible copper links. The conductor connection shall be designed as such that a removal of transformer bushings is possible without any cutting and re-welding on the IPB enclosure.

Surge arresters for each phase of IPB shall be provided near to the generator transformer. The arresters shall be mounted within cabinets which are an integrated part of the IPB enclosure.

9.22 Short Circuiting and Grounding Equipment

For testing and maintenance purposes the IPB shall be equipped with a manually operated short-circuit device (SCD). The SCD shall be directly connected at the IPB between the generator and the generator circuit breaker. The SCD must be designed for the full short circuit current for 1 s duration.

For each phase in the IPB a temporary grounding device (TGD) at the transformer side of the generator circuit breaker shall be provided. For this reason, the IPB conductors shall be equipped with a counter-contact for installation of the TGD. Access to the conductor must be achieved by covers within the enclosures which are securely bolted.

9.23 Electrical Analysis and Calculations

Ground investigation study for Earthing system design

- Load flow /short circuit and voltage drop studies carried out through internationally acceptable software.
- Harmonic study
- Protection coordination study
- Transformer sizing calculations
- Cable sizing calculation
- Lighting calculations
- Grid Code impact Studies
- Arc Flash Hazard Study For all Power System SWGRs– As per NFPA 70 E, IEEE 1584-2002, IEC 62271-200. All SWGRs shall be labelled for HRC (Hazard Risk Category) identified and provision of respective HRC PPE in all Switchgear rooms
- UPS and DC System – Battery and system sizing
- Study of Short-circuit impact on grid to identify current limiting solution
- Other calculations as required

9.24 Testing

In addition to the manufacturer's production tests, the Contractor shall carry out routine tests on completed equipment. The Owner reserves the right to witness such tests.

Routine tests on each piece of equipment shall include at least the following:

- i. Visual inspection.
- ii. Performance test.

- iii. Control loop testing
- iv. Voltage withstands test and insulation test.

Where a number of items of equipment are intended to operate together, the complete assembly shall be tested to demonstrate that the interconnected items are compatible. Design consideration should be given to the testing requirements of the Plant as defined in the OTS.

The Site acceptance tests shall include, but not be limited, to the following:

- a) All outstations are tested without connection to workstations in the CCR.
- b) Data communications to workstations are established for each outstation.
- c) Commissioning tests including, in-situ instrument calibration check to DCS control loops, DCS device drivers, protection and interlock logic, sequences, station master load control, alarms, and all other equipment supplied

10. MECHANICAL SYSTEM & EQUIPMENT REQUIREMENTS

10.1 Fuel System

The fuel gas system shall include fuel gas pressure reducing stations, compressor station, heating, knock-out drum and filter systems, separator, fuel gas piping, shut-off and safety valves, condensate tanks, and metering.

Sizing of Plant Gas Buffer Volume should consider the case of Loss of one gas booster compressor during the most stringent Fuel Gas Terminal point physical parameters (lowest pressure, highest temperature, and lowest gas molecular weight) shall not lead to Gas Turbine trip (considering GT Runback) when the Plant is operated at base load conditions (Contracted Capacity Conditions). EPC Contractor should follow OEM Fuel Gas Specifications for Gas Turbine's fuel gas requirements.

Fuel gas buffer system including compressors and tanks are not considered for instantaneous FG to Back-up Fuel changeover, EPC will liaise with the gas supplier to make sure of the conditions that will make the fuel change over successful. The gas specifications are available in Appendix AB of EPC Contract.

10.1.1 Natural Gas

The Plant shall be designed for firing natural gas (Gas).

The equipment shall include but not be limited to:

- Emergency shutdown valve
- Compressor station
- Filter Separator/Knock out drum
- Redundant Coalescing Filter/Moisture Separator (2x100%)
- Fuel gas metering (Main and Back-Up)
- Fuel gas conditioning
- Fuel gas performance heaters 1x100% for each Gas Turbine
- Dew Point Heater (if required)
- Condensate collection and disposal system

For the design and operation of the fuel Gas system, the following main constraints shall be considered:

- The fuel Gas system shall comply with the applicable standards like ASME, EN and API, as far as applicable.
- For the design, block and bleed approach shall be used.
- All the equipment shall be fitted with isolating valves to permit repair and maintenance without interrupting the Gas supply .

For each Gas Turbine, following equipment installed as related accessories will be provided as 100% x 1set, in hence, repair and maintenance without interrupting the gas supply is limited.

- GT Fuel gas flow meter and GT Fuel gas heater (Radiator type & Feed water type)

- All isolating valves shall be ball type with double block and bleed function.

For Gas supply line inside Gas Turbine package, two (2) globe valves i.e. control valve and isolation valve with bleeding valve in-between are considered.

- Emergency shut-down (ESD) valves shall be used down-stream the Gas delivery point to allow the entire Gas supply to the plant to be isolated in the event of an emergency.
- ESD valves shall be "fire-safe" in accordance with API-607 or BS-5146 and of anti-static design in accordance with BS-5351.

- Sufficient permanent installed valve connections and vents shall be provided to permit the gas system to be purged with inert gas.
- All butt welds shall be subject to 100% radiography test and all other welds shall be subject to 100% non-destructive testing. If socket welds are approved by design for small bore (DN 50 and below) pipe, 100% surface NDT shall be done i.e Magnetic Particle Testing (MPT) for ferritic steels and DPT for austenitic steels.
- The maximum flow velocity for piping and valve dimensioning shall not be higher than 25 m/s.
Inside Gas Turbine enclosure and related accessories, flow velocity is optimized as per OEM design standard.
- Buried gas pipes shall be adequately protected and insulated against the aggressive environment in which they are found

Considering the applicability of the Fuel Gas Boosting Compressors, single failure of the Fuel Gas Boosting Compressors may lead to a loss of Plant output during the transient conditions (between the time of the trip until the full load of the standby compressor) however, this trip should not cause a trip within the Power Block. In order to design the Fuel Gas Buffer System, Contractor should consider the worst Fuel Gas receiving conditions and Plant Maximum output at Reference Site Conditions and the Gas Turbines operating at maximum load. The configuration should provide no trip within Gas Turbine when a trip has occurred on a Gas Boosting Compressor at any Gas Supply conditions. EPC Contractor should consider the standby compressor startup time for Gas Turbine operating conditions. The fuel gas booster compressors shall be configured with IGVs for PR control (if required). The fuel gas booster compressors shall be 2 sections, serial oriented with interstage bypass. When the fuel gas terminal point pressure is equal or higher to reference gas pressure but lower than the gas turbine required inlet pressure, only section 2 will operate while section 1 will be fully by-passed but kept in hot standby. In case the fuel gas terminal point pressure is between minimum fuel gas terminal point pressure and reference fuel gas terminal pressure, both section 1 and section 2 will be in operation.

Gas turbines operating flow range, minimum flow considering single GT operation with minimum emission compliant load flow as well as maximum flow considering Plant Maximum output at Reference Site Conditions at new and clean / maximum GT Degradation case should be covered with selected Fuel Gas Booster Compressor flow ranges.

The plant shall include in its scope all the systems required for the provision of the gas pressure and quality required by its combustion equipment (e.g., reducing station, booster compressor, filtering, heating, fuel gas analyser (One Chromatograph for GRMS)).

An emergency shut-down ("ESD") valve shall be provided in each fuel stream to isolate the fuel supply to the plant and shall meet all the appropriate standards.

The ESD valve shall be subject to control from the CCR. Additional shutdown valves shall be located in each supply lines to the gas turbine and the auxiliary boiler.

Shut-off of the gas fuel supply shall be performed by two independently operated automatic shut-off valves; the piping between the valves shall be vented. At least one of the two shut-off valves shall be an automatic fast acting shut-off valve and one shall be a leak tight valve. One of the two shut-off valves may serve as a flow control valve. This shall not be the automatic fast acting shut-off valve which shall be provided as an independent function.

A strainer shall be installed upstream of any automatic fast acting shut-off valve at a suitable location to prevent valve malfunction due to debris entering the valve.

As a minimum each gas fuel supply shall include the following functions:

- a) manual isolation
- b) leak tight shut-off shall fail closed by permanent available energy e.g. spring force.
- c) automatic fast acting shut-off shall fail closed by permanent available energy e.g. spring force.
- d) flow control

- e) venting for depressurization between leak tight and automatic fast acting shut-off valves and
- f) venting for pipework depressurization

Custody meters shall be provided for custody metering of imported Fuel Gas.

The plant shall be designed and operated so that:

- The GTs will accept specified qualities of fuel gas within the maximum acceptable flue gas emission values are met without the need to shut down the GTs.

10.1.2 Back-up fuel

The Back-up Fuel system shall include, but not be limited to the following:

- 2x100% unloading pumps with necessary platforms suitable for 2 tankers
- Storage tanks of 24 hours full load operation capacity of EDG
- Treatment facility as required
- 2x100% forwarding pumps for the gas turbines including inlet filters
- 2 x 100 % forwarding pumps including inlet filters for emergency diesel generators, black start facility and firefighting pumps
- Drain collection tank and disposal pumps
- 2x100% Metering Facilities
- Final filters if required
- Related electrical facilities, instrumentation and control, etc.

A slope oil tank shall be provided to enable draining of all piping network.

As a minimum each liquid fuel supply shall include the following functions:

- a) manual isolation
- b) flow control
- c) automatic fast acting shut-off
- d) leak tight shut-off
- e) spill and/or drain
- f) fuel pump

Shut-off valves shall fail close by permanent available energy e.g. spring force. The fail-safe status of drain valves shall minimize any risks.

For operation during maintenance activity or by fire service personnel a valve that is capable of manual operation shall be fitted to the inlet of each gas turbine installation upstream of the automatic valves. This valve shall be clearly marked located in an accessible position and should be capable of being operated by an acceptable level of physical force.

An automatic shut-off valve shall be located outside the gas turbine package to automatically isolate the fuel supply to the gas turbine in the event of a dangerous situation. A filter/strainer shall be fitted upstream of the fuel flow control device and automatic fast acting shut-off valve at a suitable location to prevent device or valve malfunction due to debris entering the device or valve. Where liquid fuel is used suitable drain points shall be incorporated to drain off unburnt fuel from the pressure section of the casing and/or exhaust system (e.g. in the event of a flame failure on start-up). The drain points shall have valves preferably automatically operated that open on shut-down and close as part of the start sequence.

EPC Contractor shall provide necessary back-up fuel facilities required to provide instantaneous on-load change-over from Natural Gas to Back-up Fuel to ensure the Contractor Guaranteed Power Capacity on Back-up Fuel without interruption in the event of a Gas Operation Failure, trip of Gas Connection Facilities or any other scenario that would cause interruption of natural gas supply.

Contractor in its proposal shall demonstrate that it has carried out design studies to ensure that a successful fuel changeover may be achieved in the event of loss or disturbance of the fuel gas supply. If any loss of power output is predicted during fuel changeover, Contractor must specify this in the proposal.

Necessary back-up fuel facilities shall include duplicate fuel forwarding pumps, each of one hundred percent (100%) capacity for each gas turbine. Back-up fuel recirculation facilities allowing manual and automatic Back-up fuel recirculation shall be provided.

To ensure that gas pressure can be monitored in the Plant the Contractor shall install a pressure sensor on the Gas line downstream of the Gas Connection Facilities to provide data into the Plant DCS. The DCS shall be configured to provide falling gas pressure and low gas pressure alarms and incorporate suitable control logic to carry out an automatic change over from Natural Gas to Back-up Fuel. The changeover pressure should be set differently for each gas turbine to avoid any negative impact on both, Natural Gas and Back-up Fuel resulting in loss of energy generation.

Back-up fuel storage capacity for the Plant shall be increased to cover at least five (5) days back-up fuel demand for the continuous operation of the Plant based on Guaranteed Power Capacity on Back-up Fuel. If required, Demineralised Water Storage of not less than five (5) days shall be kept for water injection necessary to limit NO_x emission as per the statutory environmental regulations. Demineralised Water production rate to refill the storage tank shall be higher than the maximum water injection flow rate and buffer storage required in storage tank.

10.1.3 Metering and Settlement System

The EPC shall provide the Plant Settlement System and shall including integration with the Offtaker Central Settlement System.

The EPC shall provide the full data required for settlement software calculation Specification for Data Exchange between the Plant and Offtaker will be defined during detail stage, the Settlement System shall include the FDM and EDM.

The EPC shall also include:

Separate IT and communication room, Marshalling cubicle for all required Original Input Data, Data link between, Plant and Offtaker.

In addition to the above The EPC will be responsible to provide the Hardware-software equipment (main server-backup server-EDM server, FDM server, Metering Server, operator station –communication etc.) and software required for settlement system.

The Standard Interface Specification for Data Exchange with Plant System and integration with Offtaker's settlement system to be agreed with Offtaker in detail design stage. All measurements and data as required for the Settlement system are to be provided.

10.2 Gas Turbine

10.2.1 Gas turbine generator package(s)

The gas turbine generator package(s) must be complete packages, including but not limited to the following:

- enclosures, (if required)
- all skids,
- two stage inlet air filtration system consisting of pulse cleaning type filter stage and fine filter stage suitable for high dust laden and humid environment,
- air inlet filter system including self-cleaning pulse filter, high efficiency filter, inlet duct system, inlet support system, with silencer;
- air inlet cooling, as required,

- evaporative cooling system; if applicable;
- All exhaust gas ducts including expansion joints, noise-absorbing elements, silencer(s), guide vanes and other internals,
- Insulating of ducts, outer insulation of auxiliaries
- fogging system, if applicable;
- wet compression system, if applicable;
- compressor washing system (on-load and off-load),
- exhaust gas system with ducts and silencer,
- bypass Stacks, including diverter dampers and bypass stack, to allow simple cycle generation mode for gas turbines;
- complete SCR system upstream of bypass stack (incl. ammonia system, flue gas cooling system), if required;
- Dehumidifier for gas turbine outages;
- Maintenance covers, if applicable;
- three phase AC generator, including excitation and voltage regulation with power system stabilizer ("PSS");
- Synchronous AC generator with complete air/water or hydrogen cooling system
- Compounded or static excitation system (brushless rotating system) and AVR with power factor control,
- Generator main connection (isolated phase bus-duct or single core cables) up to generator stepup transformer and unit auxiliary transformer, including generator circuit-breaker, isolators, voltage and current transformers, surge arrestors/capacitors, etc.,
- Electrical auxiliary systems such as MV and LV switchgears, Safe AC and DC supply,
- lubrication system,
- Lube oil systems for turbine, gearbox(es) and generator, complete with all pumps, filters, heaters, coolers, controls etc.,
- Portable lube oil conditioning unit (common), or alternatively static oil purifier per unit shall be provided,
- Closed-circuit cooling water system for lube oil and generator cooling, pumps, piping etc., or alternatively, direct lube oil/air cooling system could be applied for gas turbine lube oil,
- Enclosure for the gas turbine unit in order to obtain reduced sound power level and environmental protection,
- turning gear,
- Coupling,
- Load sharing system,
- fuel compartments,
- Measuring, metering and monitoring equipment,
- Final fuel gas filter skid
- combustors and fuel gas systems for operation on the specified fuels
- fire and flammable gas detection system,
- firefighting and alarm system (individual for each GT),
- Instrumentation & Control ("I&C") system,
- turbine and generator protection system,

- transformer protection system,
- emission control system,
- continuous emission monitoring system and related instrumentation,
- synchronizing facilities,
- automatic and manually operated synchronization facilities for each generator circuit from the unit control desk and Central Control Room,
- local control compartment,
- unit related electrical facilities, including excitation, voltage regulation and start-up facilities;
- main and auxiliary transformers,
- gas turbine cooling system,
- gas turbine start-up facilities,
- all required gas conditioning equipment's, and
- fuel gas preheating (1x100% per GT) (if applicable)
- lubrication and jacking system
- Dry low NOx combustion systems and complete gas fuel system for operation on the specified fuel
- I & C system
- protection system
- Generator circuit breaker
- Generator to be provided with Online Partial Discharge measurement with remote monitoring of PD in control room
- cranes and hoists (considering the GTs are outdoor design, there is no turbine house, so electric double-beam bridge crane & mobile crane for GTGs maintenance and overhaul is not in EPC scope)..
- Gas turbine lube oil supply package shall be equipped with:
 - Sufficient Heat exchanger (minimum 2X100%) (to be designed and implemented according to study results, meeting OEM recommendations for maintain lube oil cooling and serving all worst case conditions)
 - 2 x 100% AC lube oil pumps
 - 1 x 100% DC lube oil pumps
 - Considering MPC (Monitoring and Predictive Center) (GE Predix) all the required tags shall be configured with DCS.
 - Control oil to be included in the package (if required).
- One mobile type of the compressor water washing system shall be provided (1 compressor washing system per 3 GTs)

The Gas Turbines shall be of proven heavy duty industrial design in multi-shaft configuration directly coupled to a 60 Hz Generator.

The gas turbine shall be provided with all TIL's incorporated into design and OEM shall provide the Technical Information Letters (TILs).

The Contractor shall provide gas turbines located outdoor.

The gas turbine shall comply with the emissions and noise requirements defined in OTS.

The gas turbines shall be capable of start-up, shutdown and continuous operation on natural gas over the whole range of ambient conditions expected at site. The gas turbines shall be capable of online change-over from Simple Cycle to Combined Cycle and vice-versa and from natural gas to back-up fuel and vice-versa without the need to shut down.

In case the GT is provided with fuel gas preheating, the outage of the preheater shall not lead to a GT trip.

The gas turbine shall operate over the frequency range and contribute to governor response to frequency excursions as per the grid code requirements (if any) or as per agreed with Offtaker and include inlet augmentation system to increase gas turbine output during periods of high ambient temperatures and low relative humidity (if applicable).

The gas turbine generator shall be capable of withstanding 100% load rejection to house load.

All gas turbine compartment enclosures shall be ventilated with filtered air. Ventilation shall be by means of 2 x 100% or 3 x 50% fans with automatic switch-over in case of failure of one fan.

Since GT will be purged before start-up considering NFPA requirement, OEM's standard design does not include gas detector in exhaust duct. An alarm shall be provided to warn the operator of failure of the enclosure ventilation system.

Permanent equipment (piping and valves) for both on-load and off-load compressor cleaning shall be provided for the Gas Turbine, allowing for the connection of stationary washing equipment. One (1) set of common GT compressor washing equipment shall be supplied for each Group. Permanent effluent disposal systems shall be provided, complying with the local environmental requirements. The Contractor shall detail on-load and off-load procedures for wet cleaning of the compressor. Suitable storage shall be provided for off load washings prior to disposal off site. For compressor on-load and off load washings, skid tank shall be provided.

The gas turbine shall be provided with all associated ancillary plant and systems for the safe, efficient and reliable operation of the units in combined cycle mode.

Each gas turbine shall be provided with their starting unit (SFC) from MV switchboard of its own unit / block. In case of non-availability of SFC of that unit, the gas turbine shall be capable of starting from SFC of other unit and facility for such operation shall be considered in the design. The gas turbine shall be provided with a full set of automatic control and protection equipment incorporating governor, temperature control and supervisory functions. Vibration levels shall conform with the relevant ISO standards.

The gas turbines to operate in simple cycle shall be complete with by-pass stacks, silencers, and suitable diverter dampers with seal air systems, horizontal and vertical sealing guillotine damper. By-pass stack will be designed so that there is not any hot exhaust gas leakage to atmosphere during combine cycle operation.

Associated ancillary plant and systems shall be provided to ensure the safe, efficient and reliable operation of the gas turbine in combined cycle mode. Moreover, the Contractor shall provide pulse-type on-line cleaning (with proven operational performance) of each gas turbine intake filter.

The Gas Turbine control system shall incorporate an interface with the DCS to facilitate integrated control of HRSGs and Steam Turbine (ST) for important and critical operational maneuvers.

Each GT shall also be capable of coming to rest without bearing or other damage on loss of cooling water or main AC power to its oil systems.

Each Gas Turbine shall be provided with a full set of automatic control and protection equipment incorporating governor, temperature control and turbine supervisory functions. Vibration levels shall conform to the relevant ISO standards.

Lifting gear, such as lifting beams, slings, jacks etc. required for removing complete compressor-turbine rotor, Generator rotor, the upper portion of casings and other heavy equipment shall be provided. (In Gas turbine package, the lifting beams for the generator rotor will not be provided because the rotor is installed or removed by using rotor saddle which will be provided as the generator special tools.

Also, slings, jacking etc. will not be provided because they are not special tools but general tools.

Means shall be provided for powered turning of the rotors for a sufficient period following normal or emergency shutdown to ensure satisfactory cooling of components. It shall not be possible to start the turning gear without adequate lubrication oil supplies.

Jacking facilities shall be provided for all turbine and generator bearings (if required).

As a backup to the powered system, means shall be provided for turning of rotors during maintenance

Since rotor of M501JAC is so heavy that hand turning is not applicable, hand turning device will not be provided, while OEM should provide means for increment rotation of turbine shaft during maintenance.

In case of black out, AC power should be supplied from EDG to AC turning device.

Gas turbine and generator units should be equipped to provide conservation conditions during outage periods.

Gas turbine package shall include online/offline oil purifier system.

Regardless of Gas turbine package own compressed air supply, the package shall be connected to plant compressed air system for reliable supply of compressed air.

10.2.1.1 Compressor Section

The axial flow compressor shall have sufficient aerodynamic flexibility to permit satisfactory operation of the turbine under all conditions, with a safe margin for blade fouling and erosion:

Bleed valves and variable geometry guide vanes shall be a feature of the design to permit more efficient part load operation and to avoid compressor surging or any other undesirable flow conditions at low compressor speeds.

All blading shall be capable of being inspected by fibre optic/borescope techniques via easily accessible, removable port plugs. Inspection holes for Borescope should be provided as per OEM standard. Some blades and nozzles in compressors without port plug are inspected with a major inspection.

10.2.1.2 Combustion System

The turbine shall have a robust combustion system designed for maximum life between inspections and repairs. The combustion chambers shall be made of material resistant to very high temperatures with the burners designed to give a stable flame through the entire operating range.

The system shall be capable of achieving the emission limits of NO_x and CO without excessive levels of combustion pressure fluctuations. Dry low NO_x combustion shall be utilized.

During the starting sequence of a gas turbine, it is essential that the presence or absence of flame is known. Hence, a reliable flame monitoring supervisory system (having triple redundancy) shall be provided.

The flame detection will be performed by software monitoring of trend of blade path temperature in the GT control system during GT startup period. Hence, the flame detectors will not be provided. While the gas turbine is in load operation, the flame loss trip will be conducted with supervising reverse power condition of the gas turbine.

10.2.1.3 Turbine

The turbine shall be of multi-stage design able to accept sudden load changes and to respond quickly to load fluctuations.

Blade and nozzle materials shall have proven capability when operating at the base load and peak load gas temperatures. It shall be possible to replace blades without having to remove any other blade (except the diametrically opposite blade) and to rebalance the rotor.

All the blades and nozzles shall be capable of being inspected by borescope via inspection plugs on the casings or some other convenient means which does not require major disassembly of the machine.

All hot gas path components shall be designed for long life between inspections and repairs.

The stator carrier ring shall be 12Cr minimum or stainless steel (if applicable).

10.2.2 Gas turbine auxiliary cooling water systems

Gas turbine auxiliary cooling water system shall be provided for each gas turbine generator unit, for cooling lubricating oil, generator and other systems, as required.

10.2.3 Fire / gas detection and protection system

The entire gas turbine generator installation, including all auxiliary enclosures, shall be protected from fire with a complete automatic fire protection system. The system shall be complete in all respects and shall include cylinders, cylinder racks, piping, valves, nozzles, detectors, fire alarm actuating system, and all other equipment necessary for the safe and efficient operation of the units.

The fire protection and detection system provided with the gas turbine will need to interface, and connect to, the main fire protection and detection system. Hazardous gas detectors and fire detectors shall be provided in enclosure and shall be designed to pick up casing leaks.

10.2.4 Performance Requirements

After construction and commissioning, a Reliability Test Run including Performance Tests is to be performed. For the Performance Test of the gas turbine, the procedures contained in ASME PTC 22 or ISO 2314 shall apply. The selected Gas Turbine shall be capable of the Continuous emission compliant operation at low loads (60% or below of rated Gas Turbine load).

10.2.5 Noise Limits

Suitable acoustic hoods/enclosures shall be provided and installed with each Gas Turbine Generator unit as required to meet the noise limits specified elsewhere.

10.2.6 Air Intake Systems and Filtration

Each Gas Turbine shall be equipped with a high efficiency, high-level air inlet filter unit and intake silencing.

The Gas Turbine intake air filtration system shall be two stage inlet air filtration system consisting of pulse cleaning type filter stage and fine filter stage suitable for plant location with high dust laden and humid. The type of filter element selected shall be capable of operating satisfactorily (i.e. delta p across filter shall be such that an acceptable margin from the surge-line shall be achieved at all times. The design operational life of self-cleaning filters and or static final filters shall be a minimum of two calendar years or 16000 gas turbine fired hours, whichever comes first. (for HEPA filter, above requirement is fulfilled. For pulse filter, filter life is dependent on the site environmental conditions and filter maintenance frequency).

The Contractor shall carry out air sampling on Site as is necessary to ensure the correct design of the intake filters and their design life. Such study shall be conducted in the dry period of the year when the concentration of airborne dust tends to be the highest due to seasonal activities and when high mist or visible fog is present with relative humidity reaching 100%. It shall be possible to operate the pulse-clean system without shutting down or reducing power output on the associated unit.

It should be ensured that any cooling system is properly designed so to provide the required cooling capacity at Reference Site Conditions.

At Project Take Over Stage, EPC to demonstrate clearly that reference site conditions and corrections are modelled and applied in a manner that shall satisfying the plant technical limits as per contractual obligation without any deration.

The selection and design of the air filtration system shall be such as to ensure that sand, water droplets, dust and other harmful contaminants in the ambient air be filtered out to a level which will not be detrimental to the life of the Gas Turbine unit under the most adverse ambient conditions at Site when the plant is operating continuously at maximum output, with the necessity for blade cleaning reduced to a minimum.

The number of access points and penetrations into the air inlet system for maintenance and inspection shall be minimized. Any door or hatch shall be capable of being securely air-tight locked, and interlocks shall be provided to prevent any attempted start of the Gas Turbine with any door or hatch not properly closed.

Particular attention shall be paid to the structural materials and paint used for the construction of the air filtration plant in view of the extensive temperature deviations and climatic conditions. Rain hoods shall be provided on any horizontal air filter inlets.

Silencing shall be provided as required in the inlet system in order to meet the specified noise limits.

The selection and design of air filtration system shall be such as to ensure that sand, water droplets, dust and other harmful contaminants in the ambient air be filtered to a level which will not be detrimental to the life of the gas turbine unit under the most adverse ambient conditions at Site when the Plant is operating continuously at maximum output with the necessity for blade cleaning reduced to a minimum.

10.2.7 Inlet Air Cooling / Fogging / Wet compression (if applicable)

The power augmentation (inlet air cooling systems or wet compression) to be provided for each Gas Turbine shall use suitable water to meet the requirements of the Gas Turbine supplier (if applicable). The water consumption by the power augmentation shall be guaranteed by the EPC Contractor. All equipment for this water supply system, e.g. pipework, valves, storage tanks and pumps (if necessary), drains, vents etc., shall be provided. Arrangements for remote monitoring and control of water quality (in particular dissolved solids) shall be provided. Temperature spread across the inlet air system shall be monitored in a single plane after each stage of cooling (Evaporative cooling / Fogging / Wet Compression). The layout of evaporative cooler piping shall facilitate leak identification and maintenance.

Temperature measurements will be provided in the following locations.

- two (2) numbers in the Ambient (upstream of evaporative cooler)
- three (3) numbers in the Downstream of evaporative cooler (upstream of wet compression)
- one (1) in the Downstream of wet compression, not used for any correction, but just for monitoring purpose

10.2.8 Fire/Gas Detection and Protection Systems

The entire Gas Turbine Generator installation, including all auxiliary enclosures, shall be protected from fire with a complete automatic fire protection system. The system shall be complete in all respects and shall include cylinders, cylinder racks, piping, valves, nozzles, detectors, fire alarm actuating system, and all other equipment necessary for the safe and efficient operation of the units.

The extinguishing agent shall be carbon dioxide (CO₂), or Clean Agent (Not CO₂), if comply to the local regulation and NFPA.

Hazardous gas detectors and fire detectors shall be provided in enclosure and shall be designed to pick up casing leaks.

The fire protection and detection system provided with the Gas Turbine shall interface and connect to the main Plant fire protection and detection system.

The distance between the pre-action sprinklers to the protected equipment shall be maintained as per manufacturer's recommendations.

10.2.9 Exhaust Ducting and Exhaust Stack

A series of test points shall be arranged in the stack at positions such that the distribution and temperature of the exhaust gas can be determined and the exhaust gas sampled in accordance with the agreed test code.

Gas tight access doors shall be provided to give access to ducts for inspection and maintenance. They shall be not less than 600 mm x 600 mm size. Interlocks shall be provided to prevent any attempted Gas Turbine start with the doors open.

The stacks shall be clad and be thermally insulated where required to protect personnel. The stacks shall be equipped with suitable earthing bosses and earthed.

A proven design of diverter flap shall be utilised so as to allow the changing of the direction of flow of the Gas Turbine exhaust gases, from going up the blast stack towards the heat recovery steam generators and vice versa.

10.2.10 Maintainability

The maintainability features incorporated in the design and construction of the Gas Turbine Generator units shall be described in the proposal. The following items shall be included:

- The formulae and numbers of charts showing the variation in recommended running hours between mandatory inspection/overhauls in relation to:
 - i. the number of emergencies trips and starts,
 - ii. firing temperature and
 - iii. fired hours on partial loads, base load and peak load.
- Equivalent hours formulae (if applicable) are to be supplied to relate the emergency trips, at various loads, and starts to the total hours between maintenance inspections.
- Identification of maintenance procedures indicating which can be performed within the enclosure, if provided, and which procedures are required to be performed with the enclosure removed.
- Details of the features of the Gas Turbine Generator unit, such as modular construction, that can reduce down-time and allow ease of maintenance, thus improving plant availability.
- The weight of individual major components, the lifting facilities required for removal, and whether the lifting facilities are part of the Gas Turbine Generator enclosure or if additional overhead lifting equipment is required. The necessary special lifting equipment shall be provided within the maintenance tools and equipment supplied.

10.2.11 Not used

10.3 Steam Turbines

10.3.1 General

The steam turbine shall be a proven design for the steam temperature and pressure capabilities of the proposed technology. The steam turbine high pressure (HP) section and intermediate/low pressure (IP/LP) section shall be designed for maximum efficiency and to guarantee the maximum reliability and performance.

Design shall ensure a long creep life, suitable for cyclic duty and fast start-up. Last stage blade materials and exhaust steam wetness shall be selected to minimize blade erosion and vibration. The ultimate and penultimate LP stages blade design shall have been proven in service.

Design and testing shall be according to international standards and include balancing and vibration tests.

The Steam Turbine shall conform to the requirements of IEC 60045-1 or EN60045-1 latest version available at the time of EPC Contract.

The steam turbine shall be provided with all TIL's incorporated into design and OEM shall provide the TIL's. The EPC Contractor shall implement all TILs until the end of warranty period. (TIL means (Technical Information Letter) provided by OEM for their specific machines.)

Steam turbine generator package located indoor consists of, but not limited to the following:

- Enclosures
- Turbine casing, rotor and blades
- Gland steam and sealing system
- Steam turbine associated piping, valves and fittings
- Automatic draining system
- Lubrication and control oil system
- Cooling systems
- Turning gear

- Three phase AC generator, including excitation and voltage regulation and all other required auxiliaries
- Synchronizing facilities
- Steam turbine related electrical facilities, including generator neutral and line connection facilities
- Steam turbine Instrumentation and control (I&C) system.
- Dual or Triple Pressure Stage turbine depending on the proposed design Emergency stop valves for live steam and HP steam complete, including the necessary blow-out provisions,
- Steam strainers for installation in the emergency stop valve or including housing for installation in the live steam and HP steam admission line,
- Hydraulically or pneumatic operated control valves at the inlet of the turbine,
- Motorised or hydraulic turning device, complete, with manual turning equipment and one
- (1) x 100% AC and one (1) x 100% DC motor
- Shaft sealing glands and gland steam system complete with gland steam condenser,
- Safety relief valve in the turbine exhaust line,
- Turbine drain system, complete,
- HP steam infeed line, complete, including stop and control valves,
- Turbine exhaust steam line (to supply duct air cooled condenser),
- Cooling system in LP section to prevent overheating at low load, e.g. spray system, if necessary,
- Complete noise and heat insulation,
- Turbine enclosure, complete, as required,
- Fire protection equipment,
- All connecting piping, bearings, fittings, safety devices, fastenings, etc.,
- Static and dynamic calculation of the complete foundations, common for steam turbine and generator, including deflection calculation due to occurring forces and moments,
- Extensive vibration monitoring facilities for a continuous diagnosis of the rotating parts and bearings of the turbine, and
- Necessary supporting steelworks, access walkways, ladders, service platforms for the complete steam turbine generators area, also covering all auxiliary equipment.

10.3.2 Condensing Steam Turbines

Due to the scarcity of water, solutions involving wet cooling towers are not feasible and air-cooled condensers will be the basis for steam turbine condensers. The steam turbine exhaust and air-cooled condenser configuration shall be as per manufacturer's standard design. The steam turbines shall be complete with and directly coupled to 60Hz generators. Steam turbines shall be provided for the Plant as defined in this OTS and shall be located indoors in a ventilated atmosphere.

The turbine shall be a reheat, multiple pressure condensing type using an air-cooled condenser and be suitable for direct coupling to a two-pole generator for power generation at 60 Hz. Designs for impulse or reaction blading, or for combinations thereof, are equally acceptable.

The steam admission to the HP turbine section shall be via suitable stop and control valves. The steam turbine governor shall be capable of controlling the steam turbine to maintain stable conditions.

10.3.3 Common Turbine Requirements

The steam turbine shall be capable of withstanding partial and full load rejection without an overspeed trip and reverse power trip.

The steam turbine shall be protected against water ingress. The recommended practices stated in ANSI/ASME TDP-1- 2013 shall be provided as a minimum.

Vacuum pumps shall be provided for air extraction from the steam turbine condenser.

Steam turbine bypass provision shall be provided by means of a pressure reduction and desuperheating station between each HRSG superheater outlet and the steam turbine condenser. To facilitate start-up of the Plant, to reduce the impact on steam turbine trips and to improve operational flexibility, e.g. during rapid load changes, steam bypass lines for HP, IP and LP steam shall be installed from each header to the cold reheater or condenser as applicable. Each bypass line shall be designed to handle the full steam flow at all operating points. The bypass shall be automatically initiated in response to high HP steam pressure or as part of an automated steam turbine start-up sequence.

The STG unit shall be able to continue operation when one of the GTs trips.

Inspection of highly stressed and back-end ST stages must be possible through adequate boroscope openings without opening the casing.

In order to protect the ST from corrosion during standstill periods, one dehumidification unit shall be provided. This dry preservation unit shall blow heated dry air through pre-determined openings to various sections of the ST and the condenser. The drying unit shall have adequate performance to reduce the moisture content of the ST hall suction air to the requirement of STG/Condenser OEM requirements. Automatic control by a hydrostat shall be provided.

A turning gear (automatically engaged and disengaged type) shall be provided to rotate the turbine rotor slowly prior to start-up and after the turbine is shutdown to allow unified warm-up or cooling, maintain eccentricity and to prevent thermal distortion of the rotor. The turning gear pinion shall be engaged only after the rotors have come to rest as indicated by the zero speed indicator and observation.

A pressure switch shall be provided in the turning gear circuit so that loss of bearing oil pressure shall automatically stop the turning gear motor.

- Manual hand turning can be applicable,
- AC Turning Motor shall be supplied

10.3.4 Blading

The blading shall be specially constructed to avoid the possibility of damage from vibration under any condition of running, including testing overspeed trips. The blading shall further be constructed to withstand the high temperature shocks met with in practice and to resist deterioration through contact with the steam conditions expected during Plant service life.

All blading shall be of proven design and shall feature the following blade strength characteristics:

- For rotating blades with a natural frequency of less than seven (7) harmonics of rotating speed (420 Hz), tuned blades shall be applied which offer sufficient deviation from the resonant frequency;
- For rotating blades with a natural frequency of greater than seven (7) harmonics, un-tuned blades may be proposed but the Contractor shall use its operating experience to ensure that such blades will have sufficient strength margin to withstand resonant vibrations.

Notwithstanding the above, equipment shall comply with any electrical Network Connection Conditions in relation to maintaining operation during network frequency excursions.

The design of blade and its root fixing as well as the material of construction shall have been used in similar applications and at similar operating conditions for a period of not less than 5 years commercial operation without serious incident.

Blading material and design shall be selected with regard to the operating conditions during all normal operation at full load and part load.

HP and IP Blading: The HP and IP stages shall consist of high-performance blades. These blades shall be designed using the latest flow analysis around the blade. The blades shall be of ISB, or integral shroud blade, i.e. shrouds and blade profile shall be manufactured from one piece, not separately manufactured and assembled.

10.3.5 Turbine Valves

The steam stop and throttle valves shall be constructed of material chosen to resist deterioration due to high steam temperatures and erosion. The valves shall be of the "fail safe" type, arranged to close under spring force and to require hydraulic pressure, under governor control, to open. The main stop valves shall be fitted with internal bypass valves or pilot valves to permit them to be opened against full steam pressure. Provision shall be made for exercising all stop valves and throttle valves with the turbine on full load.

All main inlet and control valves shall be provided with facilities for on-load testing. Positions for all valves, steam pressures and temperatures shall be brought into the DCS.

The turbine shall have two HP stop and control valves and two IP stop and control valves, HP valves shall be combined in a common casing valve body directly connected to the turbine casing and, driven by hydraulic servomechanism, IP stop and control valves shall be as per OEM design.

The steam strainer cylindrical in shape fits around the valve.

Control valve:

- The control valve with its tubular valve disk and integral valve stem slides in the bushing of the valve cover. Balancing holes in the valve disk reduce the operating force required to position the valve.

Actuation:

- The control valves and the stop valves each have dedicated electrohydraulic actuators which opens the valves against the force of a spring with the hydraulic force of the actuator. The valves are closed by spring force in the event of a power failure.
- The Steam Turbine shall be provided with fast cooling system (forced air cooling system) on the main steam control valve and on the reheat steam control valve. The logic to operate the system shall also be implemented. The system shall be designed to achieve a cool down time of less than 3 days.
- Steam turbine inlet and control valves, which are stated shall be as close as possible to the Turbine casing in order not to leave any piping which cannot be flushed, cleaned with acid and blown out.

LP Stop Valve:

- The turbine shall have one LP stop valve subject to detailed engineering shall be driven by on-off control of hydraulic servomotor.
- The turbine shall have one LP control valve which is controlled by individual hydraulic servomotor. LP stop and control valve shall be of identical design.

On Load Testing of Valves:

- Valve closing test function during on-load operation shall be provided in order to confirm the valve free from sticking, therefore full closing test shall be carried out for all valves. The valves will be tested one by one and depending on valve-sets per turbine part with full or part stroke test.

10.3.6 Steam Strainers

In order to protect the steam turbines from foreign object damage, the Contractor shall undertake boiler steam blowing with agreed target plates and install temporary steam strainers at the stop valves which shall prevent that passage of foreign objects larger than 4mm during commissioning. Following commissioning, the temporary steam strainers shall be removed by the Contractor and the stop valves shall be fitted with permanent steam strainers designed as an integral part of the turbine stop valves. Such permanent strainers shall prevent the passage of foreign objects larger than 7mm while ensuring that the valve pressure loss is not excessive. Alternatively a dynamic separator shall be installed on the elbow of the main steam(HP/IP) line in front of main valves, to remove particles and contaminants by inertial forces.

Alternatively, radial flow blade type strainers with a maximum slot width of 0,762 mm (that can withstand high differential pressure and in normal operation has a small pressure drop due to the diffuser action) may be used during commissioning and as permanent protection after suitable cleaning.

10.3.7 Lubricating Oil System

The lubricating oil system shall provide cooled and filtered oil under pressure to all of the bearings of the turbine and generator. The system shall comprise as a minimum:

- an oil reservoir,
- main and standby full duty oil pumps,
- 2x100% AC motor driven turning oil pump sufficient for extended barring and an emergency DC-driven oil pump. The main oil pump may be shaft driven or motor driven. The standby full duty pump shall be AC motor driven.
- Sufficient Heat exchanger (minimum 2X100%) (to be designed and implemented according to study results, meeting OEM recommendations for maintain lube oil cooling and serving all worst case conditions)

Lubricating oil pressure shall be automatically controlled either actively or passively.

Low lubricating oil pressure shall start the standby pumps as appropriate or, if a minimum adequate pressure is not maintained, shall trip the turbine.

Duplex filters and coolers shall be provided to allow on-line changeover and cleaning of standby equipment.

Oil temperature leaving the coolers shall be automatically controlled.

The oil reservoir shall be supplied with two 100% duty exhaust fans to maintain a depression in the oil return pipework and bearing pedestals. The exhaust from the fans shall be ducted to a safe location outside the building.

Oil leakages shall be contained and prevented from entering the surface water drainage system.

The lubricating oil system shall be common for the steam turbine and the generator. Oil piping from oil strainer/filter to bearings should be Stainless Steel material, oil piping from bearings to lube oil tank should be Carbon Steel material. To eliminate galvanic corrosion between dissimilar materials, EPC contractor should consider special gaskets, bolts and nuts for such connection as applicable.

It is required that lubrication oil supply system and control oil supply system shall be separated. Control oil system and lubrication oil system shall be supplied from separate tanks.

All oil systems shall be flushed after the completion of installation. The oil is circulated from the tank, through the associated pipe systems and coolers to the lube oil system of the steam turbine and back to the tank again. The process is performed in accordance with the prescribed terms of pressure and flow rate. Bearings are to be bypassed during the lube oil flushing by installing a temporary hose at the return line of the lube oil to the tank. The cleanliness level of the oil is analysed by using a comparison contamination kit as well as an electric particle counter measuring instrument. The oil is to meet the specified criteria for optimal system cleanliness.

10.3.8 Lubricating Oil Purifier Separator

Mobile oil purifier-separator of the centrifugal type shall be provided for the Plant. It shall be easily connectable to each of the oil reservoirs as required using permanent connecting points for supply and return flows as well as drain and or waste oil flows. The oil purifier-separator shall be capable of removing varnish. EPC Contractor is free to propose one mobile oil purifier-separator for 2 GTs and one mobile oil purifier-separator for 2 STs.

10.3.9 Control Oil System

A control oil system shall be supplied to control and actuate the main steam turbine inlet valves. Such a system may use the same oil and the lubricating oil system and, if so, no separate control oil reservoir, pumps and coolers will be necessary. If a separate system is proposed, however, the control oil system may use either mineral oil identical to the lubricating oil or synthetic fire resistant fluid and shall have a dedicated reservoir and dedicated pumps, coolers and filters.

The pressure and temperature of the oil shall be automatically controlled. Failure of the control oil system pressure shall automatically trip the turbine.

10.3.10 Jacking Oil System

A jacking oil system shall be provided to facilitate turning of the rotor for cooling down and during maintenance activities. If required, it is preferable that the jacking oil system should only be required to assist starting of the barring motor and that normal barring speed should be high enough to establish hydrodynamic lubrication at the bearings using the main or auxiliary oil pumps.

10.3.11 Barring Gear

There shall be a motor driven barring gear capable of turning the turbine and generator rotors continuously at a high enough speed to establish full hydrodynamic lubrication.

The barring gear shall be capable of automatic and remote-manual engagement, disengagement, starting and stopping. Interlocks shall be provided to prevent starting and/or engagement if the turbine rotor is turning or if there is insufficient lubricating oil pressure.

An emergency facility shall be provided for turning the turbine rotor by hand with the aid of a hand wheel or, if applicable, jacking oil.

10.3.12 Gland Steam System

The turbine shall be supplied with a fully automatic gland sealing system complete with all condensers, pipework, valves and control equipment. All remotely operated isolation and bypass valves required for remote switching of gland steam sources shall be provided. The system shall ensure that the rotor glands are sealed against air ingress and steam leakage to atmosphere at all turbine loads.

The system shall be self-sufficient in steam (i.e. it will not require constant make-up) throughout the majority of the load range.

The system shall comprise as a minimum:

- Gland steam make-up valve for low load operation
- Gland steam regulating/dump valve
- Gland steam receiver
- Gland steam condenser
- 2 x 100% condenser exhaust fans

Intermediate gland leak-offs shall be returned to an appropriate stage of the turbine.

The outermost gland leak-off shall be connected to the gland steam condenser, which shall be held under a slight depression by the exhaust fan. The discharge of the fan shall be directed to a safe location outside the turbine enclosure.

10.3.13 Turbine Bearings

The turbine and generator rotors shall be supported in a conventional two or four journal bearing arrangement with a solid coupling and a single thrust bearing. Sleeve-type or tilting-pad type bearings may be used.

The bearings shall provide adequate damping to prevent excessive resonance at critical speeds.

Main journal bearings shall be of the plain bore white-metal type, with forced lubrication. Alternative designs featuring 'lemon bore' or 'offset halves' may be considered. Bearings for small auxiliary drives such as governor, oil pump, barring gear, may be of the rolling element type.

Journal bearings shall be horizontally split and mounted on spherical pads within spherical housings to facilitate alignment between bearing and journal.

Supply of jacking oil to the journal bearings shall be made.

Each bearing shall have thermocouples to measure the temperature of the bearing metal. Each bearing shall also be supplied with at least two orthogonally mounted vibration monitors of the electronic proximity type to continuously monitor shaft radial vibration. Bearing metal temperature shall be monitored locally and remotely in DCS.

Thrust bearings shall be of the horizontally split, tilting pad type and shall incorporate main thrust and reverse thrust bearings. At least two rotor axial position monitors of the electronic proximity type shall be provided to continuously monitor shaft axial position.

Special attention shall be given to the design of bearings and supports, and to the accurate dynamic balance of the rotating parts of the turbines and generators, to minimise vibrations. The rotors shall be statically and dynamically balanced in the factory so that the maximum effective vibration magnitudes do not exceed the recommended values under steady running conditions at all loads. Means shall be provided so that, if necessary, adjustments to the balance may readily be made. Vibration limits shall comply with vibration criteria as per guideline ISO 20816-2. Both bearing and shaft vibrations shall meet requirements of Zone A. The acceptance criteria for the inspections performed during the component manufacturing must be defined before the start of the works and must be submitted to the Owner for approval.

10.3.14 LP Spray Cooling

If necessary, the design of the LP turbine shall include spray cooling to maintain acceptable last stage blade temperatures when operating at part-load conditions.

Automatic sprays shall be provided to prevent high exhaust temperature of LP turbine. When exhaust hood spray valve is opened, water from the condensate pump shall discharge to spray nozzles installed in the turbine exhaust hood chambers. Overheating of the exhaust end shall be avoided even with no load steam and full vacuum. The exhaust hood spray shall be set to control exhaust hood temperature to 90°C maximum. The exhaust hood spray will be deactivated when exhaust hood temperature falls below 60°C.

If power increase does not lead to fulfilment of the criterion ($T_{Le-0} < 100^{\circ}\text{C} \wedge T_{\text{exit steam}} < 60^{\circ}\text{C}$), the exhaust hood spray will be deactivated when the power level is greater 20% of the rated power.

10.3.15 Protective Devices

All protection and control systems shall be 100% redundant including all unit protection systems. The details shall be described in detail engineering stage with Owner's Approval.

At least the following turbine protection devices shall be included in the supply:

- Local hand trip
- Remote emergency push button
- Low bearing oil pressure trip
- Over speed protection system (duplex)
- High back pressure
- High exhaust temperature
- Differential expansion between rotor and casing

- Vibration sensor and monitoring devices

Arrangements shall be provided for on-load testing of these devices.

10.4 Condensing Systems

The condensate systems shall supply condensate from condenser to the deaerator. The system shall include condenser, vacuum pumps, condensate extraction pumps, control valves, pipe work, etc.

10.4.1 Steam Turbine Condensers

Air cooled condenser(s) will be required to let down steam from the condensing part of the steam turbine and from the LP-steam headers.

The system shall comprise at least the following equipment:

- steam duct from the steam reducing stations to condenser and steam distribution ducting,
- steam duct from turbine exhaust flange to condenser and steam distribution ducting,
- condensate collection headers,
- air moving system,
- fan drives,
- steel support structure, including maintenance platforms, access stairways and escape ladder,
- facilities for prevention of air recirculation,
- noise attenuation features/devices as required,
- all required electrical and control equipment for safe operation, monitoring and measuring, earthing and lightning protection,
- air evacuation pumps, if applicable,
- ejector condenser, drain tank and strainers at ejector steam inlets,
- condensate tank and drain,
- drain and vent pumping, piping, valves and traps with supports, and
- semi-automatic ACC outside heat transfer surface cleaning system.

The steam turbine condenser shall be complete in every respect for reliable, efficient and safe operation for a high service lifetime. The design of the air-cooled condenser shall be in accordance with the ambient conditions at Project Site and steam turbine requirements.

The condenser shall be sized to receive:

- Condense all steam for all operating ranges
- Accept the full steam turbine bypass steam when the steam turbine trips.

The capacity of the steam turbine condenser shall include a reserve to cover possible steam turbine overloads.

10.4.1.1 Air moving system

The cooling air to each heat exchanging cell shall be forced by individual axial flow fan; each fan, whose blades shall be adjustable when the fan is out of service, shall be driven by a variable speed motor.

All the components of the fan/drive unit shall be designed for continuous outdoor service over the range of air condenser loads, ambient air temperatures and air temperatures adjacent to the tube bundles.

It shall be possible to initiate a cold start of the fans during any ambient air temperature within the specified range.

All drive components shall be rated to take into account any shock loading that may occur from starting a fan which may be rotating in a reverse direction under the influence of prevailing winds or convection currents. As an alternative, a reverse rotation lock may be provided.

Each fan drive shall be individually remotely controlled from the Control Room allowing the operator to optimize the operative configuration on the basis of actual ambient condition.

10.4.1.2 ACC cleaning system

It shall be possible to clean the outside heat transfer surfaces by application of well established cleaning procedures. A semi-automatic high-pressure spray-water cleaning system shall be supplied for periodically cleaning the heat exchanger external surfaces during normal operation, keeping them free of deposits and maintaining the heat transfer coefficient of the condenser. It shall be possible for this cleaning system to be used whilst the ACC is in operation.

10.4.2 Air Extraction System

Inert gases shall be evacuated by the air extraction system. The system for air removal from the steam side of the condenser may consist of steam ejector system or water ring vacuum pump system for holding operation (2x100% units for any system). For hogging prior to start-up of the ST a water-ring vacuum pump system with one stand-by pump (e.g 4 x 33% or 5 x 25%) for parallel operation shall be applied, in case steam ejectors are provided, 1x100% steam jet air ejector can be used provided that the time required for reaching the vacuum is less than 30 minutes. If water ring pump system is selected, EPC Contractor can propose to be used as common system for holding and hogging.

ST condenser air extraction system discharge flow indicator to provide to know the air ingress quantity.

10.4.3 Condenser Extraction Pumps

3x50% or 2x100% duty centrifugal pumps, complete with motors, couplings, guards, suction barrel, all necessary steel supporting structure, cooling and lubrication systems with mechanical sealing shall be provided for each steam turbine. The CEP type (horizontal or vertical) shall be selected also considering condensate storage tank water level (height).

The pump design point shall be selected in order to cope with all the possible operating conditions.

The standby pump is intended as back-up (automatic start in case of failure of the operating pumps). In case of turbine bypass the third condenser extraction pump may need to come into operation.

The pump casing shall be arranged within the suction barrel, which is submerged in the foundations. The suction and discharge pump nozzles shall be located above the floor level and arranged so that the pump can be easily withdrawn after pipe branches have been disconnected.

The pump driving motor shall be air cooled and be flexibly coupled to the pump shaft. All bearings shall be of the removable type to permit easy replacement at site.

Adjustable rigid coupling shall be used instead of split rigid coupling.

Pumps may have either packing or mechanical seals, as determined by the application.

The pumps shall work satisfactorily and free of cavitation under all operating conditions. The available NPSH at the minimum level of the condenser hotwell and with maximum head losses shall be higher than the required NPSH at all loads. The water velocity in the independent pump suction pipework shall not exceed 1 m/s.

The characteristic curve (flow/head) will constantly decrease maintaining a uniform slope from the point of zero flow to the design point. All the identical pumps shall have the same characteristic curve.

Pump must be equipped with the necessary instruments (suction and discharge side manometer, thermometer etc.) For condensate extraction pump, modulating automatic recirculation control valves or combined recirculation/check valves will be used.

The pump impellor shall be made of a suitable grade of stainless steel with high resistance to pitting and erosion., The impellor shall be balanced statically and dynamically and be secured to the pump shaft in a manner, which permits easy maintenance and replacement at site.

Interchangeable stainless steel protecting sleeves shall be provided at the shaft seals and at the lower journal bearing.

The pump casing material shall be cast steel. The suction barrel material shall be carbon steel. A rustproof protective coating shall be applied to the surfaces of the barrel.

Pump characteristic curves and all relevant data for each of the specified operating conditions shall be submitted as well as the arrangement and foundation drawings.

All necessary auxiliaries shall be supplied for the pumps as applicable (e.g. Associated pipework and valves). For each pump a stop valve shall be provided on the suction and a non-return valve and motor operated valve on the pump discharge.

A strainer with large surface area and low-pressure losses shall be provided in the suction line of each pump.

Venting of the upper parts of each pump shall be realized by suitable connections to the main condenser.

A complete water sealing system shall be provided to seal the stuffing boxes and all parts subjected to vacuum at start-up, during standby operation and/or during normal operation. The sealing water shall be taken from the pump delivery.

A minimum flow recirculating system, common for the two or three pumps and equipped with built in cascade throttling shall be provided.

Sampling and chemical dosing points will be provided on condensate system (downstream of condensate extraction pumps).

10.5 Make-up, Deaerators and Feedwater Systems

10.5.1 System Description and Introduction

Each feed system shall be furnished with all necessary equipment required for manual and automatic control from the Central Control Room during the operations associated with Start-up, synchronising, loading and shut down of the GT/HRSG sets. Each feed system shall be designed as appropriate as to the function with either Power Block in case multiple blocks are provided. The design shall also take due consideration of all possible operating scenarios of the Plant. All materials in the feed water system shall be ferrous.

10.5.2 Make-up Water and HRSG Filling

The demineralised water storage tanks shall feed 2 x 100% make-up water supply pumps for initial filling of HRSGs, deaeration facilities, condensers and makeup to the condensate water storage tanks. The Make-up water during plant operation shall then be provided from the condensate water storage tanks by 2x 100% condensate transfer pumps.

10.5.3 Deaerators

The design, manufacture and construction of the deaerators shall be in accordance with recognised pressure vessel codes. The deaerators shall be designed to withstand full vacuum.

The deaerators shall be of the spray-tray type (although a purely spray type will be considered) and shall consist of a deaerator vessel, water storage tank, pipes, valves, ancillary equipment, instruments and controls necessary to maintain operation under all normal conditions. Equipment shall be provided to maintain the minimum deaerator pressure at all steam turbine loads.

The deaerators shall be provided with sufficient connections to accommodate feed pump suction and recirculation pipes, condensate return, LP steam heating and instrumentation. The deaerators shall be located at high level such that the feed pump NPSH requirements are satisfied.

The deaerators shall be designed to reduce the oxygen content of the feed water to not greater than 0.007mg/l, under all operating conditions above 10% feed flow, as measured by the Heat Exchanger Institute method.

The deaerated water storage capacity shall be designed on the basis of at least 10 minutes

In case of integrated deaerator type HRSG, each drum shall be 2 mins for HP drum, 5 mins for IP drum and 10 mins for LP drum from normal to low low water trip levels with feed water flow at full load; the actual time shall be agreed during the detailed design phase of the Project.

The deaerator parts including spray nozzles and splash trays shall be of stainless steel.

The EPC Contractor is free to propose the deaerator integrated with LP drum.

10.5.4 HRSG Feed water Pumps

HRSG will have natural circulation and horizontal flow type. Each pump suction pipe from the deaerators shall be designed to avoid pump cavitation under all conditions and shall adhere to applicable international code and shall include valve connections for chemical dosing as necessary.

Only proven feedwater pump designs at similar duties shall be proposed.

Minimum flow recirculation bypass line and control shall be provided.

Each HRSG shall be provided with 3x50% or 2x100% horizontal centrifugal electrically driven boiler feed pumps. Feed water pumps shall provide feed water at controlled rates via split control valves as detailed below:

- A low load valve – used during start up and low loads.
- A high load valve – used for higher load

The pump total head at rated capacity shall include a margin for wear in addition to normal design margins.

The characteristics of the pumps shall be such that the closed valve head is approximately 120% of the pressure at rated output.

The minimum flow lines and relief lines (if required) shall run individually to the deaerator or to the feedwater storage tank. The minimum flow valves shall be provided for the stable start/stop of pump while protecting the pump. For these purposes all of design conditions of the valve including actuator type shall be decided. The minimum flow capacity shall be as required by the feedwater pump vendor. The minimum flow shall be checked via control valve and monitored on DCS.

It shall be possible to start the pumps safely from cold without prior warming up or other preparatory measures.

Pumps shall be equipped with mechanical seals, single balanced type. Gland seals shall be supplied for each main pump. Seals flushing shall be provided with double magnetic separator, if applicable. The pump glands shall be arranged in such a way that the removal of the packing can be carried out with minimum disturbance.

The feedwater pumps shall be equipped with bearing temperature and vibration monitoring devices.

The discharge connection of the main pump shall be of the flanged end type.

A basket type strainer shall be fitted at pump inlet, fine mesh baskets shall be fitted for initial commissioning.

Vibration and temperature monitoring and protection system shall be installed.

The whole pumping unit shall be capable of withstanding, without detriment, reverse rotation to a speed that would occur if the pump were to stop when the differential head was at a maximum and the delivery and/or non-return valve failed to close (this would typically occur if one trips and the discharge valve remain open, while the other pump maintains operation). If the pump cannot withstand this scenario the pump must incorporate a safety system to avoid this from happening.

Pump must be equipped with the necessary instruments (suction and discharge side manometer, thermometer etc.)

Couplings and intermediate shafting will be guarded. Bedplates will be of ample proportions and stiffness to withstand the loads likely to be experienced in shipment and service.

The weight of the impeller and shafting for vertically suspended pumps will be supported by the motor thrust bearing.

Pumps that have mechanical seals will be arranged to facilitate seal removal.

Pumps may have either packing or mechanical seals, as determined by the application.

The characteristic curve (flow/head) will constantly decrease maintaining a uniform slope from the point of zero flow to the design point. All the identical pumps shall have the same characteristic curve.

The Contractor shall propose a suitable number of feed pumps having regard for availability and reliability requirements and shall be supplied with all necessary auxiliary equipment. Provided that the Contractor's proposed arrangement permits boiler drum levels to be adequately controlled by main and start-up, control valves installed at the inlet of each drum, the HRSG feed water pumps may be driven by constant speed electric motors or Variable Frequency Drive (VFD)/hydraulic coupling. All pumps shall be capable of operation either singly or in parallel whilst remaining inherently stable.

The rotating components shall be balanced to ensure minimum vibration throughout the entire speed and load range. The characteristics of the pumps shall be such that the closed valve head is approximately 120% of the pressure at rated output.

10.5.5 Auxiliary Boiler (if required)

Depending on EPC Contractor design, auxiliary boiler(s) with natural gas or back-up fuel oil might be required to provide auxiliary steam during the start-up phase of the ST.

The auxiliary boiler(s) shall be of the flame/smoke tube or water tube type.

The auxiliary boiler(s) shall have the characteristic of frequent start-up, long periods between maintenance, start-up at any time and fully automatically controlled, operated and monitored via DCMS. The cold start-up shall also be initiated and monitored by the CCR operator without any intervention from a local operator.

The entire pressure part shall rest on a steel frame in such a way that free expansion in all directions is possible. The large water volume shall enable the boiler to adapt well to extreme load fluctuation.

The outer lining of the insulation shall be made by galvanized steel sheets.

Boiler protection and burner management shall be in accordance with NFPA 85 or EN 12953-7 and EN 267.

The steam pressure at the auxiliary boiler outlet shall be a result of boiler manufacturer preoptimization considering boiler size, steam temperature and boiler capacity. The related steam pressure shall depend on the steam-cycle optimization, which would be the minimum steam pressure.

Demineralized water shall be used as boiler make-up water which is supplied from the demineralization water plant.

Because of characteristic difference between start-up boiler and normal boiler, the water of this boiler can be drained completely and facilities for conservation shall be provided.

The platform and gallery of boiler shall possess enough strength and stiffness, and stairs shall be provided at operation, monitoring or maintenance locations.

The boiler shall be provided with the necessary observation ports, manhole for maintenance and explosion vents.

The boiler shall be provided with safety valve, vent valve, exhaust piping and a motor-driven valve, silencers as necessary, etc.

Connections for measuring of parameter of steam, water, combustion fuel shall be provided. Local instruments and remote instruments (in the control room) shall be used for boiler water level measuring, and the pressure, temperature and flow rate shall have indications in the control room.

An emission monitoring system shall be provided for the auxiliary boiler stack.

The technical requirements specified above for the steam generators shall apply also for the auxiliary boiler(s) as far as applicable.

ASME stamp and NB registration is not applicable. Heat Recovery Steam Generator.

10.6 Heat Recovery Steam Generator

10.6.1 General

The steam generators shall be of proven design and be designed, manufactured, constructed and tested in accordance with relevant internationally accepted standards, e.g. ASME as well as the latest editions of all applicable local and national laws, codes and regulations in effect at the time of contract award.

The HRSG shall be an outdoor, fully weatherproof clad, with cladding/insulation where necessary. The main pressure parts shall be arranged in the form of a self-contained construction and shall be of the self-draining type.

The top (or side in the case of vertical HRSGs) galleries of each HRSG shall be provided with partial sun and weather canopies (including over the HP drum) to protect Plant operation and maintenance personnel from direct exposure to the sun. Instruments and electric panels shall also be provided with local suitable sunshades.

The HRSG shall be suitable for automatic start up, shutdown and normal operation from the Central Control Room.

The HRSG (and the steam turbine) shall be capable of sliding pressure operation to a minimum operating pressure that may be maintained in continuous operation.

The water side of each HRSG shall be cleaned after completion of erection by alkaline boiling-out with suitable chemicals in accordance with accepted chemical cleaning procedures. With respect to HRSG cleaning, the Contractor shall also provide in its offer an option price for chemical/acid cleaning in accordance with accepted chemical cleaning procedures for the consideration of the Investor.

HRSG pressure parts exposed to exhaust gas in the cold section of the HRSG shall be selected from materials to achieve a minimum design life as per the applicable international standard/code. The HRSG design shall be such that the temperature of water entering any heat transfer section shall be above the exhaust gas dew point or acid dew point as the case may be.

Walkways, ladders, stairs and platforms shall be provided wherever necessary for proper operation, inspection and maintenance. Elevator shall be provided to access top area. Interconnecting platform between HRSG Units shall be provided if only one elevator is being provided for multiple HRSG units.

Design of the HRSG shall be based on factory built modules or HARP type with as much equipment as practicable factory assembled. HRSG shall be made up vertical tube bundles which shall be top supported and attached to the steel frames and horizontal gas flow.

In the design phase of the HRSG, particular attention shall be paid to the exhaust-gas side pressure loss of the HRSG and of the stack and shall be minimised in order to realise a high efficiency factor of the gas turbine.

Each element and tube bank must contain tubes on a horizontal and vertical pitch, adequately supported and of suitable depth to enable easy cleaning. Any fins used must be solidly attached to the tube throughout the fin length.

Temperature measurements shall be provided in the transition ducting between GT and HRSG at different elevations to monitor temperature profile across the first row of tubes.

A chemical dosing system shall be provided in order to maintain the proper boiler feed water chemistry and a steam sampling system in order to monitor the steam quality.

No drain pipes or venting pipes shall be concealed in the boiler insulation.

To be able to reach the upper headers of modules, manholes and/or handholes shall be installed at the top of HRSG. Access doors between modules with a coverable manway to the bottom section (location of bottom headers and piping) to be included.

Steam drums, headers and major HRSG pressure parts shall be visually, radiographically and ultrasonically checked in accordance with the ASME codes for weld defects during all stages of manufacture and erection. The Contractor shall also undertake visual and random magnetic particle inspection (MPI) checks of small

items (e.g. stub/nozzle to header connections). As noted earlier, the Contractor shall agree with the Owner, or his nominated representatives, an inspection and testing witness and hold point programme for equipment or any other tests identified by the Owner or its representatives as requiring inspection and such a programme shall include provisions to ensure that the Owner or his nominated representatives are given adequate prior notice to attend the agreed inspections and tests. The design of each HRSG shall be such as to permit adequate access for welding and to ensure that the effective examination of all site welds is possible during fabrication and erection as well as later into the service life of the Plant.

HRSG may have supplementary firing (SF), if required. The additional power capacity achieved by SF must not exceed 10% of the Contracted Capacity.

Each completed boiler shall be subject to a hydraulic test using demineralised water and in accordance with procedures defined by the design codes and the insurers inspectors.

Design life criteria of 25 years (equivalent to greater than 200,000 hours) alternatively, design life can be done considering 300 cold starts and 1100 warm starts shall be applied to materials operating in the creep region, the most stringent requirement shall apply. The HRSG design shall allow for appropriate inspection, cleaning and repair access of pressure parts and all auxiliaries. The number of rows on single header will be selected as per supplier experience on similar HRSG applications and confirmed by fatigue and stress analysis.

All necessary installations for conservation during outages of the HRSG shall be included. For short periods wet conservation shall be used.

Variations in the temperature of the steam generator exhaust gas and the potentially associated corrosive environmental conditions shall be considered in the design of the extended surface tubing and back-end facilities such as LP economizers and preheaters which shall include for recirculation and bypassing and adequate hot/cold end corrosion protections (no damage for Acid Dew Point).

Design to consider and provide good accessibility for the inspection and repair of welding seams between individual pipe bundles and between heat exchanger pipes and collectors. Access openings and manhole will be provided at bottom on both side casing and top casing for each module.

HRSG design to provide the exhaust-gas side pressure loss of HRSG and of the stack shall be paid particular attention and shall be minimised in order to realise a high efficiency factor of the gas turbine.

A steam turbine bypass system shall be provided for each HRSG for three of pressure sections to allow fast and independent start-up and shutdown. The safety valves shall be provided for each pressure level as required by the ASME code.

The heat recovery steam generators shall be completely drainable and shall be designed so that in case of tube or bend failure, replacement of tube shall be performed without dismantling major components according to the manufacturer's repair standards.

Pipe wall thickness shall be dimensioned in accordance with DIN EN 12952 or equivalent ASME code. In this process, the Contractor shall select the next higher wall thickness from the series of standards.

The HRSGs shall be designed:

- for maximum utilization of the heat content of the gas turbine exhaust gases,
- for triple pressure steam generation and

HRSG stack shall have Continuous Emission Monitoring Systems (CEMS) and related instruments.

The HP temperature at the HRSGs' outlet shall be controlled (defined by the contractor) for optimized power generation.

The HRSG's blow down rate shall be adapted accordingly so that the guaranteed steam quality will be met by the contractor.

All drums and pressure stages of HRSG shall be equipped with a BSE system (Boiler Stress Evaluator). The BSE system shall be connected to the DCS in order to visualise the stress data for the main components (such as HP, IP, LP drums, superheater/reheater collectors, LP collectors etc.). These data serve to

determine the remaining life cycle of the critical components and the optimization of the ramp functions for start-up and/or shut-down as well as for load change processes.

Performance Data:

The Contractor shall provide a HRSG performance prediction computer model program specific for this project. This program shall be capable of running on a personal computer with the latest version of Windows operating system. The performance prediction program shall have the following inputs, and outputs, as a minimum:

Inputs:

- Gas turbine exhaust flow rate, temperature, and composition.
- Terminal steam pressures for HP steam, Hot Reheat steam, and LP steam.
- The temperature of the water being sent to the HRSG.
- The temperature of the Cold Reheat at the terminal point.

Outputs:

- Steam flow, steam temperature, and steam pressure at the terminal points for HP steam, Hot Reheat steam, and LP steam.
- Steam flow, steam temperature, and steam pressure of the IP steam.
- Stack temperature.
- HRSG efficiency (according to ASME code, calculation based on temperature differences shall not be accepted)
- Gas side static pressure drop.

The HRSG shall be foreseen with temperature measurements for both water-steam and gas side before and after each heat exchanger in order to facilitate the mapping of its performances.

Steam and Water Analysis System (SWAS) shall be installed for each HRSG separately, the SWAS system shall be designed for continuous monitoring and control of water and steam purity in the HRSG.

Heat recovery steam generator shall be provided separately for each gas turbine generator, which consists of following systems and components, but not limited to the following;

- Gas turbine exhaust gas duct system
- Pressure parts and steam drums
- Insulation and cladding
- HRSG associated piping, valves and fittings
- Vent and drain systems
- Blow-down system
- Flue gas ducts
- Main stack
- HRSG related electrical facilities
- HRSG related Instrumentation and control (I&C) system
- HRSG steel structures and platforms
- Provision of Online Heat Flux measurement

10.6.2 HRSG Drums and Headers

The steam collecting drums (long, slim design) shall be designed in such a way that they withstand high temperature gradients resulting from fast load changes. Greatest care shall be given in thermal and mechanical design of the drums.

The drums shall be fusion welded and provided with all necessary internal distributors, baffles, shields, and internal piping.

Each HRSG evaporative section shall be provided with steam drum(s), each of adequate size to ensure stable operation under all normal load conditions and other single item trip conditions. The steam drums shall be sized to enable start up without operating any drain valves, except the one at the outlet of the superheater. Drum internals shall include steam and water separators to limit the mechanical carry over. The moisture carryover from the drum to the first stage of the superheater will be less than 0.1%. The moisture carryover after the first stage of the superheater should not be detectable.

Internal feed-water distribution along the drum length shall be provided. Internal distribution headers for the HRSG chemical feed and saturated steam sample connections shall be provided.

All drums shall be fitted with leak free hydra step level indicators for both local visible indication and remote level indication.

The drums shall be equipped with steam separators for maintaining the required steam quality. The chemical balance of the steam circuit shall be controlled via the steam collecting drum. Suitable connections for a continuous blowdown and an intermittent blowdown shall be installed. Chemicals shall be added in such a way that a uniform distribution of the chemicals is ensured by means of an internal distributing pipe over the entire length of the drum or by feeding them in the feed water line directly before the inlet into the drum.

Internal distribution headers for the water and saturated steam sample connections shall be provided. All openings including down comers, steam outlets, manhole covers, and instrument and gauge connections shall be covered to provide a moisture-proof seal during shipment.

Hinged manholes shall be provided in both heads of all drums as per ASME PG-32.3 criteria. An insulated steel cover shall be provided to cover each manhole.

The steam drum shall be of large diameter to handle water level fluctuates during quick start-up without excessive dumping of water. At start-up, no water carryover from the drum is allowed.

As far as the dimensioning of the drums is concerned, load rejection for isolated operation also shall be considered.

Distribution and collection manifolds and headers shall be designed to provide uniform water or steam flow to all passes of the tube bank. Manifolds shall allow free expansion of tube banks.

All tube ends, header and other pressure part opening shall be cleaned and free of scale and foreign matter and shall be suitably sealed to prevent the entrance of foreign materials during shipment and during the expected storage period until erection. These parts shall not be sealed at the time of treatment, in order to provide proper inspection and complete removal of all cleaning medium.

All stand pipes or branch stubs shall be welded to the headers in the workshops.

Test pressure connections, test thermometer wells, temperature recorder connections, sampling connections etc. required for complete testing of economizer, superheater or reheater performance, and for all permanent instrumentation shall be provided. Inspection stubs shall be provided on large bore pipes and headers for borescope observations.

Addition of oxygen provision shall be considered in feed water downstream of FFLP (feed forward low pressure drum) or upstream of SALP (standalone low pressure drum).

HRSG drums shall be furnished with hinged manholes according to ASME standards. An insulated steel cover has to be provided to cover each manhole.

10.6.3 Superheater, Reheater, Evaporator and Economiser Banks

All HRSG sections shall consist of vertically arranged finned tube banks located within the steam generator casing.

Each element and tube bank shall contain tubes on a horizontal and vertical pitch, adequately supported and of shallow depth to enable easy cleaning. Any fins used shall be solidly attached to the tube throughout

the fin length. Tubes sheet clearance as per HRSG OEM design should be between 10 - 12 ft (not more than this) apart but can be increased or decreased according to CFD studies.

All superheater, evaporator and economiser tubes shall be solid drawn seamless and not electric resistance welded.

The superheater and reheater tube metal design shall include provision for all operational and system imbalances.

HP superheaters and the reheaters shall be provided with a spray type steam attemperator for superheater and reheater steam temperature control. The steam temperatures on the turbine inlets may not exceed the values specified by the turbine manufacturer.

The attemperator spray water control valves and automatic block valves, all temperature sensing elements, transducers, and controls for controlling steam temperature shall be furnished under these specifications.

If there is a low point, valves and fittings for automatic drainage shall be installed.

The spray coolers shall be designed and installed in such a way that the steam and water flows cannot result in inadmissible vibratory stress.

All pressure parts shall be designed and constructed to withstand the forces and moments applied through the HRSG pipework under all service conditions.

All the HRSG heating surfaces shall be designed considering the requirement for off-load water washing and CO₂ blasting and a number of local drain points.

The design shall incorporate measures to prevent sagging and vibrations occurring in the tube banks.

The design economizer outlet feed water temperature shall be sufficiently below the saturation temperature at the design operating pressure to prevent steaming throughout the entire operating range. The economizer shall be designed to allow recirculation of feed water in the economizer, only if needed to prevent steaming at low loads.

The low-pressure economizer design temperature shall be high enough to prevent corrosion of the economizer surfaces due to condensation of the corrosive flue gas components.

For superheaters and reheaters, Gr91 should only be used for steam temperature up to (including) 585°C, for steam temperatures higher than 585°C Gr304H shall be used.

EPC Contractor shall provide details of all the HRSG pressure parts materials by indicating the respective maximum design pressure and temperature for each grade and type of material for Owner review and approval before start of any material procurement and fabrication.

All heat transfer surfaces have to consist of compact banks of spiral and serrated finned.

Adequate tube spacing have to be provided to avoid plugging and fouling and to allow for proper cleaning and maintenance.

All alloy tubes have to be seamless and truly circular (carbon steel tubes have to be also seamless), free from blisters, scale and mechanical defects. The tubes have to be made of suitable materials and with a suitable wall thickness.

Tube ends to be field welded (including any factory installed stubs in drum or headers) have to be carefully machined to suit the weld joint design used and have to be adequately protected from damage due to shipping, handling or rusting. Tube-to-tube-header weld joints have to be full penetration welds on superheaters and reheaters. Tube-to-tube-header weld joints may be full strength welds for the remaining heating surface.

Distribution and collection manifolds and headers have to be designed to provide uniform water or steam flow to all passes of the tube bank. Manifolds have to allow free expansion of tube banks. Tube headers have to be arranged to permit replacement of tubes and cleaning of both internal (chemical cleaning during commissioning) and external surfaces.

Connections between tube banks and manifolds or headers have to be made outside the gas path.

All tube ends, header and other pressure part opening have to be cleaned and free of scale and foreign matter and have to be suitably sealed to prevent the entrance of foreign materials during shipment and during the expected storage period until erection. These parts have to not be sealed at the time of treatment, in order to provide proper inspection and complete removal of all cleaning medium.

The minimum water inlet temperature to the condensate preheater have to be selected to be sufficient to prevent corrosion of the economizer and preheater surfaces due to condensation in the flue gas.

The high, intermediate and low pressure superheaters and the reheater (if applicable) have to be of the vertically/horizontally arranged finned tube type with all piping connections between the sections of welded construction. The superheaters and the reheaters have to be drainable and access have to be provided such that individual elements or tubes can be removed or repaired with minimum disturbance.

HRSG will be designed and shall have places where temperature measuring devices can be installed in order to be able to measure between-module and bundle temperature across the width of the path on either side of the HRSG.

10.6.4 Steam Temperature and Output Control

Steam temperature control equipment shall be provided to ensure that the HRSG metal temperatures and steam temperature at the steam turbine do not exceed the design rated conditions under all methods of Plant operation. Regulation of the final steam temperature by the control of desuperheater water injection will be the prime method of control.

Design shall be done in such a way that even in the event of low differential pressures between the injected water and the superheated steam as well, a complete atomization of the injected feed water has to be ensured so that no drops can get into the turbine – a sufficiently long evaporation section have to be provided. By means of a suitable design, it has to be ensured that no water that has not been atomized yet can get into the steam pipe at any time when the control valve is closed.

If there is a low point, valves and fittings for automatic drainage have to be installed.

The spray coolers have to be designed and installed in such a way that the steam and water flows cannot result in inadmissible vibratory stress.

Each HRSG steam output shall be controlled in response to signals from the Central Control Room DCS. The steam temperature control loop of each HRSG shall respond accordingly.

10.6.5 Valves and Accessories

Boiler and superheater and reheater safety valves for all steam circuits shall be provided and shall be specified in accordance with design code recommendations. Each safety valve shall be fitted with its own vent-pipe, silencer and suitable drain and shall be designed and constructed to prevent back-pressure from interfering with the proper operation of the valve. Alternatively, the Contractor may propose a combined relief system with silencers in accordance with ASME B31.1 provided that discharge piping is designed with sufficient flow area to prevent the blow-out of steam or other fluid and that the Contractor provides the Owner or his nominated representative with its design calculations during the detailed design phase of the Project; the Owner's preference is, however, for a combined relief/silencer system for HP superheater safety relief valves to be separate from a combined relief/silencer system for HP drum safety relief valves. The safety relief valves for deaerators shall be arranged separately from a combined safety relief systems due to the lower pressure. IP Safety valves shall be arranged in a similar manner.

All safety valves have to be designed to avoid undue stresses and damage to the pressure systems for all operational conditions, including dynamic procedures.

Pressure relief arrangements shall be provided on the HP steam main before each boiler stop valve for the purpose of venting steam during start-up and pressure excursions; an electrically operated relief valve (ERV) shall be provided. In addition a suitably arranged start-up bypass system shall be used provided that such an arrangement allows the correct warming of individual HRSG pressure parts before the HRSG is connected to any common headers and that the DCS system will accommodate the use of the bypass system for HRSG and ST start-up purposes and pressure excursions (a start-up vent valve may also be provided for initial start-up of the HRSG) and enables independent HRSG operation during start up, load changes, shutdown and trip conditions. In the case of an electrically operated relief valve, this valve may discharge

into the silencer of one of the safety relief valves as long as this does not impair the operation of the safety relief valve or may discharge into the blow down tank. The relieving capacity of this valve shall not be included in the overall safety valve relief capacity of the HRSG. Safety valve silencers shall feature protection to prevent operational and maintenance personnel from exposure to high temperature steam.

The following quick start-up vent arrangement can be considered in order to avoid using an electrically operated relief valve:

- Each Start Up vent will be designed as follow:
- HP motorized relief valve if it is needed
- HRH start-up vent sized for 30% base load flow at 100% base load pressure (Quick Opening)
- LP start-up vent sized for 100% base load flow at 100% base load pressure (Quick Opening)

One electric motor operated stop valve and one non-return valve, together with one motor operated drain valves shall be installed in all main steam lines except LP. The electric motor operated valve shall each be provided with a smaller electric motor operated bypass valve to enable downstream warming and pressure equalization. The electric motor operated valve which is a high pressure (PN>63 bar) valve of 150 mm bore and above may be provided with a smaller electric motor operated bypass valve connected to its valve body.

HP steam from each HRSG shall be routed to a steam header feeding the steam turbines.

Double isolation valves operating as master/martyr shall be provided on all boiler emptying, venting and drain lines when the design pressure is greater or equal than 40 barg or the design temperature is greater or equal than 425°C. Main vent and drain valves which are associated with the automatic Plant start-up and shutdown shall be motorised for remote operation. In the case where motor operated drain/vent valves are provided, there will be one motorised valve and one manual valve in series.

All pressurised main drains in the boiler area shall be connected to the HRSG blowdown vessel/tank.

Drain and vent valves shall be N+1 (If operating pressure is greater than 4 MPa).

10.6.6 Feed Water Control Stations

The control of feed water flow to each HRSG shall be achieved by a feed regulating valve capable of controlling between 0% and 100% flow (to be confirmed by OEM). In case OEM cannot confirm the capability, then the HP feed water control configuration shall be 2 x 100% + 1x30%.

HRSG Drums and Feedwater tank level control valve shall be N+1.

10.6.7 Supplementary Firing System (if applicable)

A supplementary firing system shall preferably utilise grid burners but register type will be considered.

The burners shall be designed to provide maximum efficiency and uniform heat distribution to the HRSG firing natural gas only. The burner system shall be of a proven design and include equipment necessary for its safe, efficient and reliable operation.

The automatic fuel control system shall ensure satisfactory and safe combustion under all HRSG normal modes of operation.

All ignitors and flame monitors shall be capable of being maintained or replaced whilst the unit is online.

For Supplementary firing system, Dry Low NO_x burners shall be provided, with maximum 47.7mg NO_x/ MJ (LHV).

10.6.8 Blowdown and Drains Systems

Complete blowdown, drains and vents systems shall be supplied with each HRSG, including a blowdown vessel designed to BS5500, ASME VIII or technically equivalent international standard so that each HRSG can be fully drained and vented. The blowdown water shall be discharged from the blowdown vessel outlet to waste after being suitably cooled (if required) and any flash steam shall be discharged to atmosphere via

a vent in a safe location. Each blowdown tank shall also accept HRSG emptying and other such drains. A silencer shall be provided if applicable.

Blowdown and drain facilities which shall be designed to operate on both intermittent and continuous basis. The blowdown lines shall be designed for periodic flows of up to 3% mass flow (at MCR) as well as continuous flows of between 0.5% and 1.0% of MCR.

10.6.9 HRSG Drum Local Water Level Gauges and Remote Indicators

All HRSG drums shall each be furnished with two fully (online) maintainable local water level gauges at each end of the drum. These shall be of the bicolour type.

Remote boiler water level indication and recording shall be provided for all the steam drums and these shall include a temperature and pressure compensating mechanism.

Dedicated drum level measurements shall be provided for drum level control which will also use steam flow and feed flow meters and temperature and pressure compensation appropriate to maintaining control of level within the defined range under all operating conditions and transients in normal operation.

10.6.10 Stacks and Dampers

Each HRSG shall have a single flue stack of sufficient height and diameter such that it is self-sustaining, i.e. its lift due to temperature differential shall be equal to or greater than the sum of all losses and shall comply with relevant regulations.

The stack shall be provided with all necessary steel structures, silencer, stairways, staggered access, ladders, with safety cage and resting platforms minimum every 8 m, holes for gas sampling, pressure, temperature, walkways and platforms and with all necessary emission measuring devices, access/observation doors, drains, grounding and lightning installations and aircraft warning equipment.

The HRSG stack shall be provided with blinded flange connection to be ready for future carbon capture connection.

The efflux velocity shall not be less than 15 m/sec at normal operating loads.

The stacks shall be carried out completely with circular or hexagonal shaped reinforced concrete foundations supported by piles (if needed as per EPC Geotech report) as per the structural analysis.

Each HRSG unit bypass/ main stack shall have common Continuous Emission Monitoring Systems (CEMS) and related instruments and safe access to CEMS.

Each bypass stack shall have a single flue stack of sufficient height and diameter such that it is self-sustaining, i.e. its lift due to temperature differential shall be equal to or greater than the sum of all losses and shall comply with relevant regulations.

The by-pass stack dampers shall be of single blade construction and be designed to avoid distortion problems and ensure tightness and minimal leakage of all dampers throughout the life of the Plant. The dampers shall be equipped with seal air fans systems.

The by-pass stack damper blade shall be of material suitable for the structural, load carrying components under the specified project operation temperatures, the internal liner on the diverter blade upstream shall be made from at least Grade 321 stainless steel and shall have a minimum thickness of 3 mm (in the horizontal section and blanking plate).

The by-pass stack damper shall be arranged as a diverter damper such that when the stack is closed the HRSG duct is open and shall be provided with durable remote indication of open/closed positions.

The by-pass stack damper material selection shall be done in order to achieve:

- Longer lifetime
- High resistance to load changes
- Handle high temperature corrosion effects

- Take into account long-term creep, fatigue effects, gas turbine load cases and specified cycles

For the bidding purpose, Contractor shall consider 60 m and 45 m height for main stack and bypass respectively from ground level.

The minimum height of the stacks shall be as per the environmental requirements and the maximum stack height shall be approved by the Competent Authority.

Adequate distribution of flue gas entering the first bank (reheater or superheater) of the HRSG shall be maintained. Adaptation or manipulation of flue gas flow distribution utilising baffle plates etc shall be portrayed utilising modelling (i.e. CFD).

10.6.11 Refractory Materials/Casings and Insulation/Structure

The use of refractory materials, if required by the design, shall be limited to an absolute minimum, and restricted to the area of supplementary firing ducts.

The HRSG structure shall be designed to give adequate support to the HRSG body under the most adverse operating conditions taking into consideration expansion and wind loading.

The casing have to fulfil the following requirements:

- Gas tightness
- Temperature resistance against high exhaust gas temperatures
- Sufficient elasticity to withstand the temperature gradients in the start-up and shut-down phases and during load change processes of the gas turbine
- The flue gas duct shall be designed for unlimited alternating expansion stresses in the range between the lowest ambient temperature and the maximum operating temperature. Possibly installed elements (e.g. deflector plates) may not limit the frequency of temperature changes and shall withstand hot exhaust gas passed into the cold plant. They shall be vibration-free and heat-elastic.
- Internally insulated ducts and HRSG casing shall be lined with shop-installed, blanket insulation with steel liners. Carbon steel liner shall be provided where gas temperature is less than 400 deg.C. There shall be no gaps in the liner that would expose the internal insulation to the exhaust gas. The insulation/liner system shall be designed to support normal live loads during access to the interior of the HRSG without damage. Liners (including field installed liners) shall be installed such that the edge of the downstream liner is under the upstream liner.
- Special care shall be taken when determining liner thickness, stud diameter, and stud spacing for velocity higher than 27.4 m/sec or high turbulence areas.
- For internal insulation, insulation linings shall be cover with 12 gauge (2,6 mm) or thicker steel metal liner in areas where the average gas velocity is greater than or equal 45 m/sec. Where the average gas velocity is less than 45 m/sec 14 gauge (2,0 mm) or thicker steel metal liner shall be used on the floor, and 16 gauge (1,6 mm) or thicker steel metal liner shall be used on the side walls and the roof. Studs shall be attached to the casing by the shielded metal arc weld (SMAW) process or a stud gun. Internal lining subject to personnel walking during maintenance periods (lower horizontal surfaces) shall be 14 gauge (2,0 mm) or thicker. Insulation joints shall be staggered throughout multiple layers.
- Robust duct liner, duct casing, and foundation supports.
- A design that permits the raising and lowering of panels and headers to gain access to repair leaks. Gas and steam side access opening locations and sizes also should be specified. An enhanced access to individual module panels and to the tube-to-header welds
- The expansion joints in the ductwork shall be the highest quality available in the market at the time of Contract signing and subject to the approval of the Owner. The expansion joints in the inlet and outlet ductwork have to be demonstrated to be fit for purpose
- To be able to reach the upper headers of modules, manholes and/or handholes have to installed at the top of HRSGs,

- The size of the manholes may follow the HRSG manufacturer's standards, but the dimensions shall be not less than 600mm x 600mm. Openings used for inspection at regular intervals have to be provided with quick-locks, all others with screw connections. Section reinforcements are designed in accordance with the ASME Code.

10.6.12 Chemical Sampling and Dosing

The Plant shall be equipped with a system for the continuous extraction and monitoring of selected samples from the HRSG steam and water circuit. The selection of sample locations and analysers shall ensure adequate observation and control of steam and water chemistry limits. The measurements from the analysers shall be recorded on the power station DCS. A manual sampling facility shall be provided for each sample.

The Plant shall be equipped with chemical dosing systems to control the boiler and feedwater chemistry.

Each chemical dosing system shall comprise all equipment for the safe, reliable and efficient operation of the dosing systems, including bunds, segregated chemical drains, appropriate ventilation and emergency eye wash and deluge shower units.

Usage of AVT (O) or OT (oxygenated treatment) shall be preferred instead of AVT (R).

Sampling systems should have redundant sample coolers.

Steam quality shall be kept and controlled through commissioning and also during steam blowing. Analysers shall not be bypassed during steam blowing.

10.6.13 Selective Catalytic Reduction

Refer to section 4.2.2.1 of OTS Part A.

SCR system shall be provided for HRSG to meet the NO_x emission limitations specified in this OTS and EIA study recommendations.

Same shall be based on aqueous ammonia system complete with all sub-systems such as unloading, storage, forwarding etc. with necessary safety requirements.

The detailed description of the system offered along with schematics shall be submitted as part of proposal for Owner's review.

10.6.13.1 Catalyst

While the main component of the SCR process is the catalyst, the full system shall contain a variety of components and subsystems. The catalyst shall be housed in a large reactor. If required, flue gas shall be directed to and from the reactor by large flues that may contain control and shut-off dampers as well as flow distribution devices and static mixers, depending upon project-specific requirements. The reactor shall be a gas-tight steel enclosure that permits the flue gas and reagent to contact the catalyst for the required residence time to achieve design performance. The reactor structure shall include internal stiffeners and a catalyst support system that also permits rapid loading and unloading of catalyst materials. The design shall include a sealing system to avoid flue gas bypass and catalyst cleaning devices if required (soot blowers and/or sonic horns). The system will be designed with proper gaps/clearances to guarantee no risk of plugging and easy maintenance. The reactor shall be configured for horizontal flue gas flow.

Catalyst formulation, pitch and sizing are the keys to successful SCR system application and performance. Although catalyst formulations are proprietary, the most common components in steam generation applications use titanium dioxide with small amounts of vanadium, molybdenum, tungsten, or a combination of other active reagents. Formulation shall be matched to the flue gas conditions.

Catalyst types

Catalyst types and materials typically fall into one of three categories:

- base metal
- zeolite and

- precious metal

Most steam generation experience has been with base metal catalysts. These are typically described as titania-vanadia and can be provided as plate type, honeycomb type, or corrugated fibre type. All three are similar in performance characteristics and can be supplied in a homogeneous construction or as a coated active layer on a base material. The plate type catalyst shall be manufactured from a roll of stainless steel expanded metal substrate. The homogeneous catalyst shall be paste-pressed into and around the substrate to form a catalyst plate of 0.9 to 1.0 mm thickness. After cutting to length, correctly sized undulations, to set the catalyst pitch, shall be formed onto each plate prior to final processing and drying. The completed plates shall then be assembled into encased blocks; multiple blocks shall be packaged into a metal- frame shipping module. Homogeneous honeycomb catalyst shall be formed in an extrusion process that produces a catalyst log approximately 152 x 152 mm in cross section with the selected honeycomb cell pitch and cut to proper length. After final processing and drying, the logs shall be assembled and packaged into metal-frame shipping modules.

Reagent

The reagent (aqueous ammonia) subsystems shall include truck unloading, storage, vaporization or hydrolysis equipment, flow control, dilution air supply and mixing, and reagent distribution and injection components. Each subsystem shall contain piping valves, instrumentation, controls and safety equipment for appropriate ammonia handling and material compatibility. The dosing pumps shall be designed for 25% concentration aqueous ammonia, however flexibility to work with different concentrations such as 20%, 29% shall be provided. A critical part of SCR system design is integration into the overall HRSG system. This is particularly important because the SCR system design must accommodate swings in flue gas temperature while maintaining the desired performance, without affecting HRSG operation. Overall HRSG pressure drop including SCR will be considered for the design. The ideal gas temperature for NO_x reduction ranges from 371°C to 399°C and can be optimized further during SCR detail design.

10.6.13.2 Ammonia reagents and control

The reagent to be used in this project shall be aqueous ammonia.

Ammonia slip at NO_x guarantee conditions (15% O₂, dry) shall be 2ppm average slip during the operation period of 25 years.

10.6.13.3 Aqueous ammonia reagent systems

Aqueous ammonia is a diluted form of ammonia with NH₃ dissolved in water to form NH₄OH.

Aqueous ammonia is typically available at concentrations up to a maximum of 29.4% solubility limit at ambient conditions of 30°C and 1atm by weight of NH₃ equivalent. This concentration is also frequently described as 26°Bé, where degree baumé is an alternate measurement of density. The storage tanks shall be horizontal/vertical and shall be rated at 138 to 345 kPa. Tank material shall be stainless steel to avoid startup problems associated with rust. Storage system design shall also incorporate a containment basin in case of a ruptured line or vessel. The ammonia storage capacity shall be for 21 days of operation considering Operation Point 1 (RSC conditions with plant operating at full load).

10.6.13.4 Ammonia flow controls

For the ammonia reaction with the NO_x, one mole of ammonia is required to reduce one mole of NO_x and thus the required injection mole ratio of ammonia to NO_x is equal to the required NO_x reduction plus the ammonia slip value (ppm) divided by the inlet NO_x concentration (ppm). The control system logic shall use a feed forward and feedback input with this equation, along with a flue gas flow input, to provide the ammonia flow control demand. The feed forward input shall be determined by either an inlet NO_x analyzer or a curve-generated value for repeatable HRSG conditions. The feedback input can use the output of an SCR outlet analyzer or the stack continuous emissions monitor (CEM). Because of the hazardous nature of the ammonia there shall be several interlocks that will interrupt ammonia flow upon upset conditions.

10.6.13.5 Distribution requirements

SCR reactor performance is influenced by the distribution of velocity, temperature and reacting gas components through the catalyst bed. The degree of a given profile uniformity can be generally described

by the coefficient of variation (Cv), commonly reported as the percentage of the profile standard deviation from the arithmetic mean value.

$$Cv = 100\%$$

SCR reactor distribution goals for flow and molar ratio are commonly stipulated in this coefficient of variation format. A temperature distribution criterion is the exception by typically being expressed as a minimum and maximum ($\pm$) deviation about the arithmetic mean. The goal is to achieve the most uniform profile possible at the SCR catalyst inlet for NO_x concentration, ammonia to NO_x molar ratio, velocity and temperature. The degree of uniformity required for each parameter, coupled with the physical space and arrangement limitations presented, drives the level of complexity and intricacy of a successful design. Numerical or physical flow model studies shall be used to determine the necessary distribution devices to achieve the desired uniformities.

10.7 Plant Main Cooling

Refer to sections 1.2.3.7 and 4.2.7 of OTS Part A and section 6.2.1 of this Part B (Air Cooled Condenser).

10.8 N.A.

10.9 - N.A.

10.10 Water Treatment Plant

10.10.1 General

A water treatment plant shall be supplied to provide all the demineralised water requirements of the Plant including, but not necessarily limited to, evaporative coolers, wet compression if applied and make-up for boiler blowdown.

The water treatment plant shall treat product water from the desalination plant by means of ion exchange System (regeneration unit and neutralisation units to be marked as required) with most advanced features with proven demonstrative commercial scale plant including regeneration unit and neutralisation units.

The demineralisation plant shall have 3 x 50% streams. Each 50% stream shall be sized such that two streams running simultaneously provide, as a minimum, a net supply rate equal to 1.5% of the steam generation flowrate of the HRSGs in addition to any other plant demands.

For demineralization plant, three (3) x 50% Mixed Bed Polishers having more than 24 hrs operations cycle between regeneration or CIP will be provided when the feed water TDS is 20 mg/L. During normal operation when gas firing, two (2) MBP will be on operation simultaneously, the others will be on regeneration or stand-by EPC Contractor is free to propose Electrodeionization (EDI) for final polishing instead of Mixed Bed Polishers.

Three (3) MBP feed pumps will be provided. During normal operation, two (2) feed pumps will be operated, one(1) will be regeneration or stand-by.

The system shall be complete with a regeneration or CPI system consisting of sodium hydroxide and either hydrochloric or sulphuric acid storage facilities, assorted chemical feed pumps, and mixing agitators, neutralisation basin and associated waste water pumps.

The demineralisation system shall be designed to treat water with a maximum conductivity of 0.1 μ S/cm (@25°C) and the maximum content in the permeate from the 2nd Pass of SWRO desalination plant will be limited to 20 mg/l of Total Dissolved Solids (TDS) for all the feed water conditions given in the OTS.

The make-up water system, incl. its storage tank, shall be designed sufficient to cover a black start of at least one desalination unit. The make-up water storage tank shall have a minimum storage capacity of at least 24 hours continuous plant operation at full base load with fuel Gas.

The system shall be arranged for automatic control of operation, regeneration of the mixed bed ion exchange resins and neutralisation of the wastewater.

The guaranteed quality of the demineralised water product at the outlet of the mixed bed shall be as a minimum as per Electric Power Research Institute, or EPRI, USA:

- Specific conductivity (Continuous) $\mu\text{S}/\text{cm}$ (@25°C) ≤ 0.1
- Sodium as Na^+ (Continuous) mg/l ≤ 0.003
- Chloride as Cl^- (Commissioning or Troubleshooting) mg/l ≤ 0.003
- Sulphate as SO_4^{2-} (Commissioning or Troubleshooting) mg/l ≤ 0.003
- Total silica as SiO_2 (Continuous) mg/l ≤ 0.010
- Total Organic Carbon (TOC) (Continuous or Grab sample) mg/l ≤ 0.3

10.10.2 Vessel Construction

The ion exchange vessels shall be of welded mild steel construction. In case of the ion exchange vessels shall be rubber lined in an approved manner with a minimum rubber thickness of 3 mm.

Ion exchange vessels and bulk chemical storage tanks shall be designed to BS 5500, or ASME 8, or an equivalent internationally recognised Standard. The demineralised water tank(s) shall comply with an acceptable internationally recognized Standard.

The maximum linear velocity of water through the ion exchange vessels shall be 40 metres per hour. All of the internal metallic fittings of the vessel shall be manufactured from stainless steel.

The vessels shall include sight windows to enable the resins within the vessel to be observed. The sight glasses shall be at the resin interface when separated and at the resin upper surface. Any other sight glasses required for operational reasons shall be provided.

A resin trap shall be fitted to the outlet of each vessel, which shall incorporate a differential pressure gauge.

10.10.3 Demineralised Water Tanks

Treated water reserve storage tanks shall be provided with a minimum usable capacity of 48 hours of normal usage at full load. Two (2) x 50% Demineralized water tanks shall be provided, each with minimum 24 hrs storage of normal usage at full load when Gas firing. All necessary valves, fittings, pipes, etc. required to integrate the storage tanks into the system shall be provided.

Demin tanks shall be provided to allow for outages, maintenance and tanks shall be interconnected by pipework. The tank water shall be protected by filter and CO_2 absorber as appropriate means to prevent the water coming in contact with atmospheric CO_2 .

Demineralized tanks will have absorbers to prevent CO_2 ingress.

10.10.4 Regeneration/ CIP Equipment

All chemical handling equipment shall be bundled such that any chemical spillage is contained.

Sulphuric or hydrochloric acid and sodium hydroxide shall be used as the regenerants. A bulk storage tank for each regenerant shall be provided with a minimum capacity of 7 days' consumption at maximum Plant output. Each storage tank shall be equipped with a bund having a volume of at least 110% of the capacity of the tank. If hydrochloric acid is used, the acid bulk storage tank shall be equipped with adequate fume extraction facilities. The CIP equipment shall be supplied with permanent piping.

10.10.5 Effluent

The existing effluent sumps and effluent control system shall be used for effluent disposal. The pH of the effluent shall be adjusted if necessary to between pH 6 and pH 9 prior to discharge of the effluent. The control of this discharge shall be co-ordinated into the overall plant effluent control system.

10.10.6 Control

Regeneration/CIP inhibition and control signals shall be co-ordinated between the two streams so that no operational conflicts occur.

An automatically resetting throughput meter shall determine regeneration of the mixed bed unit. This meter shall adjust the volume to be treated by the unit in accordance with the quality of the distillate feed.

water using a pro-rata calculation comparing the actual quality of the feed-water with the design quality of the feed-water.

A continuously recording conductivity meter shall be installed on the product output from the unit. This same position shall provide a source for a manual grab sample.

The Water Treatment Plant will be supplied with proprietary, purpose-designed systems for control, monitoring and protection. Such systems will be designed to function independently of each other but will be interfaced to the DCS for supervisory level to allow co-ordinated operation of the Facility.

10.11 N.A.

10.12 Hydrogen Generation Plant (if applicable)

The H₂ generation plant including all required auxiliaries shall be sized for plant normal consumption plus 10% margin. The plant shall have storage for 24 hours of normal consumption of all generators or the required amount for 4 times filling of the largest turbine generator installed. Storage shall be provided by using either storage tanks or H₂ bottle clusters (each cluster has 20 cylinders).

Particular attention shall be paid to the explosion risk in the design and construction of the plant. All electrical equipment shall be explosion-proof; this applies also to lighting and power outlet sockets. All components of the entire H₂ plant including ladders, platforms, etc. shall be earthed.

The feed water for the electrolyser shall be supplied from the demineralization plant. Cooling water shall be taken from the station closed circuit cooling water system.

Where components are in contact with hydrogen under pressure the resistance of the material to hydrogen embrittlement shall be considered.

Pumps, compressors, electrolysis units, etc. shall be provided with stand-by units. Purging lines with Nitrogen shall be provided for each equipment in case of shut down or maintenance of the plant.

The H₂ generation plant shall be housed in a separate building.

A control system shall be provided which enables that H₂ generation and filling shall be operated in a fully automatic manner.

Hydrogen generation plant shall be equipped with firefighting system as per NFPA standard. Hydrogen seal oil units shall be complied to requirements of NFPA850, Cl:7.7.4.4 (i.e. automatic sprinkler system or foam system).

The plant shall be described in detail by the EPC Contractor supported by flow diagrams.

10.13 Wastewater Treatment System

Rumah and Nairyah Plant will be designed with Zero Liquid Discharge (ZLD) which means that no emissions of liquid effluents other than uncontaminated storm water will be discharged to the environment. Hence, all wastewaters will be directed to an evaporation pond where the accumulating sludge will regularly be disposed.

The EPC Contractor shall present a concept for the wastewater treatment plants and demonstrate the major wastewater treatment steps including the corresponding maximum and normal flows.

10.13.1 Sanitary Wastewater

Sanitary wastewater drainage from the entire plant area shall be treated in a sanitary treatment plant.

Sanitary wastewater shall, to the extent possible, be plant internally re-used after its treatment for irrigation purposes. Its sludge shall be used, if applicable, as fertilizer.

For this re-use, the PME's and NCEC's General Environmental Regulations and Rules for Implementation apply.

The sanitary wastewater system shall consist of a piping system with inspection manholes every 60 m and at each change of direction. Lifting stations shall be provided as necessary.

10.13.2 Other Wastewater

Other waste waters shall be collected for treatment. Rainwater or any surface water flows from areas which are likely to be contaminated in any way shall also be handled as wastewater for treatment. All treated wastewaters should be re-utilized with means of demineralization and re-used within the Plant consumers.

The wastewater streams must be separated according to their origin and/or type of pollution and the type of treatment that they require. Thus, as a minimum requirement, the following separate treatment systems shall be provided:

- **Oily wastewater treatment system:**

All oily wastewaters shall be pre-treated at least by oil separators and further adequate and suitable treatment steps in order to meet the requirements of Water Treatment System; and

- **Industrial and process wastewater system:**

All industrial and process related wastewater such as from (chemical) cleaning, from ion exchange regeneration, from rinsing and flushing processes, from chemical storage areas and workshops etc. must be treated in an applicable wastewater treatment plant, that is adequately and suitably designed to neutralize and detoxify the particular wastewater and to meet the requirements of the water treatment system.

Wastewater drainage to the respective wastewater treatment plants shall be using pipes made of customary and proven materials in adequate diameters for the likely maximum quantity of wastewater flow in the pipes. In the case of differential height constraints, pumping stations must be installed in adequate numbers. In the case of using cesspools, intermediate storage is allowed. Cesspools are to be emptied by vacuum loaders, which are also part of the scope of supply.

Following the pre-treatment of the waste waters, the filtrate shall be routed to the Water Treatment System for further treatment, demineralization and re-use of the water as per the process requirements mentioned above. The reject water of the Water Treatment Plant will be eliminated in means of Evaporation Ponds. In order to optimize the Project lifecycle costs, EPC Contractor may utilize forced evaporation systems for the non-utilized waste waters. Selection and operation of such systems should not lead to any environmental risks in terms of spreading of the Dissolved solids within the neighbouring areas around the evaporation pond and EPC Contractor will be requested to demonstrate the performance of such systems during the Defect Liability Period.

Wastewater from GT washing processes shall be collected separately and suitable arrangement shall be provided to pump the same to road tankers for disposing the same externally at defined disposal areas.

10.13.3 Storm Water

Clean and uncontaminated rainwater from the entire plant Site including rainwater from roofs and the paved areas of the plant shall be collected, guided and discharged in plant storm water drainage system. The plant clean storm water drainage system shall be capable of carrying corresponding discharge of flows from the offsite area at its upstream end. The system shall consist of piping system with inspection manholes every 60 m and at each change of direction. The dirty or contaminated rainwater shall be collected and treated in a separate system and shall be disposed to evaporation pond. Dedicated distribution to ensure for uninterrupted service & instrument air supply with interconnection in case of emergency backup.

If Owner allows, Storm water can be re-used in the process plants or infiltrated to the soil at the plant area. In case the Contractor shall considers an overflow of the storm water, an appropriately designed infiltration pond shall be implemented at the site. Bidder has to prove that this separate pond is able to infiltrate the storm water and the permeability of the soil enables seepage of the storm water without harm to any adjacent installations.

For this purpose, the Contractor shall consider for his design additional treatment facilities to increase the water recovery from the treatment plant. The optimization of water re-use and evaporation is subject to the Owner's approval.

10.13.4 Evaporation pond(s) (if applicable)

The evaporation pond(s) shall be sized sufficiently in order to accommodate all Plant effluents and considering the evaporation rate at site during all seasons of the year.

All effluents shall be treated by neutralization, oil separation and solids removal before final discharge to the evaporation pond, where applicable.

The ponds shall be sectionalized and accessible by wheel loaders to allow cleaning and maintenance of the pond(s) during operation.

The Bidder shall provide the basic design of the evaporation pond(s) with the proposal to as well explain the operation during rainy conditions.

Waste water pond complete area bottom as well sides shall be provided by HDPE liner & if maintenance is required concrete tiles shall be provided with access steps & ramp.

However, for Storm water pond shall be provided gravel filling at bottom and sides to be provided with a HDPE liner with concrete tiles with O&M accesses.

The evaporation pond shall be divided in two chambers. Lifting station, vehicular access as required shall be provided.

10.14 Chemical Storage and Dosing

10.14.1 Chemical Storage and Dosing

- A) EPC shall equip the Plant with centralised chemical storage area(s).
- B) The capacity of each chemical should have a volume that is equivalent to 7 days of full load operation as minimum inventory under normal operation dosing. For the Long lead chemicals, the storage capacity shall be 30 days.
- C) The preparation tank and dosing tank for each chemical shall have a capacity of 7 days under normal operation dosing rate.
- D) The chemical storage area shall be either indoor or sheltered with a roof.
- E) The tanks and pumps shall follow the redundancy concept as defined in the other section of this document.
- F) Chemicals shall be transferred by filling pumps to local dosing tanks, where required, each with a volume equivalent to at least 48 hours operation at Plant full output. The lime slurry tanks comprise of 2 x 100% and shall run continuously.
- G) The chemical storage tanks shall be bunded (each chemical separately) to ensure that an accidental leakage is retained within the bund. The bund shall be of sufficient volume to contain 110% of the entire contents of the contained tanks. Contractor shall propose a system to prevent incorrect or incompatible chemicals delivered into the wrong storage vessel.
- H) EPC shall provide an air conditioning room for the chemical truck drivers to wait during chemical unloading and out of their protective clothing before and after they discharge the chemicals in the storage tanks.
- I) Storage facilities for hazardous chemicals shall be subject to review and comment by the Owner at the design stage of the Project.
- J) The chemicals supplied for this Plant shall be of food grade /non-carcinogenic. All chemicals used in the project shall be in line with the EIA study recommendations.

10.14.2 Chemical Dosing System

The chemical dosing systems required for the process shall include but not limited to:

- A. Hypochlorite for shock chlorination of the feed water;
- B. Antiscalant;
- C. De-Chlorination system;

- D. Acid dosing;
- E. Caustic dosing;
- F. RO Cleaning In place cleaning;
- G. RO product water sterilization;
- H. Coagulant;
- I. Polymer;
- J. Biocide.

10.15 Plant Potable Water System

The water quality shall be in accordance with the latest revisions of all relevant laws and regulations of the country, as well as with the latest revision of the relevant World Health Organization (WHO) guidelines. The potable water capacity shall be sufficient to serve all operational and operation personnel requirements under all normal operation circumstances and according to best practices, plus a 20% safety margin.

Any material used for the potable water system and storage tanks shall meet the requirements of the WHO in regard to harmful effects on the drinking water.

Paints (manufacturer and type) and internal coatings of potable water steel tank(s) shall be subject to Owner's Engineer approval.

Plant potable water system shall include but not be limited to potable water storage tanks, chemicals dosing units, pumps (2x100%), pipework, valves, fittings, instrumentation, controls and other associated accessories for distributing potable water. The Potable water from remineralisation system shall be stored in Potable Water storage tank of suitable capacity and shall be distributed to all consumer points through Potable Water distribution network within Power plant facility.

Potable water lines and discharge header shall be of High Density Poly Ethylene (HDPE) type. QA/QC must be in place to inspect the quality.

10.16 Compressed Air Systems

10.16.1 General

The Facility shall be provided with compressed air to supply instrument air and service air required throughout the respective Plant Facility. The equipment shall be designed with appropriate redundancy for the Plant Facility and shall consist of heavy-duty air compressors.

Compressed air system for instrument air and service air requirements consisting of air compressors, air receivers, air driers, air filters, piping, valves, fittings, instrumentation, controls, etc.

3x50% compressors shall be provided for Power Plant. The instrument air system shall consist of 2x100% filtering skid and 2x100% twin tower heatless regeneration type air dryers. Two air receiver tanks each of minimum 15 m³ capacity shall be provided. Compressed air system pipeline shall include automatically water trap with proper distance and the line shall be designed with proper slope so to avoid any water accumulation in the system.

The air release valves in the standpipe system header shall be installed at the highest point of the piping.

The compressed air mains shall be routed to ensure integrity of supply to minimise the possibility of damage. In this context the air distribution system shall be based on ring main systems. Piping material shall be stainless steel for instrument air. Plastic piping on any part of the system is not acceptable.

The compressed air system will be designed according to ISO 1217 at reference conditions. The grades will be suitable for compressed air used in process control instruments. The kind of air quality (dirt /water / oil) in a facility of this type is classified according to ISO 8573-1.

The instrument air distribution system shall normally operate independently, but to rationalise on the compressor installation, a valve and protected interconnection may be made upstream of the dryers, to

allow the service air supply to supplement the instrument air supply, in the event of failure of more than one instrument air compressor.

An air distribution ring should be provided for each compressed air facility.

The service air system shall include hose stations and fixed connections for air tools.

10.16.2 System Configuration

The compressors will supply air to a common sectioned header. From this header instrument air shall pass to the driers before entering the distribution pipework. Service air shall be supplied upstream of the driers once the air has been cooled.

The system will be designed such that the failure of a single component will not result in the loss of compressed instrument air supply. Dedicated distribution to ensure for uninterrupted Service & instrument air supply with interconnection in case of emergency backup.

10.16.3 Compressors

The compressors shall be heavy duty, water or air cooled, oil-free with carbon rings, driven by electric motors and provided with air-cooled intercoolers.

Each compressor unit shall be capable of continuous operation under the specified service condition, without injurious heating and shall operate without objectionable noise or vibration.

The Contractor shall provide intake filter and silencer, necessary gauges and all accessories and controls required for a complete air compressor unit to operate satisfactorily. Factory-set relief valves shall also be provided to protect the compressor in case of automatic control failure.

10.16.4 After Coolers

Three water or air-cooled after coolers with condensate separators shall be provided, each capable of cooling the rated compressed air capacity of one compressor to a temperature of not more than 10°C higher than the incoming air temperature. Water separator and drain with automatic trap and bypass shall be provided.

10.16.5 Air Driers and Filters

The compressed air shall be sufficiently dry to prevent condensation occurring in the control air system under any operating conditions of temperature and pressure.

Two (2) 100% twin tower desiccant heatless air dryers together with outlet filters shall be provided. The air dryers shall dry the air to a dew point of at least -40°C.

All filters for the control air shall be provided as required. Suspended particles within the air shall not be larger than 5 microns.

10.16.6 Air Receivers

Each air receiver shall have sufficient capacity to enable the Plant to be shut down in a safe and controllable manner and enable standby compressor plant to come into operational service without reduction in air supply.

All air receivers shall be designed, assembly and inspection in accordance with ASME Boiler & Pressure Vessel Code Section VIII and shall be provided with factory-set safety valves, pressure gauges and bottom tap drains.

10.17 Fire Protection and Detection Systems

10.17.1 General

The design, materials, manufacturing, installation and testing of the fire and explosion prevention as well as the firefighting equipment shall be selected based on local codes and standards and comply with requirements of the relevant Government Authority.

The Codes and Standards of the National Fire Protection Association (NFPA) latest edition shall apply in case the above-mentioned codes and standards do not cover all subjects concerning the fire detecting and firefighting systems or are less stringent than NFPA Codes and Standards.

The Plant shall be protected against fire by a comprehensive fire detection, alarm, and protection system. The fire detection system shall provide the means to detect fires, provide visual and audible alarms both locally and remotely, and, where required, actuate the firefighting system.

Fire protection for the power plant shall mainly comprise civil structures (fire areas, fire walls, etc.) fire detection and fire alarm system, fire water storage and pumps, fire main ring system with outdoor hydrants, indoor hydrants, extinguishing systems (sprinkler, water spray, foam, human safe gas, etc. as applicable) and portable fire extinguishers.

Fire alarm and detection, fire protection and firefighting system for the entire Plant complete with all necessary piping, pumps, valves, fittings, control panels, etc. shall be provided in accordance with NFPA standards and the local guidelines. Firefighting system shall be provided with flow meter to undertake pumps performance test. All fire detection and protection equipment and components shall be UL listed or FM approved and shall be suitable for use in fire detection and protection purposes.

Hydro test of the complete ring shall be performed during commissioning for fire water piping.

Surge test shall be conducted in pre commissioning and commissioning of the piping.

Use of compatible soil for refilling and compacting of the firewater piping for underground piping.

Underground firewater piping shall be checked for thrust blockings so to prevent water hammering effect.

Required reserve storage capacity for fire water system shall be made available in the clarified water storage tank / sump.

The design and installation of the fire protection and alarming systems shall be in accordance with the recommendation of the NFPA standards and shall be approved by an insurance company. Some standards are listed below as good engineering practices and with any additional requirements of local authorities and shall comply with the local regulations in respect of supplier, installer and system approval.

- NFPA 1: Uniform Fire Code;
- NFPA 5000: Building Construction and Safety Code®;
- NFPA 101: Life Safety Code®;
- NFPA 850: Recommended Practice for Fire Protection for Electric Generating Plants and High Voltage Direct Current Converter Stations;
- NFPA 15: Standard for Water Spray Fixed Systems for Fire Protection;
- NFPA 14: Standard for the Installation of Standpipe and Hose Systems;
- NFPA 30-Flammable and Combustible Liquids Code Handbook
- NFPA 20: Standard for the Installation of Stationary Pumps for Fire Protection;
- NFPA 13: Standard for the Installation of Sprinkler Systems;
- NFPA 72: National Fire Alarm and Signaling Code®;
- NFPA 70: National Electrical Code®;
- NFPA 2001: Standard on Clean Agent Fire Extinguishing Systems;

For Fire water Tank, requirements of NFPA 850 shall be strictly complied for designing and calculations

The main hydrant ring shall be distributed to multiple zones for delegate isolation (to rectify any leakage without isolating the entire ring).

In particular the recommendations of NFPA 850, and the particular standards referenced therein, shall be followed, with the recommendations being interpreted as requirements and "should" being interpreted as "shall".

The works shall include but not be limited to:

- Underground hydrant ring main system (along with hydro and complete system integrity testing)
- Fire pumps
- Exterior hydrant outlets with equipment cabinets
- Fixed sprinkler and spray water systems
- Foam systems
- Environmentally friendly fire suppressant gas systems
- Portable fire extinguishers
- Fire and gas detection systems with fire alarm panels
- Audible and visual alarms
- Fire alarm central control panel including a schematic diagram based on a site layout plan and located in the CCR
- Repeater panel in the gatehouse
- Site wide manual call points (smash glass alarms)
- Fire signs
- Hose reels
- Provision and supply of Safety and eye wash showers as per OSHA instruction STD 01-08-002
- for all plant areas and rooms like chemical handling, battery room, Lab and all others

A central fire brigade station shall be constructed as per local regulations and equipped with all facilities for the fire brigade as per relevant regulations. Supply of mobile fire equipment, consisting of at least:

- One (1) truck ladder;
- Two (2) tankers;
- Special mobile vehicles;
- Three (3) foam water trailers;
- Ambulance car; and
- Auxiliary equipment.

10.17.2 Fire Protection

10.17.2.1 Underground Hydrant Ring Main System

Water supplies for all the firefighting services on the Facility Site shall be provided via a common system of firefighting pump house and underground hydrant ring mains.

The firefighting water reserve shall be sufficient to provide firewater to the largest risk area plus 500 GPM for a period of 2 hours with the necessary fire hoses as per NFPA. The water shall be potable water from independent fire water tank. The tank shall provide water to the firefighting pumps, which shall comprise a 100% duty electrically driven fire pump and a 100% duty diesel driven pump. The pressure of the ring main hydrant system shall be maintained by suitably sized 2x100% electrically driven jockey pump.

The pumps shall supply water to an outdoor buried ring main system for the Plant, which shall extend to all areas of the Site, with extensions into all the main buildings and complete with all necessary ancillaries to enable any fire hazard location to be covered by two hydrants. It shall provide water supplies to all fixed firefighting installations as well as to the fire hydrants and hose reels.

All fire pumps shall be provided with proper recirculating line PRV and flow meters to test the pumps as per NFPA-25. Fuel level of diesel pump shall be monitored in control room.

Provision shall be made for future extensions without the need to shut down fire protection to operating plant. A blanked off section isolating valve shall be provided at each future tie-in point.

Two water supply connections from pipework following physically different routes, from the water storage tank into the firefighting pump house at different locations, shall be provided and similarly, the supply connections to the hydrant ring main shall exit from the pump house at two different locations and shall connect to the hydrant ring main at two separate locations. The system shall be arranged such that either water supply route and connection can be used, via either fire pump, discharging via either ring main connection, to supply firefighting water to any risk area via alternative routes. Easily accessible section/isolating valves shall be provided throughout the system to achieve this objective.

An adequate number of vent and drain valves shall also be provided to permit the venting and draining of the complete system without the splitting of flanges or similar practises being necessary.

The below ground pipework shall be nonferrous and shall be manufactured from materials such as GRP, FRP or HDPE, selected to ensure long term durability under the test/design pressure and site conditions. Underground firewater pipping shall be checked for thrust blockings so to prevent water hammering effect. Above ground pipework shall be hot dip galvanised in accordance with appropriate standards.

Weatherproof hose/equipment cabinets shall be provided and located in the same manner, and containing the same equipment, as those provided for the existing plant.

10.17.2.2 Fixed Fire Protection

Fire protection shall be provided per the standards referenced herein. Where it does not contravene these requirements, the fire protection shall accord in all respects with any existing fire protection provisions.

In addition, the Contractor shall comply with the provisions described below.

All deluge valves, whether automatic or otherwise, shall be located at a safe distance from the fire risk on an exit route so that they can be operated manually if required.

The deluge valve assemblies shall be fitted with an alarm bypass test connections for a water flow alarm switch and shall be made on the water supply side of the system and provided with control valve and drain for alarm piping as per the requirements of NFPA13, Cl:8.17.1.3. This arrangement enables the end user to carry out the manual testing (dry test) of the system without discharging the water into the protected area. Therefore, it shall be installed in portion of alarm connections/bypass test connections in the deluge valve trim assembly which will get activated and provide the signal of water flow equal or in excess of discharge from one sprinkler/nozzle.

Environmentally friendly fire suppression systems (e.g. FM200) shall be provided for the internal protection of the GIS substation relay and protection panels, and the CCR cubicles and panels. Clearly visible signs shall be posted at each area to warn personnel that the particular room/area is protected by a gas flooding system.

As per the NFPA-70, Cl:760.133, the fire alarm circuit cables shall not be placed in any other cable, cable tray, compartment, enclosure, manhole, raceway, outlet box or electric light, unless it is separated by barriers or spacing required by NFPA-70 based on specific application type.

The water spray nozzles in STG building shall be installed in a manner without any obstruction to its spray pattern on the protected equipment.

For transformer deluge valve assemblies, all automatic water spray systems shall be provided with a local alarm complying with requirements of NFPA 15, Cl:6.4.1.1

For administration building, sprinklers to be installed in all parts except the areas which can be omitted as permitted by NFPA13.

10.17.2.3 Portable Extinguishers

Hand-held and portable extinguishers shall be provided. The extinguishing agent shall be selected to suit the particular fire hazard. The extinguishers shall be provided at standpipe and hose stations and other key locations in the buildings and in any area normally occupied by personnel.

Painted markings and signs shall be provided to direct attention to the location of extinguishers and to their use.

10.17.3 Fire and Gas Detection

The fire detection system shall cover the entire Facility.

Fuel gas detection shall cover areas of the Plant where the risk of damage due to gas accumulation is considered to be significant and as required by local regulations.

The master panel shall be mounted in the CCR with repeat indication in the gatehouse. Repeat panels shall also be provided to cover individual buildings where necessary.

A complete self-monitoring, fully addressable fire detection and alarm system shall be provided for the power/desalination plant.

The alarms in the protection zones, interfaces with the DCMS and local repeater panels shall be provided for information and protection of the operators. The system shall interface with the other local fire detection protection systems, as appropriate, to inform the control room operators of fire elsewhere on the site.

The preferred format of the indicator and repeater panel mimics is a schematic diagram based on a site layout plan.

Audible alarms shall be installed in all locations and shall have a pronounced range and volume distinct from the background noise. All siren/fire alarms shall be of adequate output level to be audible above the normal working conditions.

A zone-wise thermal type detection system shall be supplied to alarm and shut off the oil supply to any burner front involved in an external fire.

10.17.3.1 Fire Detectors

The fire detection equipment shall comprise heat, smoke or gas detecting devices to suit the risk. The fire detectors shall be located at strategic positions adjacent to the plant to be protected and also in suitable positions in the general station areas. Smoke detectors shall be spaced as per NFPA -72.

Linear fire detectors shall be provided on each major cable tray run areas without ventilation.

The air heater fire detection system shall comprise a series of temperature detectors/thermocouples installed in the air outlet duct from the air heater(s). The detectors shall be arranged to detect abnormally high temperatures and/or high rates of change of temperature in air leaving the air heater(s). Fire protection shall be arranged for manual initiation.

Air duct detectors to sense the presence of combustion products in air handling systems shall be installed in the ventilating and air conditioning duct systems. They shall shut down fans and actuate smoke dampers or other controlled equipment upon detection of combustion products so as to prevent re-circulation of smoke.

10.17.3.2 Manual Call Points

Manual call points shall be provided at all exits and stairwells and at locations not greater than 50 metres apart in turbine areas, HRSG areas (at each level) and all other buildings. A striker shall be provided adjacent to all call points to facilitate breaking the glass. As per NFPA72, Cl:17.14.8.4, the manual pull stations shall be located within 1.5m of each exit doorway.

10.17.4 Auxiliary Fire Protection Equipment

The Contractor shall provide auxiliary fire protection equipment in accordance with local regulations including, but not limited to fire-resistant clothing, compressed air breathing apparatus, fire blankets, goggles, helmets, stretchers, portable gas detectors, etc. in such quantities as are compatible with power plant requirements.

The sprinkler system in the warehouse area shall be designed based on the hazard classification for flammable & combustible liquids and complying with requirements of NFPA - 13.

10.17.5 Fire and Exit Signs

A complete system of warning instruction and directional signs for safe and orderly evacuation of the buildings shall be included and shall be inscribed in Arabic and English or be pictorial.

Fire exit doors shall comply with NFPA 850. Wirings of exit signs shall comply with requirements of NFPA 70, article 700 so to provide maximum availability in case of fire.

As per OSHA's guidelines, "exit doors must be unlocked. Employees must always be able to open an exit route door from the inside without keys, tools, or special knowledge. "Construction materials used to separate an exit from other parts of the workplace must have a one-hour fire resistance-rating if the exit connects three or fewer stories and a two-hour fire resistance-rating if the exit connects four or more stories.

The signs shall include DC illuminated exit signs indicating the nearest safe exit from the buildings.

10.17.6 Hose Reels

Hose reels connected to the plant fire main shall be provided at strategic locations within the Plant and its buildings.

10.18 HVAC Systems and Design Conditions

HVAC systems shall be designed, installed and commissioned in accordance with the latest applicable codes of the following:

- ASHRAE American Society of Heating, Refrigerating and Air Conditioning Engineers.
- SMACNA Sheet Metal and Air Conditioning Contractors National Association.
- NFPA National Fire Protection Association.
- ARI American Refrigeration Institute.
- ANSI American National Standards Institute.
- AABC Associated Air Balance Council.

All systems shall be designed and installed to provide acceptable environmental conditions for each area. The conditions of temperature, humidity, air movement and air filtration shall be controlled as required in the respective areas in summer and winter.

HVAC systems serving critical areas such as Control Rooms shall be provided with duty/standby air handling and mechanical cooling plant to ensure continuity of operation.

HVAC systems shall be arranged to shut down in the event of a fire alarm, seal supply and extract ducts as required for the application of clean agent fire suppression systems and start up smoke control pressurization systems. Provision shall be made in all plant areas for the venting and clearance of smoke and fumes in the event of fire, by means of roof vents or extract fans.

All systems and components shall be suitable for the design life of the Plant, and be designed for minimum life cycle cost and complexity, consistent with functionality, ease of maintenance and reliability.

For all general cases the HVAC systems the ambient design conditions shall be as ASHRAE recommendations to suite the site conditions. The HVAC systems shall be designed to obtain the internal conditions and system redundancy as indicated in the section 4.2.15.17 of the OTS Part A. Central air-conditioning systems having humidity control may be provided with air handling units for fresh air supply. These units shall be provided with cooling coils and steam humidifiers to handle the full dehumidification/humidification load for the central system, with dew point control. Fresh air-cooling coils shall have 10 per cent margin on heat transfer surface to cater for periods of abnormally high humidity. Face air velocity shall not exceed 2.0 m/sec.

Humidification shall be by electric powered steam generators with proportional control and steam lance injection.

The sound levels in plant and occupied areas shall not exceed the following, with HVAC systems only in operation:

- Switchgear Rooms/Relay Rooms NC50

- Control Rooms/Offices/Prayer Rooms NC30
- Workshops/Mess Rooms/Toilets/Corridors NC45.

Sound attenuators shall be provided in duct systems, as required, to achieve the above sound levels.

All HVAC units and systems shall be provided with local control panels for local manual/automatic control. The operation status such as alarms, humidity, temperature, fault indication, status indication of each air conditioning system shall be indicated and available at the local control panel and any individual fault signal results in a group signal, which shall be transmitted to the central control room.

In addition the following criteria shall be considered in the design of the HVAC system:

- (i) A general lighting level dissipation of 33 watts/m²
- (ii) Sensible and latent heat gains for occupancy should be allowed on the basis of one person per 10m².
- (iii) A minimum quantity of fresh air should be provided at the rate of 0.00125m³/s per m² of floor area or 12.5 l/s/person, whichever is the greater and this should be shown in the calculations. All equipment gains within each room should be allowed for in determining Plant capacities.
- (iv) Re-circulated air should be filtered. Air from areas where pungent fumes are likely to be present should not be re-circulated.

Battery rooms should be air conditioned by self-contained air-conditioning units. The design of the air conditioning system shall avoid recirculation of air back into the battery room. These units should be of the non-recirculating type and sized to provide the internal conditions previously specified with a minimum of five air changes per hour. Duplicate extract fans should be provided, mounted at high level and positioned such that accumulations of hydrogen cannot occur. The fan units should be suitable for long life in the acidic environment.

10.18.1 Fresh Air Requirements

Fresh air shall be supplied for the following purposes:

- Fresh air for occupied areas, minimum 15 litres/second/per person;
- Free cooling during periods of low ambient temperatures;
- Make-up air for exhaust ventilation systems;
- Building pressurization to exclude dust. Minimum fresh air supply shall not be less than 5 per cent of supply air volume.

Fresh air supplies shall be obtained from areas not subject to contamination from fumes. Air intakes shall be located at maximum available height, to reduce dust load.

Fresh air intakes on large systems shall have inertial sand filters with bleed fans and access sections. Small systems with fixed amounts of fresh air shall be provided with sand louvres.

All supply systems shall have two-stage air filtration, comprising washable primary panel filters and secondary bag filters, efficiency as required for the areas served.

10.18.2 Mechanical Cooling

Complete cooling systems shall be provided. The units and complete installation shall comply with ASHRAE 15 – Safety Code for Mechanical Refrigeration – and ASHRAE Guideline 3 for Refrigerant Leaks, Recovery, Handling and Storage Requirements.

Critical air-conditioning systems shall be provided with 100 per cent standby plant. Critical air conditioning systems are those which are essential for the continued operation of the buildings and allow continued occupancy of essential working areas.

HCFC refrigerants shall not be used.

10.18.3 Ductwork

Ductwork shall be fabricated, installed and tested in accordance with the latest SMACNA Standards. General supply and extract system ductwork shall be of galvanized sheet steel construction. Irrespective of pressure/velocity classification, ductwork at low level in plant rooms and in areas subject to mechanical damage shall be constructed (as a minimum) to medium pressure standard.

All duct systems shall be complete with dampers to facilitate balancing and fire dampers. Splitter dampers will not be accepted. Access doors shall be provided as required for inspection/maintenance of dampers, fire dampers, instruments etc. Final connections to ceiling-mounted supply and extract terminals may be carried out in insulated flexible ductwork.

Extract system ductwork from battery rooms and chemical areas shall be of PVC construction.

High-pressure ductwork shall be pressure/leak tested. Low and medium pressure ductwork need not be tested, except where it is located outdoors.

The following ductwork shall be insulated:

- Fresh air ductwork
- Low Temperature supply and extract ductwork.
- Medium temperature supply ductwork.
- Ductwork in Electrical rooms, where it is subject to condensation.

Fire dampers to UL Standards and NFPA-90A shall be provided on ductwork systems passing through firewalls and floors, and as required for compartmentation of gas fire suppression systems.

Dampers shall be operated by fusible link and (re-settable) damper motor, suitable for actuation through the fire detection system and fire damper control panel. Smoke dampers for clean agent fire suppression systems shall be motor operated/solenoid actuated for closing and opening purposes. Damper status shall be indicated on the control panel mounted adjacent to the Fire Control panel.

10.18.4 Air Filtration

All fresh air intakes on large central systems shall be provided with inertial type sand filters, with access sections behind. Small systems employing fixed volumes of fresh air shall be provided with sand louvres.

Primary and fresh air filters on all systems shall be permanent metal panel washable type, dust spot efficiency 20 per cent. A complete set of spare cells shall be provided for cleaning, along with coating fluids. Washing facilities shall be provided, with service water being used for cleaning.

Secondary air filters on medium temperature systems serving plant areas shall be replaceable bag filters, dust spot efficiency 50 per cent. Secondary air filters on low temperature systems serving offices/control rooms/electronic areas shall be replaceable bag filters, dust spot efficiency 90 per cent.

All bag filters shall be supported in proprietary mounting frames for front or side withdrawal. The Contractor shall supply a sufficient number of replacement cells to last during the test periods and during the Defects Liability Period, with one complete set being provided at that the end of the Defects Liability Period.

10.18.5 Power Supplies

The Contractor shall include for all power and control cabling and containment, MCC and controls for the complete HVAC systems.

Dual redundant power supplies (normal ac and safe ac) shall be provided for systems requiring 100 per cent standby plant, including mechanical cooling.

10.19 Electric Overhead Travelling Cranes / Lifting Equipment

Electric overhead travelling cranes shall be provided for ease of dismantling and maintenance of major plant items in each Power Unit. The cranes shall be capable of lifting and accurately positioning all loads within their specified rating.

The cranes shall be designed, manufactured and tested to an internationally recognised standard suitable for their intended application.

For guidance purposes, the following maximum crane speeds for the main turbine and other cranes should be considered:

- Hoisting 4.0m/min
- Trolley Travel 10.0m/min
- Bridge Travel 25.0m/min
- Creep speeds 5% of operating speeds

The cranes shall be capable of inching in all directions of motion at a speed of approximately one tenth of the maximum full load speed.

All plant and plant components requiring maintenance in excess of 0.5 tons shall be provided with a means of lifting by the use of cranes as defined above, electrically/manual operated hoists with runway beams or mobile cranes. Below 0.5t, lifting arrangement such as hoists, chain-pulley, monorails to be provided. Access for installation and lifting of the equipment below 0.5t shall be ensured.

The use of mobile cranes shall be justified by the Contractor and agreed with the Owner. In any event mobile cranes shall not be considered for lifts at a height of greater than 15m above the level at which the mobile crane is standing. Where the maintenance procedures prepared by the Contractor require more than three simultaneous lifts anywhere on the Facility permanent lifting facilities shall be provided for the additional lifts. Mobile cranes are not to be supplied under the scope of Contractor.

Runway beams shall be permanently installed above the plant requiring lifting devices, manual hoists may be transferred between runway beams.

All associated lifting beams and slings shall be provided.

The pump screening and stop log gate areas shall be serviced by a single permanently installed gantry crane operating on rails.

Risk study to consider also Crane equipment during design stage.

10.20 Mobile Plant and Equipment

Except where stated otherwise in this specification, mobile plant such as fork-lift trucks, cranes, sewage tankers etc required for the operation and maintenance of the Facility shall be provided by the Owner to a recommended list provided by the Contractor. All mobile plant required for EPC purposes, such as construction and commissioning, shall be provided under the scope of the EPC Contract.

10.21 Laboratory

The Contractor shall provide a fully equipped laboratory suitable for carrying out all testing requirements for the day-to-day operation of the Facility including, but not limited to:

- HRSG feed water.
- Steam.
- Drain water, including hydrocarbon content.
- Fuel testing.
- Demin water.
- Desalinated product water.
- Moisture analysis for lubricating and transformer oils.

The laboratory shall also contain suitable office accommodation for chemists (1 office having 3-people occupancy).

The fully air-conditioned laboratory shall be equipped with counter flanges for demineralized water and for potable water supply from the local distribution systems. Operator desk and respective furniture shall be included along with suitable storage facilities.

Chemical and analysis of wastewater shall be collected and drained separately.

The Contractor will present the list of laboratory test and measurement devices and furniture to the owner for approval. Laboratory analysis benches will have a suction fan to absorb toxic gases.

There will be separate and suitable storage cabinets to prevent chemicals from interacting. All chemicals to be used in the laboratory will have traceability with the certificate of analysis.

Analysis devices that will meet ISO standards and are compatible with international certification will be selected.

Analysis devices suitable for field measurements will be available in the laboratory (PH, Conductivity Meter, SDI, Hydrometer, Validation Devices).

Automatic sprinkler system protection shall be required for all new laboratories handling chemicals as per NFPA13, Cl:22.8.1.

11. OWNER'S MINIMUM INSPECTIONS & TESTING ON FACTORY AND SITE QUALITY ACCEPTANCE TESTS

All shop/factory inspection and test activities shall be separately indicated in Contractor's Inspection and Test Plan. The Contractor must deliver to the Owner a detailed plan of the equipment that will be subjected to a Factory Acceptance Test before the shipment to site. The Owner reserves the right to include on that list the equipment that must be subjected to this type of test and the applicable hold points for the Owner.

The general approach shall be that all equipment and materials are tested and inspected in the shop or factory to the maximum extent possible and where technically feasible. The extent of testing in the shop shall be indicated in the Inspection and Test Plan. The Inspection and Test Plan, relevant checklists, and formats available at the workshop shall be included in the EPC Contractors Quality Plan. After the Purchase Orders between the Contractor and the different Suppliers and sub-contractors are signed the Owner must be informed about the schedule of manufacturing, and inspections and receive the applicable Inspection and Test Plan applicable for the manufacturing/assembly works.

11.1 NDT Inspection Program

Refer to Owners Specific Technical requirements related to welding qualification and inspections program.

11.2 Tests on Mechanical Equipment in Manufacturer's Works

The tests shall be arranged to represent working conditions as closely as possible. Type Tests, when called for shall be made on equipment, which has previously passed its routine tests. Where Type Tests have been carried out under previous contracts on equipment similar in all essential respects to the equipment included in the EPC Contract, the Owner may waive the Type Tests on production of complete test records relating to the equipment concerned.

All instruments used shall have valid calibration certificates, and calibration data shall be made available to the Owner on demand. All the inspection, measuring and test equipment required for inspections and tests shall be provided by the Contractor.

11.2.1 GT and auxiliary equipment

The GTs shall be tested separately at the Contractor's workshop or at any other suitable location approved by the Owner. The tests shall be carried out to ensure smooth and satisfactory operating of the Unit. The following tests shall be carried out as a minimum:

- All critical raw materials like compressor/Turbine castings, rotor forgings, blades/wheels/buckets casting/forging raw material test certificates shall include the detailed foundry stage material specification required Heat treatment details either directly in the material MTC's and/or as a separate report.
- All the casting material used for compressor mid/inner casing, turbine shell, diffuser casing, buckets/blades shall be examined for 100% RT/UT at rough machined condition and 100% LPI/MPI on final machined conditions.
- All the forging materials used for rotors and wheels shall be tested for 100% UT at rough machined condition and 100%LPI/MPI on final machined condition.

- Run out check and High-speed dynamic balancing test of the rotor to be conducted at factory.
- Parting plane gap measurement for Turbine and compressor casing
- All the base frame welding shall be tested for 100% RT/UT & MPI
- Final assembly compressor and turbine rotor clearance and dimensional checks to be offered for Owner witness
- Preliminary checks and settings of all equipment and system
- Starting sequence control equipment set and operation checks.
- Manual shut down sequences, emergency shut down and tripping sequence checks. Other trips and alarms shall be subject to simulated tests.
- The GT rotor with blades shall be operated, by a test drive at high speed; balance test of GT rotor with the blades will be done up to 111% of the rated speed in accordance with Supplier's standard and
- Operation and setting of overspeed trips shall be checked and adjusted as required.

The operating characteristics of the following equipment and systems shall be checked:

- Ignition and combustion systems
- Turbine cooling
- Lube oil system
- Barring gear
- Starting equipment
- Control
- Protection
- Motors, pumps and heaters
- Performance load test
- Solenoid valves, stop valves, control valves and relief valves.
- Thermocouples, thermometers, and gauges; and
- Pressure switches, temperature switches and limit switches

Insulation tests shall be carried out on all control panels. Prior to test, electronic instruments and similar items shall be isolated to prevent damage. The insulation resistance of the remainder of the equipment shall be measured at a minimum of 500 V during manufacture. After reconnection of all equipment the circuits shall be checked. Function/sequence tests shall be carried out which simulate as far as practicable the performance of the turbines. These tests shall include all automatic sequencing, protective gear and alarms.

11.2.2 Steam Turbine and Auxiliary Equipment

The following Factory tests shall be carried out on the ST at the Manufacturer's works prior to dispatch to Site:

- Hydraulic test of casing (except LP casing) and piping, hydraulic tests for the turbine casing shall comply with IEC Recommendation and OEM's procedures
- All the HP & IP Casting shall be checked for 100% UT on rough machined condition and MT after final machined condition, weld repairs shall be documented with defectogram. All the heat treatment details, including the weld PWHT after the casting repairs shall be part of the raw material test documents.

- Hardness test on WC6, WC9 and C12A grade castings after the weld repair and PWHT shall be carried out and same shall be part of the Raw material test documents.
- All the forging raw material used for rotor/shafts, blades etc, shall be checked with 100% UT at rough machined condition and 100%MT on final machined condition.
- Residual stress measurement of the rotors forging after the final quality heat treatment shall be conducted to check the effectiveness of the implemented heat treatment.
- All the stationary and moving blades shall be checked for hardness, MPI and PMI
- Dynamic balance of rotor carried out with dynamic balance checked over the complete speed range up to maximum allowable overspeed and including critical speeds. The balancing weight and its distribution shall be properly recorded.
- Overspeed test: The turbine rotor shall be subjected to an overspeed test for a period of at two minutes at a minimum speed of two percent (2%) above the maximum calculated overspeed that would occur by the sole action of the overspeed trip device.
- HP, IP Casing assembly with Rotor – Clearance between the Radial and axial alignment of the inner casing and out casing to be checked and the same shall be performed as part of site assembly process for LP casing..
- Radial and axial blade/fins clearances in the assembled turbine shall be checked
- Casing bolts identification and tightening procedure verification

The no-load mechanical test at Site shall be performed as follows:

- Detailed test procedures shall be submitted by the Contractor and are subject to the Owner's approval
- Initial steam conditions and operating vacuum shall be as close to the rated figures as practicable
- When sufficiently warmed up a one-hour test run shall be made at maximum speed. Bearing temperatures and vibration readings shall be recorded
- Vibration readings shall be taken over the whole speed range at intervals of 100 rpm starting at idle speed and proceeding to maximum speed and vice versa, with special attention given to readings at rated speed. Every individual test shall last at least one minute. Testing shall not be performed close to the lower critical speeds
- Governor response shall be checked; and
- Overspeed and all protective devices shall be checked and adjusted where necessary.

11.2.3 Pipework

The supply, fabrication and erection of all pipework shall comply with the requirements of ASME B31.1, ASME B31.3, or alternate code as approved by the Owner. Pipework for the GT shall comply with International Standards. All butt welds shall be full penetration welds with complete fusion between weld runs and base metal. The following tests shall be carried out on welds (both Shop welds and Site welds):

- All the Pipe and fitting material for the critical piping and HRSG piping shall be only procured from the well-known manufacturers, approved by the Owner. The hardness testing and material heat treatment shall be clearly specified in the inspection program, which shall also to be witnessed by the Owner approved third party inspection services at the mills or at the fabrication workshop.
- EPC Contractor shall use the proper elbows and fittings for all the piping system, particularly for the high temperature steam turbine drain system. If, any site/shop manufactured bends are used, then same shall be post forming heat treated (PFHT) as per clause 129.3 of ASME B31.1 code as applicable.
- Visual examination of all welds to the requirements of ASME Boiler & Pressure Vessel Code Section IX relevant piping design codes.

- Ultrasonic or Radiographic and surface flow examination on all pipe butt welds and branch welds to the requirements of ASME Boiler & Pressure Vessel Code Section IX relevant piping design codes. The Owner reserves the right to elect one of the two methodologies of inspection depending of the welds to be inspected. However, for the Butt welds of thickness of 60 mm and above shall be tested either by X-Ray or by Phased Array Ultrasonic Testing (PAUT) method instead of Gamma rays due to its limitation. If PAUT method is selected, demonstration block as same as welding & material being tested is necessary in addition to code required standard calibration blocks.
- 10 Percent (10%) magnetic particle examination of all fillet welds.
- Hydrostatic test to the requirements of ASME Boiler & Pressure Vessel Code Section IX relevant piping design codes as applicable.
- The minimum requirements in terms of type of inspections and percentage of inspections must always as minimum fulfil the requirements of the Design, Manufacturing and Inspection Code. Furthermore, turbine lube oil, control oil and jacking oil piping systems butt welds shall be 100% RT tested, irrespective of the design code requirements. All the fillet and attachment welds shall be 100% PT tested.
- Over the minimum NDT requirement as specified in the design, Manufacturing and Inspection code as above, EPC Contractor shall conduct 10% minimum volumetric NDT per welder per system in order to evaluate the in-process weld repair rate for each welder.
- All the variable and constant load hangers supplied for the plant piping system shall be tested for 100% performance testing at the manufacturing shop as per MSS SP-58. Springs used for the hangers shall be subjected to scragging, cramping test and load vs deflection tests.
- All the pipe and fitting supplied to site shall be clearly traceable with respect to the heat number and original raw material test certificates.
- EPC Contractor to prepare the welder wise/system wise/material wise weld rejection rates and welder performance monitoring report for all the welders involved either in the manufacturing and/or site erection activities and same shall be submitted for Owner review and approval on weekly basis.
- EPC Contractor shall include or depute one qualified welding Engineer for every 20 welders working at site and/or manufacturing shop. The Contractor/sub-contractor welding Engineer and/or deployed NDT company shall share the real time NDT results on daily basis to the Owner for review.

Piping and piping support inside GTG package is designed and fabricated and tested in accordance with the specified code and standards, or equivalent (JIS code).

Weld defect acceptance levels shall be in accordance with ASME Boiler & Pressure Vessel Code except that lack of root fusion or incomplete root penetrant will not be permitted.

11.2.4 Structural Steelwork

Welding and inspection in both shop and Site shall be in accordance with AWS Standard or equivalent for Building Construction. Where only partial inspection is required the location of the joints selected for testing shall be agreed by the Owner. Should any of these welds prove to be defective on inspection, the number of welds to be tested in that system shall be twice the number originally specified. Should any of the second increment welds prove to be defective, then all the welds in that system shall be tested. Irrespective of the code and drawing requirement, any butt welds in the load bearing members of the HRSG and Steam Turbine building structures shall be tested for 100% volumetric NDT.

11.2.5 Castings, Forgings and Wrought Products

All castings, forgings and wrought products shall be subjected by the manufacturer to chemical analysis, material tests and inspections in accordance with the appropriate American Standard except where otherwise agreed by the Owner. Inspection should be scheduled in order to detect defects as early as possible, to avoid unnecessary delay in manufacture and delivery of the Plant. In case of a rejection

influencing the delivery time, written and certified notice must be given in due time to the Owner. The Owner reserves the right to examine and witness acceptance tests, prior to and following weld repairs and subsequent post weld heat treatment, mechanical tests etc, at the material supplier's works. Weld repair procedures are subject to approval of the Owner.

HP and IP casing of steam turbine are manufactured and tested according to EN. For the HP & IP rotor forging material, OEM maker standard is acceptable. However, during the project execution stage, EPC Contractor/OEM to provide the actual material details including the forging raw material test certificates from the forging supplier.

11.2.6 Dearator and Pressure Vessels

All pressure vessels shall be designed, manufactured and tested in accordance with ASME Boiler & Pressure Vessel Code or other accepted international Code. The vessels will be subject to inspection and certification by the Owner's Insurers. The Owner reserves the right to verify or witness any activity required by the Technical Specification. Pressure vessels for the GT shall comply either with International, ASME Standards or European Pressure Vessel Guideline.

11.2.7 HRSG Plant

The boiler Plant shall be designed, manufactured, and inspected in accordance with ASME Boiler & Pressure Vessel Code Section 1 Power Boilers. The boiler Plant will be subject to inspection and certification by the Owner's Insurers. The Owner reserves the right to verify or witness any activity required by the Technical Specification. Irrespective of the ASME Section requirements, all the tubular parts butt welds either in the manufacturing shop or at erection site of the HRSG shall be subjected to minimum of 20 % of volumetric NDT and Grade.91/92 material welding shall be tested for 100% volumetric NDT. All the welding involved in the flue gas ducts shall be subjected 10% MPI/LPI as applicable for the material of construction. Insulation of any pressure parts and duct works at site, shall only be conducted after the satisfactory NDT examination and Hydrotest as applicable.

The supplier shall qualify all bends involved in the tube banks with a proper mock-up qualification including the post forming/bending heat treatment requirements as mentioned in PG-19 & 20 of ASME Section 1 code as applicable. Copy of such mock up qualification records shall be part of EOMR/Manufacturing Quality Dossier.

Additionally, the list of tight bends (less radius) which requires PFHT (post forming heat treatment) as per ASME Section 1, PG-19 & 20 shall be provided for Owner review. The ovality of hot bends shall be within the 5% and the thickness difference between the extrados and intrados of these bends shall be within the 10% of wall thickness.

Supplier shall design and manufacture the HRSG pressure parts with a very minimum number of dissimilar welding requirements. Supplier shall complete all the dissimilar welding at the manufacturing shop instead of site welding. Any direct welding of high temperature stainless steel tube material with CSEF (Creep strength ferritic steel material) of grades. T91/T92 is not acceptable. Instead, supplier shall design the same with intermediate transition tube material of 'Inconel grades' or the 'buttering' of the Gr. T91/92 weld edges with Inconel filler material shall be considered to avoid the PWHT of stainless-steel tube material.

Furthermore, while designing such dissimilar material welding, supplier shall ensure that these welds should not fail at the weld fusion line cracks during the operation phase of the plant due to differential expansion of the different material at all the loads HRSG is subjected to.

Any Austenitic stainless-steel tubes of 'H' grades (like TP304H, S303432/Super304H, TP347H/HFG) used in the final SH & RH modules of the HRSG shall be welded with the matching process parameter filler wire like ER304H/ER308H in accordance with qualified and Owner approved WPS & PQR documents. The usage of 'L' grades filler material for any high temperature application is prohibited.

11.2.8 Valves

All castings and forgings shall comply with the requirements of this Technical Specification and ASME Boiler & Pressure Vessel Code B16.34, ASME B31.1 or other international accepted Code, as

applicable to the service of the valves. Main steam and HRH system Steam Turbine inlet stop, and control valves casting shall be checked for RT/UT in rough machined condition and MT after final machined conditions. All the major and minor weld repairs shall be documented with defectogram and same shall be part of manufacturing quality dossier. All valves shall be subject to hydraulic test at 1.5 times the PN rating of the valve. All the valves shall be back seat (as applicable) and seat leakage tests shall be conducted at the manufacturing workshop as per the applicable design and testing codes. In addition, valves supplied with butt weld ends shall be subject to ASME B16.34 or radiographic or ultrasonic examination for a length of 75 mm from the prepared weld tip and magnetic particle examination of the weld tip profile PT examination after welding will be applied. Valves for the GT shall comply with ASME Standards or equivalent Japanese or EN Standards.

All the valves casting of pressure class 900 and above shall be tested for 100% volumetric examination on body and bonnet. All the valves forging of pressure class 1500 & above, shall be tested for 100% UT examination. All the valves seat face, hard faced/stellated area shall be 100% LPI tested after final machining.

All the control valves shall be performance tested to verify Hysteresis, Repeatability, dead band, linearity, Sensitivity, accuracy, operating time, fail safe position, flow direction, Cv test for one size/per type etc. Type test certificates of standard Cv & trims shall be submitted for Owner review. In absence of any type test certificates or for project specific designs, Cv tests shall be witnessed Owner or its representative.

Safety Valves and Safety relief valves shall be subjected to 100% Pop up test at the manufacturing shop, in additional to standard body and seat leakage tests.

11.2.9 Pumps, Axial and Radial fans and Compressors

All castings and forgings shall comply with the requirements of this Technical Specification. In addition, the radii, changes in section and butt weld ends of the castings shall be subject to PT examination unless otherwise agreed by the Owner. Castings shall be subject to hydraulic test at 1.5 times the design pressure.

The high-pressure feedwater pumps impellers casting shall be volumetric tested after rough machining at the foundry and/or pump manufacturing shop. The NDT/RT sketch for the impeller shall be provided along with pump manufacturing ITPL.

Pump and compressor rotors shall be subject to ultrasonic examination in the rough machined condition and magnetic particle or dye penetrating examination in the final machined condition unless otherwise agreed by the Owner based on the pump manufacturer standard inspection procedure. Dynamic balance shall be carried out on the completed rotor as per Gr2.5 of ISO 1940 and performance tests carried out on the completed pumps. If selected OEM is able to technically justify the reason of inability to conduct the reference test in its premise, other QA methods will be discussed and applied, acting reasonably by both parties. Pumps, fans and compressors shall be factory tested with their contract motors or test motor, however Owner needs to approve for any test motor. The main Feed water pumps and Condensate extraction pumps shall be tested for NPSH test (one per type), in additional to the code required performance testing.

11.2.10 Air Ejectors (if applicable)

Each air ejector shall be tested to demonstrate compliance with the air extraction capacity required by HEI Recommended Practice for the design of Surface Type Steam Condensing Plant.

11.2.11 Motor Driven Auxiliaries

Running test will be done as per the manufacturer's standard.

11.2.12 Cranes

Principal items to be examined for compliance with the requirements of this Owner Technical Specification are main girders, end carriages and crab structure, hoist drums, hooks, ropes, slings and shackles. All the butt welds in tension flanges, web plates and rope drums shall be tested for 100% RT. All the butt welds and fillet welds between the tension flanges and webs shall be tested for 100% LPT/MPI. All the gear box supplied shall be submitted with relevant stage wise manufacturing document including the run test report as per the design and manufacturing code. All the assembled cranes shall be tested for no load, full load and

overload test at the manufacturing shop and as well as at erection site and these tests shall be witnessed by Owner approved third party inspection agency.

11.3 Tests on Electrical Equipment in Manufacturer's Works

The equipment provided is subject to inspection and tests by the Owner during the course of manufacture and on completion in order to ensure compliance with this Technical Specification. The Contractor shall give the Owner at least 15 days notice of materials for inspection or test. For tests of long duration at least one month notice shall be given.

The Contractor shall provide to the Owner as soon as practicable after tests have been witnessed three copies of the relevant test certificates. These shall contain details of each test performed. Records, results and calculations of all electrical tests shall be provided.

All tests as per Inspection and Test Plan (ITP) shall be carried out in the presence of the Owner unless otherwise agreed in writing. The Contractor will regard witness test as to be waived unless the Owner will advise his intention of witness test three (3) days prior to its test date. All instruments used shall have valid calibration certificates, and calibration data shall be made available to the Owner on demand. All the inspection, measuring and test equipment required for inspections and tests shall be provided by the Contractor.

Where type tests are called for these may be omitted at the discretion of the Owner only if the Contractor can provide certified test results for an identical or essentially similar equipment which satisfies the Owner. Such certified results shall be presented to the Owner for consideration no later than six months after the Contract date.

Six copies of test certificate detailing all the works tests performed shall be issued by the Contractor to the Owner within two months of completion of works tests on each item of Plant or equipment.

11.3.1 Generators

Inspection and Testing During Manufacture

The following tests, at minimum shall be performed, as applicable, during the course of manufacture:

- Rotor forging and rotor end winding retaining rings. An approved series of non-destructive tests.
- Overspeed: At 1.2 pu rated speed for 2 minutes on complete rotor at ambient temperature. Vibrations of shaft and bearings shall be measured before, during and after the overspeed test.
- Oil coolers and other fluid/water coolers. Hydraulic pressure test of 1.5 x working pressure applied to both sides of each cooler.
- Cooler tubes: Hydraulic pressure test of 1.5 x working pressure applied to each individual tube, unless otherwise approved in writing by the Owner.
- Rotor winding: High voltage tests as appropriate to the stage of manufacture.
- Stator core: An approved test at approximately rated flux (specifically for gas turbine generator at 80% of rated flux with calculation of 100% value) at rated frequency on the core before winding to verify the efficacy of the interlaminar insulation and correct assembly (Magnetic Polarisation test). EL-CID test is acceptable.
- Stator winding: Tests in accordance with IEC60894, IEC 60034-27-3 and other IEC 60034 series standards, including sample test (Type test) and stator end windings performance with Bump testing. Bump Test on end winding shall be performed on the Stator of project specific generators, applicable to all units under project scope.
- Generator casing: (hydrogen cooled machine). Hydraulic pressure test at 1.5 pu highest working pressure. For Steam turbine generator, air leak test at x 1.25 pu rated pressure is acceptable.
- Gas coolers: (hydrogen cooled machine). Hydraulic pressure test of 1.5 x working pressure applied to the water side.

11.3.1.1 Routine tests

The following tests, except for the ones marked as "Type Test" shall be carried out on all the generators. For tests mentioned as type test, these shall be applied to the first generator of each type once complete:

- Generator casing: (hydrogen cooled machine). Leakage test at rated gas pressure.
- Winding resistances: The stator and rotor winding resistances and relevant cold state temperatures shall be measured.
- Phase sequence: The phase sequence shall be checked and permanently marked.
- No-load saturation characteristic: The no-load saturation characteristic shall be measured in accordance with IEC 60034-4 or ANSI/IEEE Standard 115. A certified type test report duly signed by client of previous other project is acceptable.
- No-load heat run: The stabilised temperature rises, temperatures and losses at rated voltage and rated speed shall be measured. A certified type test report duly signed by client of previous other project is acceptable.
- Short circuit characteristic (Type Test): The sustained three-phase short circuit characteristic shall be measured in accordance with IEC 60034-4 or ANSI/IEEE Standard 115.
- The direct-axis synchronous reactance and the short circuit ratio shall be determined from the results of the no-load saturation and sustained three-phase short circuit tests. A certified type test report duly signed by client of previous other project is acceptable.
- Short circuit heat runs (Type Test): The stabilised temperature rises, temperatures and losses at rated current and rated speed shall be measured.
- Unexcited run (Type Test): The stabilised temperature rises, temperatures and losses with the generator unexcited at rated speed shall be measured.
- Temperature rises and temperatures (Type Test): The temperature rises and temperatures at rated output of the generator shall be predicted from the results of the no-load and short-circuit heat runs and the unexcited run by an approved method.
- Vibration: Vibration of all bearings shall be measured during the running tests. A certified type test report duly signed by client of previous other project is acceptable. During the commissioning test of generator, vibration level shall be complied with international code requirement confirming the type test reference value.
- High voltage tests: The stator shall be tested in accordance with IEC 60034-1 or ANSI/IEEE Standard 115.
- High voltage tests: The rotor shall be tested at standstill in accordance with IEC 60034-1 or ANSI/IEEE Standard 115.
- Insulation resistance: The insulation resistance of both the stator and rotor windings shall be measured before and after the respective high voltage tests.
- Rotor impedance: For Gas turbine, the rotor impedance to an alternating voltage shall be measured before and after the rotor's dynamic balancing and over speed test.
- Rotor impedance: For Steam turbine, the rotor impedance to an alternating voltage shall be measured before the running tests.
- Dielectric measurements: The dielectric loss angle (tan delta), capacitance to earth and dielectric loss energy characteristics shall be measured for each phase of the stator windings.

11.3.1.2 Type tests

The following tests shall be carried out on the first generator:

i) Voltage waveform and harmonic analysis.

The wave forms of the generator phase and line voltages at rated speed and voltage on open circuit shall be recorded and the telephone harmonic factor (THF), or total harmonic distortion (THD) for Steam turbine, measured in accordance with IEC 60034-4.

ii) **Sudden three-phase short-circuit tests.**

The sudden three-phase short-circuit tests shall be carried out in accordance with IEC 60034-4. The tests shall be made with the generator excited to give 0.3, 0.5 and 0.7 pu rated voltage on no load.

Specifically for Steam turbine, OEM is required to engage an independent 3rd party to validate the test result of 0.15 p.u. value and validate the extrapolation to 0.3, 0.5, 0.7 and 1.0 p.u. (including unsaturated and saturated values).

Specifically for gas turbine generator, test shall be performed at 0.15 pu for unsaturated value whereas the saturated values shall be calculated using the extrapolation method in line with IEC 60034. The values at 0.3, 0.5 and 0.7 pu shall be calculated and shared for Owner's review and approval.

Saturated and unsaturated values of the direct-axis sub-transient reactance and direct-axis transient reactance together with the direct-axis sub-transient short-circuit time constant, the direct-axis transient short-circuit time constant, and the armature short-circuit time constant shall be determined from the tests.

After the short circuit tests the machine shall be inspected to check that the windings and support structure have not been damaged or distorted.

iii) **Negative-sequence reactance.**

The negative-sequence reactance shall be determined from the line to line sustained short-circuit test in accordance with IEC 60034-4.

iv) **Zero-sequence reactance.**

The zero-sequence reactance shall be determined from the application of single-phase voltage to the three phases in accordance with IEC 60034-4.

v) **Losses and efficiency**

Tests shall be performed to determine generator losses including those associated with the excitation system and equipment where for excitation system the individual losses for each of its components shall be derived from their individual design value and type test report, where mandated by relevance standard

The efficiency of the generator shall be predicted from the results of the above tests.

Losses and efficiencies shall be stated for 0.25, 0.5, 0.75 and 1.0 pu generator rated output.

11.3.2 Generator Excitation Equipment

11.3.2.1 General

Type and routine tests shall be carried out on all the excitation equipment and excitation control equipment. The type tests shall include environmental tests. Substitute certificates tests may be accepted, subject to Owner's approval.

11.3.2.2 Thyristor Rectifier Type Tests

The excitation rectifier shall be type tested in accordance with IEC 60146. The type test certificate as per IEC shall be submitted for owner review.

The light load test shall be performed at a voltage equivalent to that existing at the excitation transformer secondary terminals when the generator terminal voltage is at its maximum steady state value as defined in the schedules.

Current and temperature rise tests shall be performed at the excitation system rated current as defined in the Technical Specification. Cooling conditions during the temperature rise test shall represent site conditions. During these tests the current equivalent to field forcing conditions shall be applied for 10 seconds at agreed intervals.

The ability of the rectifier to withstand the field current induced by a sudden short circuit at the generator terminals at full load shall be demonstrated.

The insulation test voltage applied to the equipment shall not be less than the test voltage applied to the generator field winding insulation.

Test details and methods shall be agreed with and approved by the Owner.

11.3.2.3 Thyristor Rectifier Routine Tests

Routine tests in accordance with IEC 60146 shall be performed on each rectifier equipment.

The insulation test voltage shall not be less than the test voltage applied to the generator field winding insulation.

Test details and methods shall be agreed with and approved by the Owner.

11.3.2.4 Excitation Transformer Type Tests

Transformers for the thyristor rectifier equipment shall be type tested in accordance with IEC 60076 and IEC 60146. The type test report as per IEC shall be submitted for Owner's review.

Cooling conditions for the transformer shall be equivalent to conditions, which will exist in service.

Test details and test methods shall be agreed with and approved by the Owner.

11.3.2.5 Excitation Transformer Routine Tests

Transformers for the thyristor rectifier equipment shall be subjected to routine tests in accordance with IEC 60076 and IEC 60146. Test details and test methods shall be agreed with and approved by the Owner.

11.3.2.6 Field Circuit Breaker Type Tests

The type test report as per IEC shall be submitted for Owner's review.

Type test certificate/report from manufacturer is acceptable if the tests are performed at an ISO 17025 (lab accreditation standard) certified test lab.

For these tests the field circuit breaker shall be mounted as in service. Where necessary the tests shall be carried out in a direct current short circuit testing laboratory. The tests performed shall include the following:

- a) demonstration that the circuit breaker can interrupt a direct current equal to the maximum direct component of field current resulting from a short circuit at the generator output terminals at rated load; the restriking voltage shall be equal to the direct voltage equivalent of the rms ac insulation test voltage for the field winding plus the excitation system ceiling voltage
- b) verification that the time between energising the trip coil circuit and final arc extinction does not exceed the specified value
- c) demonstration that the discharge circuit contacts can close onto a circuit carrying a direct current equivalent to the worst case possible in service, and that these contacts do not bounce
- d) verification that the interval between main and discharge contact operation is sufficient to ensure satisfactory operation of both circuits
- e) demonstration of circuit breaker closing
- f) demonstration of correct operation of all auxiliary switches
- g) demonstration that the circuit breaker can complete at least 1500 mechanical close-open cycles without the need for adjustment or maintenance; and
- h) dielectric tests.

If in addition to field suppression duty the circuit breaker also has to interrupt the current due to a short circuit across the connections between the rectifier and the generator field winding terminals then the following test shall be included:

- a) demonstration that the circuit breaker can interrupt the maximum current due to a short circuit across the rectifier terminals, and that short circuit current is limited to a value which will not damage the rectifier.

In case, the type tests are not previously performed or performed tests do not fulfil all the requirements as outlined above, the tests and the manner of their execution shall be agreed with and approved by the Owner.

11.3.2.7 Field Circuit Breaker Routine Tests

Routine tests shall be performed on each circuit breaker to demonstrate correct mechanical operation, electrical control circuits, auxiliary circuits and high voltage withstand. The tests shall be agreed with and approved by the Owner.

11.3.2.8 Type Tests on Excitation Control Equipment

Type tests shall be carried out on the first complete excitation control equipment to prove compliance with the Technical Specification. These tests, and the manner of their execution shall be agreed with, and approved by the Owner, and shall include at least the following:

- a) Range and rate of travel of reference voltage setting device.
- b) Static and dynamic characteristics of the automatic voltage regulator. (shall be performed as routine test at factory for each unit)
- c) Operation of limiters. (shall be performed as routine test at factory for each unit)
- d) Sensitivity of excitation system to frequency variations within specified operating range.
- e) Range and sensitivity of manual regulator.
- f) Demonstration of diode (or Thyristor bridge) failure detector. (shall be performed as routine test at factory for each unit)
- g) Demonstration of all alarm and protection circuits including changeover from automatic to manual control.
- h) Thermal cycle and soak tests.
- i) Demonstration that control circuits are immune to interference

11.3.2.9 Routine Tests on Excitation Control Equipment

An approved series of tests shall be carried out to prove the correct assembly and functioning of the equipment and its compliance with the Technical Specification.

11.3.3 General Test Requirements for Electronic Equipment for Generating Plant

Electrical interference type tests. The fully assembled and operating control and regulating equipment shall be subjected to simulated electrical interference testing in accordance with International standards as specified below. On completion of these tests agreed functional tests shall be performed.

Test apparatus and methods of testing shall be agreed between the Owner and the Contractor.

Acceptance criteria. There shall be no malfunction during or immediately following the test being performed.

If a component failure occurs during a test the component shall be replaced and the test repeated on the whole equipment before proceeding to the next test in the series. If the module containing the failed component is replaced then the test series shall start again from the beginning. If there is a second failure of an identical component the equipment shall be rejected. If a second, but different component fails it may be replaced and retested as above. If further component fails then the equipment shall be rejected.

The equipment shall also be tested for immunity to radio interference. The test shall be made with the equipment operating normally and cubicle doors closed. If the equipment can be operated with its cubicle doors open the test shall be repeated in this condition. A portable transceiver or comparable apparatus shall be used operating over the range 10 to 500 MHz to give a field strength of 10 V/m at the equipment

on test. The following table gives the approximate distance between transmitting aerial and test object related to transmitted power.

Type tests on electronic control system modules. One module or fully assembled printed circuit card of each type shall be subjected to tests in the categories listed below.

- a) performance test to verify compliance with design intentions and Technical Specification
- b) demonstration of satisfactory operation during specified power supply variations and interruptions
- c) environmental tests in accordance with the following IEC standards:

60068-2-2	Test B	Dry heat
60068-2-6	Test Fc	Vibration
60068-2-32	Test Ed	Drop

Substitute certificate tests for electronic equipment for excitation and static starter may be accepted, subject to Owner's approval.

11.3.3.1 Thermal stability and short time failures

The following tests shall be carried out on each equipment to demonstrate reliable and repeatable operation, and as far as practical, to eliminate weak components.

- a) Thermal cycle test.

The complete electronic equipment shall be subjected to two thermal cycles. The ambient temperature of the equipment shall be raised from the test room ambient to a steady 55°C and held for at least 1 hour. The temperature shall then be allowed to fall to test room ambient. When the equipment has reached a steady temperature the cycle shall be repeated.

The equipment shall be energised for the duration of the test. Agreed functional tests shall be carried out at normal ambient and at 55°C to demonstrate that there is no significant change in performance, either temporary or permanent, due to the effects of temperature.

- b) Soak test.

The complete equipment shall be energised for at least 5 Days before dispatch. The hours can be accumulated during other specified tests with the balance made up as necessary.

During the test the failure rate for semiconductor components shall not exceed 1 percent for components which have not been previously burnt in. Unless warranted by the number of identical components used in the equipment there shall be no second failure of an identical component. Failures of passive components and of burnt in semiconductor components shall not exceed the mean time between failure figures for the equipment.

Test details and, where appropriate, also the test methods shall be agreed with and approved by the Owner.

11.3.4 Generator Switchgear

Type and Routine Tests shall be carried out generally in accordance with the requirements of IEC 62271-37-013 and with an X/R ratio appropriate to generator circuit switching.

11.3.5 Main Connections

11.3.5.1 Type Tests

The main connections shall be type tested design to IEEE C37.23. A complete section of each type of main connection equipment incorporating an approved selection of fittings, connections etc, with at least one right angle bend and a vertical section shall be erected as in-service for the purpose of these tests. Tests shall be carried out as follows:

- a) Test of fault current carrying capability. The rated fault current carrying capacity, as stated shall be proven by means of short circuit tests.

- b) Temperature rise. The temperature rise shall be measured at points to be selected by the Engineer and shall, unless otherwise approved, be by means of maximum reading thermometer fixed to the conductor in an approved manner.
- c) Conductivity. The resistance of the lengths of conductors containing joints, flexible connections and connectors when measured, and expressed as a percentage of equivalent lengths without joints, flexible connections or connectors, shall not exceed the figures stated.
- d) For the resistance tests a current of not less than 100 amperes dc shall be used. The electrodes used for measuring the voltage drops shall be placed 250 mm away from the nearest part of the joints, flexible connections or connectors. The distance between electrodes used to measure the resistance of conductors containing the joints, flexible connections and connectors.
- e) Power frequency test. The connection section shall be subjected to an ac high voltage test at the value stated for one minute. The test shall be carried out at 60 Hz and the rms voltage shall be measured in an approved manner. Each phase shall be tested separately with the other two phases earthed. The insulation resistance shall be measured before and after the HV test in an approved manner.
- f) All auxiliary equipment (current and voltage transformers, surge arresters, neutral grounding resistors, enclosures, and control panels etc., as applicable) shall be type tested to relevant IEC/IEEE standards.

Routine tests. The extent of and programme for routine tests for main connection as well as it's auxiliary equipment (current and voltage transformers, surge arresters, neutral grounding resistors, enclosures, and control panels etc., as applicable) shall be according to relevant IEC/IEEE standards and shall be agreed with the Owner.

11.3.6 Neutral Earthing Equipment

Type tests and routine tests shall be agreed with the Owner.

11.3.7 Motors

Routine and Type Tests shall be carried out as defined in IEC 60034.

11.3.8 Switchgear

11.3.8.1 Type Tests

The Contractor shall produce a certificate from an independent testing station to demonstrate that his equipment (and its component parts) has been fully tested in accordance with the appropriate IEC Standards and the specified ratings. In the absence of satisfactory evidence, type tests shall be carried out by the Contractor for complete assemblies (IEC 62271 series for MV switchgear and IEC 61439 series for low voltage switchgear), as well as for any non-type tested component to its relevant IEC standard.

11.3.8.2 Routine Tests

The components of each switchgear system shall be routine tested, in accordance with the relevant IEC standard, at the appropriate manufacturer's works whereas complete switchgear assemblies shall be subjected to routine tests as outlined in relevant IEC standards (IEC 62271 series for MV switchgear and IEC 61439 series for low voltage switchgear).

11.3.9 Transformers

11.3.9.1 Type Tests

The Contractor shall provide evidence of type tests for transformers of similar design, rating and construction as specified in IEC60076 for Power Transformers and / or IEEE C57.12.00, 2015 - General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers, IEC 61869-2 (Current Transformer) or IEC 61869-3/5 (Voltage Transformer). In the absence of satisfactory evidence type tests shall be carried out by the Contractor. Where, some type test of current transformer, the provision of IEC 61869-2_7.2.2.205(temperature rise test), IEEE C57.13_8.6.2(short time current test) can be accepted.

11.3.9.2 Routine Tests

The transformers shall be subject to routine tests in accordance with IEC60076 and / or IEEE C57.12.00, 2015 - General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers, IEC 61869-2 (Current Transformer) or IEC 61869-3/5 (Voltage Transformer).

11.3.10 DC Supplies Equipment

DC batteries and charger panels shall be type and routine tested to relevant IEC standards in addition to following performance test to be performed on a sample battery.

Performance tests. One battery shall be subjected to the emergency discharge and recharge time requirements of the Technical Specification. Voltage/time, current/time and specific gravity/time curves shall be plotted throughout the recharge period.

The automatic and manual voltage control system and alarm and indication functions of the chargers shall be demonstrated in accordance with Inspection and Test Procedure agreed in advance with the Owner.

11.3.11 Cranes

Electrical tests on the cranes in manufacturer's works shall include:

- a) thyristor and rectifier assembly
- b) control gear and wiring
- c) contactors
- d) motors

11.3.12 Protection Equipment

11.3.12.1 Routine Tests

All relays shall be subjected to routine tests at the manufacturer's works to confirm that they comply with the claimed performance and design limits.

For measuring relays (i.e. relays which have a defined setting of the input and/or characteristic quantity subjected to accuracy requirements, e.g. current, time, etc) these routine tests shall include as a minimum the following:

- a) Measurement of the assigned error(s) under reference conditions, i.e. measuring accuracy and operating time characteristics.
- b) Measurement of the resetting ratios.
- c) Dielectric tests as specified in Clause 6 of IEC 60255-5, the test voltage being 2 kV rms. All normally open output contacts of all relays shall withstand a test voltage of 1 kV rms.

For all-or-nothing relays, the routine tests shall include a check of relay operation and resetting, together with the dielectric tests described above.

Unless otherwise agreed with the Owner, all unit protection schemes using either biased differential, current balance or voltage balance principles shall be subjected to heavy current conjunctive tests using the actual current transformer windings which will be used in service. Tests shall be made to prove operating sensitivity, time of operation and to demonstrate stability of the protection under the worst transient external fault conditions. Tests will only be waived if the manufacturer is able to produce type test results for an identical scheme. In this case it will be sufficient to prove that individual component characteristics are identical, e.g. current transformers are of the same design, have the same magnetization characteristics, knee-point voltage and secondary resistance.

Each circulating current protection scheme shall fulfil the following routine testing requirements:

- a) Each current transformer, which must be of the low reactance type, shall be individually tested for turns ratio, secondary winding resistance and excitation characteristic up to a secondary voltage equal to 120 percent of the "knee-point" voltage.

- b) The VA consumption at operation of current operated relays shall be measured and shall not exceed the maximum value declared by the manufacturer.
- c) The operating current of voltage relays shall be measured and shall not exceed the maximum value declared by the manufacturer.

11.3.12.2 Type Tests

Approved type tests shall be carried out in the manufacturer's works on each type of protective system. The protection system shall be type tested but type testing of cabinet itself shall not be required. During the tests, ancillary equipment shall be erected and connected so as to reproduce service conditions as closely as possible. The main purpose of these tests shall be to determine the performance of the protection for the range of system conditions, which will be encountered, by the protection in practice, and to determine all the appropriate application parameters. The test condition shall be as agreed by the Owner.

Where type tests have been carried out under previous contracts on protective equipment similar in all essential respects to the equipment included in the Contract, the Owner may waive the type tests on production of complete test records which he approves, relating to the equipment concerned. Each set of test records shall include a full statement of the performance claims, e.g. performance under reference conditions, effect of influencing the quantities, steady state and dynamic stability for unit protection schemes, current and voltage transformer requirements, etc and full details of tests performed on representative samples of production equipment to demonstrate that the performance claims have been met.

For circulating current protective schemes using high impedance relays, calculation of the predicted performance will be accepted in lieu of type testing of each individual scheme. Type test results must, however, be available for inspection by the Owner to show that heavy current conjunctive tests have been performed on a representative scheme to prove the stability and operating time performance. Details of the accepted method of calculation are given below.

11.3.12.3 Definitions

a) **Low-reactance current transformer.** All current transformers are in line with IEC 61869-2. A current transformer shall be deemed to have low-reactance performance, without test of further proof, provided:

- i. that the core is of the jointless ring type;
- ii. that the secondary turns are substantially evenly distributed along the whole length of the magnetic circuit except that a circumferential spacing not exceeding 19 mm on the outer periphery or subtending an angle not exceeding 30°, whichever is the greater, shall be permissible between the two ends of the winding;
- iii. that the primary conductor passes through the approximate centre of the core aperture, and
- iv. that there is normal spacing between adjacent phase conductors.

Unless otherwise indicated, post-type current transformers satisfying requirements i. and ii. above shall be deemed to be of the low-reactance type without test or further proof.

b) **High impedance relay circuit.** This is defined as a relay or relay circuit whose setting voltage is not less than the calculated maximum voltage which can appear across its terminals under the assigned maximum through-fault condition, it being assumed that:

- i. there is a complete saturation of the current transformer having the highest loop-resistance including its own secondary winding resistance between it and the relay circuit terminals;
- ii. the primary winding of this current transformer carries the assigned maximum through fault current and

- iii. the remaining current transformers transform their primary currents perfectly.

The main parameters of the relay, or relay circuit are:

- i. the setting of the relay which depending on the type of relay concerned, may be a percent, current or voltage value;
 - ii. the value of the stabilising resistor (applicable only when the relay is of the type calibrated in terms of current);
 - iii. the values of any shunt resistance, linear or non-linear used for fault setting adjustment and/or peak secondary voltage limitation.
- c) **Relay terminals.** These are the terminals on the relay to which the internally mounted coil current components are connected. The circuit between these terminals thus excludes any externally mounted components such as series stabilising resistors and linear and non-linear shunt resistors, which may be provided.
- d) **Relay circuit terminals.** These are the points at which the relay circuit leads join the leads which connect the current transformers in parallel. The relay circuit thus indicates all the externally mounted components, such as stabilising resistor and linear and non-linear shunt resistors, connected as in service.
- e) **Relay circuit setting current.** This is the minimum value of current in the relay circuit, with the relay set at its prescribed setting, to produce operation of the relay.
- f) **Relay circuit setting voltage.** This is the minimum value of voltage, which must be applied between the relay circuit terminals, with the relay set at its prescribed setting, to produce operation of the relay.
- g) **Current transformer "knee-point".** The "knee-point" of a current transformer marks the upper limit of the roughly linear portion of the secondary winding excitation characteristic and is defined exactly in British practice as the point at which a 10 percent increase in rms exciting voltage produces 50 percent increase in rms exciting current. The excitation characteristic is that determined by applying sinusoidal exciting voltage to the secondary winding with the primary winding on open-circuit. The exciting voltage and current at the "knee-point" are termed the "knee-point voltage" and "knee-point exciting current" respectively.

11.3.12.4 Performance Requirements

The primary fault setting and through-fault stability limit of the equipment shall satisfy the following requirements:

- a) **Primary fault setting.** The effective setting shall be as set out in this Technical Specification.
- b) **Through fault stability limit.** The limiting current for through phase or earth fault shall not be less than that appropriate to the specific application as specified in this Technical Specification.

11.3.12.5 Declaration of Prescribed Relay Circuit Parameters

In submitting equipment under these terms, the manufacturer shall provide the following information:

11.3.12.5.1 Current calibrated relays.

- i. the nominal current settings of the relay together with the maximum VA consumption of the relay at those settings;
- ii. the prescribed current setting (i.e. the setting to be used with the particular equipment concerned) and
- iii. the design values, with tolerances, of all external ancillary apparatus.

11.3.12.5.2 Voltage calibrated relays

- i. the nominal voltage settings of the relay with all series-connected ancillary apparatus (internal or external) in circuit, together with the maximum relay operating current at those settings;
- ii. the prescribed voltage setting (i.e. the setting to be used with the particular equipment concerned) and
- iii. the design values, with tolerances, of any shunt connected ancillary apparatus.

11.3.12.6 Acceptance Conditions for Equipment Submitted on Basis of Calculated Performance

All relevant details of the protected transformer or system and the protection equipment, including secondary lead lengths and corresponding resistance values, shall be given in the appropriate test report form together with the results of the calculations specified. The test report shall also include a schematic diagram showing the electrical connections of the equipment, the CT ratios and secondary winding resistance values, all lead resistance values and the relevant relay or relay circuit parameters.

To be acceptable the equipment must satisfy the following requirements:

- a) The through-fault stability limit calculated shall not be less than the value stipulated in this Technical Specification.
- b) The primary fault setting calculated shall not exceed the limit given in this Technical Specification.
- c) No current transformer of the set shall have a "knee-point" voltage less than twice the relay circuit setting voltage.
- d) The peak secondary voltage developed with maximum internal fault current as obtained by test or calculation shall not exceed 3000 volts.
- e) No current transformer of the set shall have a percent turns-ratio error greater than twenty-five times the ratio of the calculated primary fault setting to the assigned maximum through-fault current.

11.3.13 Instrument Transformers

Current and voltage (potential) transformers shall be type and routine tested in accordance with the requirements of IEC 61869-2 and 61869-3 respectively.

The current transformers shall be wired to short-circuiting, screw type terminal blocks mounted in the transformer control panel.

11.3.14 UPS equipment

Type tests and routine tests to relevant IEC standards shall be performed for complete UPS system including DC batteries.

11.3.15 Power Cables

Power cables shall be type, routine and sample tested in accordance with the requirements of IEC 60502 and IEC 60811. Wiring cables shall be in accordance with IEC 60227.

11.3.16 Lighting and Small Power Installation

The lighting and small power installation, component parts thereof and all accessories shall be tested in accordance with the relevant standard.

For special application like sockets, lighting and heaters, earth leakage circuit breakers with residual current rating of maximum 30mA shall be installed.

Storage batteries used as source of power of emergency lights, shall be of suitable rating and capacity to supply and maintain the total load for a minimum period of 1½ hours without the voltage applied to the load falling below 87.5% of normal as per NFPA70 article 700.

11.3.17 Metering and Instruments

Indicating and recording instruments shall be type and routine tested in accordance with the requirements of IEC 60051 and 60258 respectively. Integrating meters shall be type and routine tested in accordance with the requirements of IEC 62052-11, IEC 62053-11, IEC 62053-21 and IEC 62053-22 as appropriate. Transducers associated with metering shall be tested in accordance with the requirements of IEC 60688.

11.3.18 Synchronising Equipment

Automatic synchronising and check synchronising equipment shall be tested to demonstrate compliance with the specified requirements. Test details and methods shall be agreed with and approved by the Owner.

11.3.19 Steam Turbine, DCS and Auxiliary Stand-Alone Control Systems

For each separate control systems, the correct operation of all inputs, outputs, control functions, monitoring functions, operator display systems, printing systems and all associated systems shall be tested in line with the requirements of IEC 62381 for FAT as well as SAT and SIT. The factory acceptance testing shall be witnessed by the client or his nominated representative and shall include, in addition to IEC 62381:

- a) high voltage testing of all circuits, as specified in the relevant standards or approved standard
- b) insulation resistance testing of all circuits

11.4 Tests on Electrical Equipment at Plant

11.4.1 Switchgear and Ancillary Equipment

11.4.1.1 Tests before Completion

These shall comprise Site testing to demonstrate complete satisfactory operation of the equipment after transport and installation generally in line with the routine tests in the manufacturer's works. EPC Contractor will submit the detailed ITP as well as test procedures for the Owner's review and approval, prior to start of the activity.

11.4.1.2 Commissioning and Setting to Work

This shall include the following:

- a) The checking of all connections, and conductivity testing of bus-bar joints.
- b) The checking and ringing through of all small wiring and multi-core cables connected to the equipment.
- c) Dielectric testing of all primary circuits on the equipment.
- d) Insulation resistance measurement of all secondary circuits on the equipment.
- e) Primary injection of all protective and metering circuits to check satisfactory operation.
- f) The checking for satisfactory adjustment and operation of all circuit breakers, switches, and mechanical and electrical interlocks.
- g) Secondary injection tests on relays and instruments to check operating characteristics.
- h) Tests to check the magnetism curve, polarity and resistance of current transformers.
- i) Final setting of all protection.
- j) Secondary injection of voltage transformer circuits.
- k) Phasing tests prior to making live.
- l) On-load checks of protection, indicating and metering circuits.

11.4.2 Transformers

11.4.2.1 Tests before Completion

These shall measure:

- a) Insulation resistance of core and windings at not less than 2000 V dc.
- b) Insulation resistance of auxiliary wiring.
- c) Proving of manual and electrical operation of each device.
- d) Operation of equipment in accordance with the overall control scheme.

11.4.2.2 Commissioning and Setting To Work

This shall include ratio and no-load current at reduced voltages on all tapplings. Vector relationship at reduced voltage. Sweep frequency response analysis (SFRA) for main transformers.

11.4.3 Battery Equipment

11.4.3.1 Tests before Completion

After erection of the batteries/chargers and switchboards the following tests shall be carried out:

- a) The checking of all connections.
- b) 500 V insulation testing of all primary and secondary circuits.
- c) Primary injection of all protection and indication circuits.
- d) The checking for satisfactory adjustment and operation of all circuit breakers, switches and mechanical and electrical interlocks.

11.4.3.2 Commissioning and Setting To Work

- a) First charging of the batteries in line with the manufacturer's recommendation while monitoring cell voltage, acid concentration and cell voltage
- b) Equalizing charge of the batteries for a period of min. 1 week but not less than the manufacturer's recommendation
- c) Discharge test of the battery with adequate testing equipment (automatic adjustment of the current as the voltage changes) in order to prove the Ah capacity

11.5 Inspection and Testing During Site Inspection

11.5.1 General

The Contractor shall be responsible for the submission to the Owner of all Plant supplied under the contract for inspection and testing during site inspection, to ensure correct erection and compliance with the specification.

During the course of erection, the Contractor shall provide access as required by the Owner for inspecting the progress of the works and checking its accuracy to any extent that may be required.

11.5.2 Mechanical Equipment

The extent of testing during erection shall include, but not be limited to, the following:

- a) Checking the accuracy and alignment of Plant erected. The accuracy shall comply with the relevant standards, the specification or the Plant manufactures requirements as may be applicable or where no requirements exist, to a standard to be agreed between the Owner and the Contractor.
- b) Checking the alignment of rotating equipment to the manufacture's requirements (under commissioning scope, to be performed after Turn Over from Construction to Commissioning)
- c) Non-destructive testing of site welds as required by the relevant standard and as detailed in this specification (under commissioning scope, to be performed after Turn Over from Construction to Commissioning)

- d) Hydrostatic testing of pipework systems at a pressure of 1.5 times the design pressure, but not less than 4.5 bar for a period of 2 hours, or at such other conditions as may be required by the pipework design code.
- e) Site fabricated tanks and vessels shall be subjected to hydrostatic tests in accordance with the relevant standards
- f) Hydrostatic tests shall be carried out with demineralized water on steam generating units in accordance with the boiler design code.

After the hydrostatic test is carried out, the complete assemble shall be drained and any non-drainable sections shall be injected with an oxygen- absorbent chemical and elements plugged.

Pressure vessels and other parts including pipework that are made up on site and are subject to pressure or vacuum under normal or abnormal working conditions, shall be subjected to a site hydraulic test at approved pressures and for approved periods in accordance with the relevant standard. Where no appropriate standard exists, the hydraulic test pressure or at such pressure to be agreed by the Owner.

Vacuum tests shall be carried out at the maximum test vacuum obtainable and condition maintained for a period of two hours with the system isolated. The Contractor shall state in the schedule of guarantees the fall in vacuum anticipated during this period but the rate of fall shall not exceed that which would occur due to the designed air leakage rate being attained.

- a) Calibration checks on all instrumentation
- b) Tests to demonstrate the correct functioning of the control loops, protective, interlocks, and alarms (under commissioning scope, to be performed after Turn Over from Construction to Commissioning)
- c) Flushing out of all pipework systems which have not been fully cleaned and sealed before shipping to site (under commissioning scope, to be performed after Turn Over from Construction to Commissioning)..
- d) Other tests as specified which have not been previously conducted

11.5.3 Electrical Equipment

The following general tests are to be carried out on electrical equipment after erection at site:

- a) Routine high voltage tests to the appropriate standards. Where no relevant standard exist, tests shall be agreed with the Owner
- b) insulation resistance tests on all electrical cables (to be performed by construction team before mechanical completion)

Tests to be performed after turn over from construction to commissioning

- a) insulation resistance tests on all electrical equipment
- b) continuity and conductivity resistance tests
- c) test operation of alarm and tripping, devices to local and remote
- d) rotational tests on all motors
- e) polarity tests on CT's and VT's
- f) oil tests
- g) grounding systems and electrode tests
- h) ratio, vector grouping and magnetizing current tests on each transformer
- i) calibration of winding and oil temperature devices
- j) vector group and phasing tests on VT circuits
- k) magnetizing current/voltage tests and winding resistance on all current transformers
- l) primary and secondary injection tests on relays, protection devices and equipment

11.5.4 Control and Instrumentation Equipment

The following general tests shall be performed on control and instrumentation equipment at the site under commissioning scope and after Turn over from construction to commissioning:

- a) functional tests of all tripping, control, alarm, and interlocking circuits
- b) the testing of all equipment in accordance with the manufactures instructions or as advised by the Owner.

11.5.5 Civil

Civil works Quality Assurance (QA) shall include the tests listed below. Suitable QA forms shall be provided by the Contractor detailing each test. The QA forms shall be submitted for Owner comment prior to commencement of the civil works.

- Soil testing of material for the roads and trenching to confirm design values. This testing shall be conducted prior to commencement of road construction or trenching
- Compaction testing of layer works for all roads, hard stand platform to be carried out by plate bearing tests or cone penetration test. The frequency to shall be as per local code, BS code, EN code or as per standard practice whichever is stringent.
- Compaction testing of layer in trench
- Confirmation of design bearing pressure of each foundation formation level carried out by plate bearing tests (minimum one test per quadrant of each foundation base) and by Cone Penetration Test to at least 1m depth (The frequency to shall be as per local code, BS code, EN code or as per standard practice whichever is stringent)
- Formwork verification
- Steel reinforcement verification in accordance with the design
- Concrete placement verification in accordance with the design
- Concrete curing and compressive testing of the proposed concrete mix design prior to use on site
- Steelwork erection verification in accordance with the design
- Pile integrity testing on 100% of piles (in case of pile foundation)
- Dynamic pile testing on 2% of piles (sampled from across the site).
- Static pile testing 2-3% of piles (sampled from across the site).
- Foundation loading tests
- Proving of all ducts by mandrel and brush

11.5.5.1 Concrete testing

Concrete samples shall be taken by the Contractor from the point of discharge of the concrete into the works.

The rate of sampling for compressive strength testing strength shall be the stringent of:

- Three per 10m³ of concrete batched on site
- One per 10m³ for the first 40 deliveries of each grade of concrete
- One per 10m³ of concrete placed in columns or cantilevers
- One per 20m³ of concrete in beams and slabs, walls and footings
- One per 40m³ of concrete in solid raft foundation
- One on each day that concrete of that grade is used
- As per the relevant International standards applicable for the project

As per the relevant local standards applicable for the project

Concrete workability shall be assessed by slump tests on all concrete deliveries to site or on every 10 m³ of concrete, if batched on site.

11.5.5.2 Steelwork

All steelwork shall be subject to the following QA checks prior to use on site:

- Verification of correct dimensions and material grade
- Visual inspection for burs, scratches, etc.
- Verification of correct galvanising thickness including record of average and minimum galvanising thickness

11.6 Mechanical Completion

11.6.1 Test Certificates

Sets of all test records, test certificates and performance curves, whether or not they have been witnessed by the Owner, shall be supplied for all tests carried out in accordance with the provisions of this Inspection and Test Plan.

Sets of all test certificates shall be endorsed with sufficient information to identify the material or equipment to which the certificates refer, and shall carry in addition to other information, at the top right corner the following reference:

- a) Owners name
- b) Project title
- c) Projects reference number

All test documentation shall be in English language.

11.6.2 Material Tests

The Contractor shall provide test pieces for the Owner to enable him to confirm the quality of the material supplied under the contract. Such test pieces shall be prepared and supplied free of charge.

If any piece fails to comply with the requirements of the appropriate specifications for the material in question, the Owner may reject the whole of the material represented by that test piece; the contractors or sub-contractors designers and metallurgists will be consulted before any material is so rejected.

If the Owner is furnished with certified particulars of tests which have been carried out for the Contractor by the suppliers of material, he may, at his own discretion, dispense with the previously mentioned tests.

Material verification (PME) survey shall be arranged to validate the installed material (for structure, panel, railing, platforms, piping & valves etc.) as per design.

11.7 Turn-Over-Packages

The mechanical completion will be completed by the submission of the "Turn-over-Packages". They will consist of the following, but not limited to:

- a) All relevant manufactures works tests and certificates on all mechanical equipment
- b) All relevant manufactures works tests and certificates on electrical equipment
- c) All relevant manufactures works tests and certificates on pipework, structural steel, castings, forging, pressure vessels, boiler Plant, valves, air ejectors, motor driven auxiliaries, cranes and other mechanical equipment tested
- d) All field fabricating tests, including all welding x-rays, non-destructive and NDE tests
- e) All electrical equipment manufacture works tests
- f) All electrical field testing related to transformers, CT's, VT's, batteries, generators, and cabling etc. as described in the Technical Specification
- g) All instrumentation and control field testing data sheets

- h) All field testing relating to switchgear classified as HV, MV and LV data sheets
- i) All protective relay field testing data sheets
- j) In addition, consolidated "Punch Lists" as described in this OTS and approved by Owner shall be included in the turn-over-package before transferring to Start-Up/Commissioning team.
- k) Units shall be on barring gear and ready for start-up on the primary fuel source, i.e. coal or steam.
- l) No turn-over-package shall be deemed turned-over if any NCR's are left out standing or not "closed-out" with the approval of the Owner.
- m) Turnover package shall include all the software backups to enable the O&M team for independent rectification of any issues related to DCS or firefighting systems. Certification of the O&M team shall also cover by EPC

11.8 Commissioning Plan

11.8.1 Overview

EPC Consortium shall deliver to the Project Company an EPC Commissioning Plan and a Turn-Over Procedure, based on Owner Commissioning Plan and Turn Over Procedure included in schedule F and schedule H, which shall be reviewed and approved by the Owner before start of Turn-Over from construction to commissioning activities. These procedures shall clearly identify at least the following points:

- Organization chart, including specification of overall Commissioning leadership role.
- Division of Responsibilities inside EPC consortium, covering all systems in the plant.
- List of Commissioning Documents, including but not reduced to:
 - Pre-Commissioning protocols and procedures
 - Commissioning protocols and procedures.
 - Functional Tests procedures.
 - Reliability and Performance Tests procedures.
 - Equipment and instruments checklist.
 - HSE procedures.
 - Chemistry monitoring and control protocol to be implemented right from the beginning of the Project starting from design, pre commissioning, commissioning, O&M and preservation. Protocol shall include review of instruments, analysers and limits.
- List of TOPs as a live document (to be under permanent update).
- Detailed full Commissioning Schedule level 4 to be updated in a monthly basis
- Weekly or 3 weekly commissioning look ahead schedule updated in a weekly basis
- Daily commissioning schedule to be updated in a daily basis

Special focus on HSE documentation is mandatory: Permit-To-Work procedure and Lock-Out Tag-Out procedure shall be submitted by EPC Consortium and approved by the Owner before start of the first turn over activity. These documents must establish the way to proceed for Construction Team and any other party not within EPC Commissioning Team at those systems already turned over to Commissioning Team. Once reviewed and approved by the Owner, all working forces onsite including PC/Owner Commissioning Team will have the obligation of fulfil the requirements defined in these procedures.

Chemistry monitoring and control protocol to be implemented right from the beginning of the Project starting from design, pre commissioning, commissioning, O&M and preservation. Protocol shall include review of instruments, analysers and limits.

11.8.2 Commissioning Schedule

EPC Consortium shall deliver to the Project Company a Detailed Commissioning Schedule level 4 Schedule six (6) months prior to the start of the commissioning phase, to be updated on monthly basis as part of monthly reporting responsibilities. The program shall include weights at each activity to enable EPC Consortium to report general Commissioning progress. The Detailed Commissioning Schedule will be used by both the EPC and the Owner Commissioning Teams as the basis to analyse Commissioning progress, test sequences, links and potential delays on Key Commissioning Activities, official commissioning plan to inform stakeholders, resources planning and projection of the successful achievement of COD

Additionally, to the Detailed Commissioning Schedule level 4, EPC Consortium shall provide:

- Weekly or 3 week commissioning look ahead schedule: to inform the Project Company and its stakeholders every Friday in a weekly basis, designating the general planned activities to be performed during the following week or weeks, per systems and disciplines.
- Daily Commissioning Schedule: to inform the Project Company and its stakeholders every day, designating the specific planned activities to be performed during next day, per systems and disciplines. It should be updated in a daily basis.

The weekly and daily schedules will be used by PC/Owner Commissioning Team as a guideline for resources organization purposes, in order to guarantee a proper supervision and follow up of all Commissioning works, as well as for reporting reasons, when requested by Off-Taker, IE, Electrical grid Operator or any other external party.

11.8.3 TOPs Requirements

As part of EPC Consortium Turn-Over Procedure, a list of Turn-Over Packages will be provided to the owner before start of any Commissioning or Pre-Commissioning activity. Documentation has to be performed based on the Owner Turn Over Procedure and Turn Over package list included in appendix K.

As a general requirement, EPC Consortium shall assure that Mechanical Completion of a system has been achieved before turn over and tests start. *Figure 1* defines the generic minimum documentation contents of each TOP, and the minimum expected schedule of submission of each item. Final closure of TOP dossier will be achieved after successful completion and approval of the corresponding Functional Test.

Turnover package shall include all the software backups to enable the O&M team for independent rectification of any issues related to DCS or firefighting systems. Certification of the O&M team shall also cover by EPC.

Document Description	Before Walkdown ECC	After Walkdown ECC	Before FT CCC	After FT CCC
[1] Reference Documentation (latest documentation, including red-lines when modifications have been applied before commissioning)				
<i>PIDs, SLDs, Layout Drawings and Data Sheets</i>	X			
<i>Functional Descriptions</i>	X			
<i>O&M Manuals including preservation requirements</i>			X	
<i>Equipment/Lines/Valves/Cables Lists</i>	X			
<i>Codes and Standards</i>	X			
<i>TOP Battery Limits Definition</i>	X			
<i>Red Line / As-Built Drawings</i>	X		X	X
[2] Construction Completion Documents to submit ECC				

<i>P&ID with system identified</i>	X			
<i>Megger and continuity cable list (Elect. & I&C)</i>	X			
<i>Instruments list</i>	X			
<i>Instruments calibration data sheet</i>	X			
<i>Supports list</i>	X			
<i>Lay out identifying system area</i>	X			
<i>Single Line Diagram</i>	X			
<i>Line and Isometric list</i>	X			
<i>Equipment list</i>	X			
<i>Cables interconnection drawings</i>	X			
<i>Construction ITPs</i>	X			
<i>NCR's between EPC and Vendors</i>	X			
<i>ECC punch list</i>		X		
<i>ECC format signed</i>		X		
[3] Pre-Commissioning				
<i>Pre-Commissioning Procedures</i>	X			
<i>calibration certificates (note 1)</i>	X			
<i>Completed and Signed Protocols</i>			X	
[4] Commissioning and Functional Tests				
<i>Commissioning Tests Procedures</i>	X			
<i>Commissioning Tests Reports</i>				X
<i>Calibration Certificates (note 1)</i>			X	
<i>Test Calculation Sheets (can be included in Commissioning tests report)-if applicable</i>				X
<i>Commissioning Completion Certificates CCC signed</i>				X
[5] Turn-Over Punchlist				
<i>Punch list</i>		X		
[6] Performance & Reliability Tests (note2)				
<i>P&R Tests Procedures</i>		X		

<i>P&R Tests reports (Including: raw data tests logs; calculation sheets in native format) –original calculation sheet in PDF format and excel sheet shown in EPC computer to the Owner</i>				X
<i>Calibration Certificates (note 1)</i>				X
<i>P&R Completion Certificates PRCC</i>				X

Figure 1: Minimum generic contents for TOP Dossiers and schedule of submittal Team

The ECC Transfer Certificate is the document which formalizes the system or equipment turn over from Construction responsibility to Commissioning responsibility, as an operational part of the plant ready for start-up tests. It has to be signed by EPC Construction representative, EPC commissioning representative and EPC QA/QC representative.

Mechanical/Electrical/Instrumentation Documents shall include all certificates, tests and protocols generated during the Construction process, including but not reduced to:

- Mechanical: ITPs, pressure tests certificates, leak tests certificates, torque certificates and alignment and coupling certificates.
- Electrical: ITPs, insulation and continuity tests and terminal torque certificates.
- Instrumentation: ITPs and calibration certificates and site laboratory verification when applicable.

For a proper management and follow up of transferences by all parties, the Turn-Over process for each system shall follow the following rules and steps:

1. EPC Consortium will define and agree battery limits for each TOP, according to its previous experiences, good engineering practices and project planning necessities.
2. EPC Consortium will prepare the TOP dossier according to contents defined on *Figure 1*: EPC Commissioning Team will submit the TOP Dossier to PC Commissioning Team **at least three days** before communication of transference walk down.
3. EPC Commissioning Team will communicate the date and time of the TOP transference walk down at **least one week in advance**. PC will acknowledge a reasonable understanding and flexibility required in case of last minute re-schedules due to unexpected commissioning events. This communication must be extended to HSE Departments of all parties involved.
4. A turn over walk down will be performed by all parties, including at least attendance of the following positions:
 - System supervisors from EPC Construction Team and EPC Commissioning Team.
 - System supervisors from PC Construction Team and PC Commissioning Team.
 - TOP Coordinators from both parties.
 - QC supervisors from both parties, if required due to existence of specific concerns and critical outstanding NCRs.
 - HSE supervisors from both parties, if required due to existence of specific concerns and critical HSE observations.
5. During the walk down, the system will be crosschecked versus the design and construction reference documentation provided in the TOP dossier in advance, being the obligation of EPC company to designate and observe all deviations or incomplete and damaged items with minor or major impact in the commence of Commissioning activities.

6. From PC side, the commissioning manager will be the contact person leading the transference and coordinating the technical comments from Team Leaders and supervisory team, ensuring the proper tracking and description by EPC Consortium staff of all punch points detected.
7. Outcome expected from the transference walk down are:
 - An exhaustive list of punch items, with assigned priorities, to be signed by all parties (EPC Construction, EPC Commissioning and PC – either Construction or Commissioning).
 - The acceptance or rejection by EPC Commissioning Team, in agreement with PC Commissioning Team, of the transference of the system for a safe and successful start of Commissioning works.
8. In case of acceptance, EPC Consortium will notify to all companies, bodies and authorities involved in any work at the project site about the transference of the specific system to Commissioning phase, identifying battery limits and supervisor in charge, and reminding about the observation of Commissioning HSE procedures (Permit-To-Work and LOTO).
9. In case of rejection, EPC Consortium shall plan a second transference attempt, being the responsibility of EPC Construction Team to correct, repair, complete or repeat the works related to the reasons for rejection.
10. After transference notification, EPC Commissioning Team will be in charge of all activities in the system, being its responsibility to clearly identify all lines and equipment within transferred TOP limits with visual warnings and signs. Any activity at that system by departments, teams or subcontractors not belonging to EPC Commissioning Team will require application according to the Permit-To-Work procedure and execution of the LO-TO procedure, being the EPC Commissioning Team accountable for the correct implementation of these rules.

11.8.4 Progress Meetings

EPC consortium and Owner Commissioning Team will arrange, lead, track and report a Weekly Commissioning Meeting with the objective of:

- Following up the general Commissioning progress and the specific activities as defined in the weekly and daily programs delivered by EPC Company or in the weekly or three weeks look ahead schedule.
- Understanding in an open forum the status of ongoing works, documentation, sequences between systems and technical problems and events associated to the safe and successful achievement of plant hand-over and COD.
- Coordination in advance of upcoming activities and inspections involved in the preparation of Functional and Reliability and Performance Tests.
- Reviewing the status of NCRs related to Commissioning works.
- Reviewing the implementation of Permit-To-Work and LO-TO procedures, as well as following up the development of Commissioning works from HSE standpoint, defining safety priorities and the spaces for improvements.

The Weekly Commissioning Meeting will be properly minuted and distributed to Owner Team and to EPC Consortium Team. Depending on the volume of points to be discussed in the meeting and on the specific weekly work load.

11.8.5 Commissioning Documents

The EPC Consortium shall deliver a document list for all Commissioning activities, including procedures and checklist such as (but not reduced to):

- Organizational procedures.
- Pre-Commissioning and Commissioning Tests procedures. All tests and start up to perform in all equipment and systems must to have a commissioning procedure. It is not allowed to perform any commissioning test or start up without out an approved procedure and relevant checklist.
- Reliability and Performance Tests procedures.
- HSE Management procedures.

- Vendor checklist for Pre-Commissioning and Commissioning field checks. All vendor tests and start up to perform in all equipment and systems must to have a commissioning procedure. It is not allowed to perform any commissioning test or start up without an approved procedure and relevant checklist.

The final content of the Commissioning Manual and the delivery schedule will meet contractual requirements as defined in the applicable contract and agreed in any case between EPC Consortium and PC based on project progress, reasonable priorities and standard commissioning practices. Specifically:

- For Commissioning Tests procedures, at least **four months before planned start** of the corresponding test.
- For Performance and Reliability Tests detailed procedures, at least **eight months before planned start** of Performance and Reliability Tests. The procedure shall include the list of instruments used for the performance test, clearly indicating if the instruments are temporary only or permanent ones.

Test procedures will be reviewed by OE with the support of the PC, sharing the information with the OE if applicable.

- Document structure of pre- commissioning procedures, commissioning procedures, Functional Test Procedures and Performance as Reliability tests Procedures as below:

No test is allowed without Commissioning procedure approved.

EPC Contractor will provide to the Owner, since the beginning of the commissioning, a monitoring station located in the control room with full access to monitor all screens and operation, historic of signals and full access to monitor the logic programmed in the power plant. The purpose is to allow the owner to monitor logic, tests and operation from his own monitoring station without problems of interference with the EPC Contractor.

The Owner will have the right to ask for Root Cause Analysis reports to the EPC Contractor in case of any power plant trip, important event, load reduction, accidents or any undesirable situation during commissioning and the EPC Contractor will have the obligation to deliver a proper RCA before restarting with the commissioning activities. RCA must include the cause of the incident, temporal and/or permanent solution to go ahead and technical reasons that it will not happen again.

EPC Contractor shall provide to the Owner when it is required, during the commissioning phase:

- Full signal list updated in excel format
- Forced signal list updated
- Any red line performed by EPC

11.8.6 Management of Punch List Items

Punch List Items generated during each transference walk down will be qualified according to the following definition of priorities:

- Priority 1: the non-completion of the point associated represents an evident clear risk and danger for the safety of persons and equipment. It means immediate rejection of the transference by the EPC Commissioning Team and immediate action by EPC Construction Team.
- Priority 2: the non-completion of the point associated represents an evident impact in normal operation of the system but it does not concern the safety of persons and equipment. System transference is allowed and actions can be implemented during Pre-Commissioning, being necessary the complete closure of the point before Commissioning of the system.
- Priority 3: the non-completion of the point associated could impact some functionality features and conditions of the system but it does not concern normal operation of the system nor safety of persons and equipment. System transference is allowed and actions can be implemented during commissioning of the system, being necessary the complete closure of the point before Initial Acceptance of the plant.
- Priority 4: Minor pending points

The outcome Punch List from each transference will be signed by all parties (EPC Construction Team, EPC Commissioning Team and PC/Owner – either Construction or Commissioning) and integrated by EPC Consortium staff into the Master Punch List, which will be delivered at least on weekly basis to the Owner;

TOPs Coordinator shall be in charge of the document from PC/Owner side, being responsible of distribution to the relevant persons in PC/Owner and to feedback the comments to EPC Consortium.

11.9 Pre-Commissioning Tests

11.9.1 Overview

Pre-Commissioning (also known as Cold Commissioning) is defined as all of the non-operating tests and adjustments performed on the systems, with the target of preparing the plant for proper Commissioning Process. It may involve individual unit operations or a process area containing several elements of the plant.

EPC Consortium Commissioning Team will perform the pre-Commissioning activities, starting from the Mechanical Completion of all systems: construction verification (crosscheck of Construction completion versus the design specifications) and Pre-Commissioning schedule will be based on the agreed TOPs schedule, in order to meet successfully project milestones. During this phase, the EPC Consortium Commissioning Team will ensure that all Pre-Commissioning works are completed, that all related paperwork is completed in the relevant TOP dossier and that appropriate signoff and completion certificates in the tracking database are completed.

Interfaces between Pre-Commissioning and construction completion checks will exist, being the duty of the EPC Consortium to organize and optimize these interfaces in order to push forward Commissioning progress.

11.9.2 Electrical Pre-Commissioning

General guideline for electrical inspections before pre-commissioning tests, and pre-commissioning works include (but is not reduced to) the testing, check of completion and proper installation of the following points. Specific guidelines for each system will be included as individual sheets in the relevant system TOP dossier.

Electrical previous inspection before pre-commissioning, it should be verified at least:

- Grounding verification for all equipment and cabinets.
- Insulation resistance tests, Voltage Hi-Pot tests (MV cables), torque tightening and sealing for cables.
- Installation and tagging of conduits and trays.
- Installation of conduits seals.
- Installation, oil filling and testing of the Main and Auxiliary Unit Transformers, as per vendor's recommendations.
- Installation and testing of surge arresters.
- Installation of indication lamps.
- Installation of equipment heaters.
- Installation of electrical heat tracing.

Electrical pre-commissioning works:

- Continuity tests, insulation tests and cold loop checks for wires.
- Insulation tests, ohm resistance tests (DC and MV motors), and hi-pot test (MV motors).
- Phase rotation of motors
- Motor run tests.
- Rotation check of motors.
- Primary tests of Turbine Generator.
- Functionality of indication lamps.
- Calibration of protective solid-state relays.
- Functionality of equipment heaters.

11.9.3 Mechanical Pre-Commissioning

General guideline for mechanical inspections before pre-commissioning tests, and pre-commissioning works include (but is not reduced to) the testing, check of completion and proper installation of the following points. Specific guidelines for each system will be included as individual sheets in the relevant system TOP dossier.

Mechanical previous inspection before pre-commissioning, it should be verified at least:

- Anchoring and grouting of equipment.
- Installation of piping supports and hangers (all set in cold position), and vents and drains.
- Pressure tests of pipelines done.
- Flushing and/or cleaning of pipelines.
- Proper installation of valves and orifice plates.
- Installation of temporary screens and strainers.
- Corresponding welding QA/QC records completed and available.
- Installation of coupling guards.
- Final alignment of motors.
- Fire Protection and Fire Detection systems installed.
- Installation of thermal insulation.
- Torque tightening of equipment and structures.
- Equipment and system areas are clean, rolled back of construction equipment.

Mechanical pre-commissioning works:

- Motors vibration tests.
- Fire Protection and Fire Detection systems functional.
- Lubrication and general maintenance of mechanical equipment up-to-date.
- Filling with process fluid.

11.9.4 I&C and DCS Pre-Commissioning

General guideline for I&C and DCS inspections before pre-commissioning tests, and pre-commissioning works include (but is not reduced to) the testing, check of completion and proper installation of the following points. Specific guidelines for each system will be included as individual sheets in the relevant system TOP dossier.

I&C and DCS previous inspection before pre-commissioning, it should be verified at least:

- Installation of device according to hook-up drawings and other design specifications.
- Installation of primary sensors is consistent with transmitters range.
- Physical separated trays for signal wiring and power cables.
- Calibration of instruments, including calibration sheet available and sticker in place.
- Verification of instruments on site laboratory.
- General visual check of each instrument.
- DCS earthed and I&C cables connected.

It is assumed that EPC Consortium will provide, maintain and calibrate all test equipment required to conduct the instrumentation checks, including a traceability system in place.

I&C and DCS pre-commissioning works:

- Programming of SMART transmitters.
- Setting of switches as per instruments specifications.

- Electrical continuity check and visual verification of instruments tags and wire labelling as per specifications, including connections between field instrument and terminal strip in I/O panel, and between terminal strip and I/O card.
- SAT tests on DCS (redundancy in servers, communications, power up of DCS, upload of programs and screens operatives, output and input cards operative)

11.10 COMMISSIONING TESTS

11.10.1 Overview

Start of Commissioning Tests is allowed only after Mechanical Completion of systems and after issuance of Erection Completion Certificate by the EPC Consortium.

The objective of Commissioning Tests is to check the correct, appropriate, efficient and safe performance of individual equipment and systems during start-up, shut-down and normal operation modes, according to:

- The corresponding Commissioning Tests Procedures.
- The instructions and recommendations from equipment vendors.

During Commissioning of systems, the EPC Commissioning Team will perform the tests specified in the corresponding test procedures to check and demonstrate that all equipment and systems, including safety systems, protection systems and redundancies are set up, operational, properly calibrated and effective. Such functionality of equipment and systems shall be proved to Owner Commissioning Team, either by means of direct operation or by means of signal simulation, during the supervision of all Commissioning works and according to the agreed tests procedures, to the schedule of works and to the technical progress meetings.

11.10.2 Power Block and BOP Commissioning Tests

The test procedures for the Power Block and Balance of Plant Systems delivered by the EPC Consortium shall provide full coverage to start-up, shutdown and functionality (including full performance of the control loops and protections and trips settings) for the equipment and systems related to, at least, the following:

- **Pumps.**
 - Verification of rotation sense
 - Verification of vibrations, temperature and speed according to vendor requirements and standards with motor and pump decoupled and coupled
 - Verification of rate current, peak start current and voltage according to vendor requirements and standards with motor and pump decoupled and coupled
- **Compressed Air System.**
 - Verification of instruments calibration
 - Upload programs and configuration in PLC
 - Loop checks
 - Setting of alarms and trips
 - Control logic test
 - Interlock tests
 - Communications with DCS
 - Dryers tests
 - Functionality tests
- **HRS**
 - Verification of instruments calibration
 - Leakage tests

- Upload programs and configuration in PLC
- Loop checks
- Setting of alarms and trips
- Dome control level tests (high, medium and low pressure)
- Control logic test
- Interlock tests (specially safety interlocks)
- Communications with DCS
- Verification of thermal expansion
- Tests on motorized drainage valves and vents
- Calibration of safety valves in real with steam
- Functionality tests on subsystems as feeding water system, condensate water system, air gas control system, water/steam control system, motorized valves control system
- Functionality, capacity and Performance HRSG test
- Verification test Combustion conditions are within design limits, Combustion tuning to manage heat flux and achieve better distribution of fuel and flow. Heat flux shall be controlled and recorded
- Verify if it is possible to keep normally and safe the steam pressure and temperature and dome levels between rated limits and under load demand changes and disturbances like as following:
 - 1- Ramp of 1% /minute (from 25%to 50%, from 50% to 75% and from 75% to 100%)
 - 2- Ramp of 3% /minute (from 25%to 50%, from 50% to 75% and from 75% to 100%)
 - 3- Ramp of 5% /minute (from 25%to 50%, from 50% to 75% and from 75% to 100%)
 - 4- Run back due of high temperature of main steam (high pressure) from 100% to 50%
 - 5- Run back due of high temperature of main steam (intermediate pressure) from 100% to 50%
 - 6- Validation of parameters of gas turbine according to ISO2314-1989
- **Safety valves calibration**
 - Verification of vendor calibration certificates
 - Verification on site with real conditions and steam and adjustments and recalibration if necessary
- **HRSG Diverter**
 - Position switches calibration
 - Local and remote functionality verification open and close
 - Local and remote signals to DCS
 - Control logic test
 - Interlock tests
- **Blowing of main steam pipes**
 - After cleaning of HRSG which can be alkaline boiling or complete acid cleaning, the blowing out of superheaters, HP, IP, reheat and LP steam lines, HP, IP and LP by-pass lines shall be performed by the Contractor in order to remove any foreign material remaining in subject parts of the circuit. Steam blow out procedures shall also cover the auxiliary steam and gland steam lines.
 - During this process, the unit shall be operated in accordance with the recommended procedure, which is to be also accepted by the Owner, and with control systems and protective interlocks functioning as required.

- EPC blowing steam procedure shall have clear cleaning acceptance criteria. The procedure should include the methodology applicable, and the cleaning factor calculations for application during the blowing works. Air blowing for steam pipes will not be accepted. For the blowing process, it must be considered a silencer in order to mitigate the noise contamination.
- During the blowing, it should be taken records and analysis of dome levels, feeding water flow, steam temperature and pressure, etc... for calculation and verification of cleaning factor.
- **Blowing of main gas pipes to gas turbine**
 - Require to EPC gas pipes cleaning steam procedure with clear cleaning acceptance criteria of cleaning degree in gas pipes in order to be agreed with the Owner.
- **Raw Water Service System**
 - Verification of instruments calibration
 - Loop checks with DCS
 - Pumps tests
 - Setting of alarms and trips
 - Control logic test
 - Interlock tests
 - Functionality tests
- **Auxiliary cooling system**
 - Verification of instruments calibration
 - Loop checks with DCS
 - Pumps tests
 - Setting of alarms and trips
 - Control logic test
 - Interlock tests
 - Functionality tests
- **Sampling system**
 - Setting and calibration of pH, conductivity, silica, sodium, phosphates, hydrazine, oxygen, steam purity analyzers
 - Alarm verification
 - Interlocks of sampling tank
 - Interlocks of isothermal tank
 - Calibration of sampling panel
 - Communication tests to DCS and chemical laboratory if applicable
 - Functionality test
- **Gas station**
 - Verification of instruments calibration
 - Test on gas chromatograph and moisture and Sulfur analyser
 - Tests on gas measurement circuit
 - Tests on gas heater
 - Tests on control valves, solenoid valves

- Loop checks
- Setting of alarms and trips
- Control logic test
- Interlock tests
- Moisture separator tests
- Coalescent filter tests
- Duplex filter tests
- Functionality tests
- **Fire Fighting System**
 - Verification of instruments calibration
 - Loop checks with control System
 - Setting of alarms and trips
 - Control logic test
 - Interlock tests
 - Pumps tests
 - Flow measure tests
 - CO2 or FM200 functionality test (test shall be conducted specific to actual installation, piping arrangements, circuit components, features and control, etc. rather than a simple general test)
 - The air release valves in the standpipe system header shall be installed at the highest point of the piping
 - Alarms and detectors tests
 - Functionality tests (deluge system, foam system and hydrants tests)
- **Gas and Steam turbine**
 - Verification of instruments
 - Loop check
 - Interlock tests
 - Logic tests
 - Final shaft alignment and record of measurements
 - Record of Rotor lift and Lift oil pressure during commissioning.
 - Generator leak tests with air of hydrogen (what is applicable)
 - Record of performance at no load, base load, 25%, 50%, 75% and 100% load
 - Tests of oil sealing system control
- **Gas turbine air intake**
 - Verification of instruments
 - Interlock tests
 - Verify filter differential pressure
 - Adjustment of air intake heating gates
- **Gas turbine washing compressor system**
 - Verification of instruments

- Loop check
- Interlock tests
- Logic tests
- Off line and on-line washing compressor according to the vendor procedure
- Control of washing time
- Control of water quality used for washing
- Tests of water washing pressure and temperature
- **Gas turbine air compressor**
 - Verification of instruments
 - Loop check
 - Interlock tests
 - Logic tests
 - Cooling system tests
 - Air dryers tests
 - Functionality tests
- **Gas turbine gas heater**
 - Verification of instruments
 - Loop check
 - Interlock tests
 - Logic tests
 - Functionality tests
 - Combustion adjustments and emissions measurement
- **Lubrication oil systems (Gas and Steam turbine)**
 - Verification of instruments
 - Loop check
 - Interlock tests
 - Logic tests
 - Main pump, auxiliary pump and emergency pump tests, in lubrication oil system, lift oil system and emergency lift oil system
 - Oil cooler tests
 - Oil gas extraction system test in oil tank
 - Centrifuge oil cleaning system
 - Electro-hydraulic control
 - Turning gear
- **Gas and Steam Turbine protection and monitoring system**
 - Primary elements verification of calibration
 - Loop tests
 - Logic and interlock tests

- Turbine trips tests (manual trip from local and remote, over speed, both Electrical & Mechanical Trip shall be at FSNL, bearing high temperature, bearings high vibrations, bearings oil lubrication low pressure, control fluid low pressure, failure of 2 speed detectors, failure of electrical feeding to trip circuit.

Gas turbine trip functions

GT OEM will provide dual redundant Electrical over speed trip devices as per OEM standard design, and Mechanical over speed trip device cannot be provided.

The high bearing metal temperature will not be applied to GT trip function as per OEM standard design. The high temperature alarm will be provided. The bearing metal temperature is expected to increase gradually, and there is enough time for an operator to take proper action such as the unit shutdown after alarm acknowledgement.

The shaft vibration (not bearing vibration) high alarm and trip functions will be provided.

Over speed Trip interlock test will be carried out after Gas turbine reaches full speed no load.

Regarding other trip tests, simulation interlock test will be carried out at Gas Turbine shut down condition.

Electric over speed tip devices shall be approved by SIL study during the execution stage.

- **Gas and steam Turbine Cooling generator system (hydrogen, water or air) and oil seal generator system**
 - Primary elements verification of calibration
 - Loop tests
 - Functionality test
- **Gas and Steam Turbine Speed governor**
 - No load tests (Start sequence, Response to speed steps, Natural oscillations, trip for over speed, synchronization range)
 - On load tests (Power statism, load limiter and load variator, Load rejection at 100% load in regulation mode, Island operation, tests limiter for high temperature in exhaust gases, load variation ramp)
- **Steam Turbine Electro-Hydraulic Control Turbine System (EHC). If applicable.**
 - Speed Control, acceleration control, speed regulation, speed equalizer
 - Load control, Load-frequency control, load governor change and load limit, pressure regulator, Load redactor for vacuum lose and/or turbine auxiliaries, power/load unbalance, valves flow pressure limiter, on off valves control, steam valves control, etc.
- **Air Cooled Condenser**
 - Primary elements verification of calibration
 - Loop tests
 - Logic and interlock tests
 - Functionality tests
 - Vacuum leak test
- **HVAC system**
 - Relative moisture test
 - Temperature control cold/hot
 - Automatism control
 - Air balance

- **Hoist**
 - Movement or translation tests
 - Length and extent cable test
 - Rotation test if applicable
 - Weight capacity test
 - Overhead crane static load tests
 - Movement Load Tests
- **Demineralization plant**
 - Primary elements verification of calibration
 - Filters functionality verification
 - Loop tests
 - Logic and interlock tests
 - Pumps running, tests
 - Functionality tests

11.10.3 Electrical System Commissioning Tests

The test procedures for the Electrical System delivered by the EPC Consortium shall provide full coverage to start-up, shutdown and functionality (including full performance of the control loops and protections and trips settings) for the equipment and systems related to, at least, the following:

- **HV substation.**
 - Insulation resistance
 - Power factor
 - Contact resistance or conductivity test
 - Synchronization and open and close time measurement
 - SF6 humid residual test
 - Functionality test in remote and local
 - Interlocks and alarms tests
 - Anti-pumping protection test
 - Insulation resistance test before and after withstand test
 - Voltage withstand test.
 - Leakage test
 - Conductivity test
 - Protections tests
 - Current Transformers
 - Voltage Transformers
 - DC system and batteries
 - UPS
 - Grounding and Lightning Protection Systems.
 - Tests according to the vendor requirements

- **Generator**
 - Insulation resistance in stator before and after Hi-Pot test
 - Insulation resistance in rotor
 - Phase sequence verification
 - Hi-Pot test to stator with AC or DC
 - Hi-Pot test to rotor with AC
 - Power factor test to stator
 - Conductivity test to stator and rotor
 - Insulation resistance to screws who support the stator (if applicable)
 - Saturation test with 3-phase short circuit
 - No load Saturation test
 - Measurement of shaft voltage
 - Neutral generator transformer
 - Voltage transformation ratio
 - Insulation resistance
 - Windings Ohmical resistance measurement
- **Generator Circuit Breaker**
 - Insulation resistance
 - Power factor
 - Contact resistance or conductivity test
 - Synchronization and open and close time measurement
 - SF6 humid residual test
 - Functionality test in remote and local
 - Interlocks and alarms tests (with HVCB, Earth and all interlocks)
 - Anti-pumping protection test
- **Emergency Diesel Generator**
 - Windings stator and rotor insulation resistance.
 - Windings conductivity test in stator and rotor
 - Control circuit tests and functionality tests (operational features test for Local, Remote and auto start on under voltage)
 - Power capacity of the EDG
 - Transference test from essential services in automatic and manual
- **Power Transformers (10MW or more than 10MW):**
 - Winding insulation resistance.
 - Voltage Transformation Ratio and polarity
 - Windings conductivity test
 - Excitation current measurement
 - Measurement of dielectric loses and capacitance of Insulation

- Bushings dielectrical tests (dielectrical loses, capacitance measurement, hot collar) Tan delta
- Current transformers tests (Transformation ratio, polarity, saturation, insulation resistance)
- Oil dielectric strength
- Oil dielectric loses
- Dew point
- Leakage test
- Gas chromatography
- Oil physic chemical tests
- Functionality test to control systems like refrigeration fans, oil pumps and tap changer systems
- Electrical protection and measurement tests.
- On line oil gas monitoring system tests
- Oil DGA, Corrosive Sulphur
- Oil -PCBs
- Oil- Inhibitor content /oxidation stability – If Oil is inhibited
- Oil Furanic test
- Oil- Particle Count
- Oil – Degree of Polymerization (at time of dispatch)
- **Power Transformers (less than 10MW)**
 - Winding insulation resistance.
 - Voltage Transformation Ratio and polarity
 - Windings conductivity test
 - Excitation current measurement
 - Measurement of dielectric loses and capacitance of Insulation
 - Current transformers tests (Transformation ratio, polarity, saturation, insulation resistance)
 - Oil dielectric strength
 - Oil dielectric loses
 - Oil physic chemical tests
 - Functionality test to control systems like refrigeration fans, oil pumps and tap changer systems
 - Electrical protection and measurement tests
 - Oil DGA, Corrosive Sulphur
 - Oil -PCBs
 - Oil- Inhibitor content /oxidation stability – If Oil is inhibited
 - Oil Furanic test
 - Oil- Particle Count
 - Oil – Degree of Polymerization (at time of dispatch).
- **Distribution Transformers (dry type)**
 - Winding insulation resistance.
 - Voltage Transformation Ratio and polarity

- Windings conductivity test
- Excitation current measurement
- Measurement of dielectric losses and capacitance of Insulation
- Current transformers tests (Transformation ratio, polarity, saturation, insulation resistance)
- Electrical protection and measurement tests.
- **Measurement and protection cabinet for generator transformer and HV bay**
 - Relay Calibration tests (HV protections, transformers protection, generator protection)
 - Interlock protections tests
 - Short circuit test (phase test for measurement and protections)
 - Short circuit test (tests of protection operation with short circuit values)
 - Setting verification according to the approved coordination study
 - Protection operation test with current and voltage secondary injection
 - Alarms verification in local and remote
- **MV Cabinets:**
 - Bus bar Insulation resistance test before and after withstand test
 - Bus bar voltage withstand test.
 - Trip/close test and remote operation.
 - Conductivity test across switch contacts and bus bar joints
 - Verification of relay settings as per final approved settings aligned with project protection coordination study
 - Relay tests with secondary injection with final settings implemented
 - Voltage, sequence and frequency measure
 - Open and close breaker time measurement
 - Time – current graphic of MV motors coupled and uncoupled
 - Transference tests between bus bars in manual and automatic.
 - High Speed BUS Transfer Protection Test – Secondary injection and On Load test for relay features
- **LV cabinets**
 - Bus bar Insulation resistance test
 - Conductivity test across switch contacts and bus bar joints
 - Voltage, sequence and frequency measure
 - Verification of relay settings as per final approved settings aligned with project protection coordination study
 - Relay tests with secondary injection with final settings implemented
- **MCC**
 - Bus bar Insulation resistance test
 - Low voltage relay test
 - Thermal protection tests in starter circuits
 - Control circuit functionality test
- **IPB (Isolated Phase Bus bar)**

- Bus bar Insulation resistance test before and after withstand test
- Bus bar voltage withstand test.
- Leakage test
- Conductivity test
- **Segregated and not segregated phase bus bar**
 - Bus bar Insulation resistance test
- **Current Transformers**
 - Polarity
 - Transformation ratio
 - Insulation resistance
 - Continuity test
 - Saturation
- **Voltage Transformers**
 - Polarity
 - Transformation ratio
 - Insulation resistance
 - Continuity test
- **DC system and batteries**
 - Capacity test to the charger
 - Capacity test to the batteries, discharge test
 - Insulation resistance to the DC distribution cabinet bus bar
 - Test to measure instrument
 - Low voltage relay test
- **UPS**
 - Insulation resistance test
 - Inverter and rectifier functionality test
 - Signals to UPS supervision
 - Bypass transfer modes functionality tests
 - Voltage and frequency measurement
- **Grounding and Lightning Protection Systems.**
 - Soil resistivity
 - Continuity grid test
 - Resistance measurement of ground grid
 - A Complete grounding and earthing study report and testing survey report to set base line for project
- **Heat Tracing**
 - Functional test
 - Temperature set point test
 - Signals supervision

- **MV cables**
 - Hi-pot test
 - Insulation resistance before and after the hi-pot test
- **LV cables**
 - Insulation resistance
- **MV motors**
 - Hi-pot test
 - Insulation resistance before and after the hi-pot test
 - Windings continuity test
- **LV motors**
 - Windings Insulation resistance
- **DC motors**
 - Windings Insulation resistance
 - Field winding and stator Continuity test
- **Cathodic protection**
 - Insulation resistance
 - Natural voltage measurement
 - Functionality test
- **Synchronization**
 - Tests of manual and automatic synchronization relays with simulation software
 - Tests to the synchronization scheme in manual (increase and decrease voltage and frequency) with cables disconnected to the breaker and monitoring on clamps with multimeter
 - Tests to the synchronization scheme in automatic (increase and decrease voltage and frequency) with cables disconnected to the breaker and monitoring on clamps with multimeter
 - Real synchronization in manual without any trip and keeping the generator synchronized during 30 minutes, after finishing open the GCB and finally synchronize in automatic.
- **Generator AVR test**
 - No load tests (Determined the excitation and excitation take off times in manual and automatic, Automatic control range, Manual control range, Automatic follower, Volts/Hertz limits, Voltage steps response, Voltage disturbance response, Maximum excitation value)
 - On load tests (Minimum Excitation Limit MEL, Over Excitation Limit OEL, Reactive Power compensation, Reactive steps without PSS, Rejection of reactive positive and negatives, PSS (Power System Stabilizer) tests)

11.10.4 Control Commissioning Tests

- **PLC on package system (all PLC over the plant that is not DCS)**
 - Upload programs according to approved engineering
 - Verify redundancy
 - Communication tests with DCS
 - Loop check with instruments
 - Logic tests

- Protections, trips, signals, Monitoring tests

- **Distributed Control System Commissioning Tests**

The test procedures for the DCS delivered by the EPC Consortium shall provide full coverage to the functionality of the control system, particularly the ability to record the data needed for the Performance and Reliability Tests.

It will be generated by the DCS engineer a loop check list and a logic list in order to check all the tests from the control room, in a later stage he would check as well the functionality tests.

- Verification of internal cables, cards, feeding cables, communication and earthing
- Energization test in cabinets
- Redundancy test in income feeders and failure of energy test
- Resistance tests in electronic earthing grid and regular earthing grid
- Fan functionality tests in cabinets
- Upload software and configure
- Redundancy test in servers and availability test
- Control system Alarm tests verification
- Communication tests with local bus
- Tests with printers, operation stations, engineering station and monitoring stations
- General functionality tests (Screens, Historic, trends, alarms, reports, functions,...)
- Communication loop tests with main and redundant bus
- Communication tests between all control systems and PLC around the plant WTP, electrical substation, electrical grid operator system, Steam turbine, Gas turbine, HRSG, Gas station, weather station, chromatograph, CEMS, SCS, etc.
- Loop check with all the instruments (for that test it must be develop a loop tests list in order to control the progress during the tests)
- Logic tests (for that test it must be develop a logic tests list in order to control the progress during the tests)
- Interlock tests
- System functionality test
- **Power Plant Master control commissioning test**
 - Interlock tests (safety devices) Gas Turbine – HRSG for instance
 - Verification of protections (trips) and reports emission from DCS after trip ready
 - Tests of Automatic Generation Control (AGC)
 - Tests of Automatic Generation Trip (AGT)
 - Tests and configuration of disturbance recorder and generation of historic reports from DCS and DR

11.10.5 Instruments and Valves

Tests to be done for the I&C equipment will be as below:

- **Pressure instruments (indicators, transmitters, switches)**
 - FAT calibration certificate and/or declaration of conformity from the vendor
 - Verification of calibration in a certified laboratory on site
 - Loop check with DCS

- **Level instruments (indicators, transmitters, switches)**
 - FAT calibration certificate and/or declaration of conformity from the vendor
 - Verification of calibration in a certified laboratory on site
 - Loop check with DCS
- **Temperature instruments (indicators, transmitters, switches)**
 - FAT calibration certificate and/or declaration of conformity from the vendor
 - Verification of calibration in a certified laboratory on site
 - Loop check with DCS
- **Flow measure instruments**
 - FAT calibration certificate and/or declaration of conformity from the vendor
 - Loop check with DCS
- **Vibration instruments**
 - FAT calibration certificate and/or declaration of conformity from the vendor
 - Loop check with Bently Nevada
- **Gas Chromatograph**
 - FAT calibration certificate
 - Program upload and characterization according to applicable standards
 - Sampling circuit test
 - Regulation circuit test
 - Scavenging circuit test
 - Calibration circuit test
 - Loop check with PLC
 - Perform calibration on site with standard gas
 - Verify PCI and flow calculation
 - Communication tests
- **Gas Analysers (moisture, sulfuric acid and total sulfuric acid)**
 - FAT calibration certificate
 - Program upload and characterization according to applicable standards
 - Sampling circuit test
 - Regulation circuit test
 - Scavenging circuit test
 - Calibration circuit test
 - Loop check with PLC
 - Perform calibration on site with standard gas
 - Verify PCI and flow calculation
 - Communication tests
- **fibre Optic cables**
 - Attenuation test

- **Solenoid valves**
 - Functionality tests from local and remote
 - Valves characterization
- **Control valves**
 - Verification and calibration of valve and actuator
 - Calibration of position transmitters
 - Calibration of position switches
 - Functionality tests in local and remote
 - Check signal to DCS in local and remote
 - Valves characterization and parametrization with control system
 - Simulations of control automatization
- **Bypass valves**
 - Verification and calibration of valve and actuator
 - Calibration of position transmitters
 - Calibration of position switches
 - Functionality tests in local and remote
 - Check signal to DCS in local and remote
 - Valves characterization and parametrization with control system
 - Simulations of control automatization
- **On-off valves**
 - Verification and calibration of valve and actuator
 - Calibration of position transmitters
 - Calibration of position switches
 - Calibration of actuators
 - Open and close in local and remote
 - Signals local and remote to DCS
 - Simulations of control automatization.
 - Pressure Safety Valves
 - Trevi testing
- **Weather Stations Commissioning Tests**

The test procedures for the Weather Stations delivered by the EPC Consortium shall provide full coverage to start-up, shutdown and functionality (including full performance of the control loops) for the instruments and systems related to, at least, the following:

- End-to-end calibrations.
- Real-time consistency checks on all measurements.
- Communication with SCS and DCS, including time synchronization with DCS and meters.
- Functionality of local data logger.

11.10.6 Chemical equipment and systems

The test to be done for the Chemical equipment and systems are the one below:

- **Water Flushing:**
 - feedwater lines, spray water lines, condensate lines, shall be cleaned by water flushing and degreased with once through system only. Flushing water shall be also stored and disposed as hazardous waste.
- **Chemical cleaning of HRSG and steam pipes**
 - Alkaline cleaning. A procedure must be delivered from EPC and be approved by the Owner before starting the chemical cleaning.
 - Acid cleaning. A procedure must be delivered from EPC and be approved by the Owner before starting the chemical cleaning.
- **HRSG packing (humid or dry)**
 - EPC must send a procedure for owner approval taking in consideration that the humid packing must be done with demineralized water dosed with an oxygen redactor and pH adjustment higher than 10 and that in case of dry packing with N2 at positive pressure no less than 0.3bar. The procedure should include clearly the sequence of activities to implement during the packing, valves closed and open during the packing.
- **Chemical dosing**
 - Verification of rated current, starting current, voltage, vibration and temperature of pumps
 - Interlock tests
 - Functionality tests from local and remote
- **Water treatment plant**
 - Primary elements verification of calibration
 - Loop tests
 - Logic and interlock tests
 - Pumps running, tests
 - Conductivity measurement
 - Functionality tests
- **Wastewater plant**
 - Primary elements verification of calibration
 - Loop tests
 - Logic and interlock tests
 - Pumps running, tests
 - Functionality tests

11.10.7 System Functional Tests

System functional tests are tests done on a system or equipment when it is fully operational in order to test all functionality, automatism, protections and all functions that an operational system should have according to the logic and functional description approved by the Owner and the Owner Engineer, according to international standards, HAZOP study and good practices.

Functional test of a System or equipment has to be performed just after all commissioning tests of the system or equipment is done and approved by the Owner.

The Owner and EPC should agree a list of systems and equipment to be performed functional tests and it should include in section 8 of annexure 10 where it is written "Functionality test" as part of tests to perform, in addition an electrical functional test and DCS functional tests should be included in the Functional test list to be proposed by EPC.

All Functional tests must be witnessed by the Owner and approved. Approval of the Owner will come with a CCC (Commissioning Completion Certificate) to be delivered by the Owner. The Owner will have 7 days to deliver the CCC after the tests was witnessed.

11.10.8 CCC Certificate

EPC should write a CCC (commissioning Completion Certificate) procedure to be delivered to the Owner for approval including list of functionality tests proposed, process to follow, protocols and format of certificates to be filled 60 days before starting functional tests. It will be review by the Owner and approved, reject or commented.

11.11 Performance and Reliability Tests

Performance and Reliability tests are according to PPA Appendix 8 (Inspection Start-Up Testing Commissioning and Net Dependable Capacity Testing). This can start when all Functional tests are finished and all CCC certificates has been approved by the Owner. In addition, Performance and Reliability tests procedures has to be approved by the Owner before start P&R tests.